REV	ECN	DESCRIPTION OF REVISION	CK APPD	DATE
6	0002265654	ENGINEERING RELEASED		2013-08-22

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

SCHEM,MLB_KEPLER,J45G

8/22/2013 DVT

Schematic / PCB #’s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-0675	1	SCHEM,MLB_KEPLER,J45G	SCH	CRITICAL	
820-3787	1	PCBF,MLB_KEPLER,J45G	PCB	CRITICAL	

DRAWING
TITLE=MLB
ABBREV=ABBREV
LAST_MODIFIED=Thu Aug 22 12:19:14 2013

DRAWING TITLE: SCHEM,MLB,KEPLER,J45G

Apple Inc.

DRAWING NUMBER: 051-0675 SIZE: D

REVISION: 6.0.0

BRANCH: dvt

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
685-0177	COMMON PARTS,MLB,KEPLER,J45	J45G_COMMON
985-0181	DEV,MLB,KEPLER,J45	J45G_DEVEL:DVT
639-5245	PCBA,MLB,KEPLER,CRW_BEST,8G-HYN,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_HYNIX_1600_S,FB_2G_HYNIX_A_DIE
639-5246	PCBA,MLB,KEPLER,CRW_BEST,8G-HYN,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_HYNIX_1600_S,FB_2G_ELPIDA
639-5247	PCBA,MLB,KEPLER,CRW_BSET,8G-MIC,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_MICRON_1600_S,FB_2G_HYNIX_A_DIE
639-5248	PCBA,MLB,KEPLER,CRW_BEST,8G-MIC,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_MICRON_1600_S,FB_2G_ELPIDA
639-5249	PCBA,MLB,KEPLER,CRW_BEST,8G-ELP,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_ELPIDA_1600_S,FB_2G_HYNIX_A_DIE
639-5250	PCBA,MLB,KEPLER,CRW_BEST,8G-ELP,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_ELPIDA_1600_S,FB_2G_ELPIDA
639-5251	PCBA,MLB,KEPLER,CRW_BEST,16G-HYN,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_HYNIX_1600,FB_2G_HYNIX_A_DIE
639-5252	PCBA,MLB,KEPLER,CRW_BEST,16G-HYN,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_HYNIX_1600,FB_2G_ELPIDA
639-5253	PCBA,MLB,KEPLER,CRW_BEST,16G-MIC,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_MICRON_1600,FB_2G_HYNIX_A_DIE
639-5254	PCBA,MLB,KEPLER,CRW_BEST,16G-MIC,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_MICRON_1600,FB_2G_ELPIDA
639-5255	PCBA,MLB,KEPLER,CRW_BEST,16G-ELP,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_ELPIDA_1600,FB_2G_HYNIX_A_DIE
639-5256	PCBA,MLB,KEPLER,CRW_BEST,16G-ELP,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_ELPIDA_1600,FB_2G_ELPIDA
639-5257	PCBA,MLB,KEPLER,CRW_CTO,8G-HYN,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_HYNIX_1600_S,FB_2G_HYNIX_A_DIE
639-5258	PCBA,MLB,KEPLER,CRW_CTO,8G-HYN,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_HYNIX_1600_S,FB_2G_ELPIDA
639-5259	PCBA,MLB,KEPLER,CRW_CTO,8G-MIC,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_MICRON_1600_S,FB_2G_HYNIX_A_DIE
639-5260	PCBA,MLB,KEPLER,CRW_CTO,8G-MIC,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_MICRON_1600_S,FB_2G_ELPIDA
639-5261	PCBA,MLB,KEPLER,CRW_CTO,8G-ELP,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_ELPIDA_1600_S,FB_2G_HYNIX_A_DIE
639-5262	PCBA,MLB,KEPLER,CRW_CTO,8G-ELP,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_ELPIDA_1600_S,FB_2G_ELPIDA
639-5263	PCBA,MLB,KEPLER,CRW_CTO,16G-HYN,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_HYNIX_1600,FB_2G_HYNIX_A_DIE
639-5264	PCBA,MLB,KEPLER,CRW_CTO,16G-HYN,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_HYNIX_1600,FB_2G_ELPIDA
639-5265	PCBA,MLB,KEPLER,CRW_CTO,16G-MIC,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_MICRON_1600,FB_2G_HYNIX_A_DIE
639-5266	PCBA,MLB,KEPLER,CRW_CTO,16G-MIC,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_MICRON_1600,FB_2G_ELPIDA
639-5267	PCBA,MLB,KEPLER,CRW_CTO,16G-ELP,VR-HYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_ELPIDA_1600,FB_2G_HYNIX_A_DIE
639-5268	PCBA,MLB,KEPLER,CRW_CTO,16G-ELP,VR-ELP,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_ELPIDA_1600,FB_2G_ELPIDA
639-5478	PCBA,MLB,KEPLER,CRW_BEST,8G-HYN,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_HYNIX_1600_S,FB_4G_HYNIX
639-5479	PCBA,MLB,KEPLER,CRW_BEST,8G-MIC,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_MICRON_1600_S,FB_4G_HYNIX
639-5480	PCBA,MLB,KEPLER,CRW_BEST,8G-ELP,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_ELPIDA_1600_S,FB_4G_HYNIX
639-5481	PCBA,MLB,KEPLER,CRW_BEST,16G-HYN,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_HYNIX_1600,FB_4G_HYNIX
639-5482	PCBA,MLB,KEPLER,CRW_BEST,16G-MIC,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_MICRON_1600,FB_4G_HYNIX
639-5483	PCBA,MLB,KEPLER,CRW_BEST,16G-ELP,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:BEST,RAM:4Gb_ELPIDA_1600,FB_4G_HYNIX
639-5484	PCBA,MLB,KEPLER,CRW_CTO,8G-HYN,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_HYNIX_1600_S,FB_4G_HYNIX
639-5485	PCBA,MLB,KEPLER,CRW_CTO,8G-MIC,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_MICRON_1600_S,FB_4G_HYNIX
639-5486	PCBA,MLB,KEPLER,CRW_CTO,8G-ELP,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_ELPIDA_1600_S,FB_4G_HYNIX
639-5487	PCBA,MLB,KEPLER,CRW_CTO,16G-HYN,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_HYNIX_1600,FB_4G_HYNIX
639-5488	PCBA,MLB,KEPLER,CRW_CTO,16G-MIC,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_MICRON_1600,FB_4G_HYNIX
639-5489	PCBA,MLB,KEPLER,CRW_CTO,16G-ELP,VR-4GHYN,J45G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:4Gb_ELPIDA_1600,FB_4G_HYNIX

J45G BOM Groups

BOM GROUP	BOM OPTIONS
J45G_COMMON	ALTERNATE,COMMON,J45G_COMMON1,J45G_COMMON2,J45G_PROGPARTS,GFX_BM,ACAPS:A2
J45G_COMMON1	CPUMEM:S0,TBTHV:P15V,SKIP_5V3V3:AUDIBLE,CHGR_5V:LDO,CPUPEG:X8X8,S2_PWR:S0
J45G_COMMON2	EDP:YES,LPCPLUS_CONN:YES,LPCPLUS_R:YES,XDP,RIO_PWR:1V5,SPI:DUAL_IO,SSD_PWR_EN:GPIO,CAM_WAKE:NO
J45G_PVT	BKLT:PROD,SENSOR_NONPROD:N
J45G_PROGPARTS	SMC_PROG:PROTO4,BOOTROM_PROG:PROTO4,TBTROM:PROG,TPAD_PSOC:PROG,GFX_PROGPARTS
GFX_PROGPARTS	DPMUXMCU:PROG
J45G_DEVEL:ENG	ALTERNATE,XDP_DEBUG,S0PGOOD_ISL,DDRVREF_DAC,SENSOR_NONPROD:Y,BKLT:ENG,DBGLED,CAM_XTAL:YES,DPMUX_DEBUG
J45G_DEVEL:DVT	ALTERNATE,XDP_DEBUG,BKLT:PROD,SENSOR_NONPROD:N,DBGLED
GFX_BM	GK107:GX,DPMUX:HOCO
XDP_DEBUG	XDP_CONN,XDP_PCH

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4599	1	IC,CPU,CRW,PRQ,C0,2.0,47W,4+3E,6M,BGA	U0500	CRITICAL	CPU_CRW:BETTER
337S4600	1	IC,CPU,CRW,PRQ,C0,2.3,47W,4+3E,6M,BGA1364	U0500	CRITICAL	CPU_CRW:BEST
337S4624	1	IC,CPU,CRW,PRQ,C0,2.6,47W,4+3E,6M,BGA1364	U0500	CRITICAL	CPU_CRW:CTO
337S4542	1	IC,PCH,LPT-M, HM87,C2,SR199,PRQ,20x20mm,FCBGA	U1100	CRITICAL	
338S1247	1	IC,TBT,FR-4C,A0,PRQ,CIO,SR1JC, FCBGA288	U2800	CRITICAL	
338S1186	1	IC,BCM15700A2,S2 PCIE CMRA,8X8,208FCBGA	U3900	CRITICAL	
333S0700	1	IC,SDRAM,4GBIT,DDR3L-1600,GEMMA,96B FBGA	U4000	CRITICAL	
333S0667	16	IC,SDRAM,4GBIT,DDR3L-1600,HUMA,78P FBGA		CRITICAL	4Gb_HYNIX_1600_S
333S0624	16	IC,SDRAM,DDR3-1600,512MX8,78FBGA,C-DIE,SAMSUNG		CRITICAL	4Gb_SAMSUNG_1600_S
333S0703	16	IC,SDRAM,4GBIT,DDR3L-1600,F DIE,RS,78P		CRITICAL	4Gb_ELPIDA_1600_S
333S0660	16	IC,SDRAM,4GBIT,DDR3L-1600,V80A,78P,FBGA		CRITICAL	4Gb_MICRON_1600_S
333S0667	32	IC,SDRAM,4GBIT,DDR3L-1600,HUMA,78P FBGA		CRITICAL	4Gb_HYNIX_1600
333S0624	32	IC,SDRAM,DDR3-1600,512MX8,78FBGA,C-DIE,SAMSUNG		CRITICAL	4Gb_SAMSUNG_1600
333S0703	32	IC,SDRAM,4GBIT,DDR3L-1600,F DIE,RS,78P		CRITICAL	4Gb_ELPIDA_1600
333S0660	32	IC,SDRAM,4GBIT,DDR3L-1600,V80A,78P,FBGA		CRITICAL	4Gb_MICRON_1600
337S4256	1	IC,GPU,GK107-GT A2,BGA908	U8400	CRITICAL	GK107:GT
337S4427	1	IC,GPU,107GX,926MHZ,1.0375V,1.5V,FBGA908	U8400	CRITICAL	GK107:GX
337S4616	1	IC,GPU,CK107-762,A2,940MHz,5GBPS,1.2V,1.5V908BG	U8400	CRITICAL	GK107:GX2
333S0630	4	IC,SDRAM,GDDR5,64MX32,A-DIE,HYNIX	U8800,U8850,U8900,U8950	CRITICAL	FB_2G_HYNIX_A_DIE
333S0631	4	IC,SDRAM,GDDR5,64MX32,D-DIE,SAMSUNG	U8800,U8850,U8900,U8950	CRITICAL	FB_2G_SAMSUNG
333S0695	4	IC,SDRAM,GDDR5,2GBIT,5GBPS,170FBGA	U8800,U8850,U8900,U8950	CRITICAL	FB_2G_ELPIDA
333S0701	4	IC,SDRAM,GDDR5,2GBIT,5GBPS,170FBGA,EDW2032BBBG-6A-F	U8800,U8850,U8900,U8950	CRITICAL	FB_2G_ELPIDA_29nm
333S0734	4	IC,SDRAM,GDDR5,64MX3,HYNIX,H5GC2H24BFR-T2C	U8800,U8850,U8900,U8950	CRITICAL	FB_2G_HYNIX_29nm
333S0685	4	IC,SDRAM,GDDR5,64MX3,HYNIX,H5GC4H24MFP-T2C	U8800,U8850,U8900,U8950	CRITICAL	FB_4G_HYNIX

DRAM SPD Straps

BOM GROUP	BOM OPTIONS
RAM:2Gb_HYNIX_1600	RAMCFG3:L,RAMCFG2:L,RAMCFG1:L,RAMCFG0:L
RAM:2Gb_SAMSUNG_1600	RAMCFG3:L,RAMCFG2:L,RAMCFG1:L,RAMCFG0:H
RAM:2Gb_ELPIDA_1600	RAMCFG3:L,RAMCFG2:L,RAMCFG1:H,RAMCFG0:L
RAM:2Gb_MICRON_1600	RAMCFG3:L,RAMCFG2:L,RAMCFG1:H,RAMCFG0:H
RAM:4Gb_HYNIX_1600_S	4Gb_HYNIX_1600_S,RAMCFG3:L,RAMCFG2:H,RAMCFG1:L,RAMCFG0:L
RAM:4Gb_SAMSUNG_1600_S	4Gb_SAMSUNG_1600_S,RAMCFG3:L,RAMCFG2:H,RAMCFG1:L,RAMCFG0:H
RAM:4Gb_ELPIDA_1600_S	4Gb_ELPIDA_1600_S,RAMCFG3:L,RAMCFG2:H,RAMCFG1:H,RAMCFG0:L
RAM:4Gb_MICRON_1600_S	4Gb_MICRON_1600_S,RAMCFG3:L,RAMCFG2:H,RAMCFG1:H,RAMCFG0:H
RAM:4Gb_HYNIX_1600	4Gb_HYNIX_1600,RAMCFG3:H,RAMCFG2:L,RAMCFG1:L,RAMCFG0:L
RAM:4Gb_SAMSUNG_1600	4Gb_SAMSUNG_1600,RAMCFG3:H,RAMCFG2:L,RAMCFG1:L,RAMCFG0:H
RAM:4Gb_ELPIDA_1600	4Gb_ELPIDA_1600,RAMCFG3:H,RAMCFG2:L,RAMCFG1:H,RAMCFG0:L
RAM:4Gb_MICRON_1600	4Gb_MICRON_1600,RAMCFG3:H,RAMCFG2:L,RAMCFG1:H,RAMCFG0:H

Development/Base BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-0177	1	J45G MLB,KEPLER BASE BOM	BASE	CRITICAL	BASE_BOM
985-0181	1	J45G MLB,KEPLER, DEVEL BOM	DEVEL	CRITICAL	DEVEL_BOM

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Programmables - All builds

341S3920	1	IC,EEPROM,FALCON RIDGE (V13.1),J44/J45	U2890	CRITICAL	TBTROM:PROG
335S0915	1	EEPROM,SPI FLASH ROM,4MBit,50MHZ,USON8	U2890	CRITICAL	TBTROM:BLANK
335S0852	1	IC,GPUROM,D2,BLANK	U9101	CRITICAL	GPUROM:BLANK
341S3565	1	IC,EDP MUX-95C,(RENESAS) V3.2.8,DVB,D2	U9600	CRITICAL	DPMUXMCU:PROG
337S4313	1	IC,MCU,H8S/2113,9X9MM,TLP-145V	U9600	CRITICAL	DPMUXMCU:BLANK
341S3856	1	IC,TRKPD/KYBD CNTRLR,CU only,V225,J45	U4801	CRITICAL	TPAD_PSOC:PROG
337S4587	1	IC,TP PSOC,QFN,BLANK	U4801	CRITICAL	TPAD_PSOC:BLANK

SMC

338S1214	1	IC,SMC-A3,40MHZ/50DMIPS,SCPL FW,157BGA	U5000	CRITICAL	SMC_PROG:BASE
341S3901	1	IC,SMC-B1,SCPL,EXT,V2.16Q13,PROTO4,J45G	U5000	CRITICAL	SMC_PROG:PROTO4
341S3741	1	IC,SMC-A3,SCPL,EXT,VXXXX,PVT,J15	U5000	CRITICAL	SMC_PROG:PVT

EFI ROM

335S0807	1	IC,SPI SRL 50MHZ FLASH,64MBIT,8SOP,FUSE,L	U6100	CRITICAL	BOOTROM_PROG:BLANK
335S0812	1	64MBIT SPI SRL DUAL I/O FLSH,SOIC8,Z	U6100	CRITICAL	BOOTROM_PROG:BLANK2
341S3712	1	IC,EFI ROM(V0008)PROTO 0,J15	U6100	CRITICAL	BOOTROM_PROG:PROTO
341S3742	1	IC,EFI ROM(V0013) PROTO 1,J15	U6100	CRITICAL	BOOTROM_PROG:PROTO1
341S3890	1	IC,EFI ROM(V0100) PROTO 3-J45 & EVT-J45	U6100	CRITICAL	BOOTROM_PROG:PROTO3
341S3904	1	IC,EFI ROM(V0106) PROTO 4,J45	U6100	CRITICAL	BOOTROM_PROG:PROTO4

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1053	376S0604		ALL	Diodes alt to Fairchild
128S0311	128S0329		ALL	NEC alt to Sanyo
138S0739	138S0706		ALL	Samsung alt to Murata
197S0481	197S0480		ALL	EPSON Alt to NDK
371S0713	371S0558		ALL	DDS alt to ST
152S0461	152S1645		ALL	Cyntec alt to Vishay
376S1080	376S0820		ALL	Diodes alt to On Semi
155S0667	155S0583		ALL	Panasonic alt to TDK
376S1032	376S0855		ALL	Toshiba alt to Diodes
376S1129	376S0855		ALL	NXP alt to Diodes
376S1089	376S1128		ALL	NXP alt to Diodes
138S0681	138S0638		ALL	Taiyo Yuden alt to Samsung
128S0371	128S0376		ALL	Kemet alt to Sanyo
333S0629	333S0703		ALL	Epida Alt die
138S0803	138S0639		ALL	Samsung Alt to Murata
138S0843	138S0674		ALL	Samsung Alt to Murata
138S0846	138S0811		ALL	Samsung Alt to Murata
127S0164	127S0162		ALL	Romh Alt to Vishay
138S0732	138S0715		ALL	Romh Alt to Vishay
128S0364	128S0264		ALL	Kemet alt to Sanyo
333S0704	333S0700		ALL	ELPIDA to HYNIX U4000
353S3527	353S3528		ALL	Pericom eDP MUX
353S3526	353S3528		ALL	TI eDP MUX
197S0466	197S0464		ALL	Epson alt to NDK
311S0649	311S0541		ALL	ON alt to Toshiba (U2030,U7001)
197S0479	197S0478		ALL	Epson alt to NDK

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BOM Configuration

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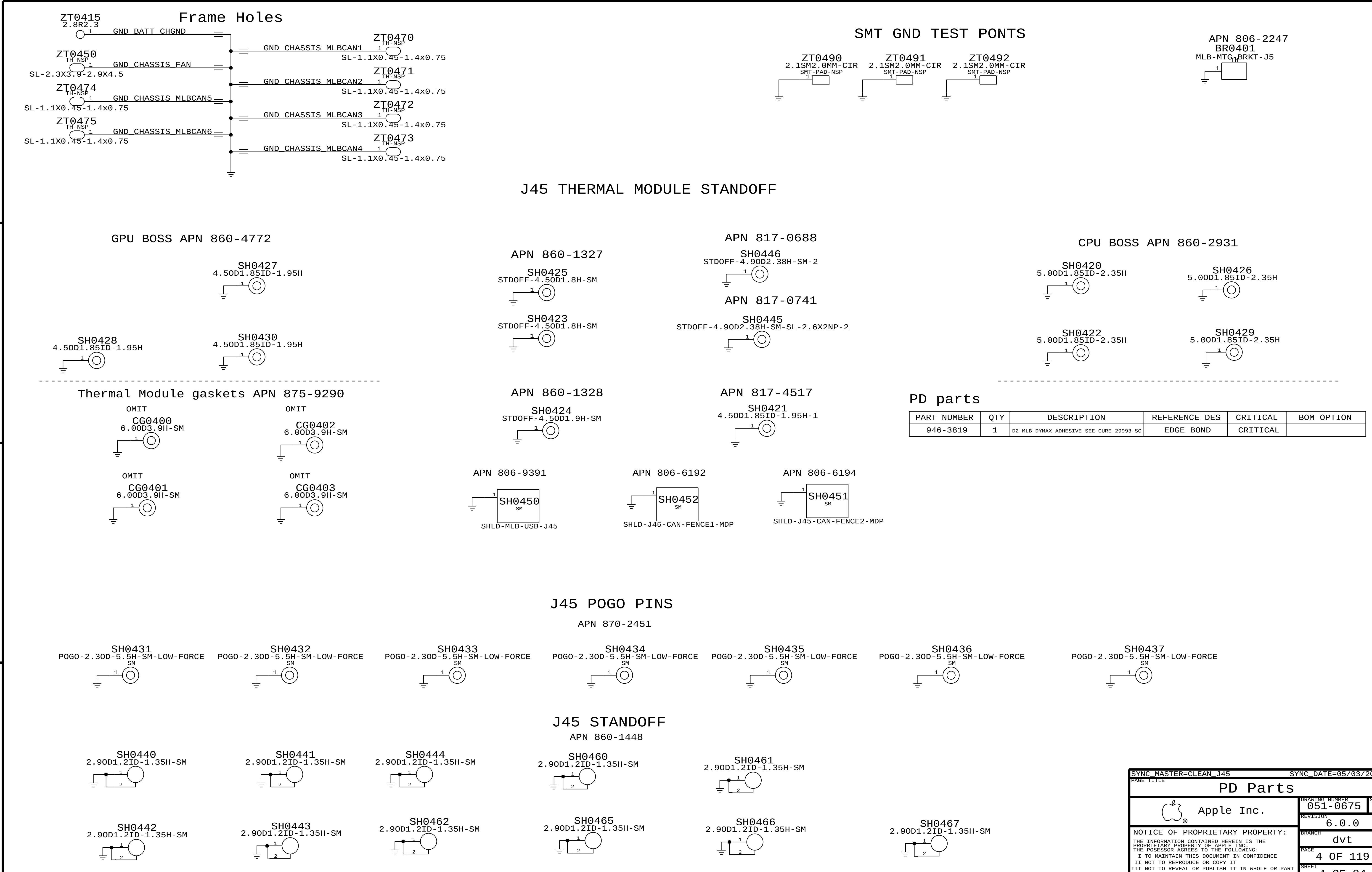

PD parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
946-3819	1	D2 MLB DYMAX ADHESIVE SEE-CURE 29993-SC	EDGE_BOND	CRITICAL	

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PD Parts

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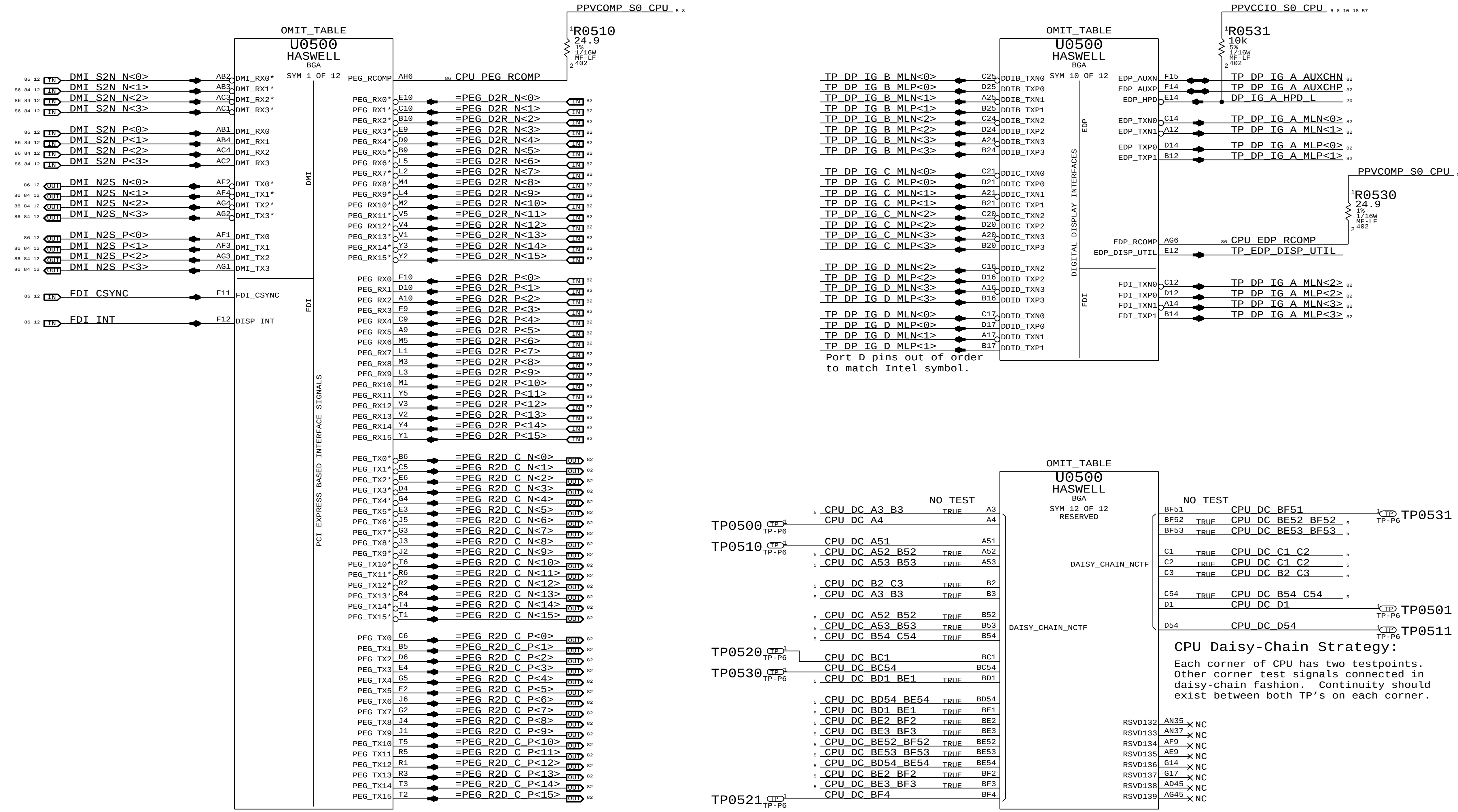

CPU Daisy-Chain Strategy:

Each corner of CPU has two testpoints.
Other corner test signals connected in
daisy-chain fashion. Continuity should
exist between both TP's on each corner.

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PAGE TITLE: CPU DMI/PEG/FDI/RSVD

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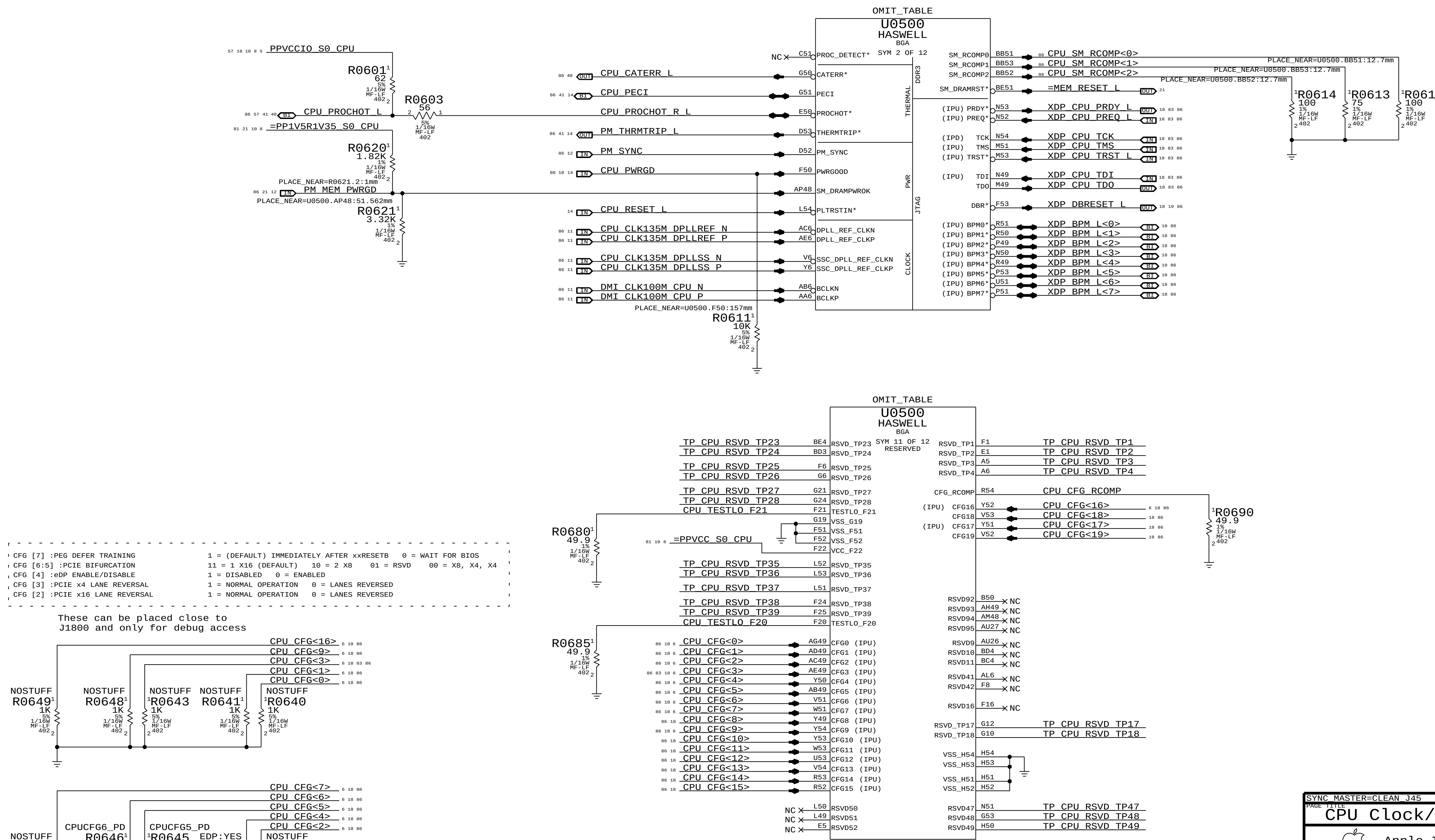

BOM GROUP	BOM OPTIONS
CPUPEG:X16	
CPUPEG:X8X8	CPUCFG5_PD
CPUPEG:X8X4X4	CPUCFG6_PD,CPUCFG5_PD

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PAGE TITLE
CPU Clock/Misc/JTAG/CFG

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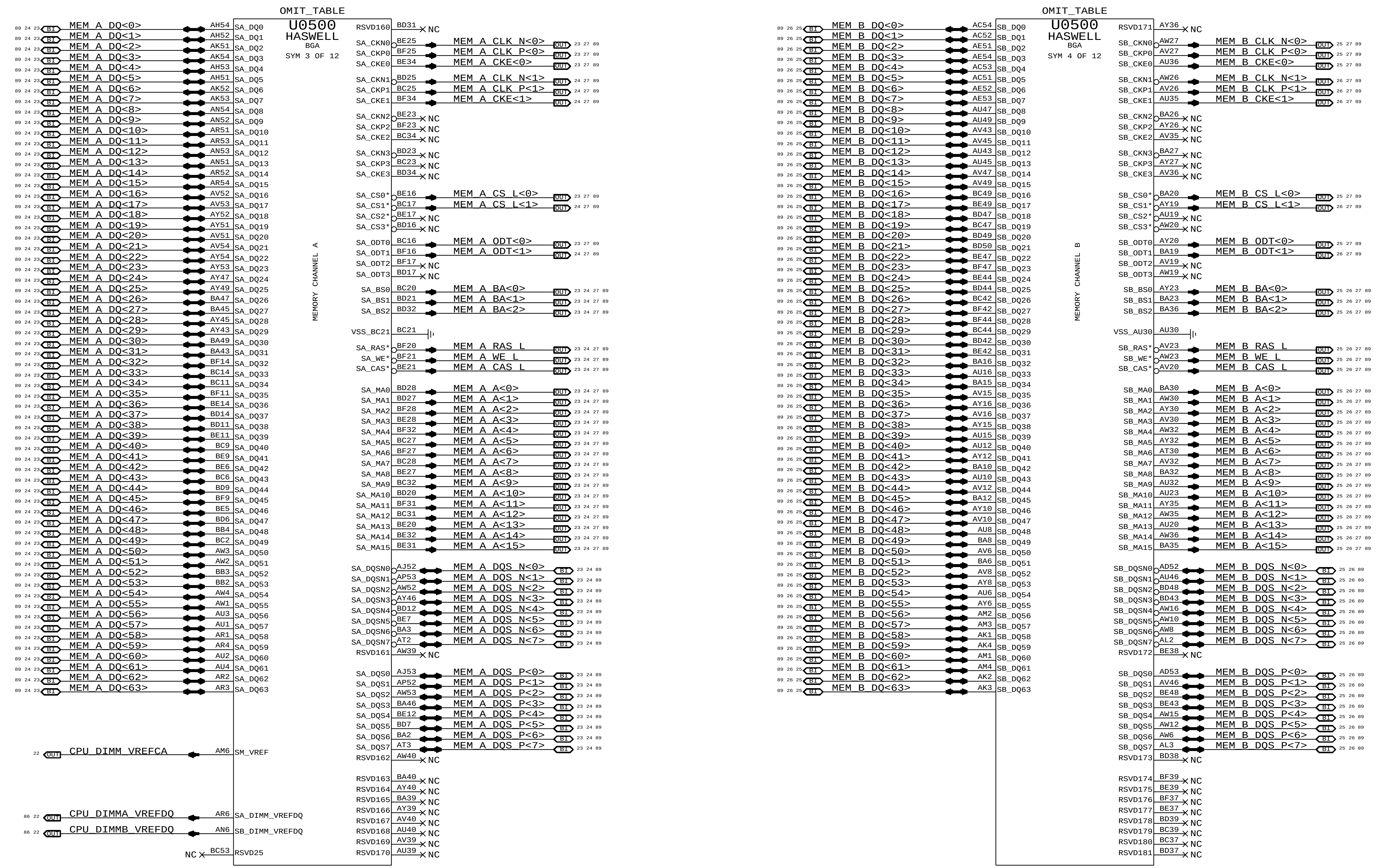

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CPU DDR3 Interfaces

Apple Inc.

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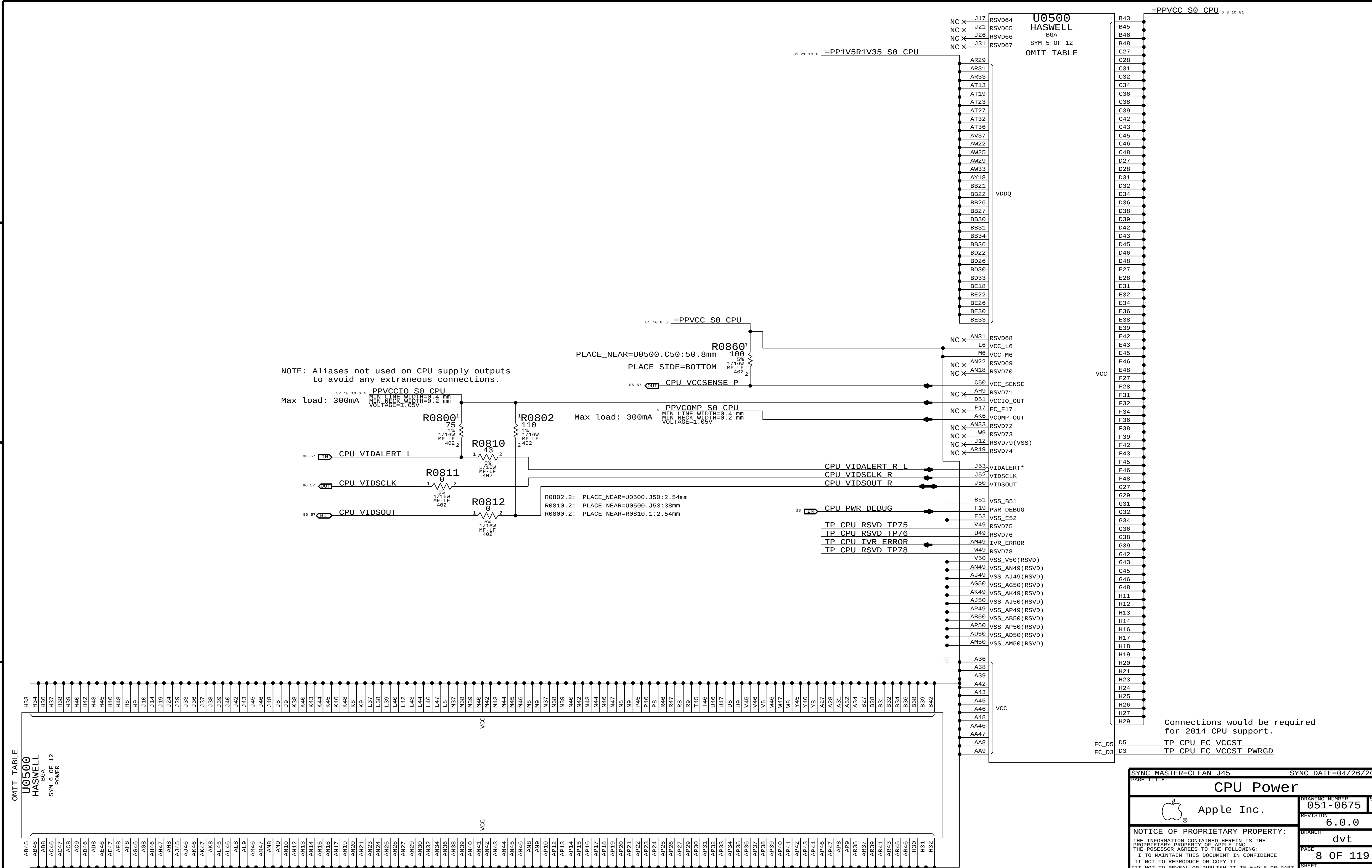

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

CPU Power

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D

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BRANCH dvt

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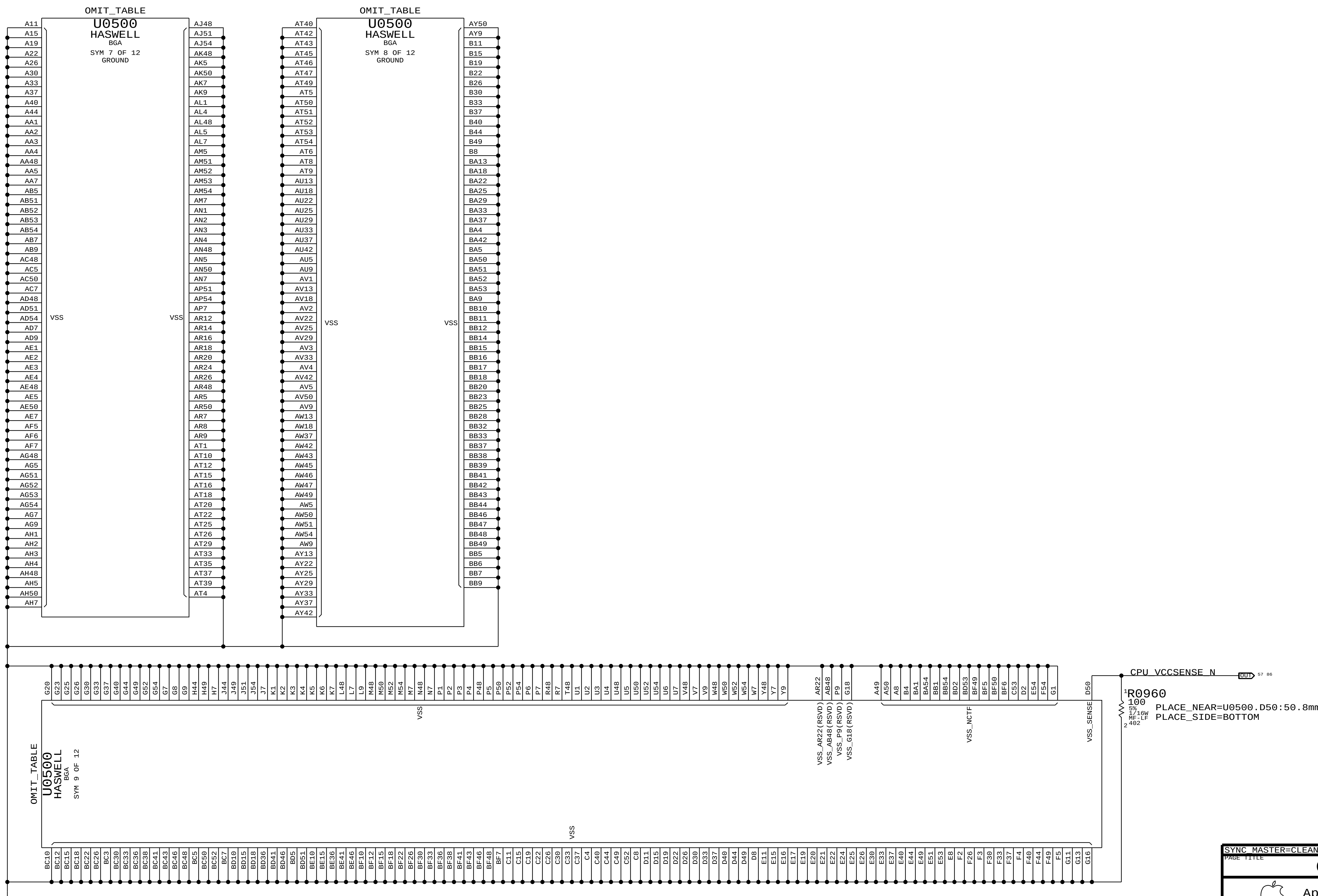

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

CPU Ground

Apple Inc.

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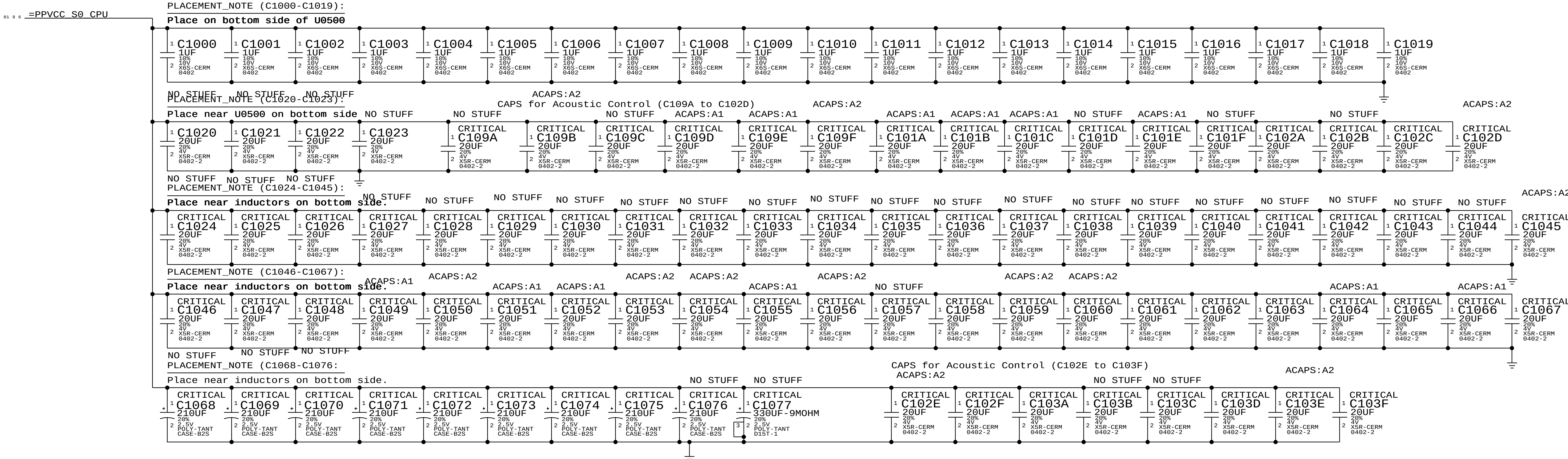
8 7 6 5 4 3 2 1

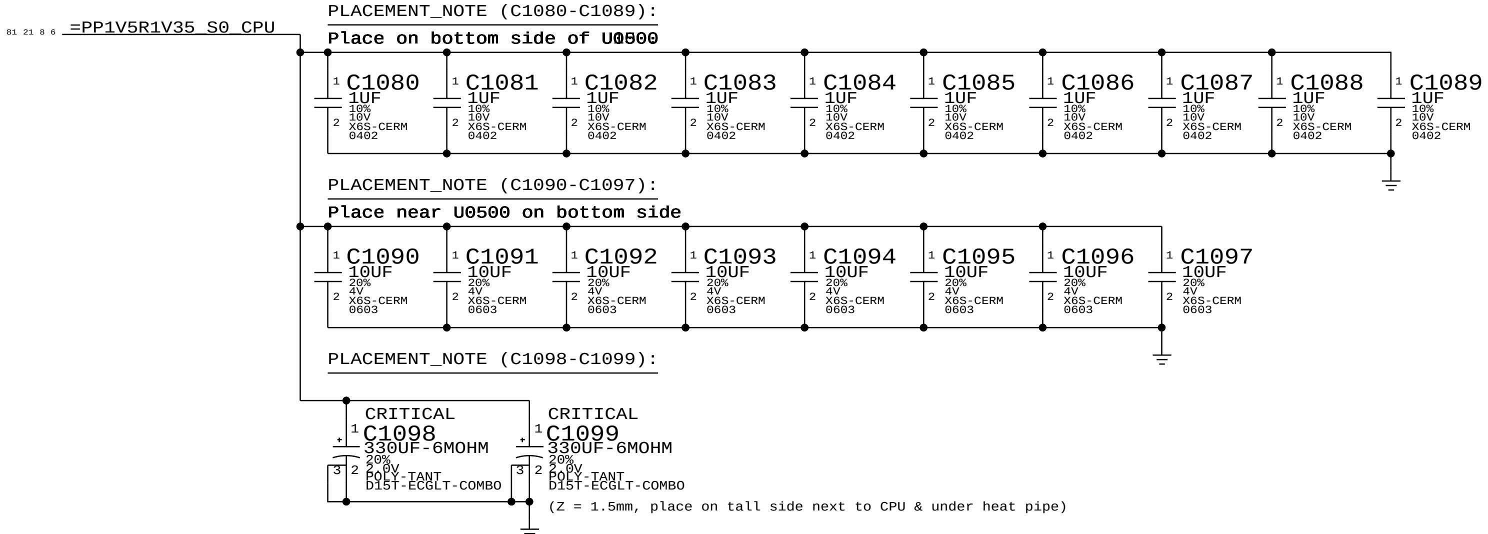
CPU VCORE Decoupling

Intel recommendation: 4x 470uF 4mOhm (3 CPU-side, 1 opposite), 20x 22uF 0805 (10 CPU-side, 10 opposite near edge, 4x 10uF 0603 (2 CPU-side, 2 opposite), 20x 1uF 0402 (under CPU)
Apple Implementation: 9x 210uF 6mOhm, 44x 10uF 0402, 4x 10uF 0402, 20x 1uF 0402



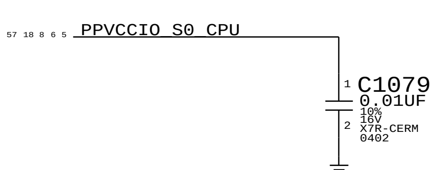
CPU VDDQ Decoupling

Intel recommendation: 2x 330uF, 8x 10uF 0603, 10x 1uF 0402
Apple Implementation: 2x 330uF, 8x 10uF 0603, 10x 1uF 0402



CPU VCCIO Decoupling

Intel recommendation: 2x 0.01uF 0402 (1 near CPU, 1 near SVID pull-ups)
Apple Implementation: 2x 0.01uF 0402 (second cap is on CPU VR page)



NOTE: Intel decoupling recommendations from Shark Bay Mobile Platform Power Delivery Design Guide (doc #487822, Rev 0.8 dated January 2012), Section 5.

SYNC_MASTER=CLEAN_J15 SYNC_DATE=05/02/2013
PAGE TITLE: CPU Decoupling
Apple Inc.
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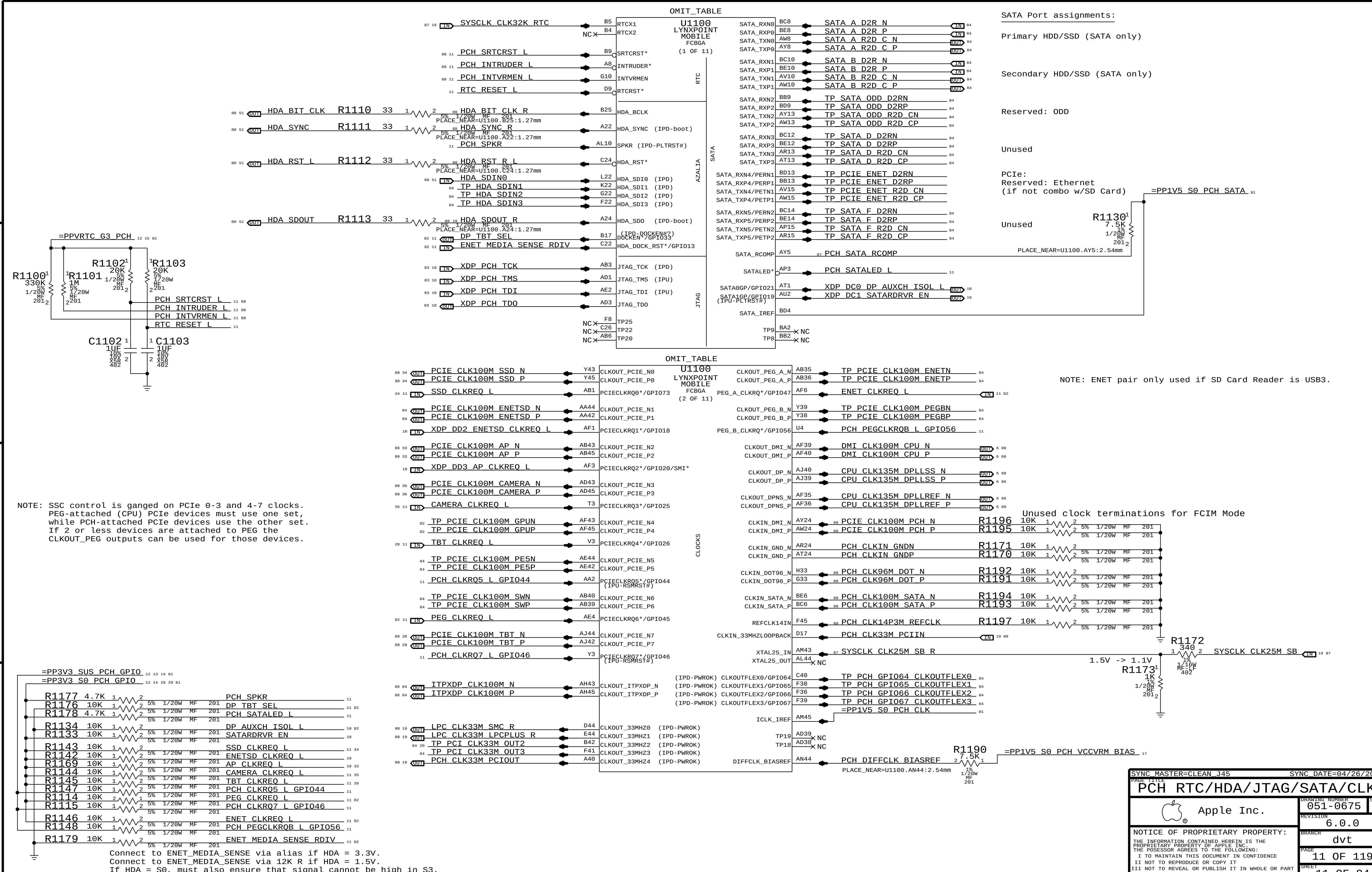

NOTE: SSC control is ganged on PCIe 0-3 and 4-7 clocks.
PEG-attached (CPU) PCIe devices must use one set,
while PCH-attached PCIe devices use the other set.
If 2 or less devices are attached to PEG the
CLKOUT_PEG outputs can be used for those devices.

Connect to ENET_MEDIA_SENSE via alias if HDA = 3.3V.
Connect to ENET_MEDIA_SENSE via 12K R if HDA = 1.5V.
If HDA = S0, must also ensure that signal cannot be high in S3.

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
PCH RTC/HDA/JTAG/SATA/CLK

Apple Inc.

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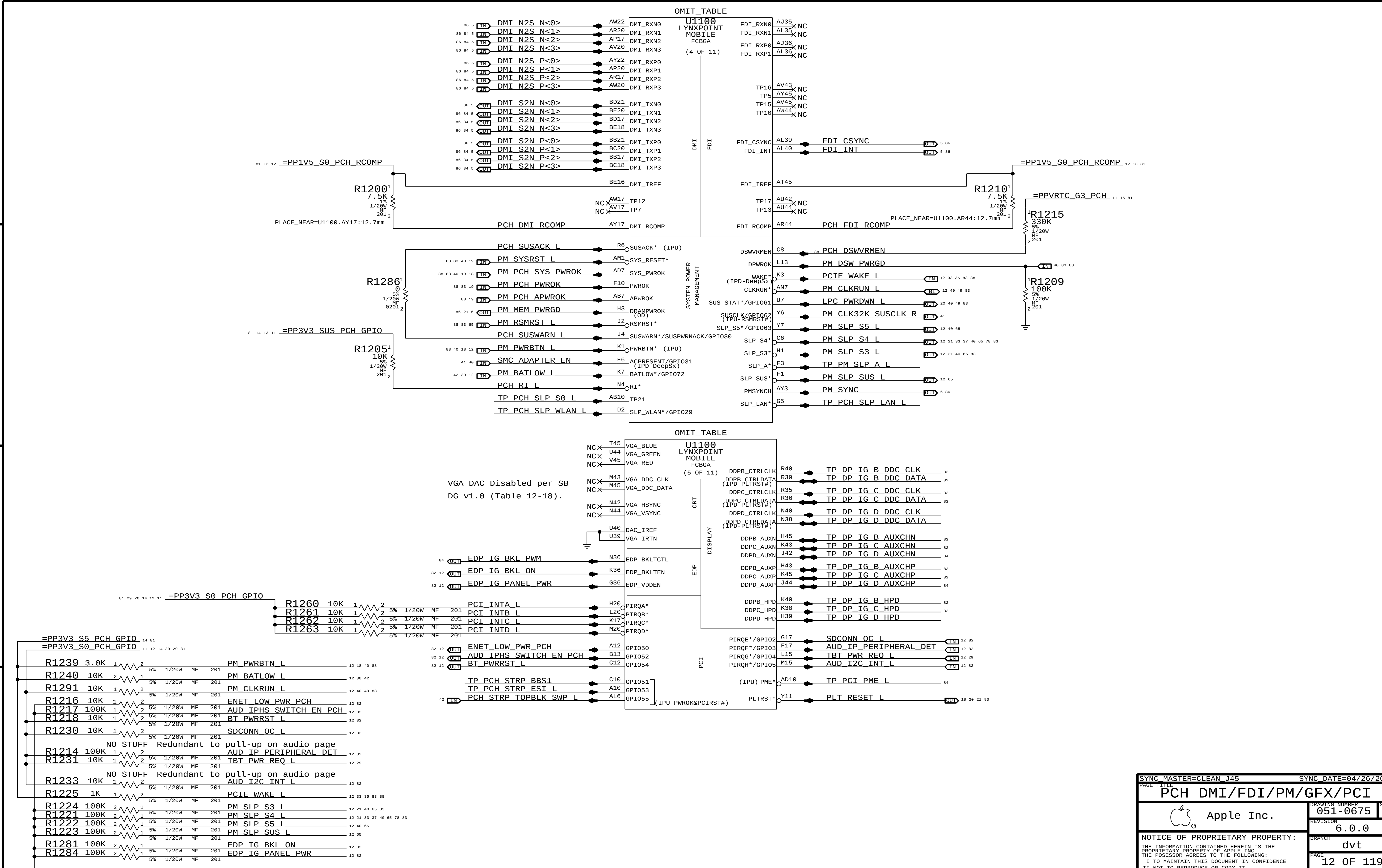

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

PCH DMI/FDI/PM/GFX/PCI

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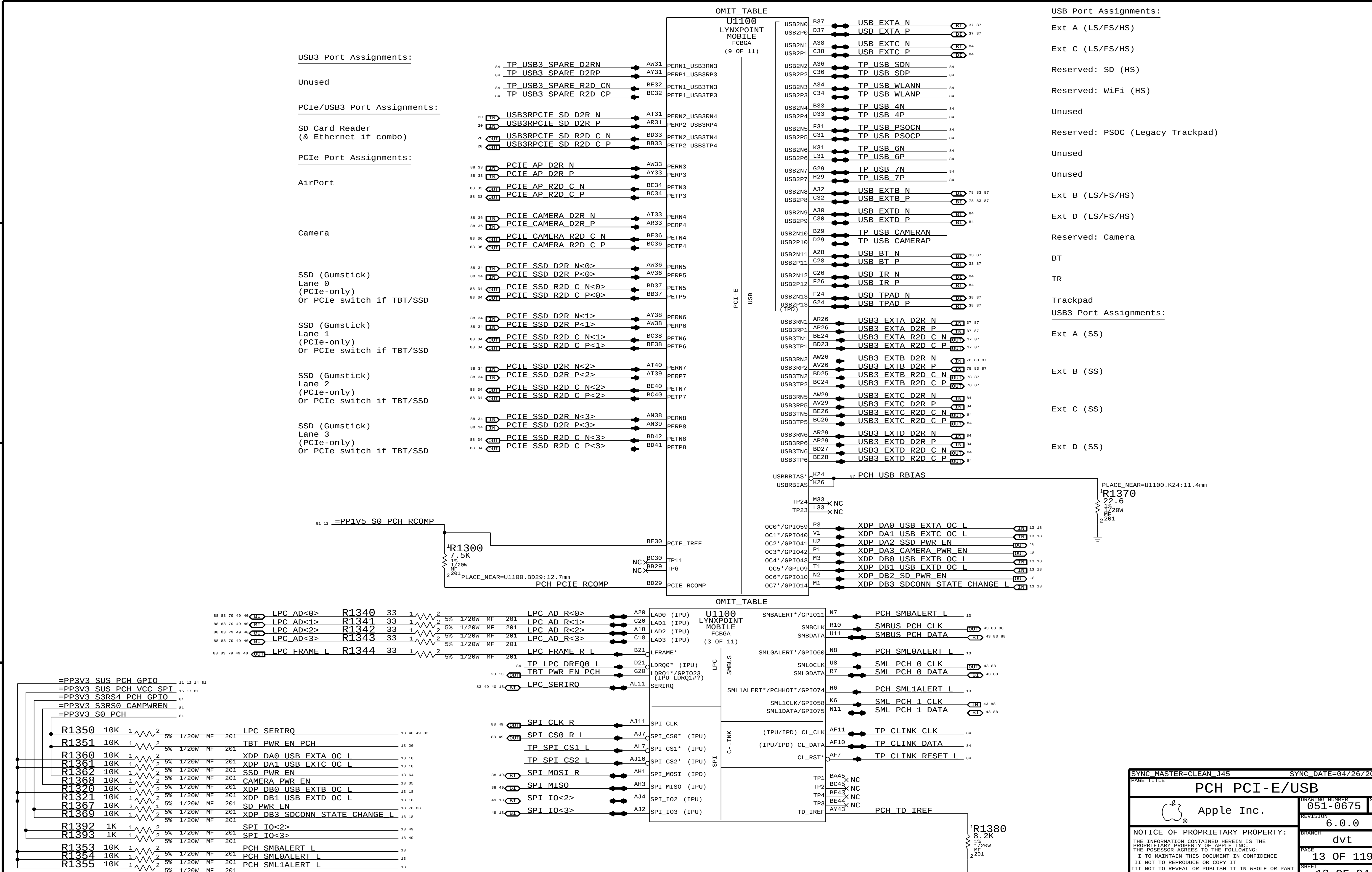

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
PCH PCI-E/USB

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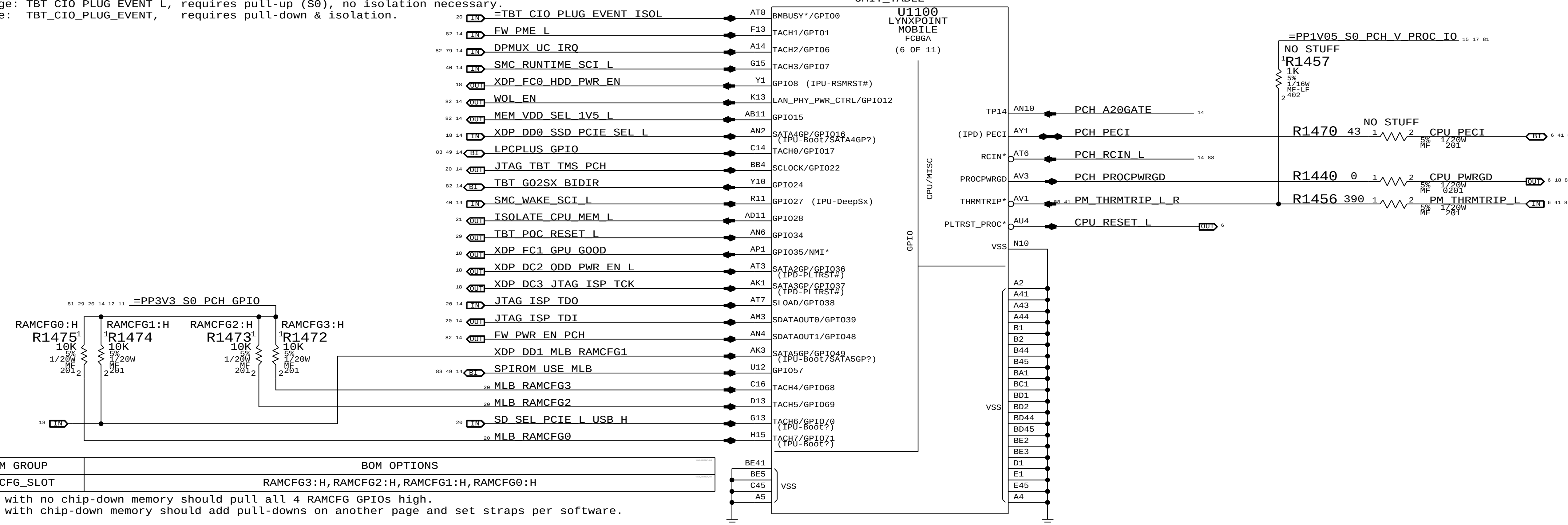
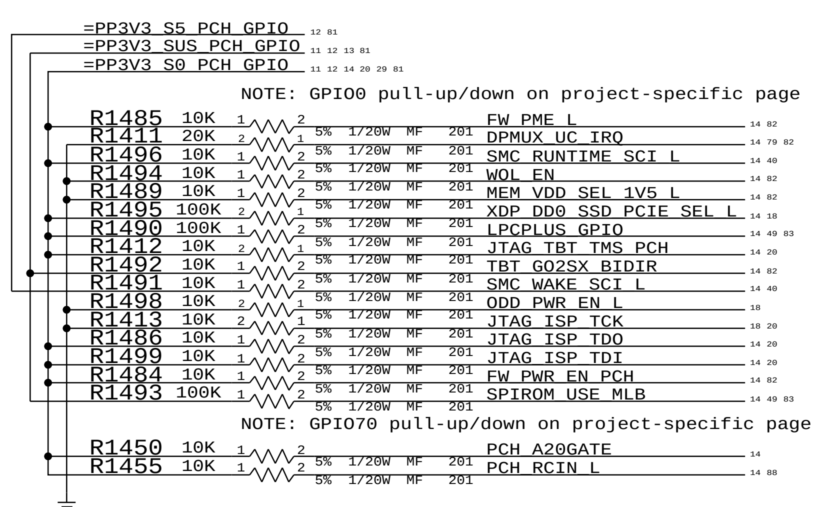
Pull-up/down on chipset support page (depends on TBT controller)
Redwood Ridge: TBT_CIO_PLUG_EVENT_L, requires pull-up (S0), no isolation necessary.
Cactus Ridge: TBT_CIO_PLUG_EVENT, requires pull-down & isolation.



BOM GROUP	BOM OPTIONS
RAMCFG_SLOT	RAMCFG3:H,RAMCFG2:H,RAMCFG1:H,RAMCFG0:H

Systems with no chip-down memory should pull all 4 RAMCFG GPIOs high.
Systems with chip-down memory should add pull-downs on another page and set straps per software.



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
PCH GPIO/MISC/NCTF

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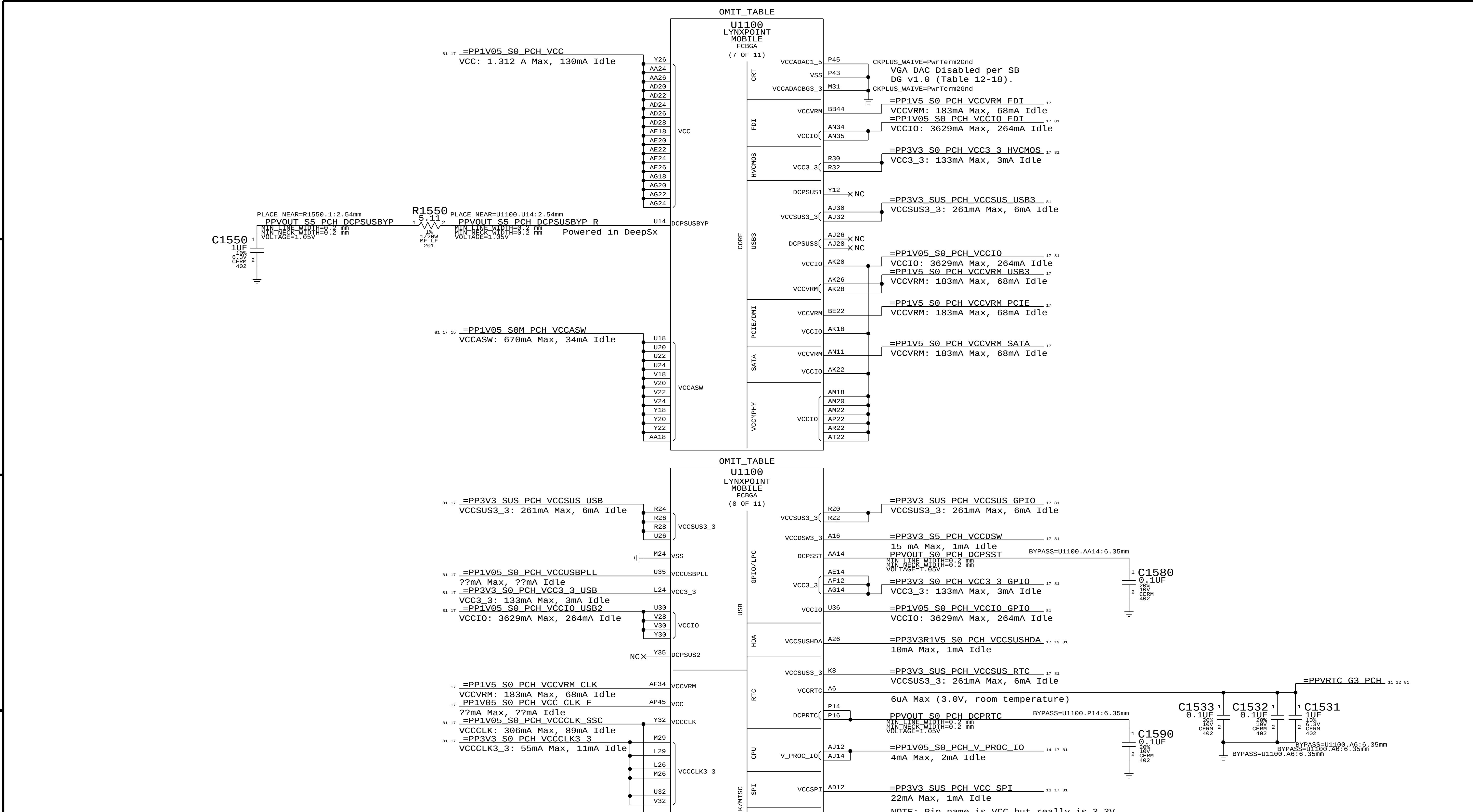

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: PCH Power

Apple Inc.

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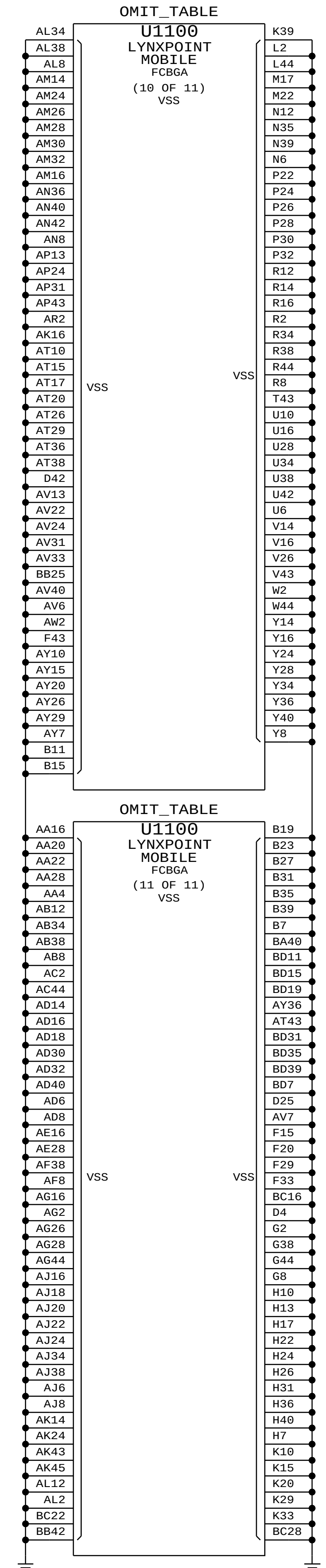
Current data from LPT EDS (doc #486708, Rev 1.0).



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

PCH Grounds

Apple Inc.

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
113S0022	1	RES,FF,0 OHM,(020OHM MAX),2A,0603	L1790		

Current data from LPT EDS (doc #486708, Rev 1.0).

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
PCH DECOUPLING

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D

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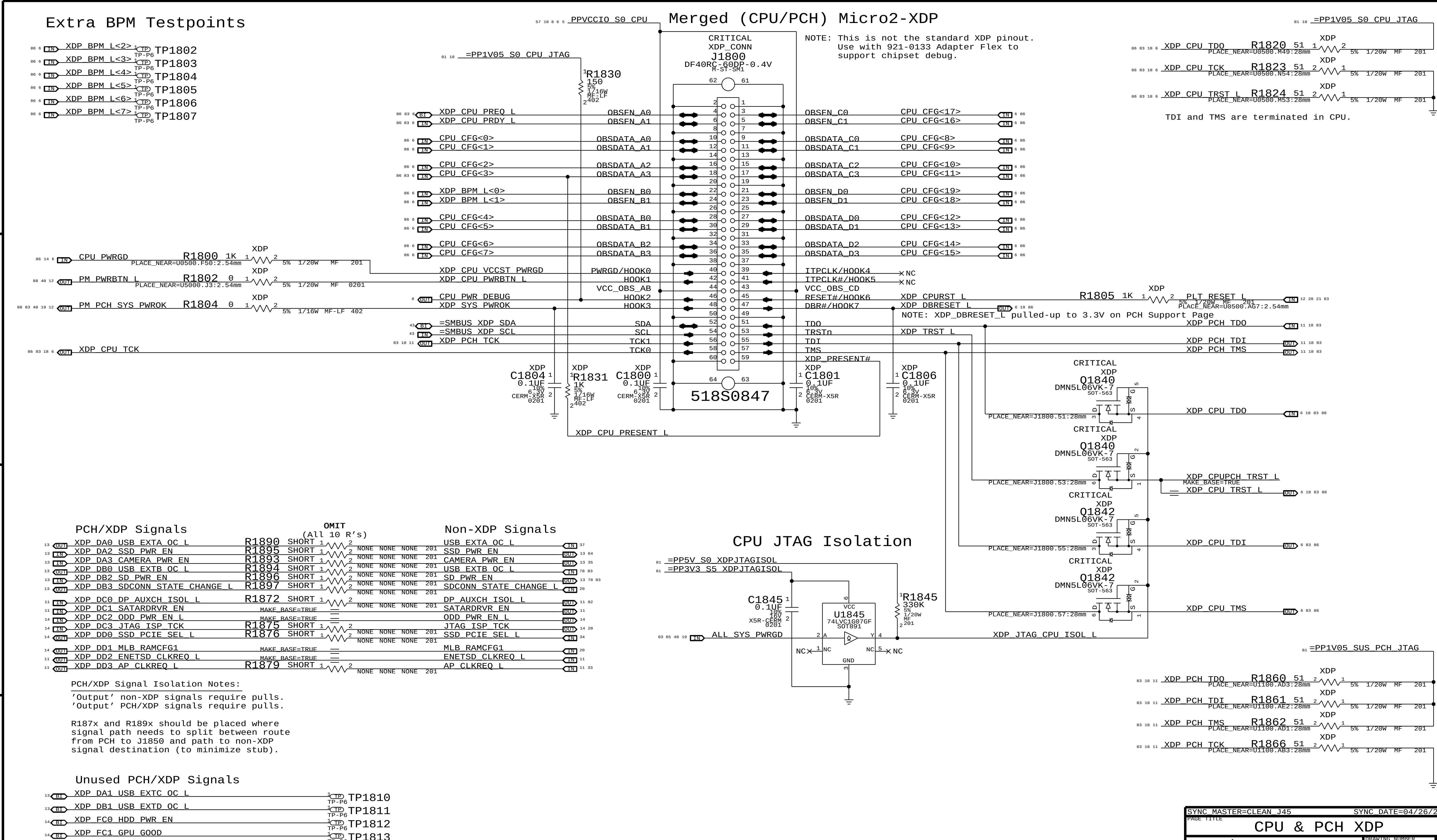

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

CPU & PCH XDP

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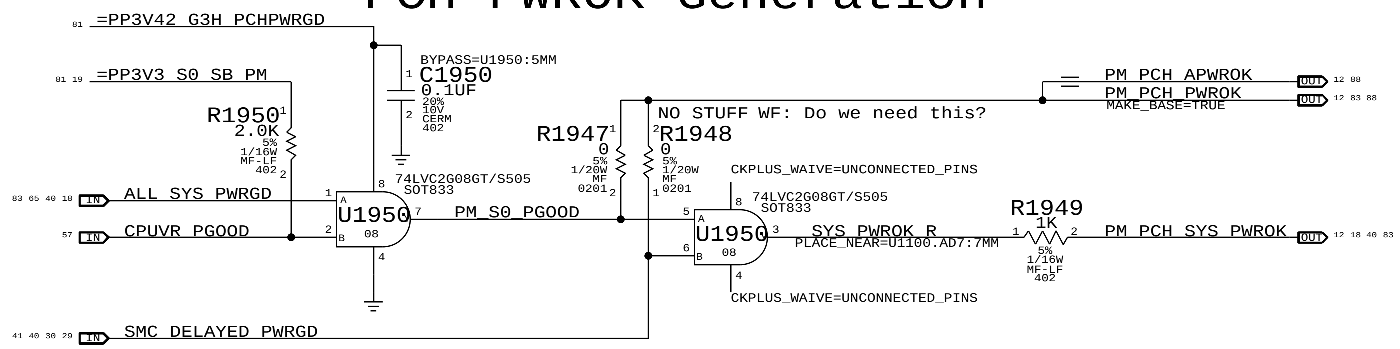
8 7 6 5 4 3 2 1

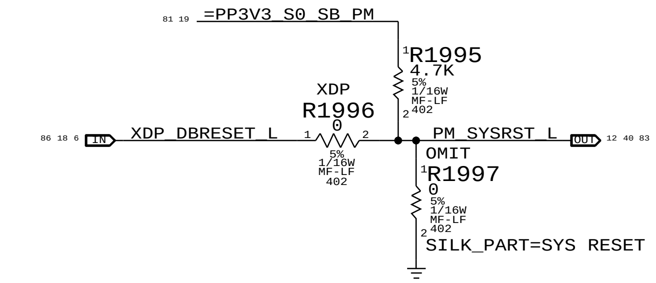
D C B A

PCH PWROK Generation



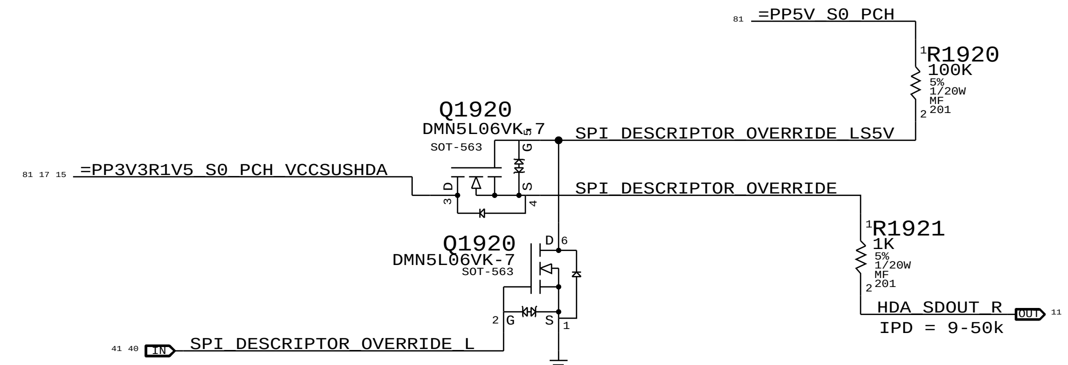
NOTE: ALL_SYS_PWRGD must remain low until at least 5ms after all rails are valid.

PCH Reset Button

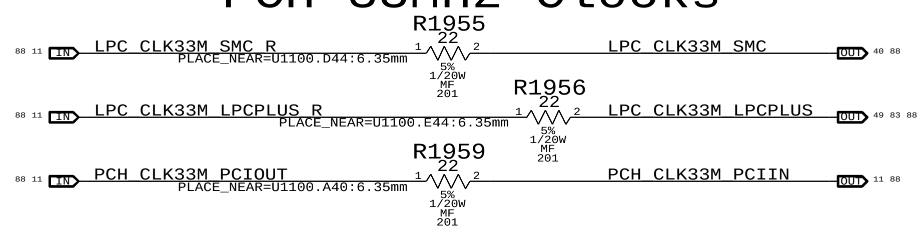


PCH ME Disable Strap

PCH uses HDA_SDO as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. Q1920 & 5V pull-up allows circuit to work regardless of HDA voltage.



PCH 33MHz Clocks



System RTC Power Source & 32kHz / 25MHz Clock Generator

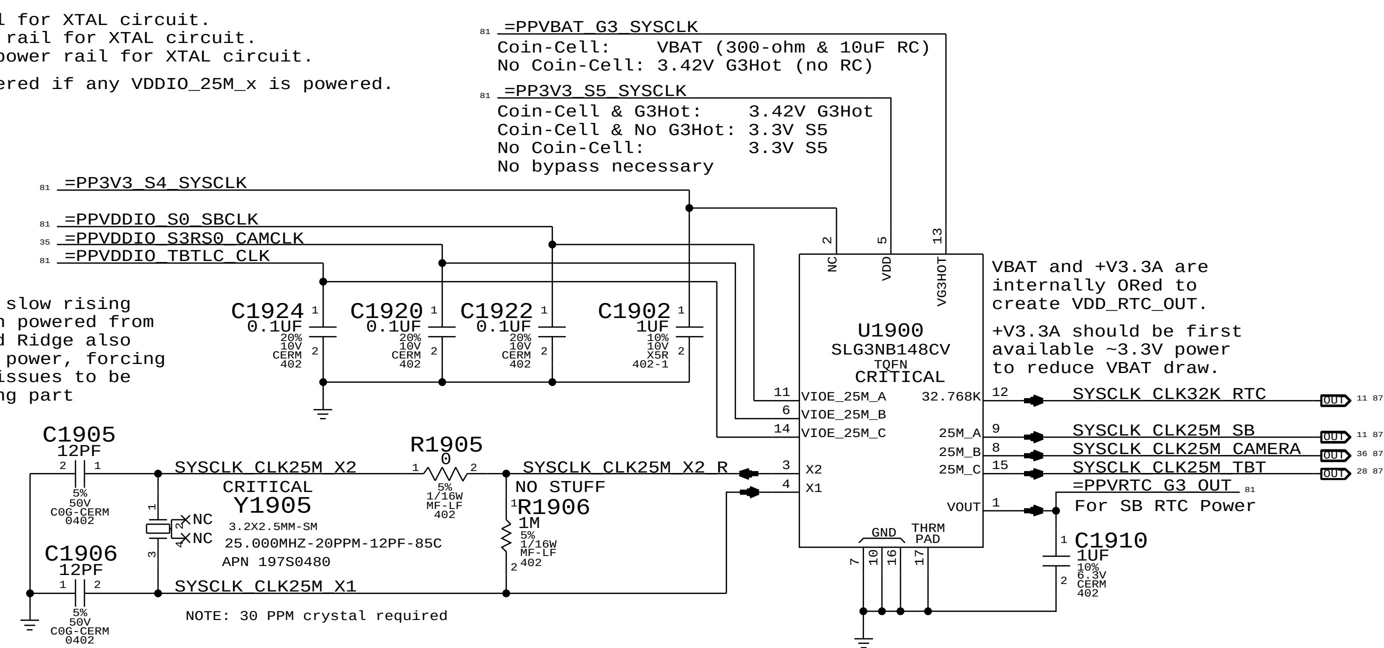
VDDIO_25M_A: SB power rail for XTAL circuit.
VDDIO_25M_B: Camera power rail for XTAL circuit.
VDDIO_25M_C: Thunderbolt power rail for XTAL circuit.

NOTE: VDD_25M must be powered if any VDDIO_25M_x is powered.

GreenClk 25MHz Power

SB XTAL Power
Camera XTAL Power
TBT XTAL Power

NOTE: SLG3NB148A provides slow rising edge on 25MHZ_B when powered from 1.2V VDDIO. Redwood Ridge also complicates VDD_25M power, forcing at least S4. Both issues to be addressed in upcoming part (SLG3NB148C).



SYNC_MASTER=CLEAN_J45	SYNC_DATE=04/26/2013
PAGE TITLE	Chipset Support
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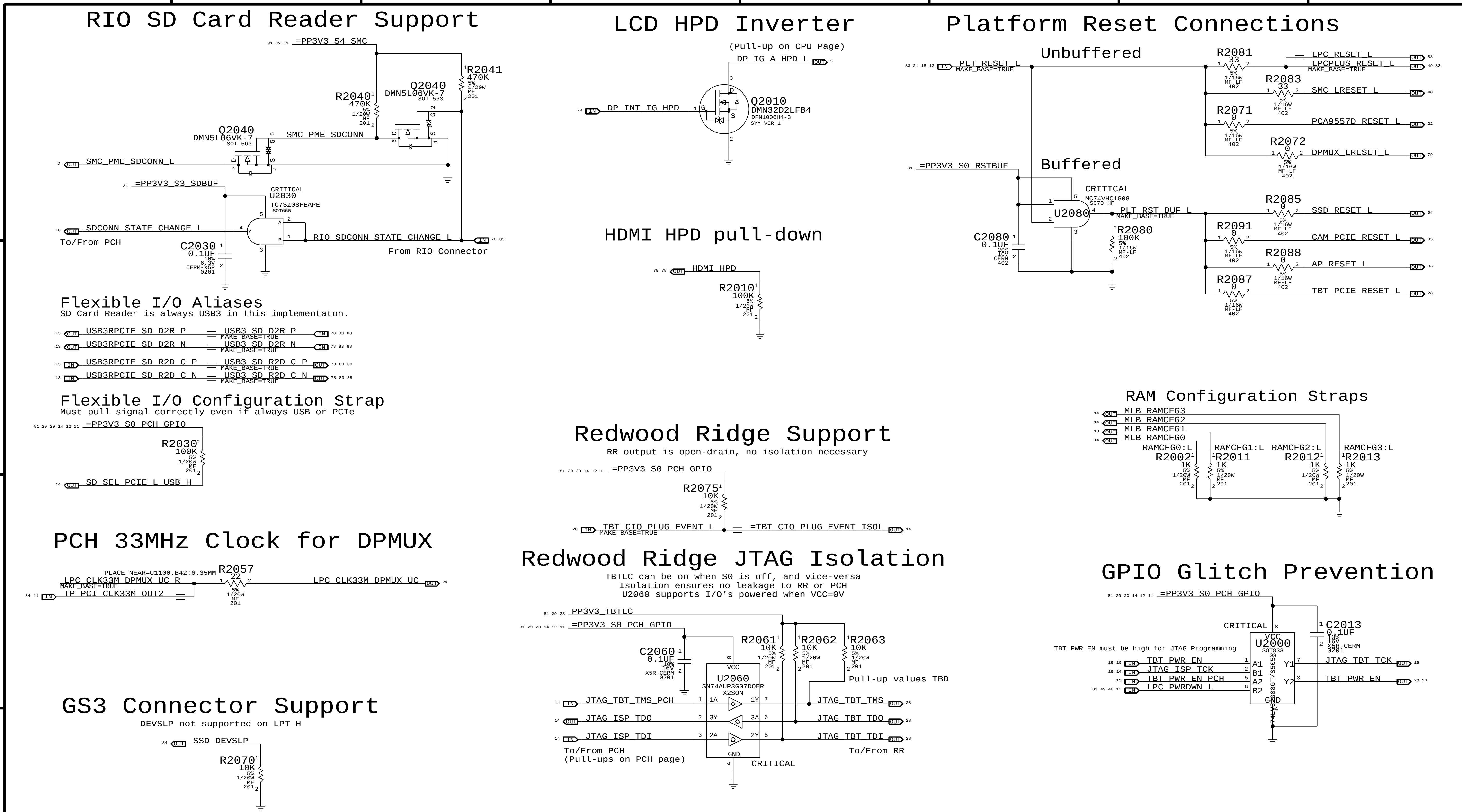

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: Project Chipset Support

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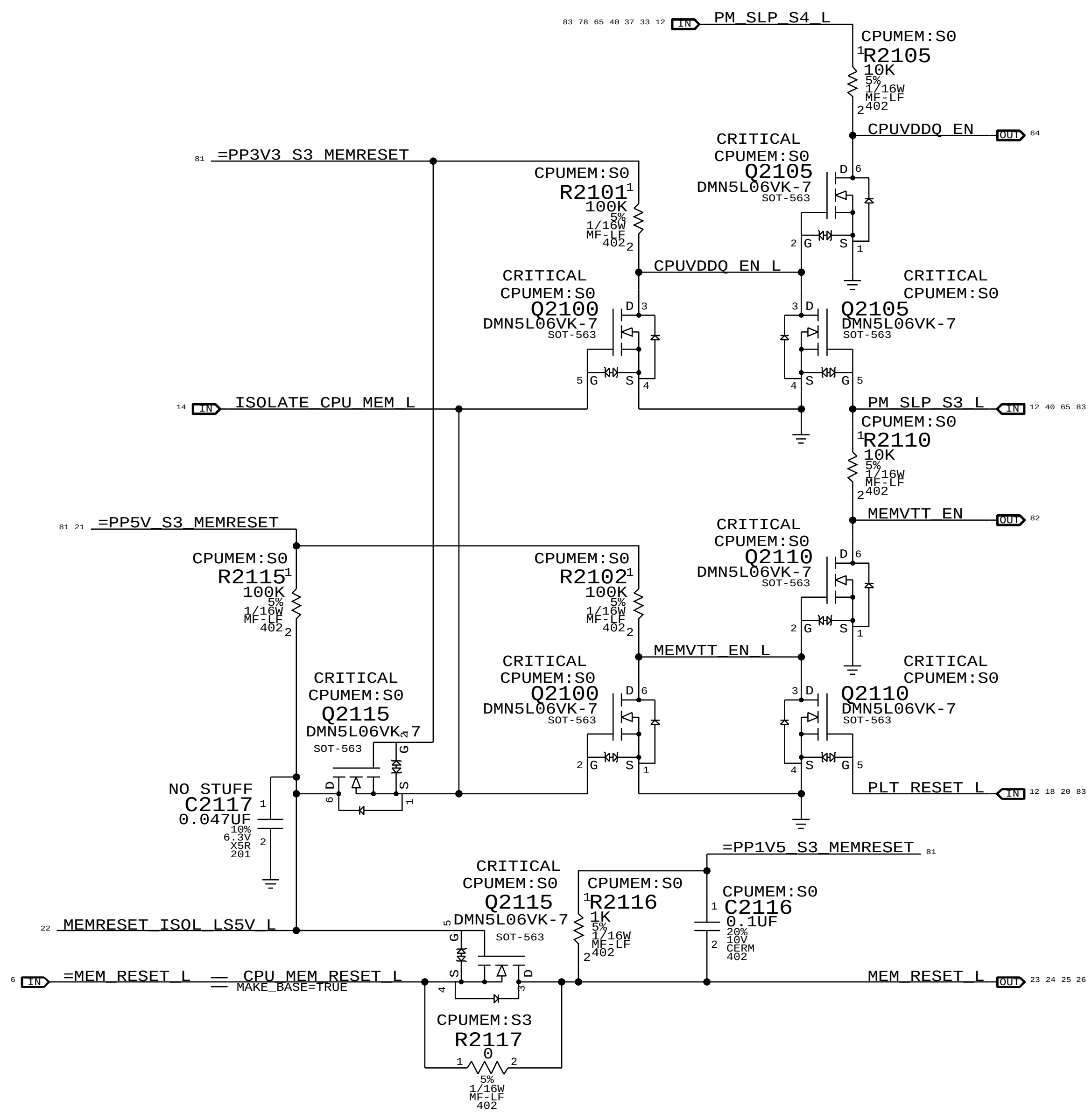
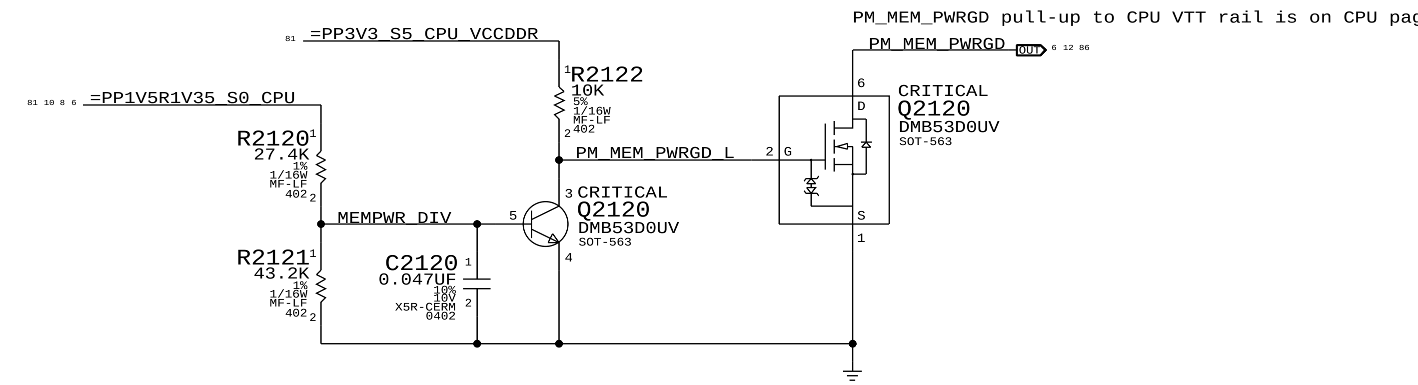
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

CPUVDDQ_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

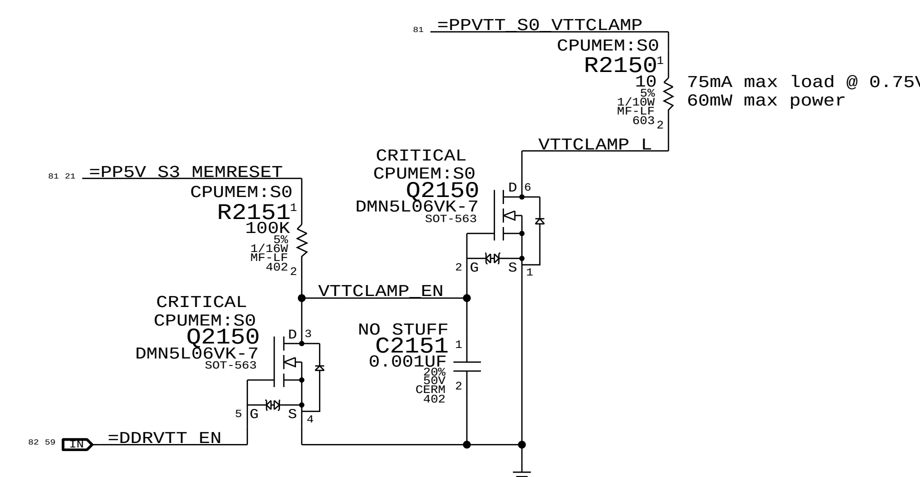


MEM S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



	Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	CPUVDDQ_EN
S0	0	1	1	1	1	1	CPU_MEM_RESET_L	1	1
to	1	0	1	1	1	1	1	1	1
	2	0	0	1	1	1	1	0	1
S3	3	0	0	0	1	X	1	0	0
	4	0	0	1	1	X	1	0	1
to	5	0	1	1	1	0 (*)	1	1	1
	6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: CPU Memory S3 Support

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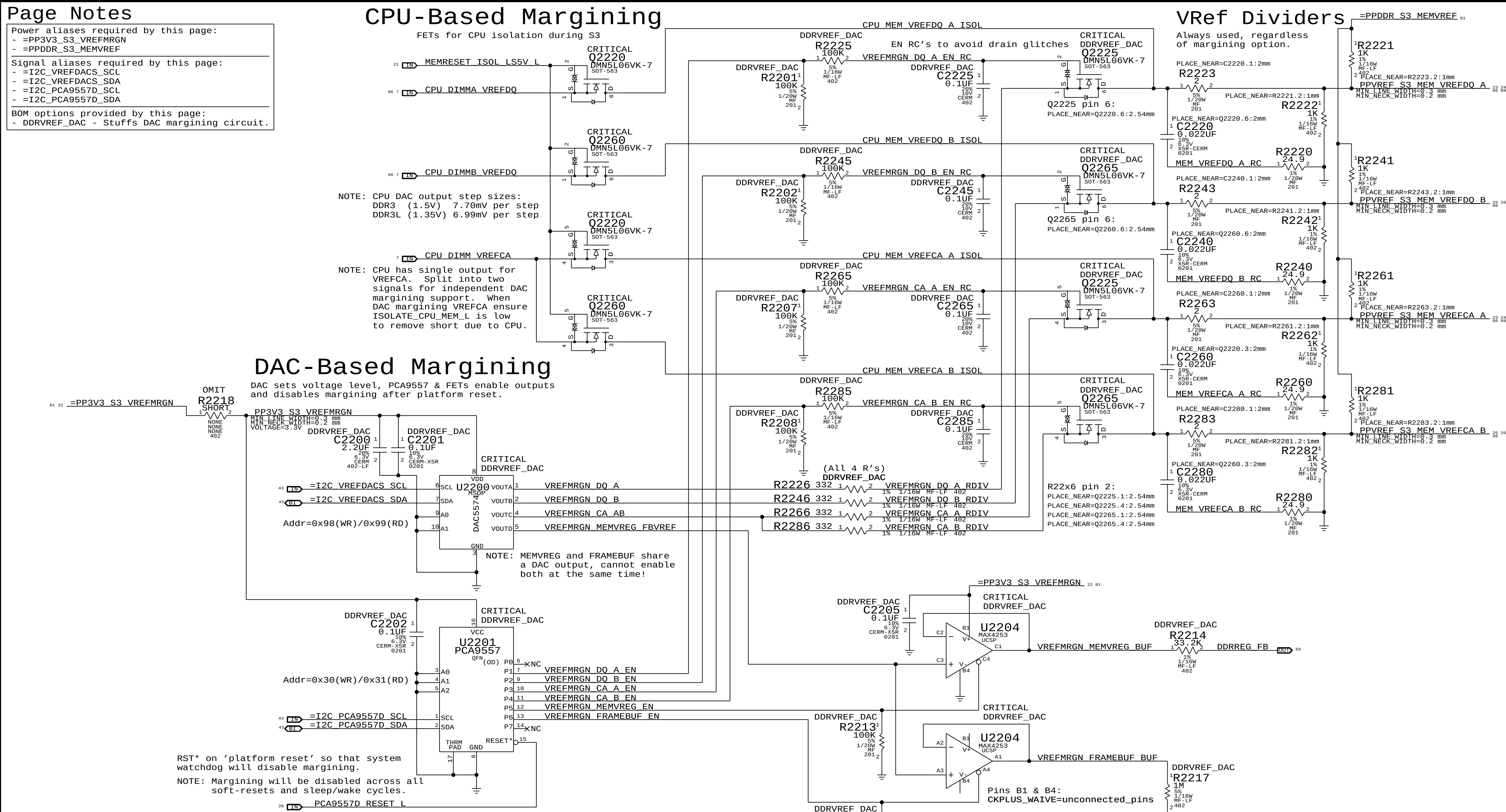
Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
- DDRVREF_DAC - Stuffs DAC margining circuit.



	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5

	DDR3 (1.5V)	DDR3L (1.35V)	DDR3 (1.5V)	DDR3L (1.35V)
Nominal value	0.750V (DAC: 0x3A = 0.747mV)	0.675V (DAC: 0x34 = 0.670mV)	1.500V (DAC: 0x74 = 1.495V)	1.343V (DAC: 0x68 = 1.341V)
Margined target:	0.300V - 1.200V (+/- 450mV)	0.275V - 1.075V (+/- 400mV)	1.200V - 1.800V (+/- 300mV)	0.950V - 1.750V (+/- 400mV)
DAC range:	0.000V - 1.508V (0x00 - 0x75)	0.000V - 1.354V (0x00 - 0x69)	0.000V - 3.004V (0x00 - 0xE9)	0.000V - 2.707V (0x00 - 0xD2)
Margined range:	0.299V - 1.206V (+/- 453mV)	0.269V - 1.083V (+/- 406mV)	1.199V - 1.801V (+/- 301mV)	0.932V - 1.760V (+/- 414mV)
VRef current:	+901uA - -911uA (- = sourced)	+811uA - -816uA (- = sourced)	+36uA - -36uA (- = sourced)	+28uA - -29uA (- = sourced)
DAC step size:	7.68mV / step @ output	7.67mV / step @ output	2.575mV / step @ output	3.923mV / step @ output

NOTE: DDR3 assumes TPS51916 supply with 10.0k/49.9k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
DDR3 VREF MARGINING

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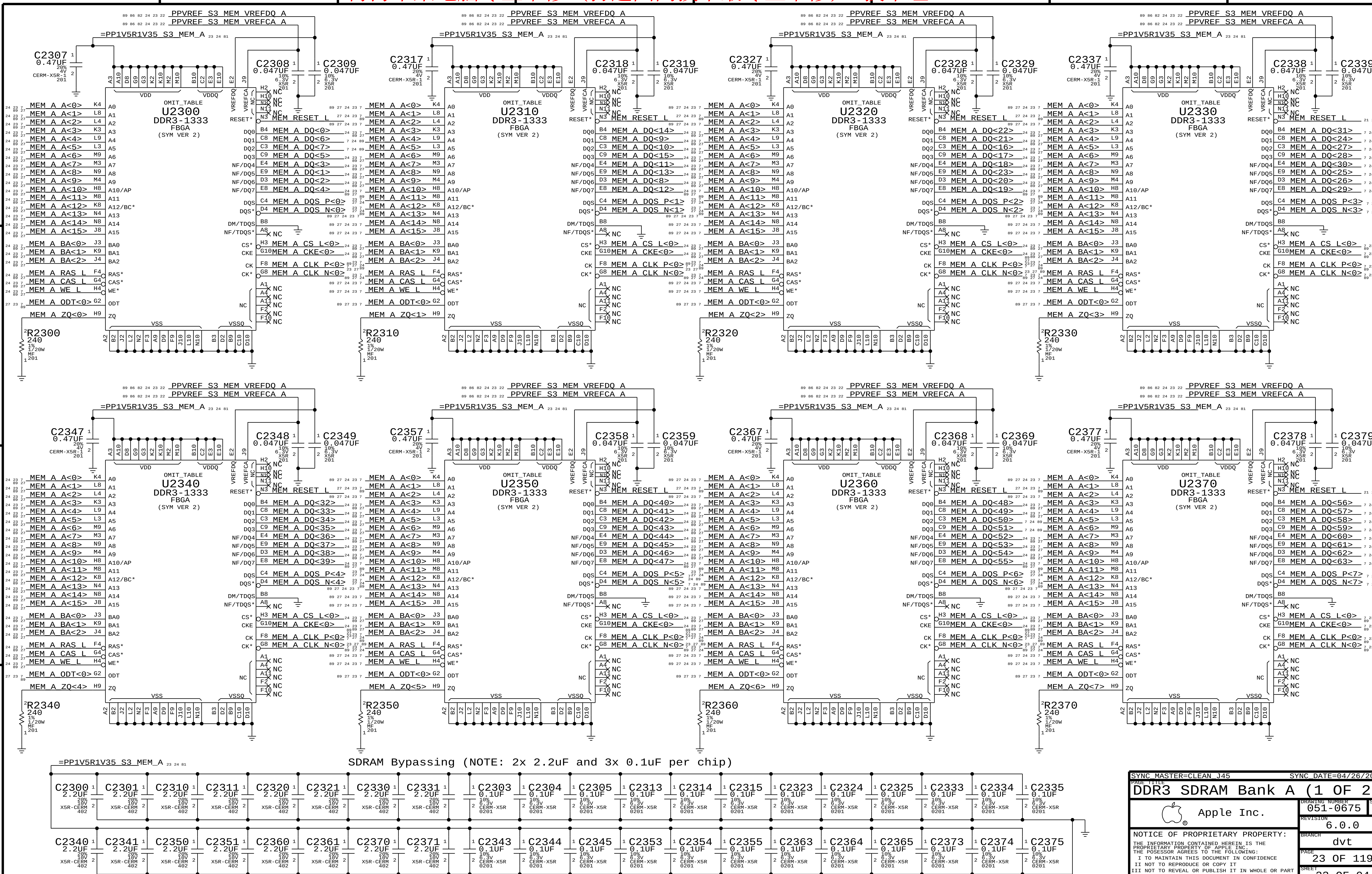

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

DDR3 SDRAM Bank A (1 OF 2)

Apple Inc.

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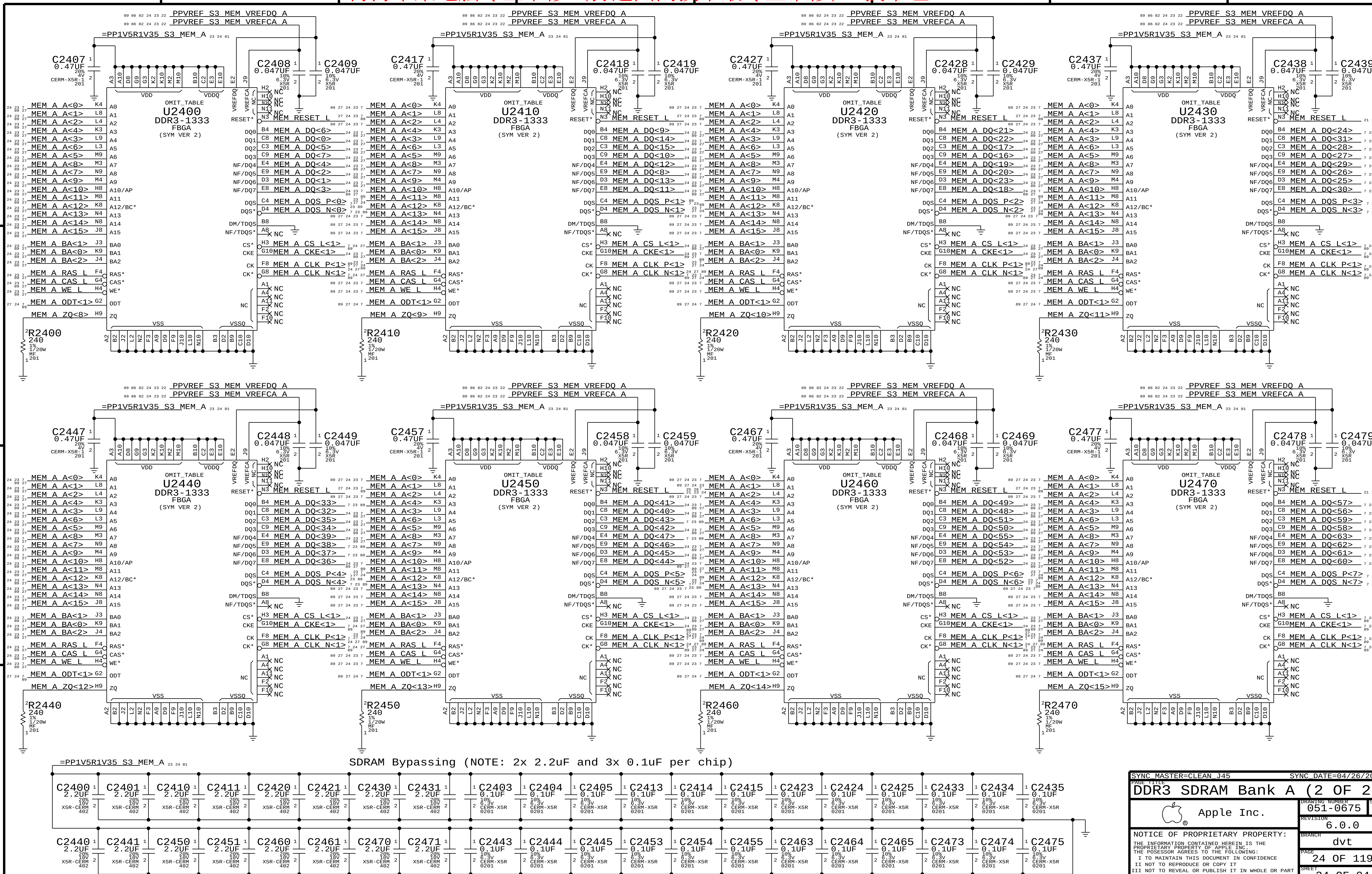

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

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DDR3 SDRAM Bank A (2 OF 2)

Apple Inc.

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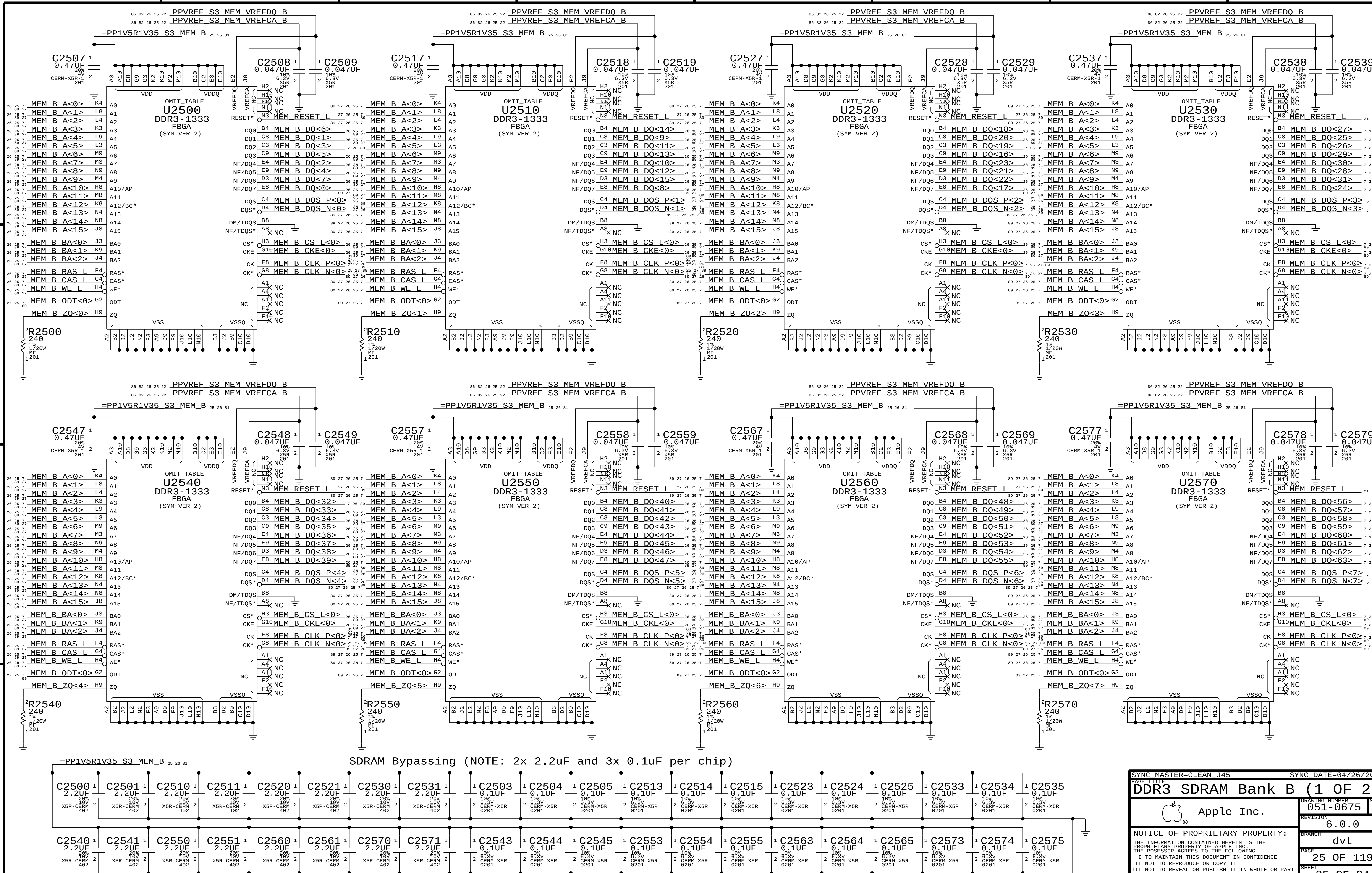
PPVREF_S3_MEM_VREFDQ_B
PPVREF_S3_MEM_VREFCA_B
=PP1V5R1V35_S3_MEM_B
OMIT_TABLE
U2500
DDR3-1333
FBGA
(SYM VER 2)
U2510
DDR3-1333
U2520
DDR3-1333
U2530
DDR3-1333
U2540
DDR3-1333
U2550
DDR3-1333
U2560
DDR3-1333
U2570
DDR3-1333
MEM_RESET_L
R2500 240 1% 1/20W MF 201
R2510
R2520
R2530
R2540
R2550
R2560
R2570
C2507 0.47UF
C2508 0.047UF
C2509 0.047UF
SDRAM Bypassing (NOTE: 2x 2.2uF and 3x 0.1uF per chip)
C2500 2.2UF
C2503 0.1UF
SYNC_MASTER=CLEAN_J45
SYNC_DATE=04/26/2013
DDR3 SDRAM Bank B (1 OF 2)
Apple Inc.
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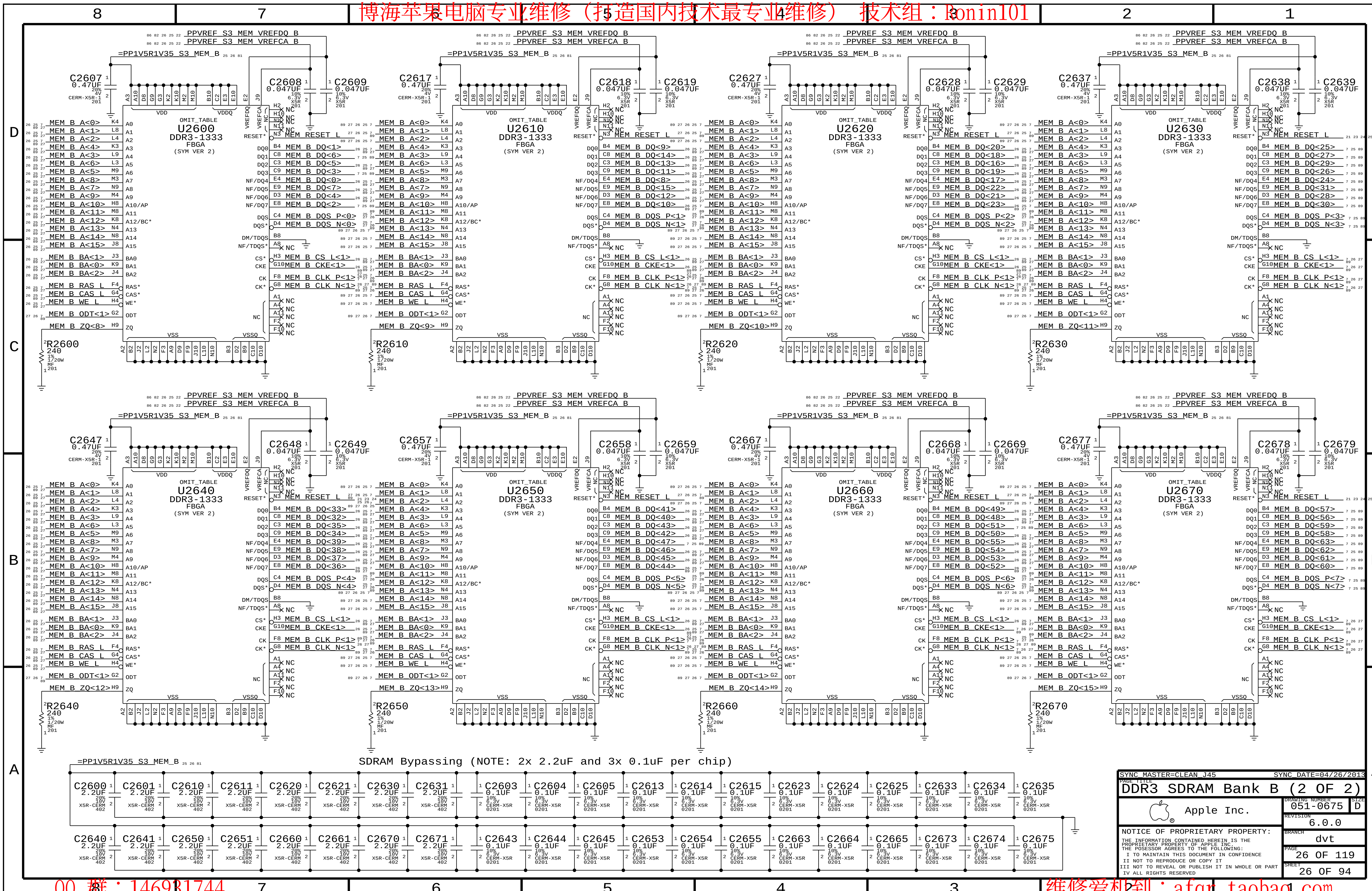
PPVREF_S3_MEM_VREFDQ_B
PPVREF_S3_MEM_VREFCA_B
=PP1V5R1V35_S3_MEM_B
OMIT_TABLE
U2600
DDR3-1333
FBGA
(SYM VER 2)
U2610
DDR3-1333
U2620
DDR3-1333
U2630
DDR3-1333
U2640
DDR3-1333
U2650
DDR3-1333
U2660
DDR3-1333
U2670
DDR3-1333
C2607 0.47UF
C2608 0.047UF
C2609 0.047UF
C2617 0.47UF
C2618 0.047UF
C2619 0.047UF
C2627 0.47UF
C2628 0.047UF
C2629 0.047UF
C2637 0.47UF
C2638 0.047UF
C2639 0.047UF
C2647 0.47UF
C2648 0.047UF
C2649 0.047UF
C2657 0.47UF
C2658 0.047UF
C2659 0.047UF
C2667 0.47UF
C2668 0.047UF
C2669 0.047UF
C2677 0.47UF
C2678 0.047UF
C2679 0.047UF
R2600 240
R2610 240
R2620 240
R2630 240
R2640 240
R2650 240
R2660 240
R2670 240
MEM_B_RESET_L
MEM_B_CS_L<1>
MEM_B_CKE<1>
MEM_B_CLK_P<1>
MEM_B_CLK_N<1>
MEM_B_RAS_L
MEM_B_CAS_L
MEM_B_WE_L
MEM_B_ODT<1>
MEM_B_ZQ<8>
MEM_B_ZQ<9>
MEM_B_ZQ<10>
MEM_B_ZQ<11>
MEM_B_ZQ<12>
MEM_B_ZQ<13>
MEM_B_ZQ<14>
MEM_B_ZQ<15>
SDRAM Bypassing (NOTE: 2x 2.2uF and 3x 0.1uF per chip)
C2600 2.2UF
C2601 2.2UF
C2610 2.2UF
C2611 2.2UF
C2620 2.2UF
C2621 2.2UF
C2630 2.2UF
C2631 2.2UF
C2603 0.1UF
C2604 0.1UF
C2605 0.1UF
C2613 0.1UF
C2614 0.1UF
C2615 0.1UF
C2623 0.1UF
C2624 0.1UF
C2625 0.1UF
C2633 0.1UF
C2634 0.1UF
C2635 0.1UF
C2640 2.2UF
C2641 2.2UF
C2650 2.2UF
C2651 2.2UF
C2660 2.2UF
C2661 2.2UF
C2670 2.2UF
C2671 2.2UF
C2643 0.1UF
C2644 0.1UF
C2645 0.1UF
C2653 0.1UF
C2654 0.1UF
C2655 0.1UF
C2663 0.1UF
C2664 0.1UF
C2665 0.1UF
C2673 0.1UF
C2674 0.1UF
C2675 0.1UF
SYNC_MASTER=CLEAN_J45
SYNC_DATE=04/26/2013
DDR3 SDRAM Bank B (2 OF 2)
Apple Inc.
DRAWING NUMBER 051-0675
SIZE D
REVISION 6.0.0
BRANCH dvt
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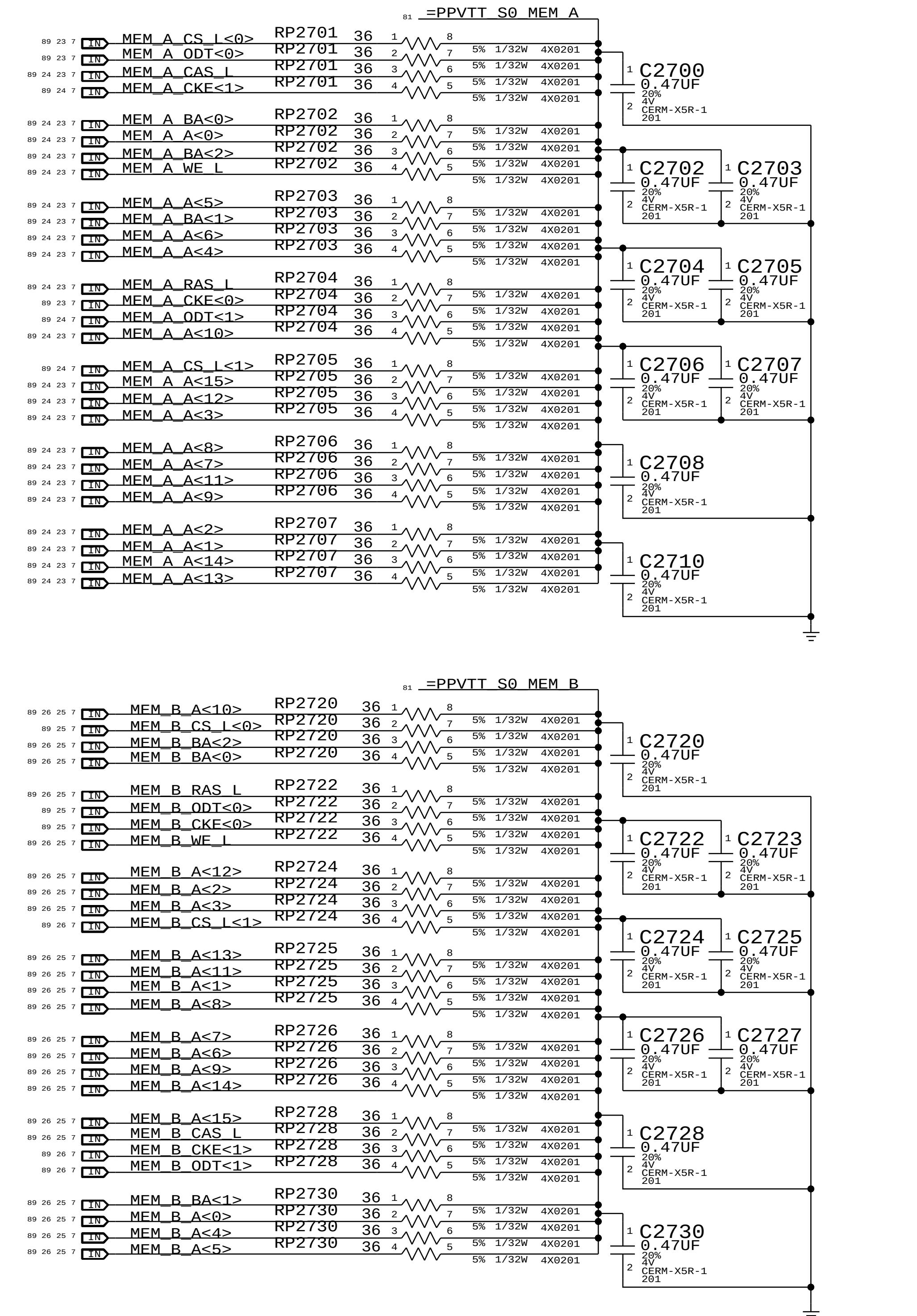
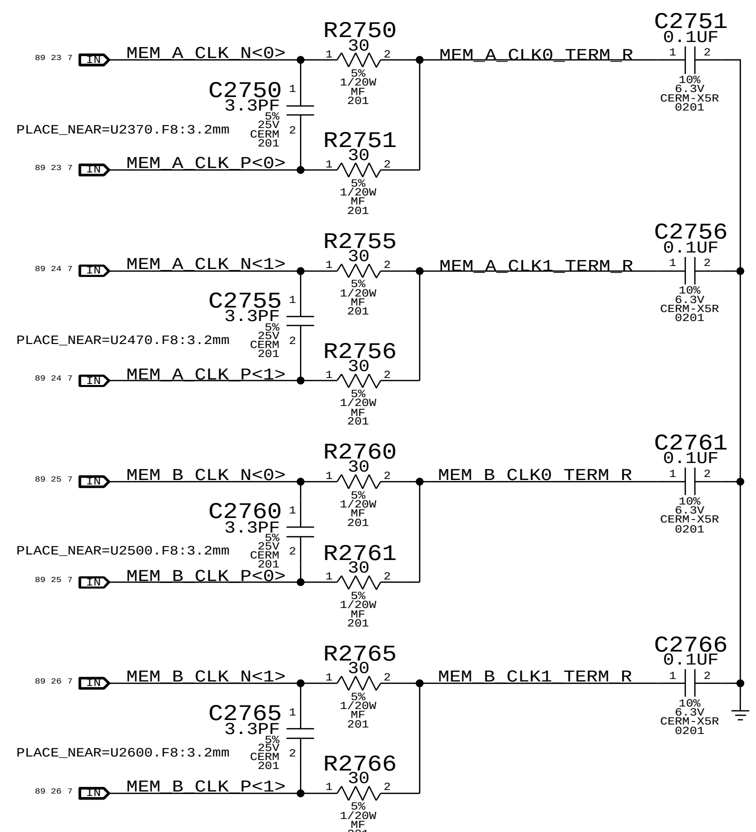
JEDEC 4.20.18 Unbuffered SODIMM Raw Card F spec recommends 36 Ohm term to VTT for CS,CKE,ODT and 36 Ohm for BA,A,RAS,CAS,WE



MEM Clock Termination

Place RC end termination after last DRAM
Place Source Cterm at neckdown at first DRAM



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

DDR3 Termination

Apple Inc.

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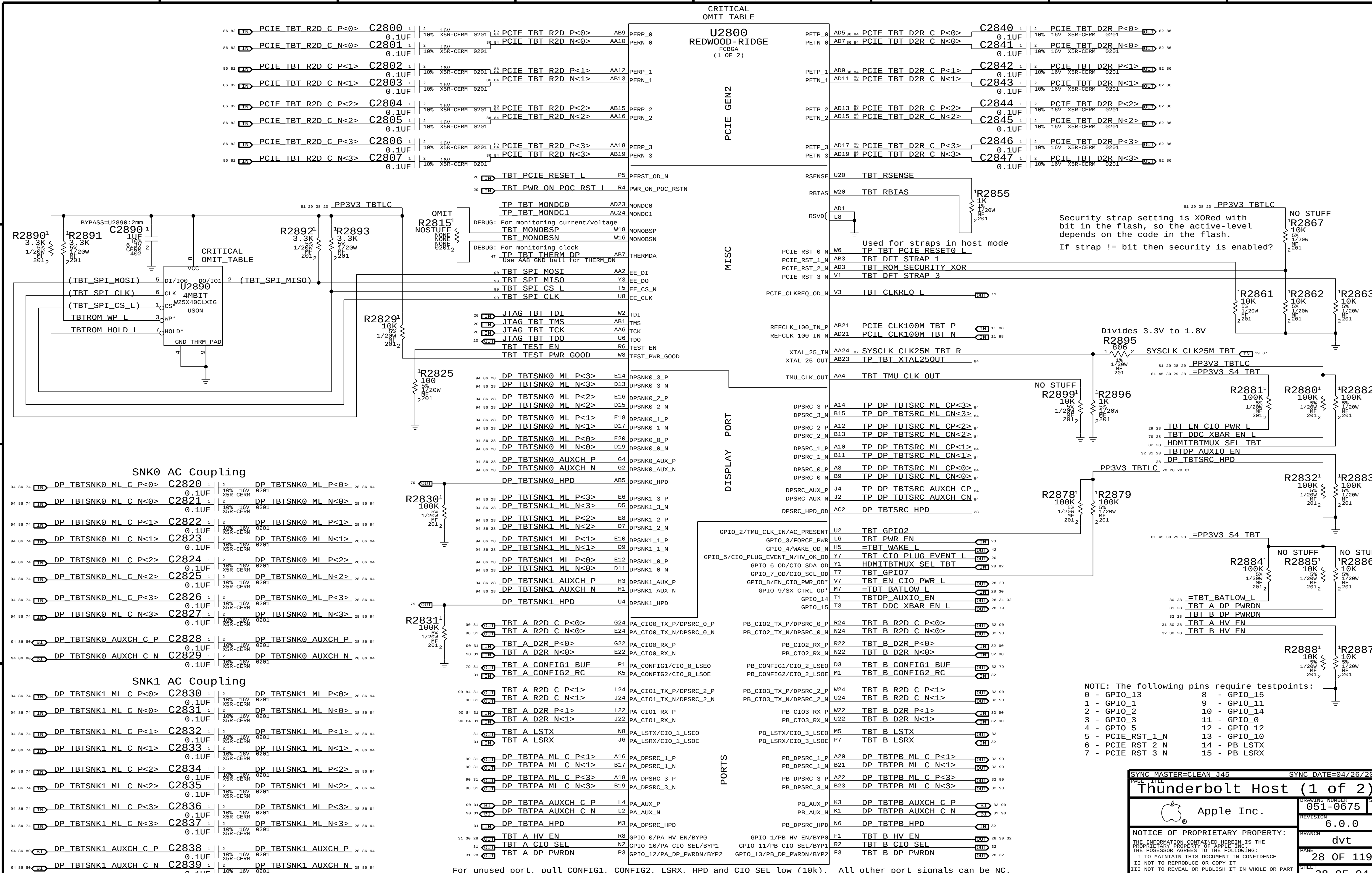
CRITICAL
OMIT_TABLE
U2800
REDWOOD-RIDGE
FCBGA
(1 OF 2)
PCIE GEN2
MISC
DISPLAY PORT
PORTS
PCIE_TBT_R2D_C_P<0>
PCIE_TBT_R2D_C_N<0>
PCIE_TBT_R2D_C_P<1>
PCIE_TBT_R2D_C_N<1>
PCIE_TBT_R2D_C_P<2>
PCIE_TBT_R2D_C_N<2>
PCIE_TBT_R2D_C_P<3>
PCIE_TBT_R2D_C_N<3>
C2800 0.1UF
C2801 0.1UF
C2802 0.1UF
C2803 0.1UF
C2804 0.1UF
C2805 0.1UF
C2806 0.1UF
C2807 0.1UF
PCIE_TBT_D2R_C_P<0>
PCIE_TBT_D2R_C_N<0>
PCIE_TBT_D2R_P<0>
PCIE_TBT_D2R_N<0>
C2840 0.1UF
C2841 0.1UF
C2842 0.1UF
C2843 0.1UF
C2844 0.1UF
C2845 0.1UF
C2846 0.1UF
C2847 0.1UF
TBT_PCIE_RESET_L
TBT_PWR_ON_POC_RST_L
TBT_RSENSE
TBT_RBIAS
R2855 1K
PP3V3_TBTLC
BYPASS=U2890:2mm
C2890 1UF
R2890 3.3K
R2891 3.3K
R2892 3.3K
R2893 3.3K
CRITICAL
OMIT_TABLE
U2890
4MBIT
W25X40CLXIG
USON
(TBT_SPI_MOSI)
(TBT_SPI_CLK)
(TBT_SPI_CS_L)
(TBT_SPI_MISO)
TBTROM_WP_L
TBTROM_HOLD_L
OMIT
R2815
NOSTUFF
DEBUG: For monitoring current/voltage
TP_TBT_MONDC0
TP_TBT_MONDC1
TBT_MONOBSP
TBT_MONOBSN
DEBUG: For monitoring clock
TP_TBT_THERM_DP
Use AA8 GND ball for THERM_DN
TBT_SPI_MOSI
TBT_SPI_MISO
TBT_SPI_CS_L
TBT_SPI_CLK
JTAG_TBT_TDI
JTAG_TBT_TMS
JTAG_TBT_TCK
JTAG_TBT_TDO
TBT_TEST_EN
TBT_TEST_PWR_GOOD
R2829 10K
R2825 100
Used for straps in host mode
TP_TBT_PCIE_RESET0_L
TBT_DFT_STRAP_1
TBT_ROM_SECURITY_XOR
TBT_DFT_STRAP_3
TBT_CLKREQ_L
Security strap setting is XORed with bit in the flash, so the active-level depends on the code in the flash.
If strap != bit then security is enabled?
NO STUFF
R2867 10K
R2861 10K
R2862 10K
R2863 10K
PCIE_CLK100M_TBT_P
PCIE_CLK100M_TBT_N
Divides 3.3V to 1.8V
R2895 806
SYSCLK_CLK25M_TBT_R
SYSCLK_CLK25M_TBT
TP_TBT_XTAL25OUT
TBT_TMU_CLK_OUT
=PP3V3_S4_TBT
NO STUFF
R2899 10K
R2896 1K
R2881 100K
R2880 100K
R2882 100K
TBT_EN_CIO_PWR_L
TBT_DDC_XBAR_EN_L
HDMITBTMUX_SEL_TBT
TBTDP_AUXIO_EN
DP_TBTSRC_HPD
R2832 100K
R2883 100K
R2878 100K
R2879 100K
SNK0 AC Coupling
SNK1 AC Coupling
DP_TBTSNK0_ML_C_P<0>
DP_TBTSNK0_ML_P<0>
C2820 0.1UF
C2821 0.1UF
C2822 0.1UF
C2823 0.1UF
C2824 0.1UF
C2825 0.1UF
C2826 0.1UF
C2827 0.1UF
C2828 0.1UF
C2829 0.1UF
DP_TBTSNK0_AUXCH_C_P
DP_TBTSNK0_AUXCH_C_N
C2830 0.1UF
C2831 0.1UF
C2832 0.1UF
C2833 0.1UF
C2834 0.1UF
C2835 0.1UF
C2836 0.1UF
C2837 0.1UF
C2838 0.1UF
C2839 0.1UF
DP_TBTSNK1_AUXCH_C_P
DP_TBTSNK1_AUXCH_C_N
DP_TBTSNK0_HPD
DP_TBTSNK1_HPD
R2830 100K
R2831 100K
TP_DP_TBTSRC_ML_CP<3>
TP_DP_TBTSRC_AUXCH_CP
TP_DP_TBTSRC_AUXCH_CN
TBT_GPIO2
TBT_PWR_EN
=TBT_WAKE_L
TBT_CIO_PLUG_EVENT_L
HDMITBTMUX_SEL_TBT
TBT_GPIO7
TBT_EN_CIO_PWR_L
=TBT_BATLOW_L
TBTDP_AUXIO_EN
TBT_DDC_XBAR_EN_L
NO STUFF
R2884 100K
R2885 10K
R2886 10K
TBT_A_DP_PWRDN
TBT_B_DP_PWRDN
TBT_A_HV_EN
TBT_B_HV_EN
R2888 10K
R2887 10K
TBT_A_R2D_C_P<0>
TBT_A_D2R_P<0>
TBT_A_CONFIG1_BUF
TBT_A_CONFIG2_RC
TBT_A_LSTX
TBT_A_LSRX
DP_TBTPA_ML_C_P<1>
DP_TBTPA_AUXCH_C_P
DP_TBTPA_HPD
TBT_A_CIO_SEL
TBT_B_R2D_C_P<0>
TBT_B_CONFIG1_BUF
TBT_B_CONFIG2_RC
TBT_B_LSTX
TBT_B_LSRX
DP_TBTPB_ML_C_P<1>
DP_TBTPB_AUXCH_C_P
DP_TBTPB_HPD
TBT_B_CIO_SEL
For unused port, pull CONFIG1, CONFIG2, LSRX, HPD and CIO_SEL low (10k). All other port signals can be NC.
NOTE: The following pins require testpoints:
0 - GPIO_13   8 - GPIO_15
1 - GPIO_1   9 - GPIO_11
2 - GPIO_2   10 - GPIO_14
3 - GPIO_3   11 - GPIO_0
4 - GPIO_5   12 - GPIO_12
5 - PCIE_RST_1_N   13 - GPIO_10
6 - PCIE_RST_2_N   14 - PB_LSTX
7 - PCIE_RST_3_N   15 - PB_LSRX
SYNC_MASTER=CLEAN_J45   SYNC_DATE=04/26/2013
PAGE TITLE
Thunderbolt Host (1 of 2)
Apple Inc.
DRAWING NUMBER 051-0675   SIZE D
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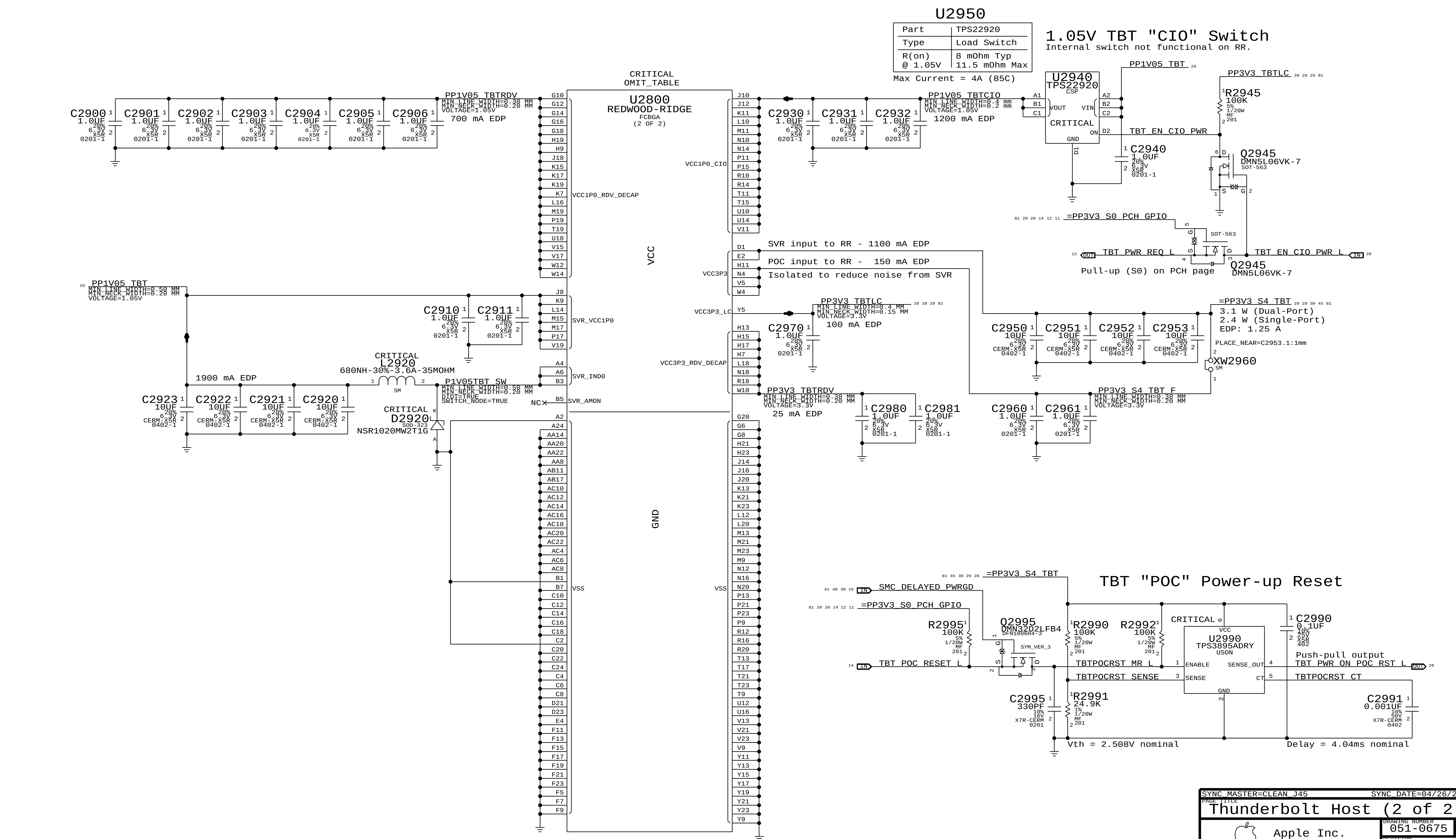

EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER: 051-0675 SIZE: D

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BRANCH: dvt

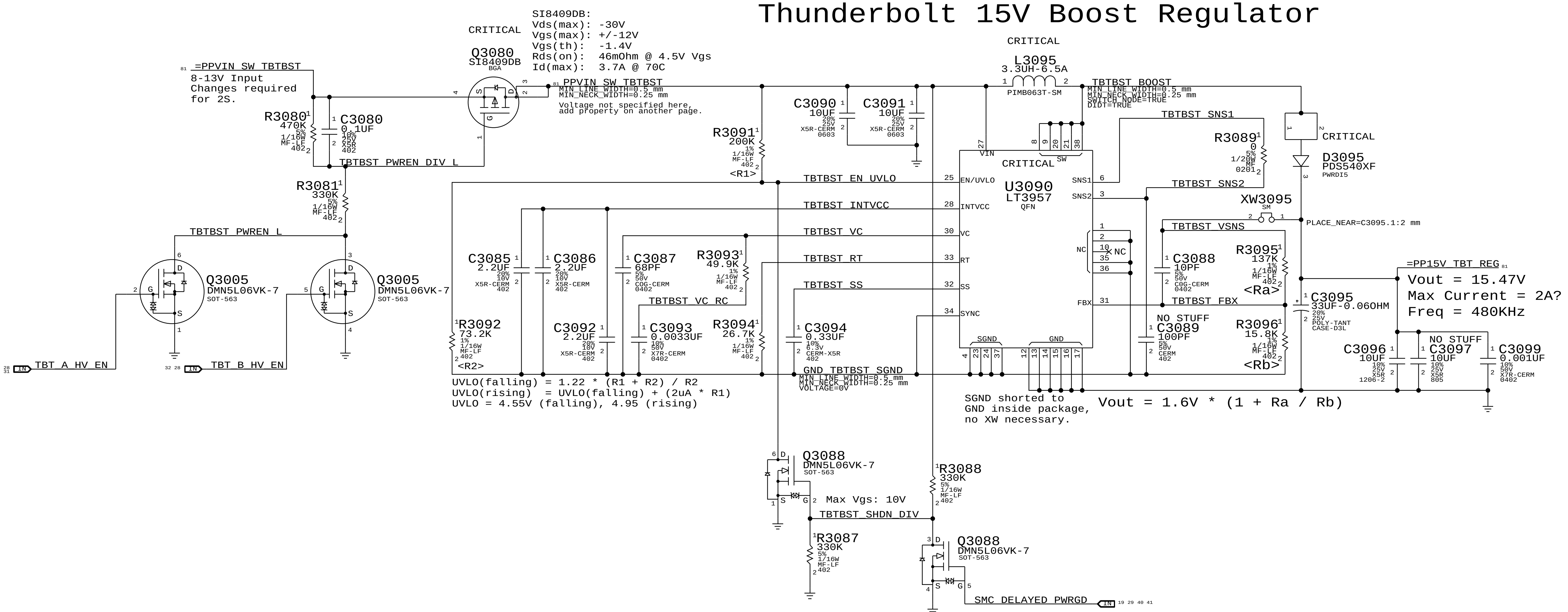
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)

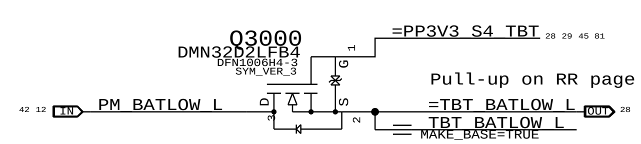
Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

Thunderbolt 15V Boost Regulator



BATLOW# Isolation



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: Thunderbolt Mobile Support

Apple Inc.

DRAWING NUMBER: 051-0675 SIZE: D

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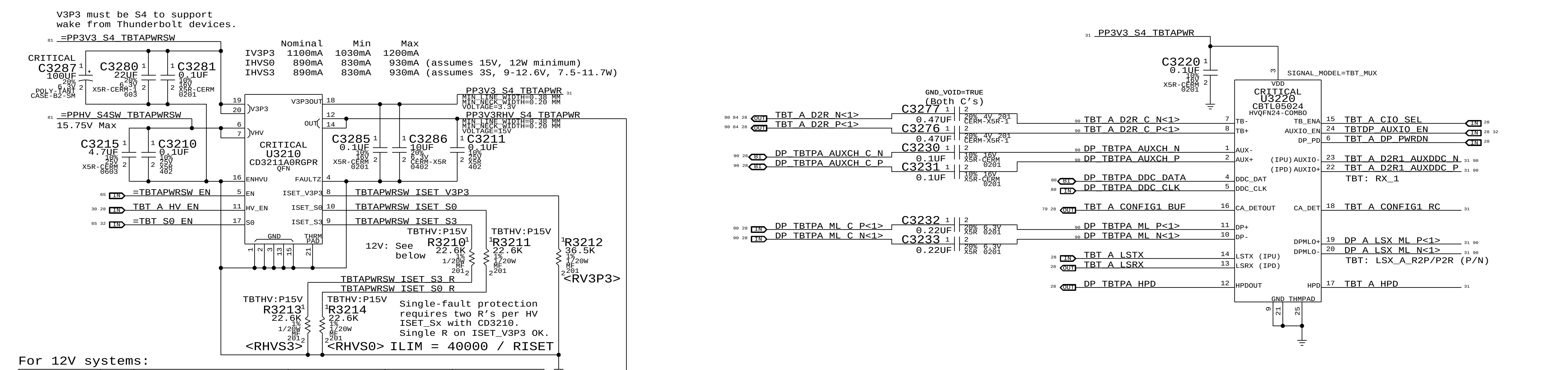
3.3V/HV Power MUX

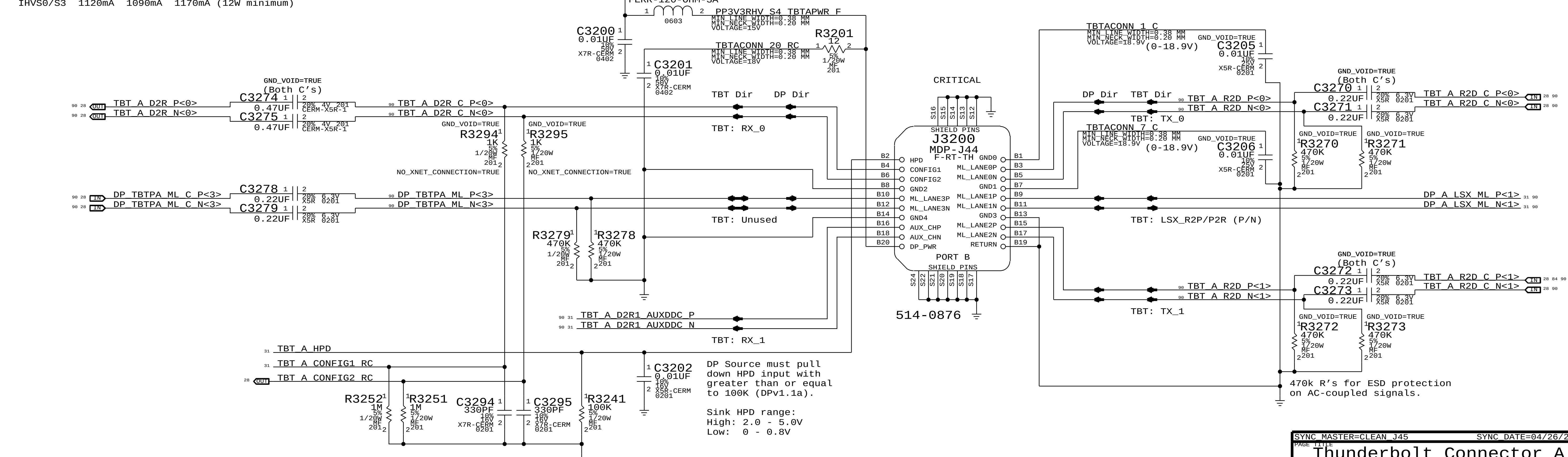


For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3210,R3213		TBTHV:P12V
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3211,R3214		TBTHV:P12V

Nominal Min Max
IHVS0/S3 1120mA 1090mA 1170mA (12W minimum)

Thunderbolt Connector A



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: Thunderbolt Connector A

Apple Inc.

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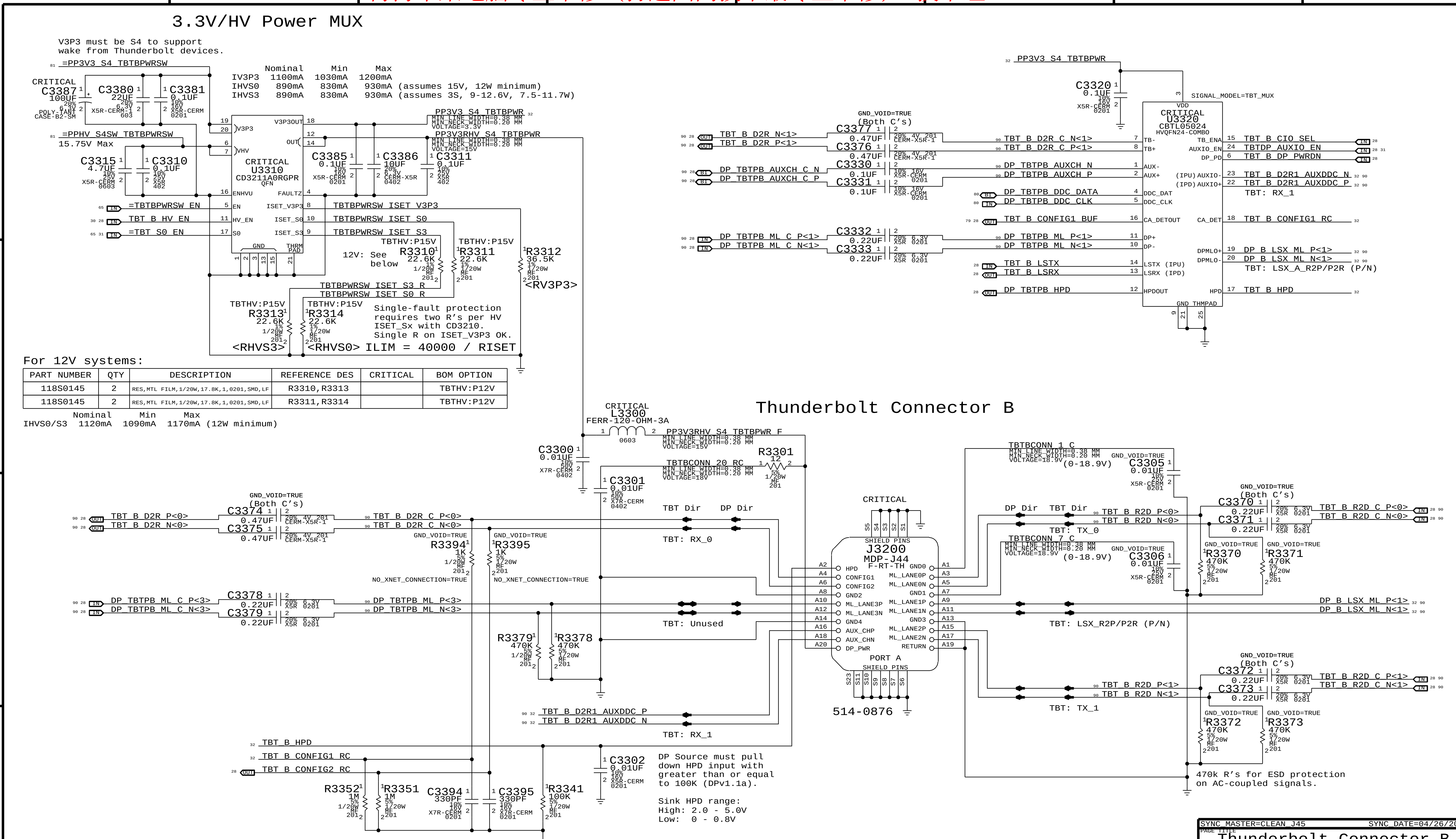
3.3V/HV Power MUX

V3P3 must be S4 to support
wake from Thunderbolt devices.

	Nominal	Min	Max	
IV3P3	1100mA	1030mA	1200mA	
IHVS0	890mA	830mA	930mA	(assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA	(assumes 3S, 9-12.6V, 7.5-11.7W)



For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3310,R3313		TBTHV:P12V
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3311,R3314		TBTHV:P12V

	Nominal	Min	Max	
IHVS0/S3	1120mA	1090mA	1170mA	(12W minimum)

Thunderbolt Connector B

DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

470k R's for ESD protection on AC-coupled signals.

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: Thunderbolt Connector B

Apple Inc.

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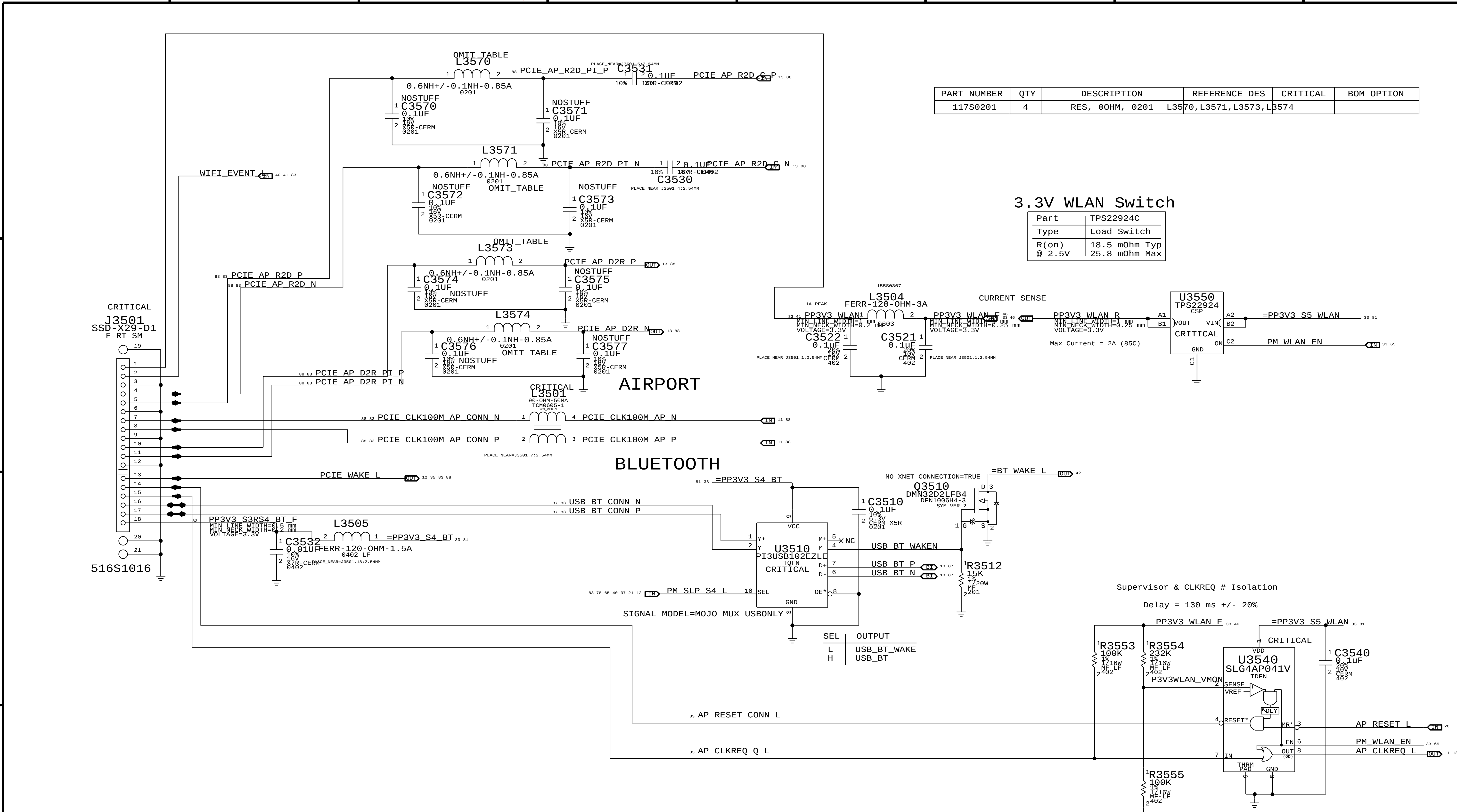

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0201	4	RES, 0OHM, 0201	L3570,L3571,L3573,L3574		

3.3V WLAN Switch

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: X29C CONNECTOR

Apple Inc.

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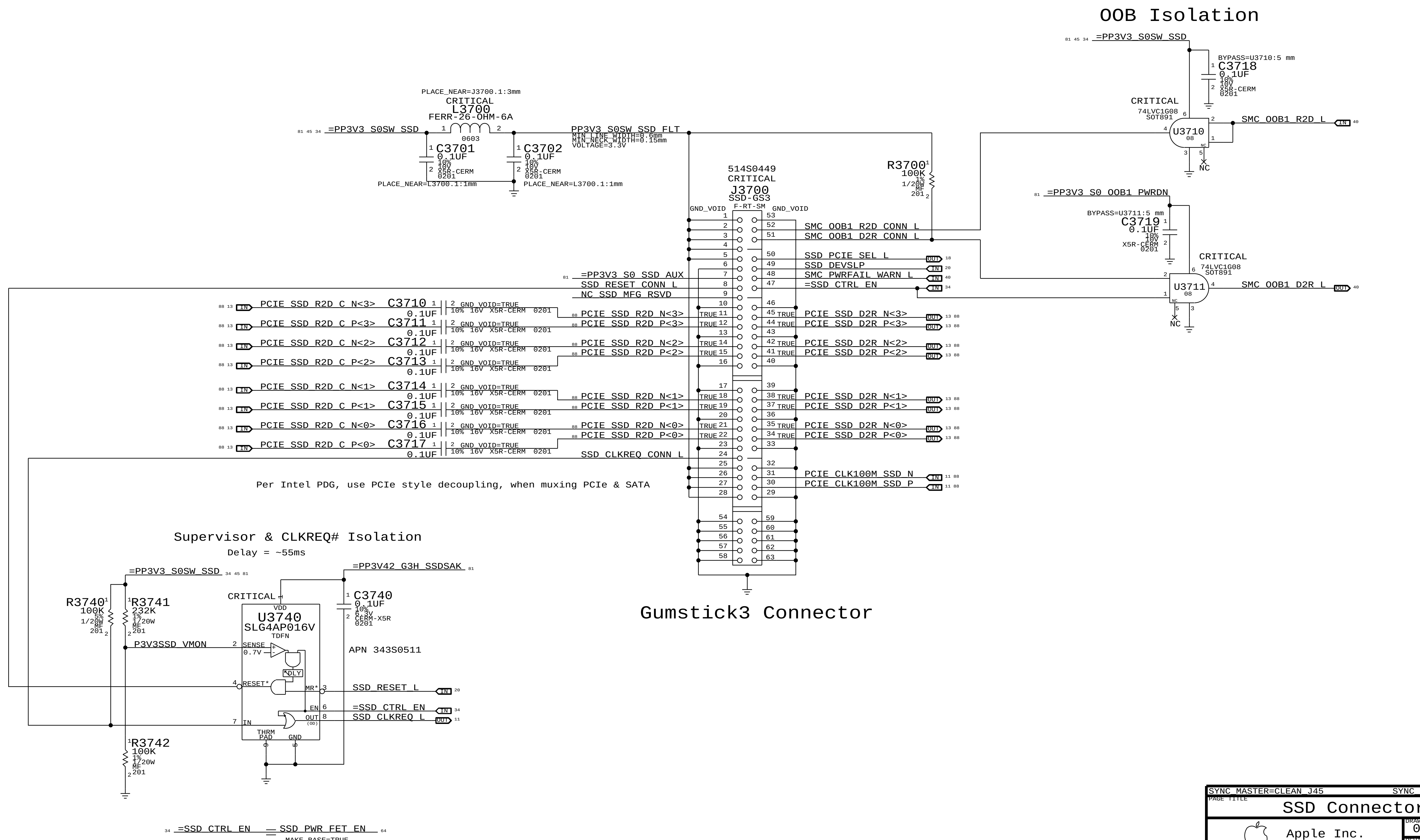

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SSD Connector

Apple Inc.

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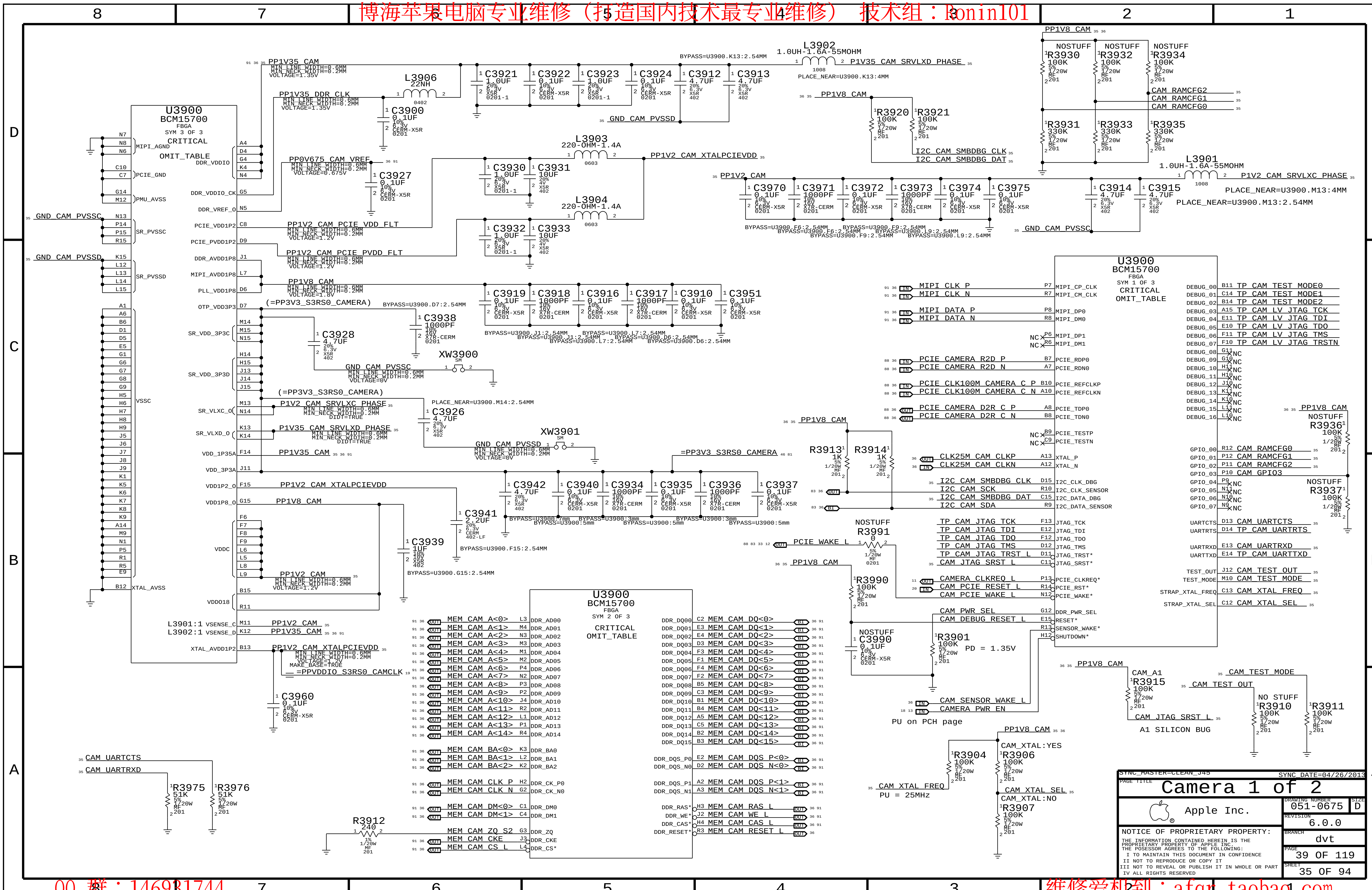
U3900
BCM15700
FBGA
SYM 3 OF 3
CRITICAL
OMIT_TABLE
U3900
BCM15700
FBGA
SYM 1 OF 3
CRITICAL
OMIT_TABLE
U3900
BCM15700
FBGA
SYM 2 OF 3
CRITICAL
OMIT_TABLE
PP1V35_CAM
PP1V35_DDR_CLK
PP0V675_CAM_VREF
PP1V2_CAM_PCIE_VDD_FLT
PP1V2_CAM_PCIE_PVDD_FLT
PP1V8_CAM
PP1V2_CAM_XTALPCIEVDD
P1V35_CAM_SRVLXD_PHASE
P1V2_CAM_SRVLXC_PHASE
GND_CAM_PVSSC
GND_CAM_PVSSD
=PP3V3_S3RS0_CAMERA
=PPVDDIO_S3RS0_CAMCLK
PD = 1.35V
A1 SILICON BUG
PU on PCH page
PU = 25MHz
CAM_XTAL:YES
CAM_XTAL:NO
L3901:1 VSENSE_C
L3902:1 VSENSE_D
SYNC_MASTER=CLEAN_J45
SYNC_DATE=04/26/2013
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Camera 1 of 2
Apple Inc.
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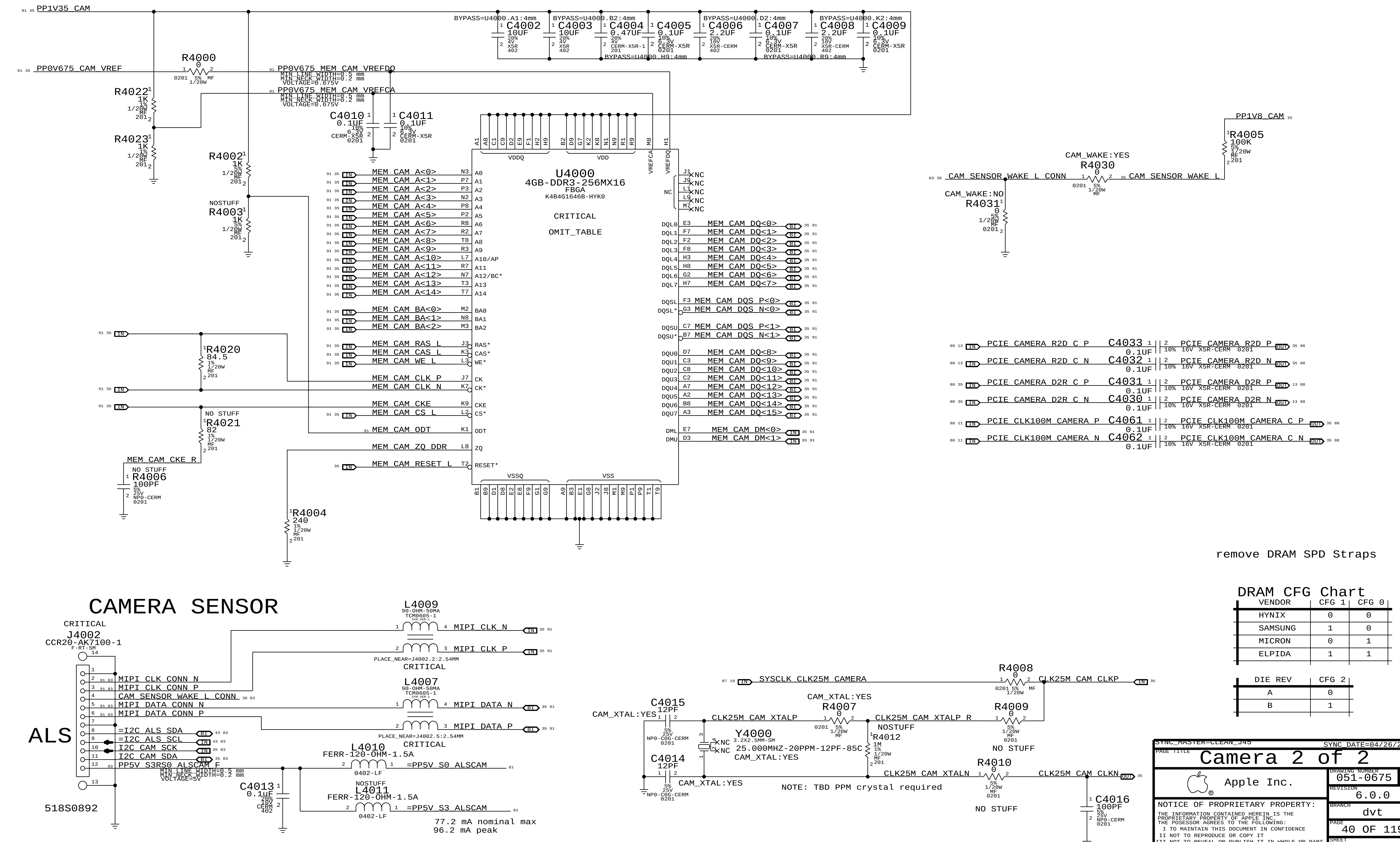

DRAM CFG Chart

VENDOR	CFG 1	CFG 0
HYNIX	0	0
SAMSUNG	1	0
MICRON	0	1
ELPIDA	1	1

DIE REV	CFG 2
A	0
B	1

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: Camera 2 of 2

Apple Inc.

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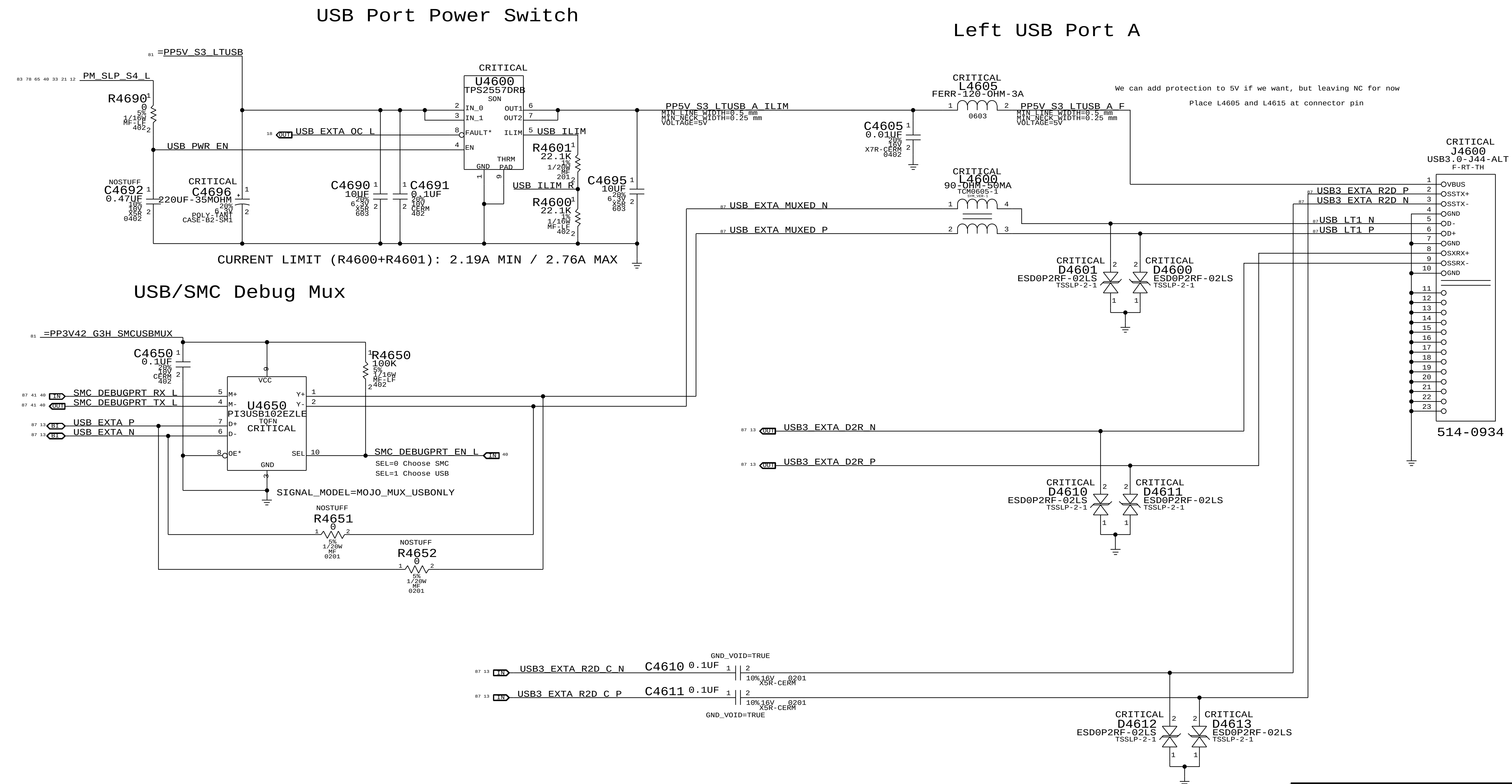

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: USB 3.0 CONNECTORS

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BRANCH: dvt

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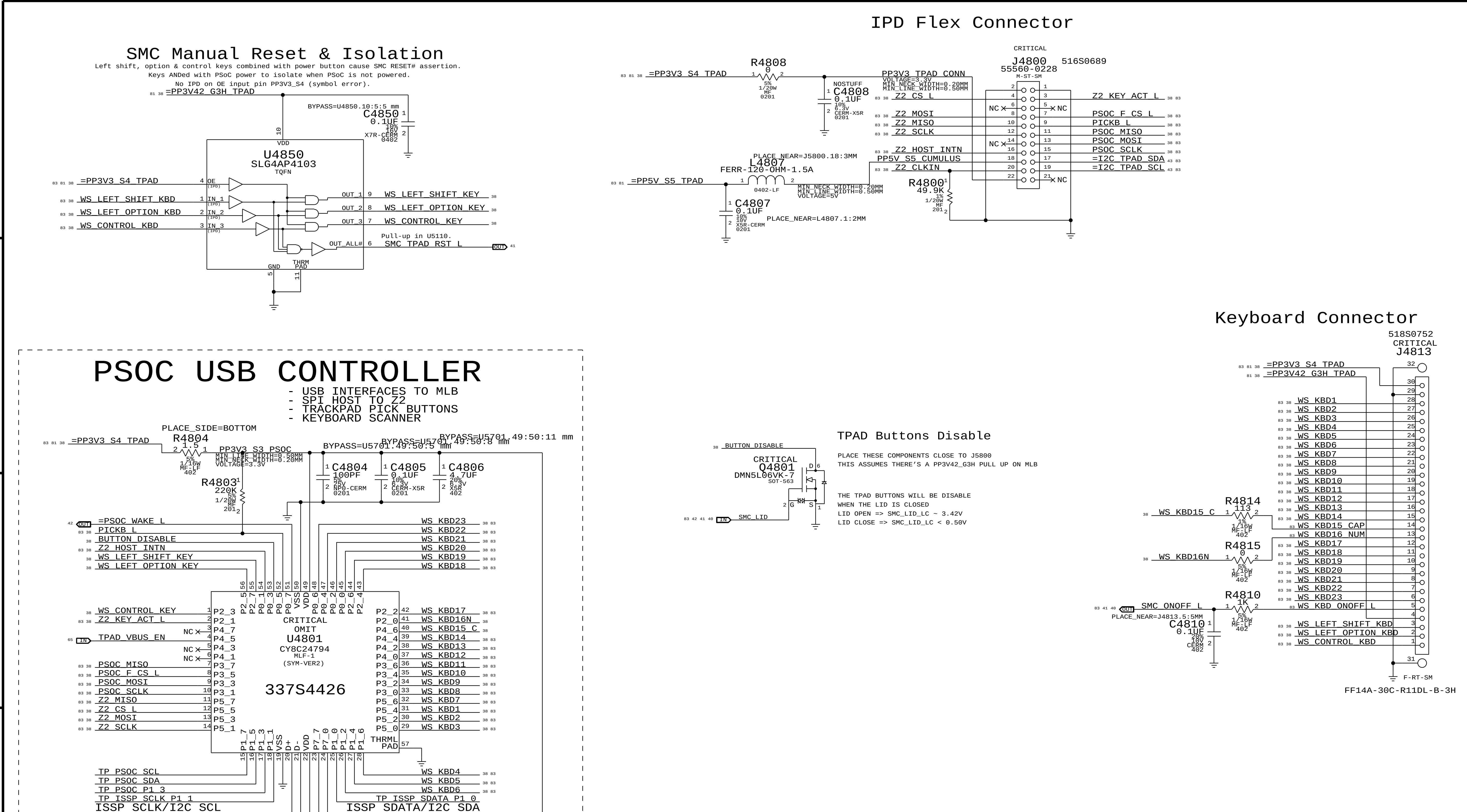

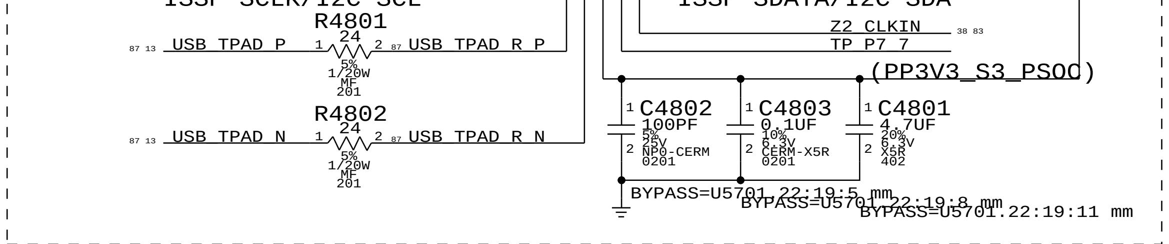

IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA	2.55 KOHM	0.0255 V	0.255E-6 W
		80UA		0.204 V	16.32E-6 W
3V3 LDO	VDD	60MA (MAX)	10 OHM	0.6 V	36E-3 W
	VOUT	60MA (MAX)	0.2 OHM	0.012 V	0.72E-3 W
PSOC	VDD	8MA (TYP)	1.5 OHM	0.012 V	96E-6 W
		14MA (MAX)		0.021 V	294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
KEYBOARD/TRACKPAD (1 OF 2)

Apple Inc.

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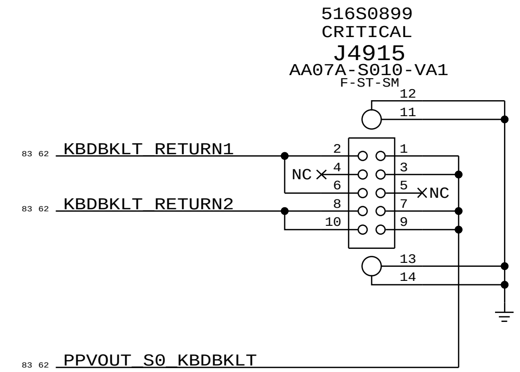
Keyboard Backlight Connector



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
KEYBOARD/TRACKPAD (2 OF 2)

Apple Inc.

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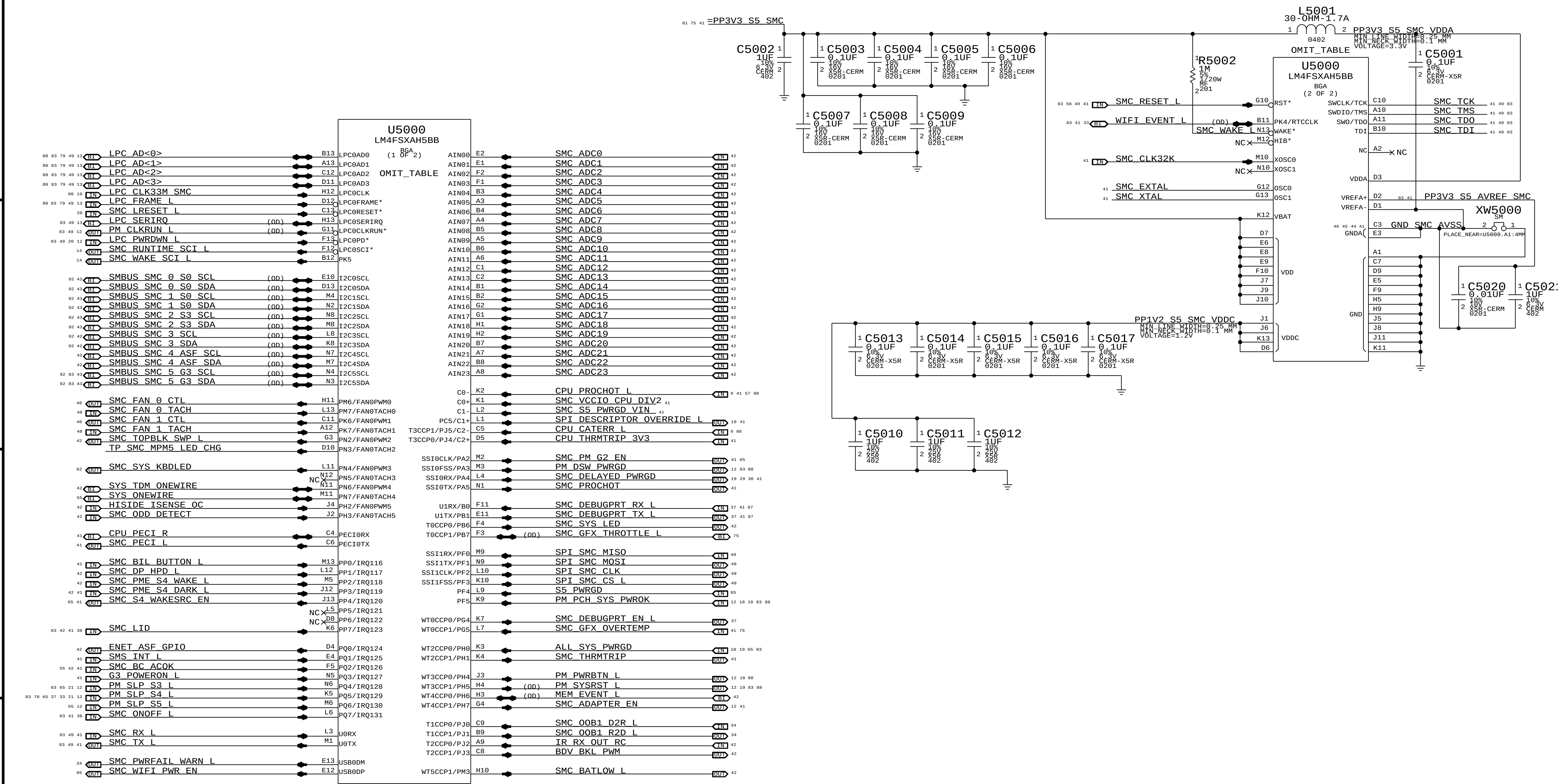
NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.



NOTE: SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: SMC

Apple Inc.

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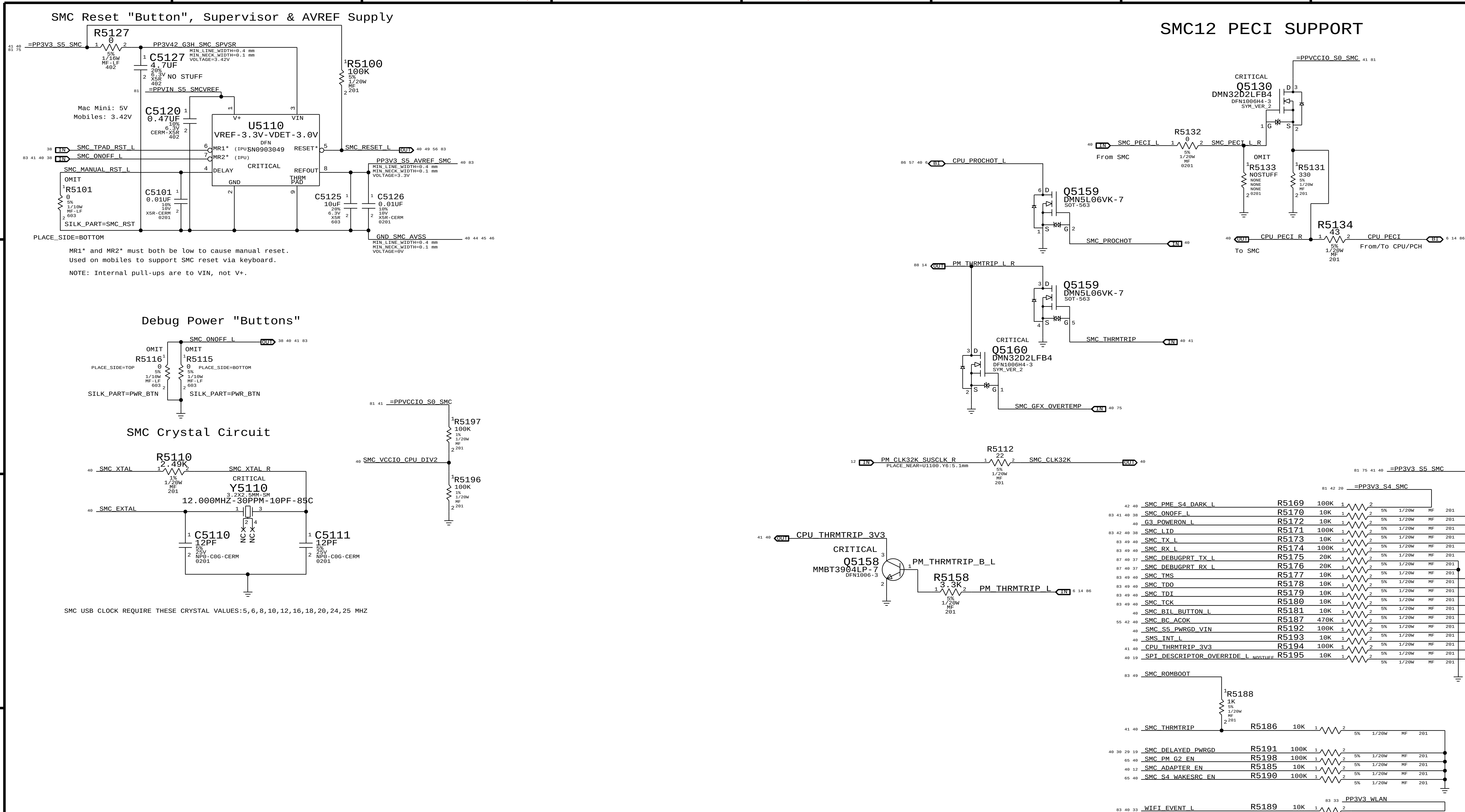

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

SMC Shared Support

Apple Inc.

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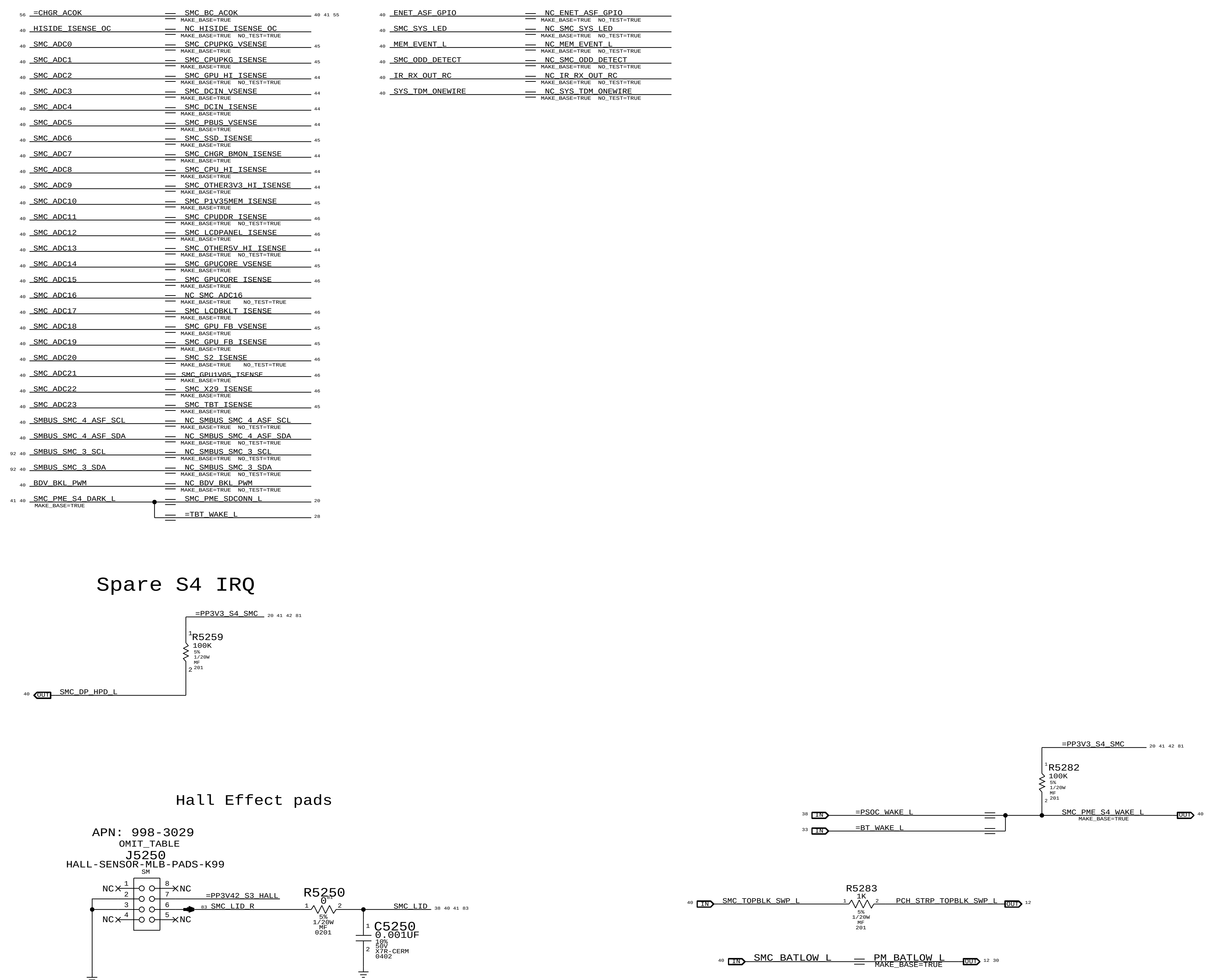

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
607-6811	1	SUBASSY,PCBA HALL EFFECT,K99	J5250	CRITICAL	

PAGE TITLE
SMC Project Support

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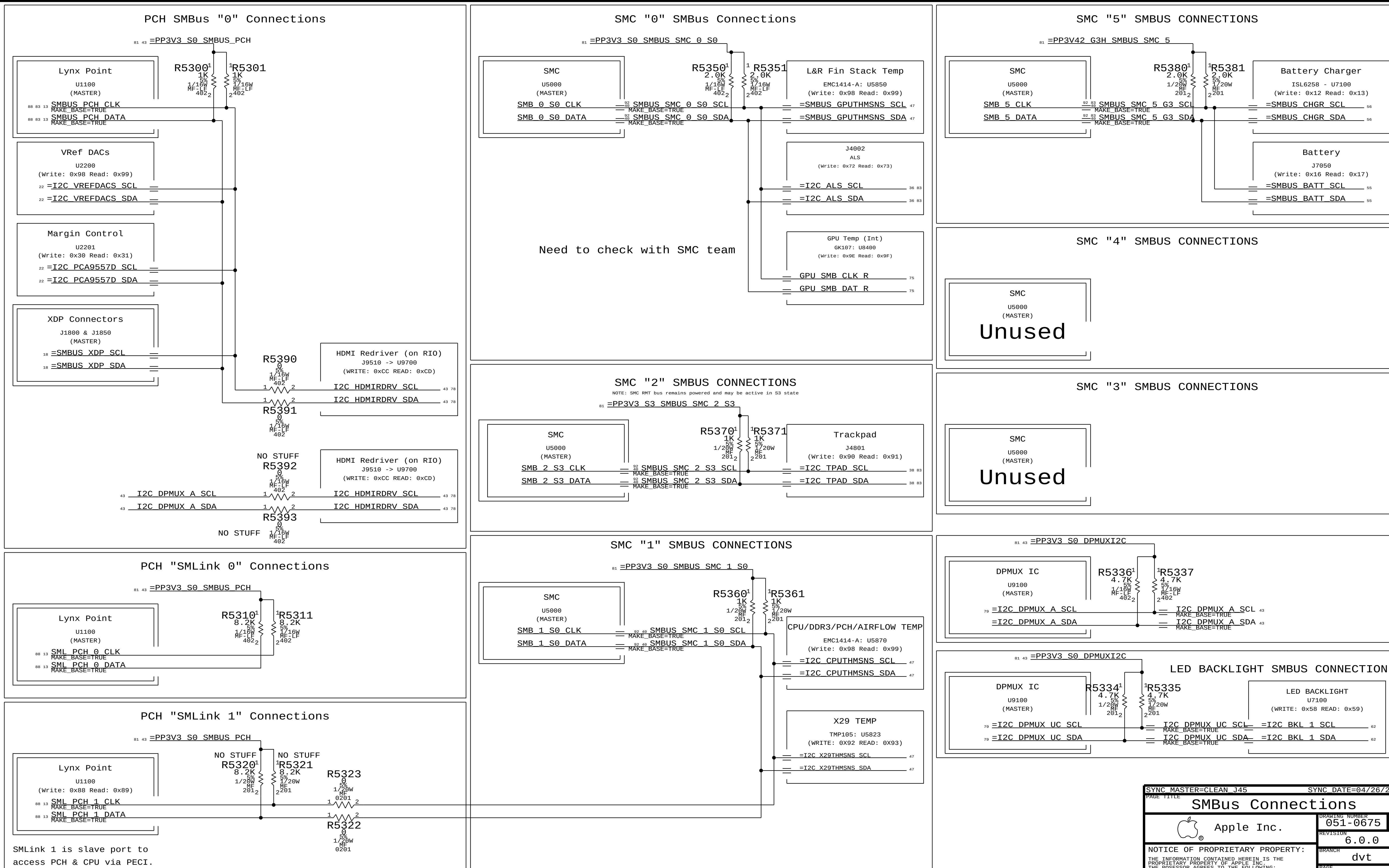

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: SMBus Connections

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DRAWING NUMBER: 051-0675 SIZE: D

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BRANCH: dvt

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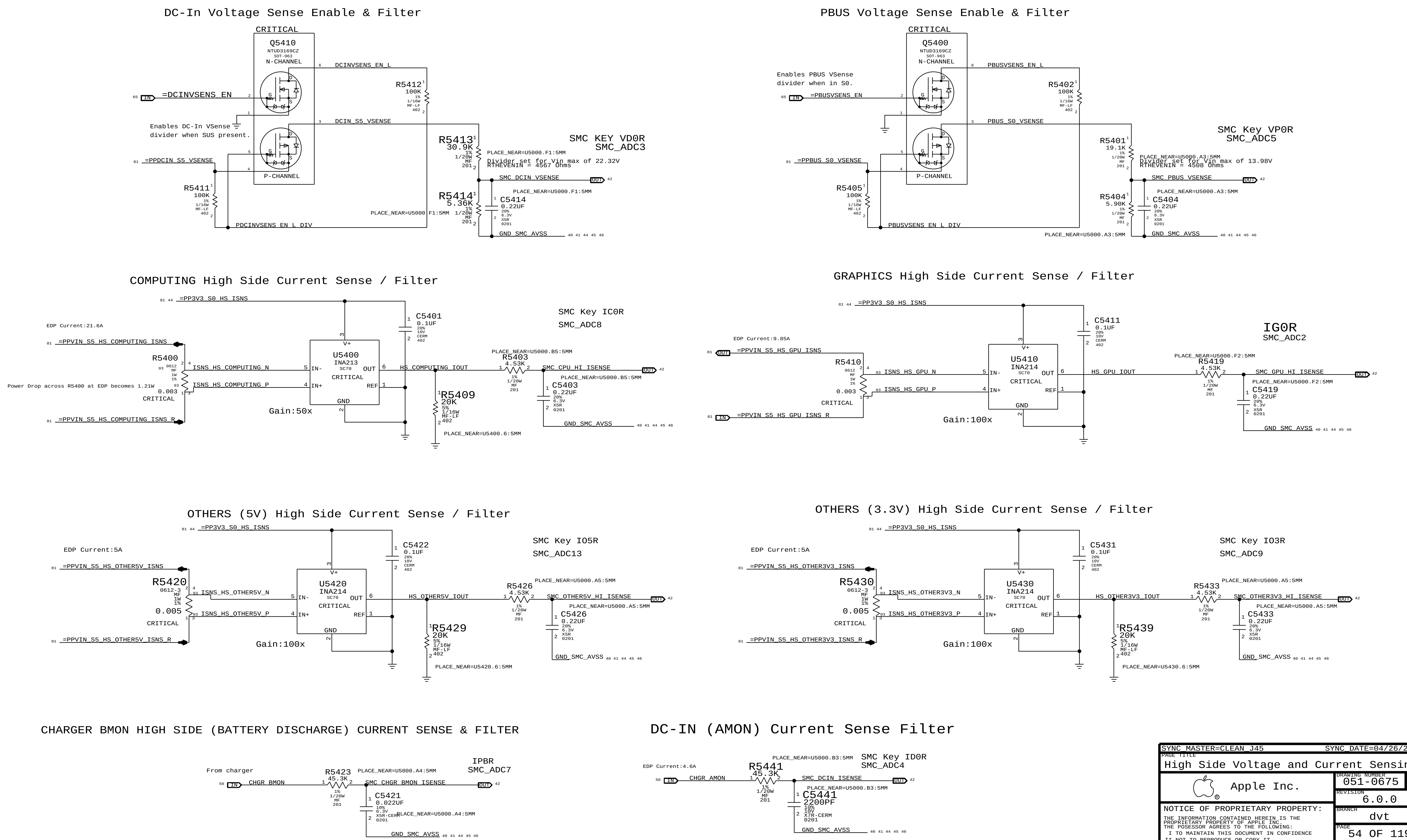


SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
High Side Voltage and Current Sensing

Apple Inc.

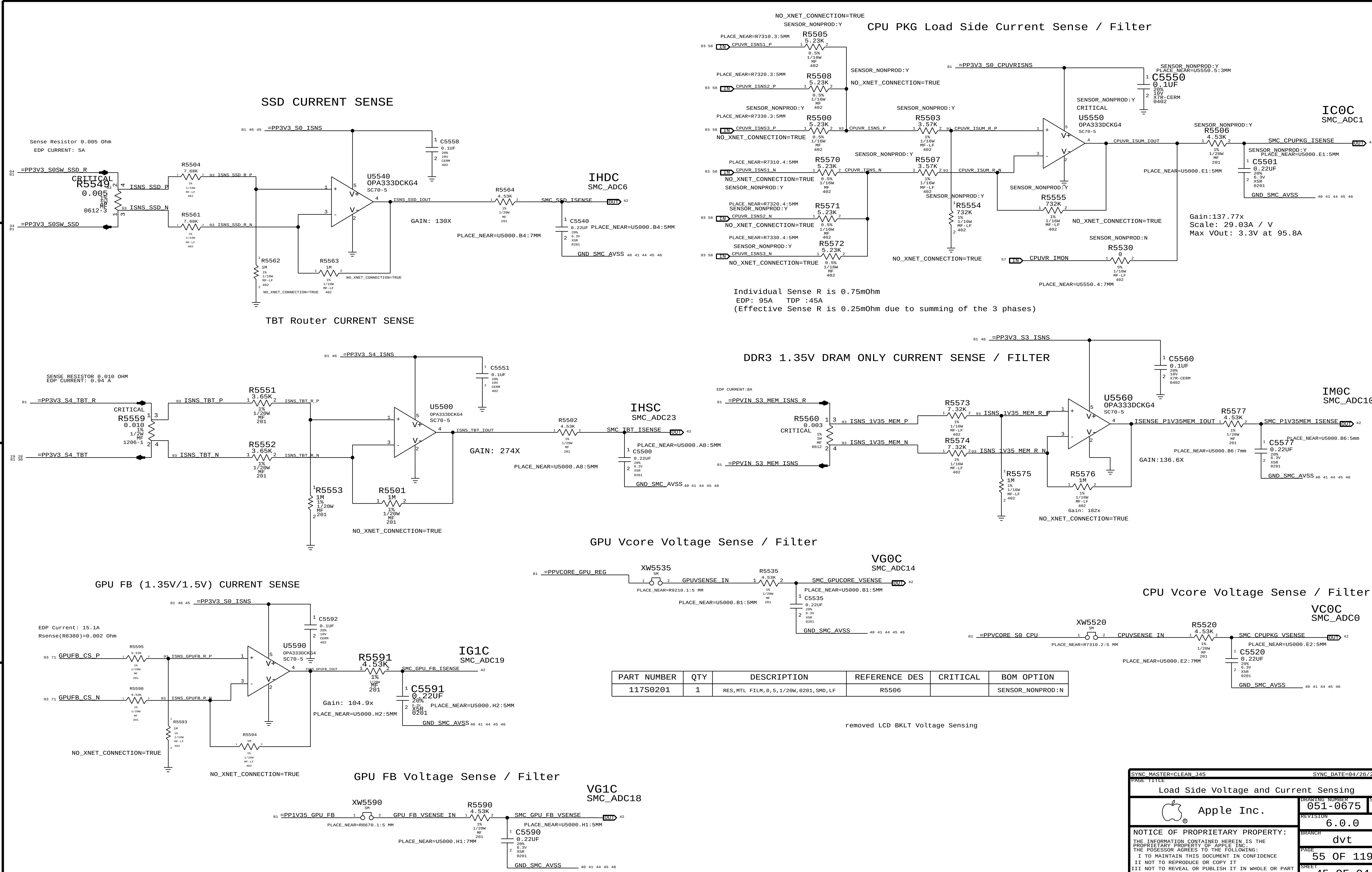
DRAWING NUMBER 051-0675 SIZE D

REVISION 6.0.0

BRANCH dvt

PAGE 54 OF 119

SHEET 44 OF 94



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0201	1	RES,MTL FILM,0,5,1/20W,0201,SMD,LF	R5506		SENSOR_NONPROD:N

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: Load Side Voltage and Current Sensing

Apple Inc.

DRAWING NUMBER: 051-0675 SIZE: D

REVISION: 6.0.0

BRANCH: dvt

PAGE: 55 OF 119

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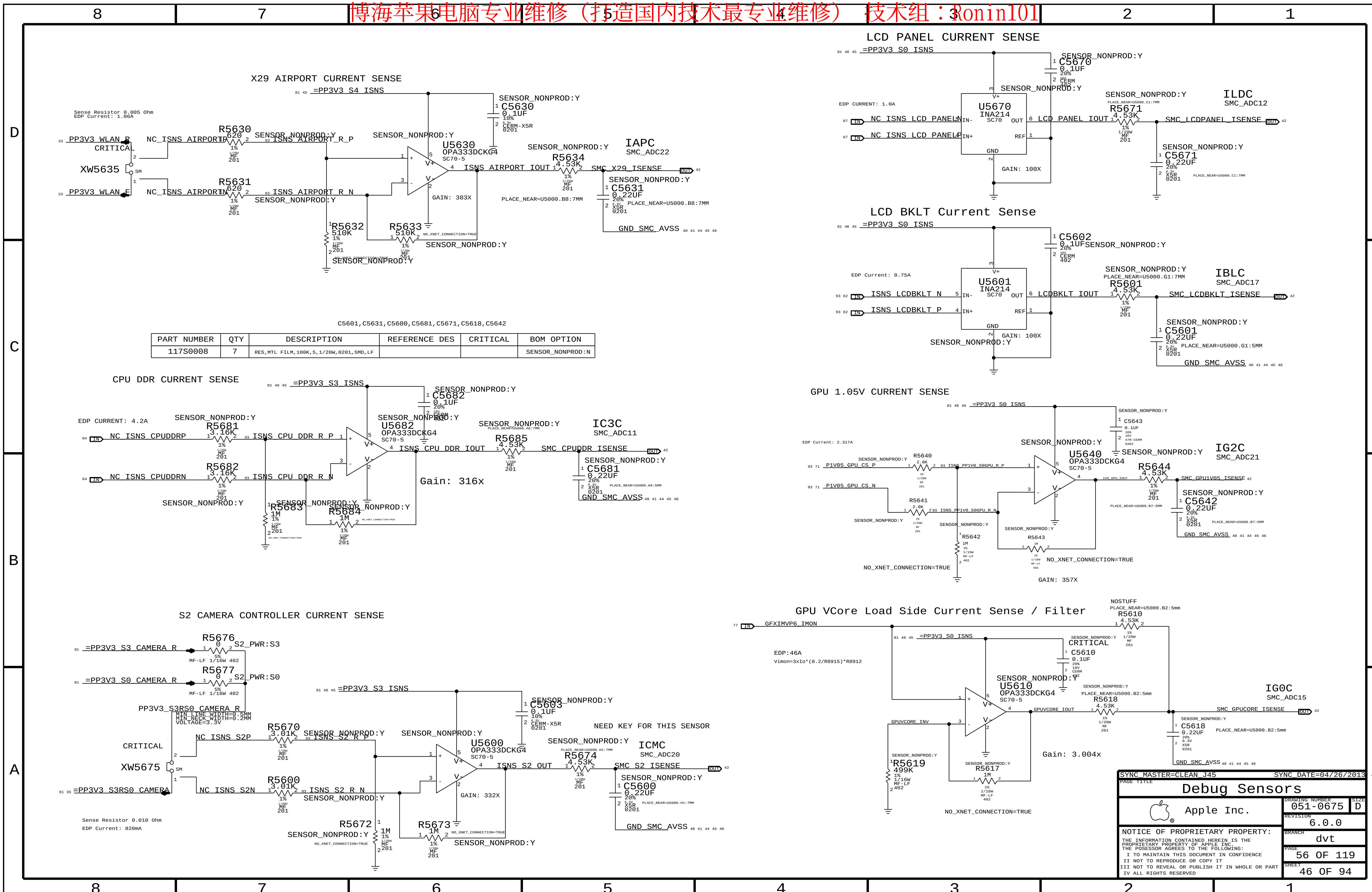

C5601,C5631,C5600,C5681,C5671,C5618,C5642

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	7	RES,MTL FILM,100K,5,1/20W,0201,SMD,LF			SENSOR_NONPROD:N

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
Debug Sensors

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D
REVISION 6.0.0
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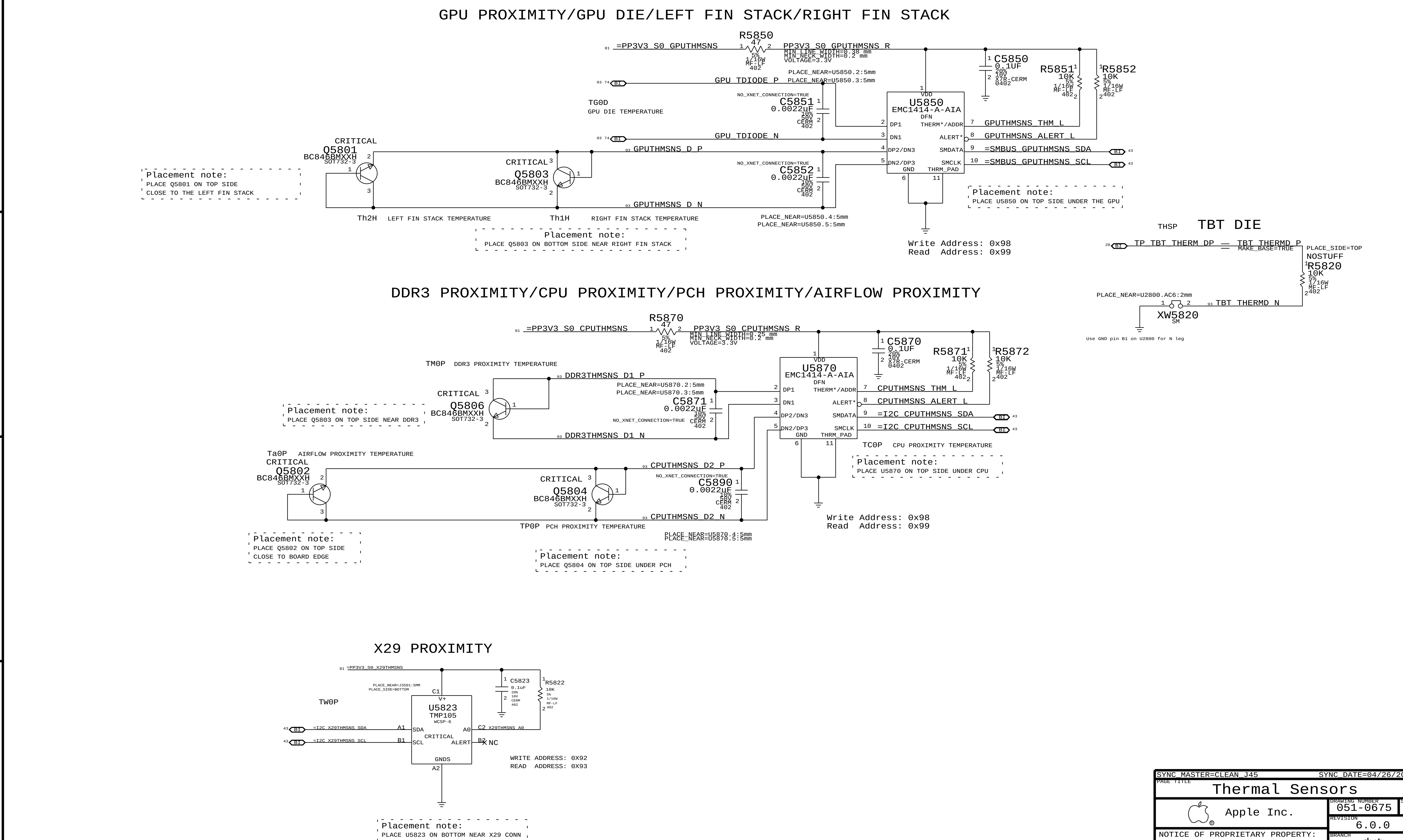

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

Thermal Sensors

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D

REVISION 6.0.0

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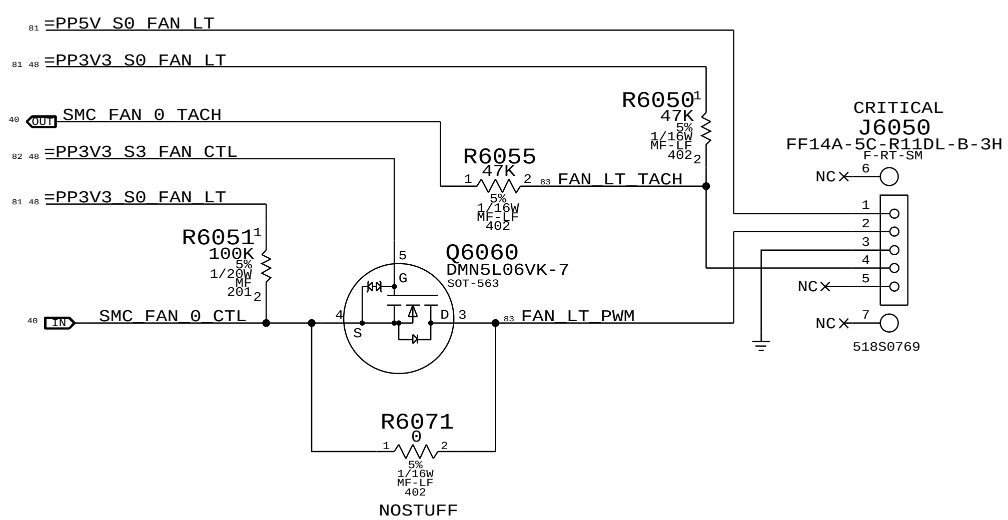
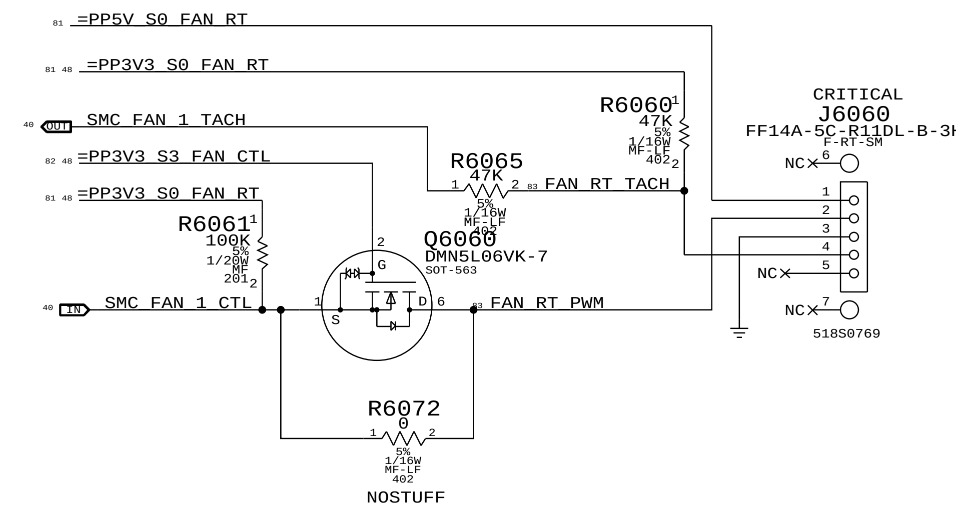
SHEET 47 OF 94

Left Fan



Right Fan



SYNC_MASTER=CLEAN_J45 SYNC_DATE=06/23/2013

PAGE TITLE: Fan Connectors

Apple Inc.

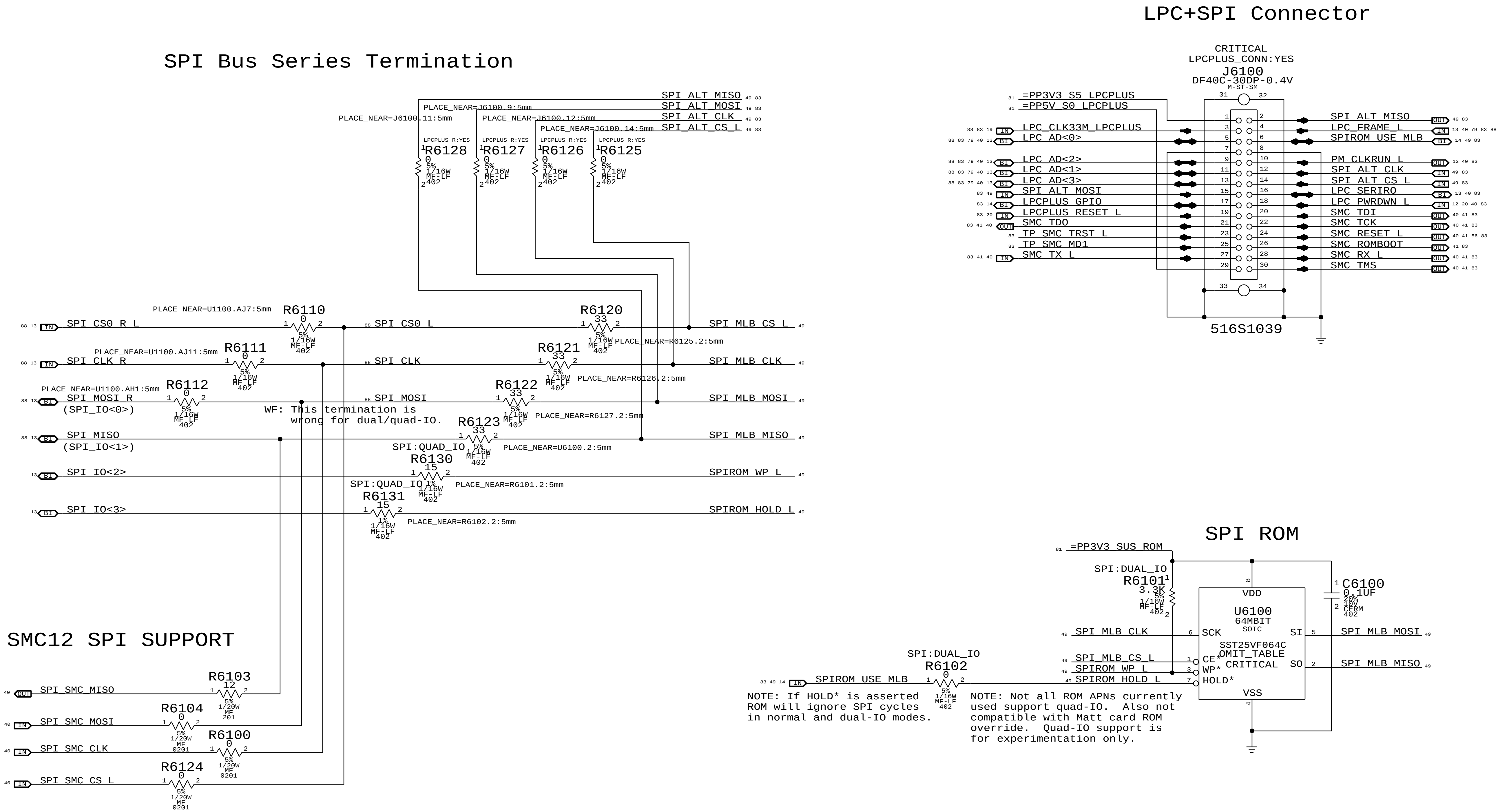
DRAWING NUMBER: 051-0675 SIZE: D

REVISION: 6.0.0

BRANCH: dvt

PAGE: 60 OF 119

SHEET: 48 OF 94



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: SPI ROM / LPC+SPI Conn.

Apple Inc.

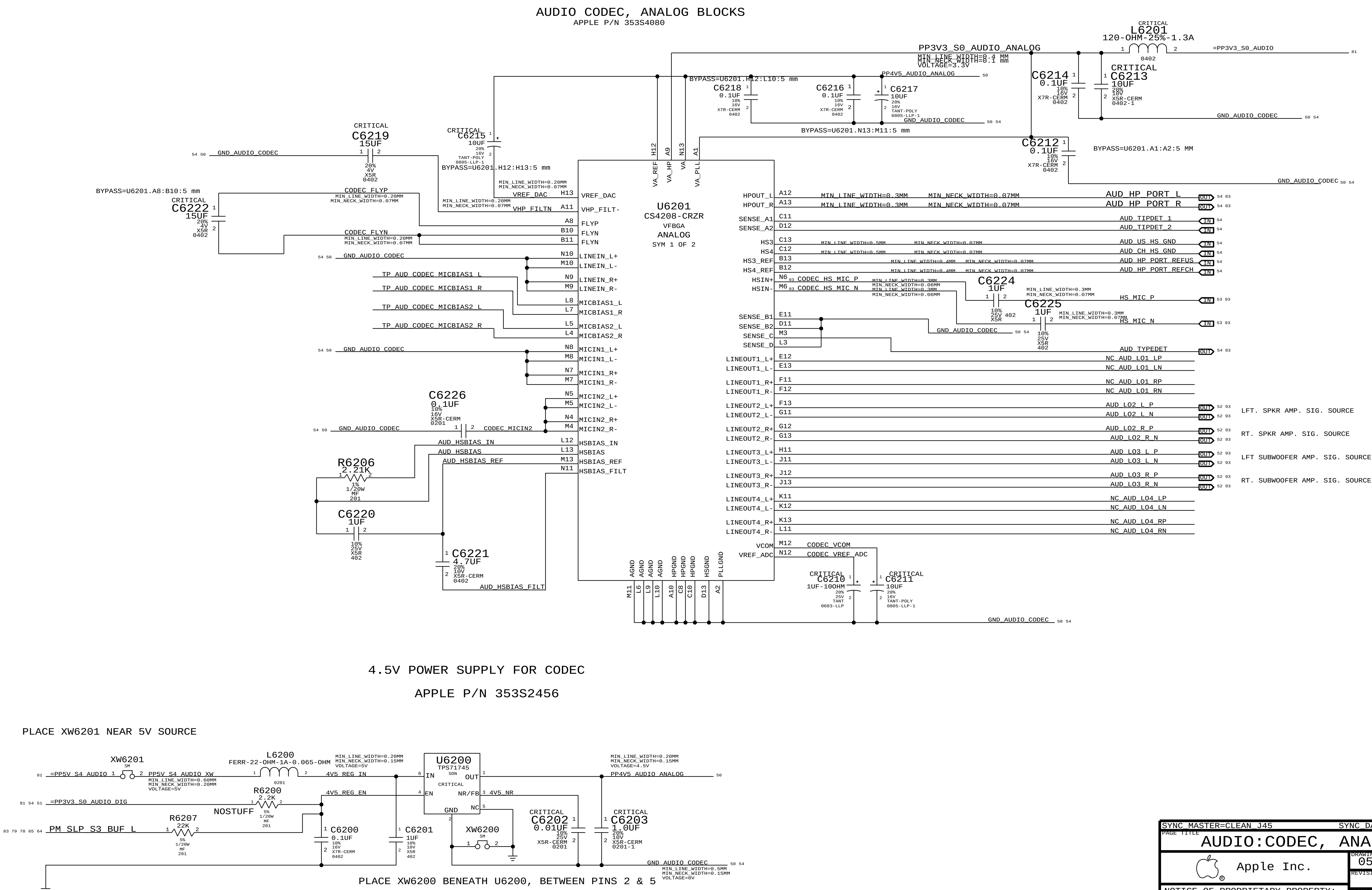
DRAWING NUMBER: 051-0675 SIZE: D

REVISION: 6.0.0

BRANCH: dvt

PAGE: 61 OF 119

SHEET: 49 OF 94



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
AUDIO:CODEC, ANALOG

Apple Inc.

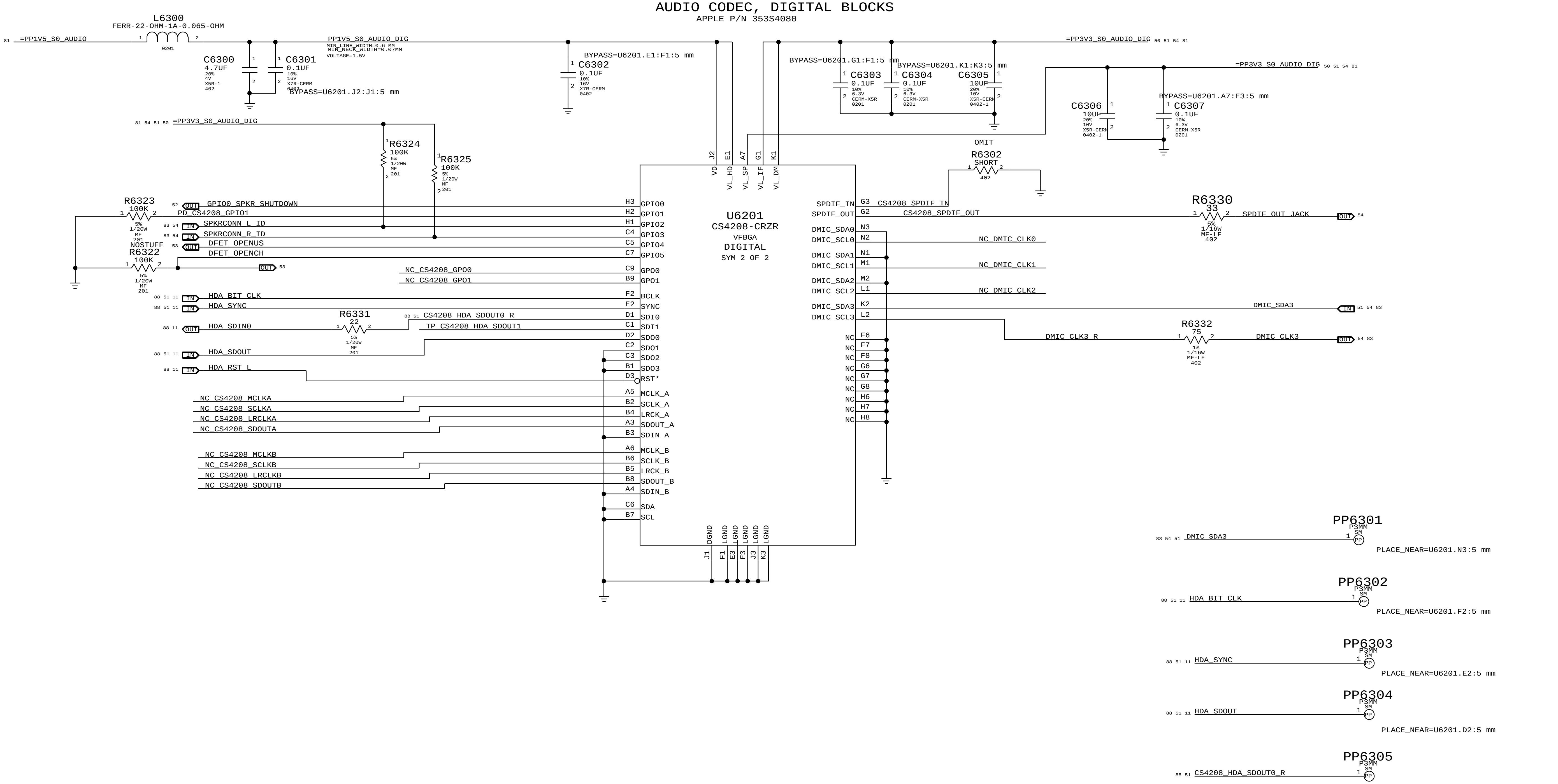
DRAWING NUMBER 051-0675 SIZE D

REVISION 6.0.0

BRANCH dvt

AUDIO CODEC, DIGITAL BLOCKS

APPLE P/N 353S4080



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

AUDIO:CODEC, DIGITAL

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D

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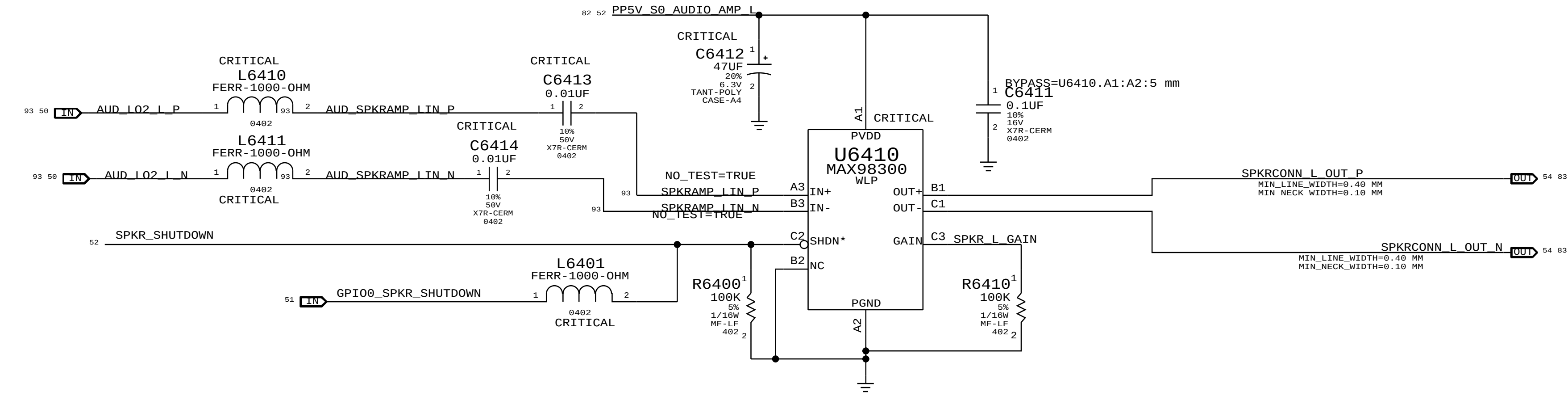
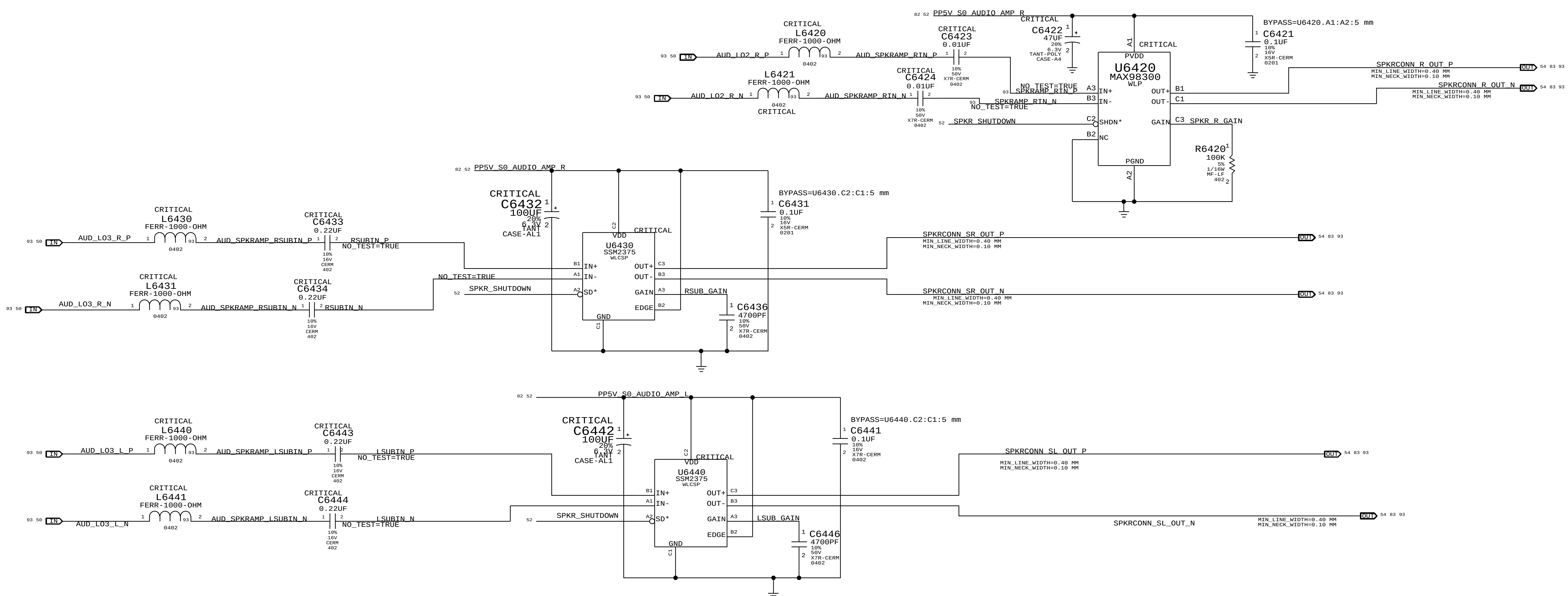
4X MONO SPEAKER AMPLIFIERS (MAX98300 & SSM2375)
APN: 353S2888 & 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = NOM 569 HZ
1ST ORDER FC (SUB) = NOM 9 HZ





SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
AUDIO: SPEAKER AMP

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D
REVISION 6.0.0
BRANCH dvt
PAGE 64 OF 119
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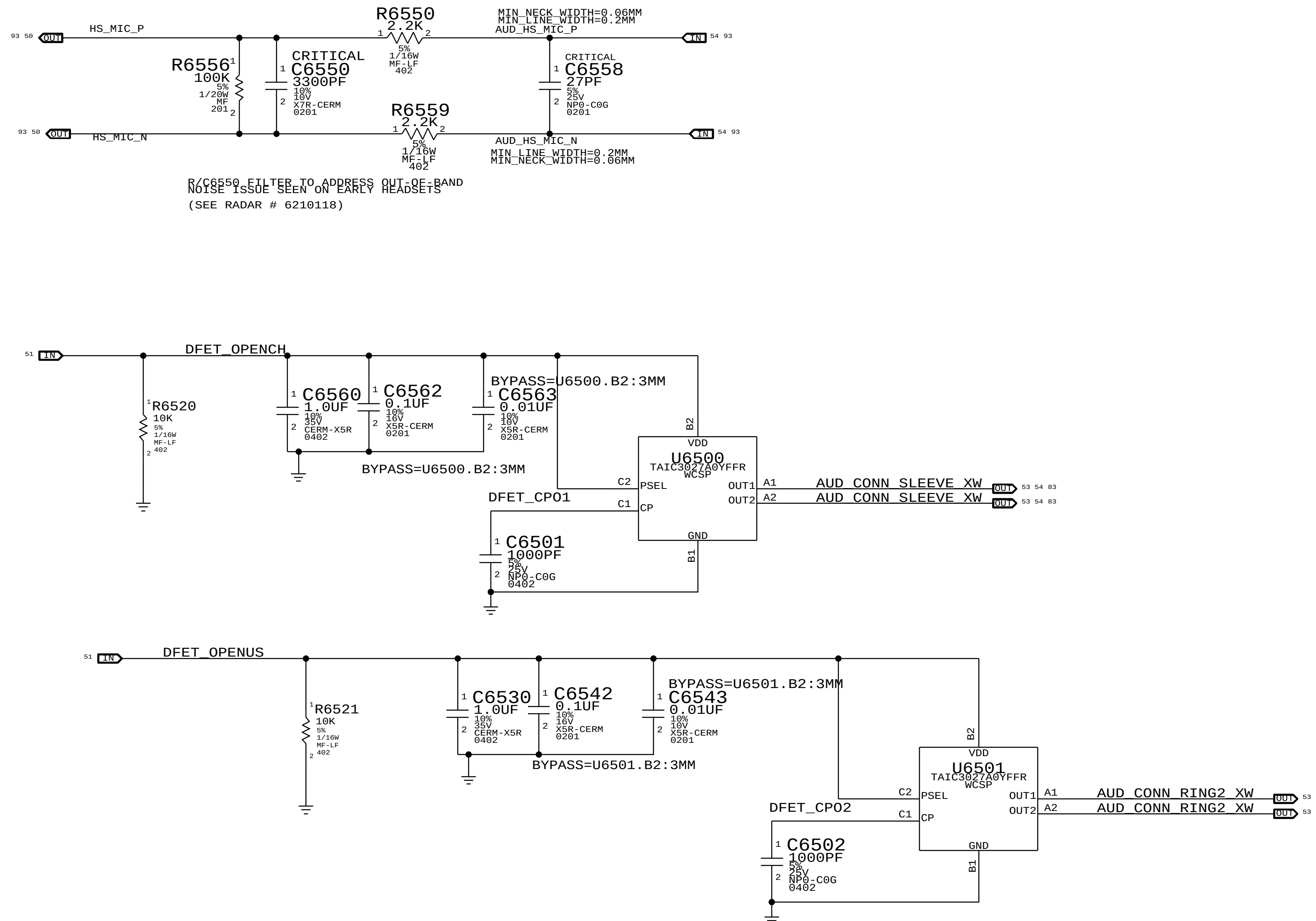


SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

AUDIO: JACK

DRAWING NUMBER	051-0675
SIZE	D
REVISION	6.0.0
BRANCH	dvt
PAGE	65 OF 119
SHEET	53 OF 94

Apple Inc.

CODEC OUTPUT SIGNAL PATHS

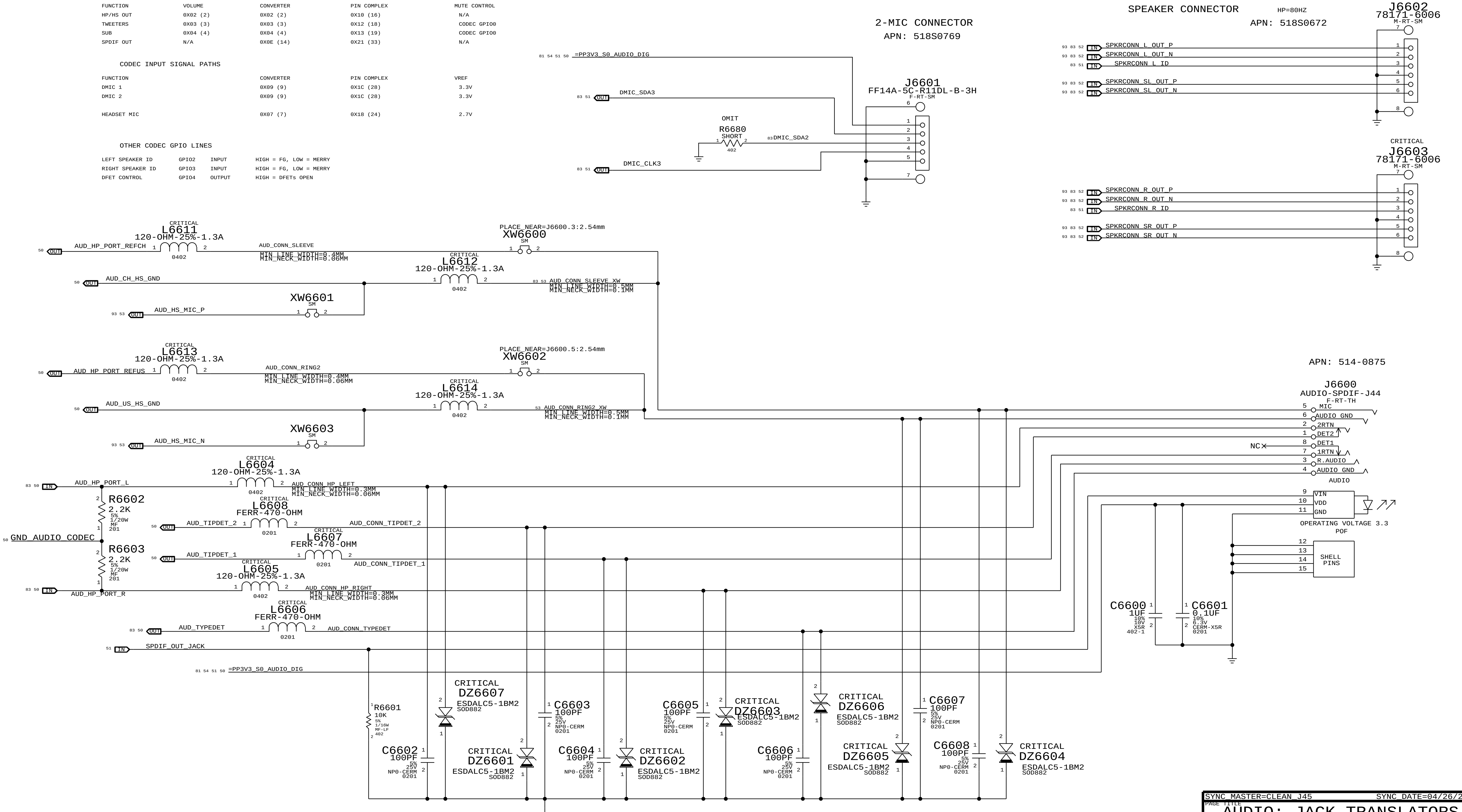
FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL
HP/HS OUT	0X02 (2)	0X02 (2)	0X10 (16)	N/A
TWEETERS	0X03 (3)	0X03 (3)	0X12 (18)	CODEC GPIO0
SUB	0X04 (4)	0X04 (4)	0X13 (19)	CODEC GPIO0
SPDIF OUT	N/A	0X0E (14)	0X21 (33)	N/A

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF
DMIC 1	0X09 (9)	0X1C (28)	3.3V
DMIC 2	0X09 (9)	0X1C (28)	3.3V
HEADSET MIC	0X07 (7)	0X18 (24)	2.7V

OTHER CODEC GPIO LINES

LEFT SPEAKER ID	GPIO2	INPUT	HIGH = FG, LOW = MERRY
RIGHT SPEAKER ID	GPIO3	INPUT	HIGH = FG, LOW = MERRY
DFET CONTROL	GPIO4	OUTPUT	HIGH = DFETs OPEN



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

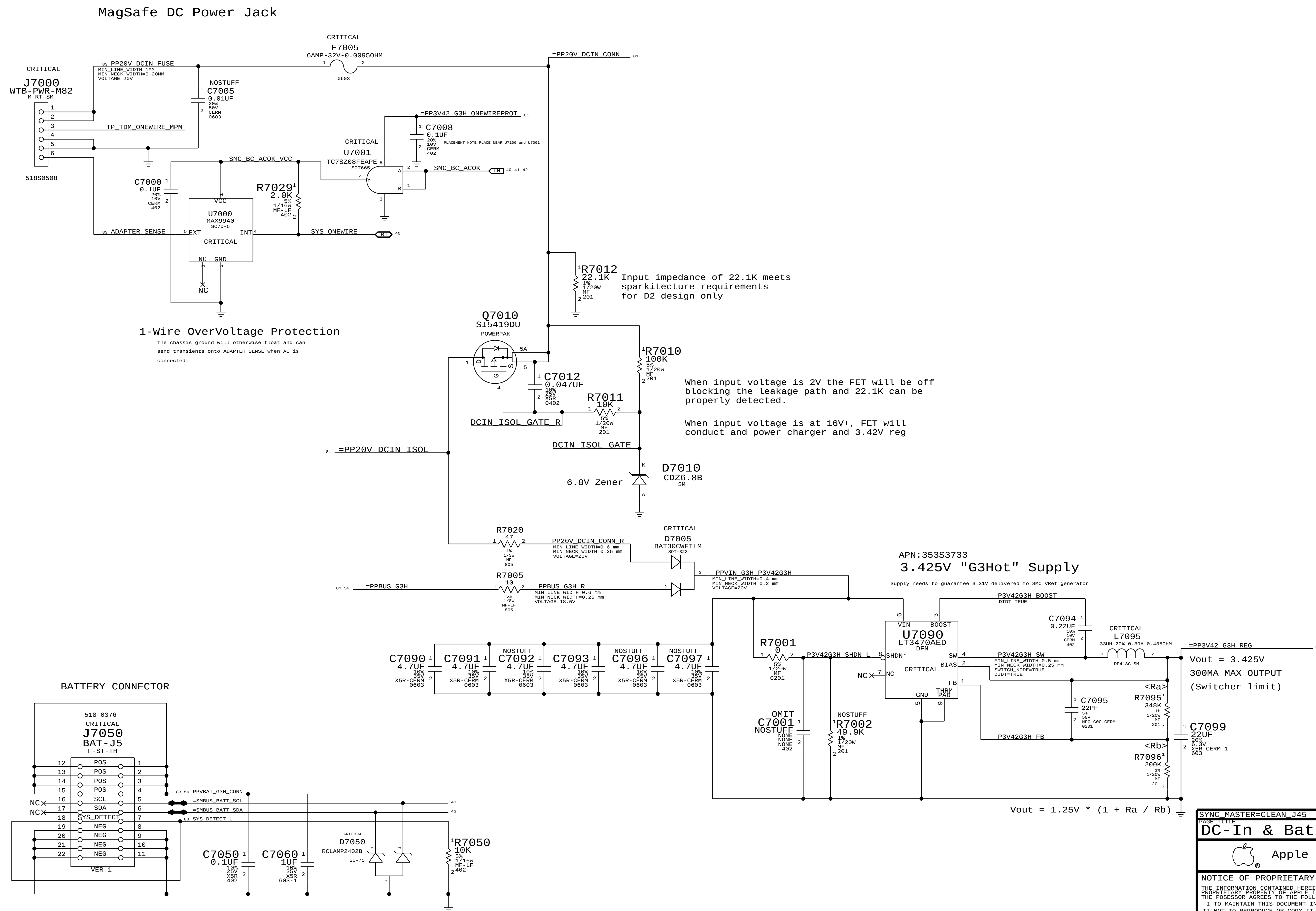
PAGE TITLE: AUDIO: JACK TRANSLATORS

Apple Inc.

DRAWING NUMBER: 051-0675 SIZE: D

REVISION: 6.0.0

BRANCH: dvt



SYNC_MASTER=CLEAN_J45	SYNC_DATE=04/26/2013
PAGE TITLE	DC-In & Battery Connectors
DRAWING NUMBER	051-0675
SIZE	D
REVISION	6.0.0
BRANCH	dvt
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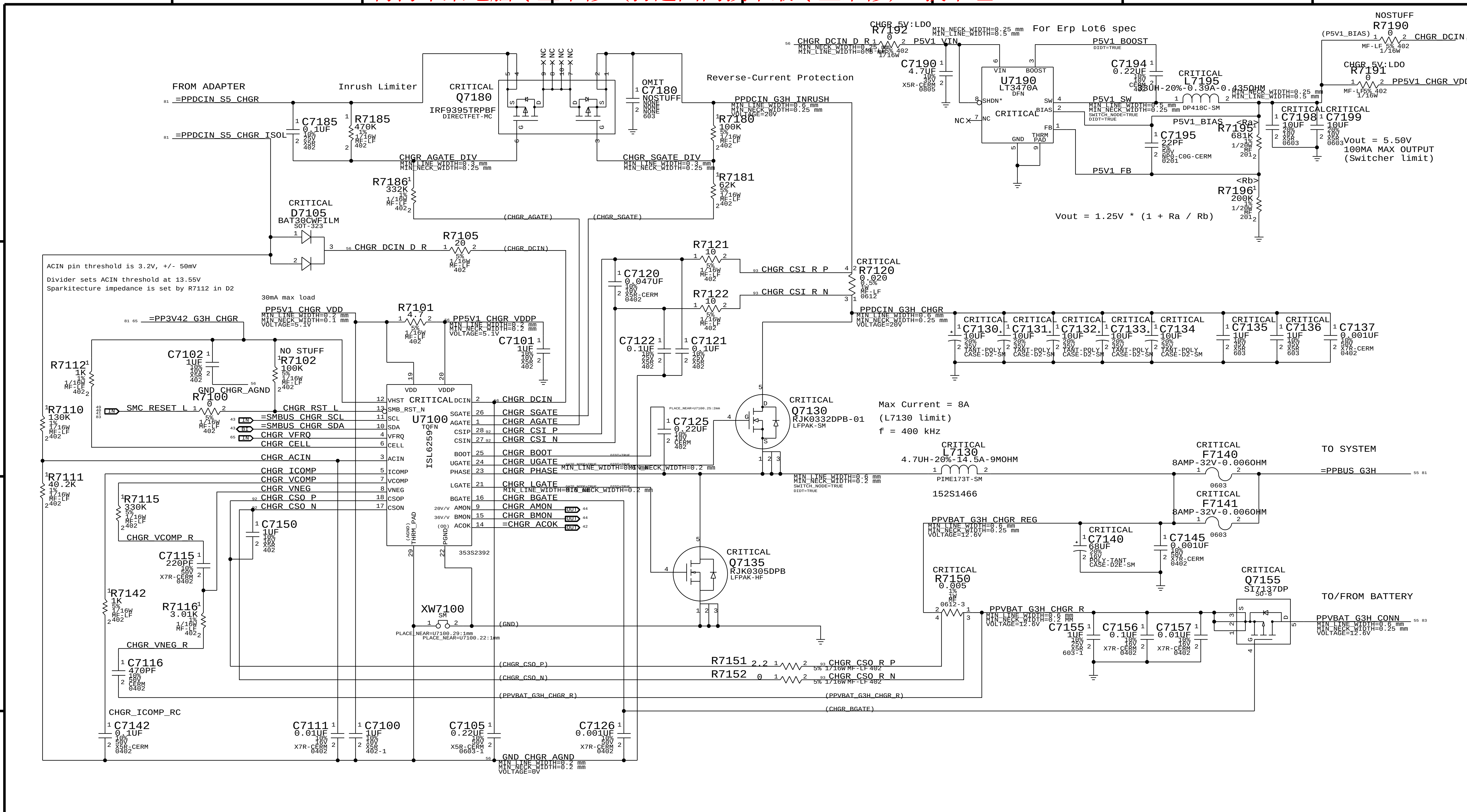
Apple Inc.



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE
PBus Supply & Battery Charger

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D

REVISION 6.0.0

BRANCH dvt

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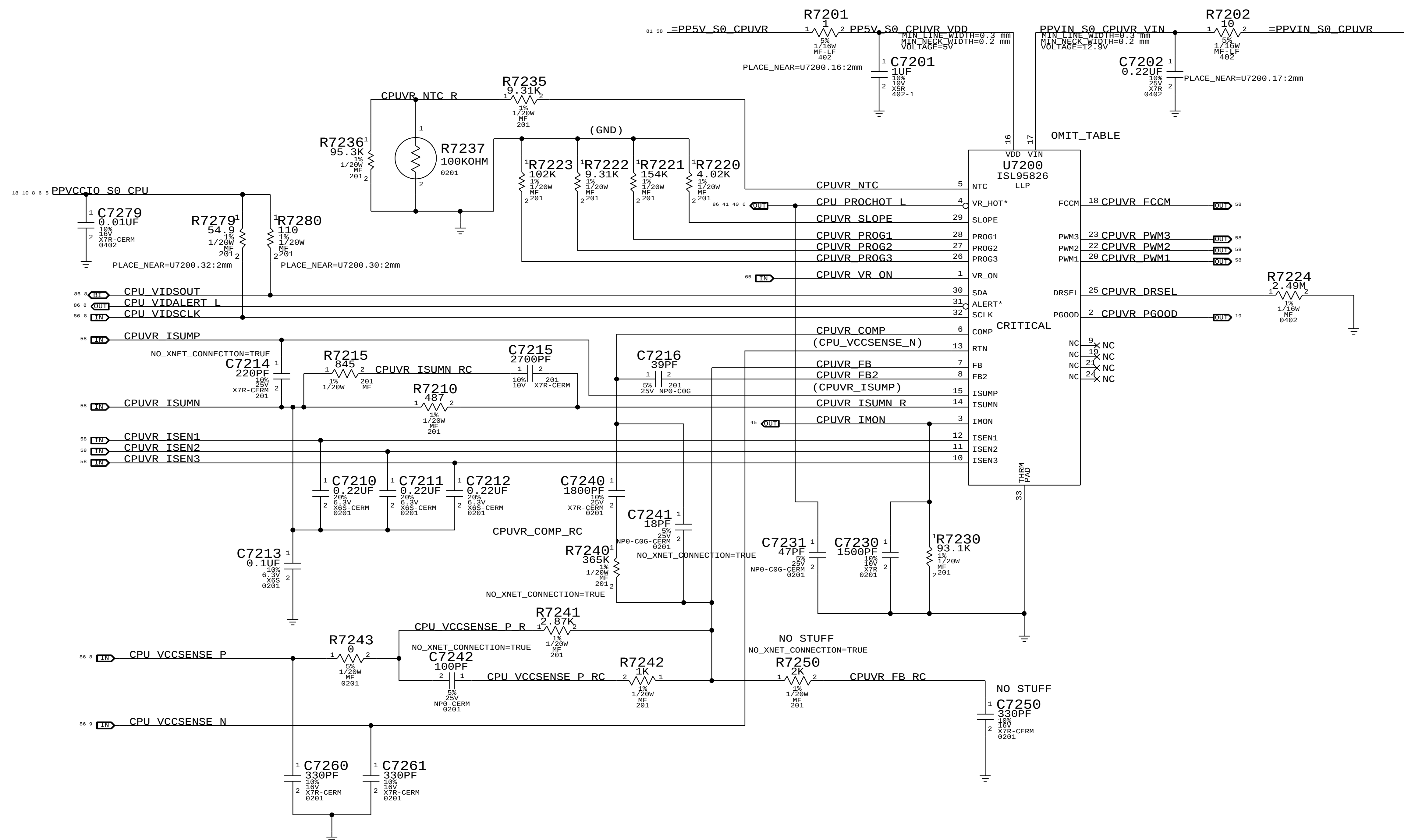
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S4170	1	IC,ISL95826R6200,PWM,PGOOD,SCREEN,32P,QFN	U7200	CRITICAL	



SYNC_MASTER=CLEAN_J45	SYNC_DATE=04/26/2013
PAGE TITLE	CPU VR12.5 VCC Regulator IC
Apple Inc.	DRAWING NUMBER 051-0675 SIZE D
	REVISION 6.0.0
	BRANCH dvt
	PAGE 72 OF 119
	SHEET 57 OF 94

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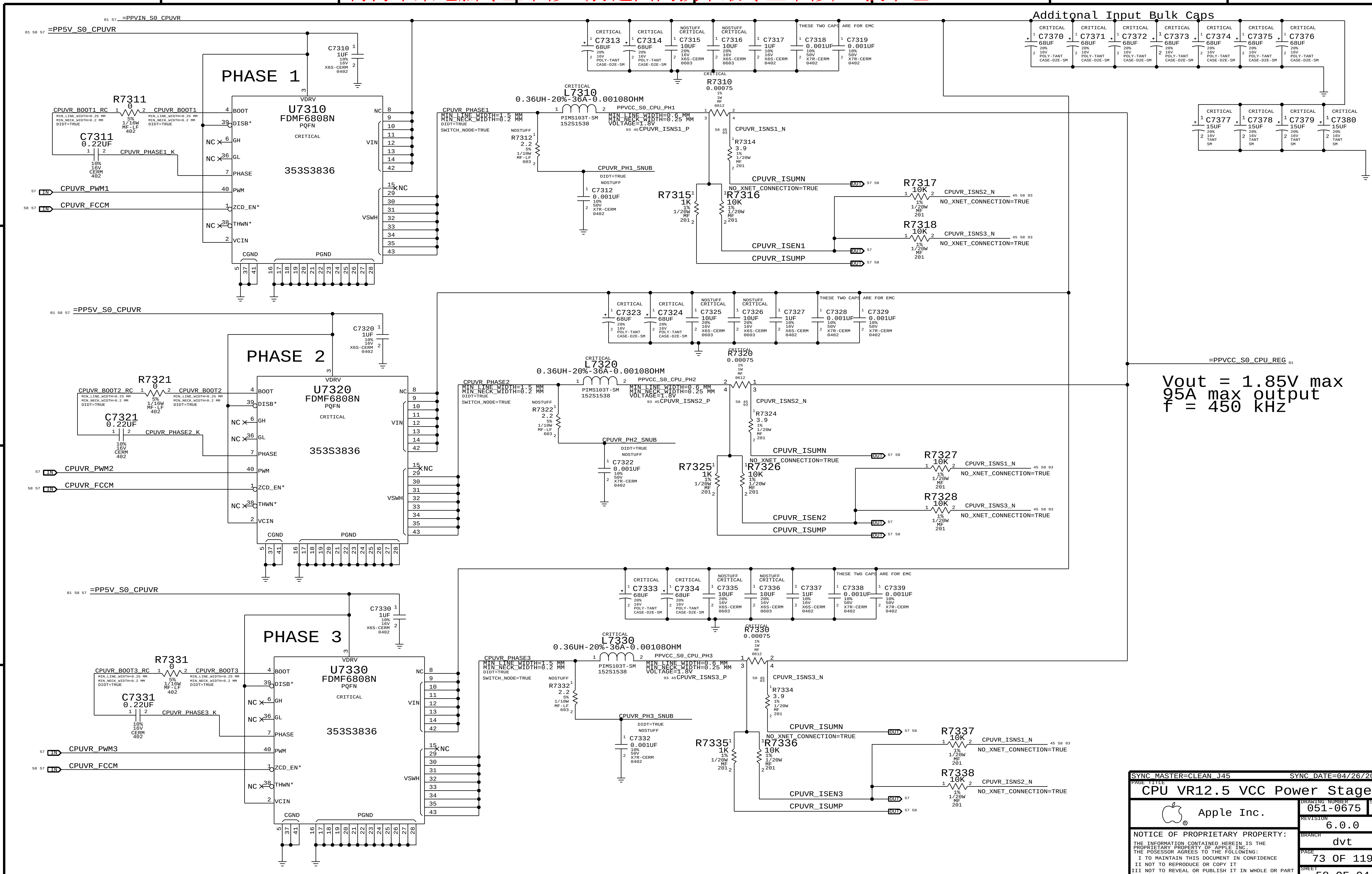

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

CPU VR12.5 VCC Power Stage

Apple Inc.

DRAWING NUMBER	SIZE
051-0675	D

REVISION 6.0.0

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SHEET 58 OF 94

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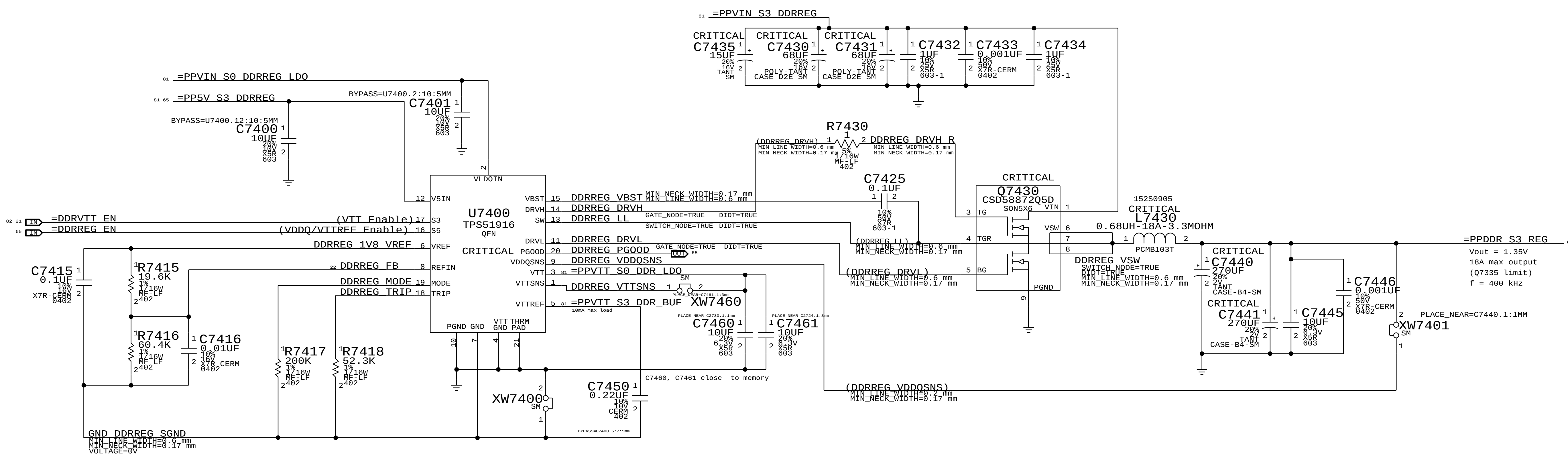
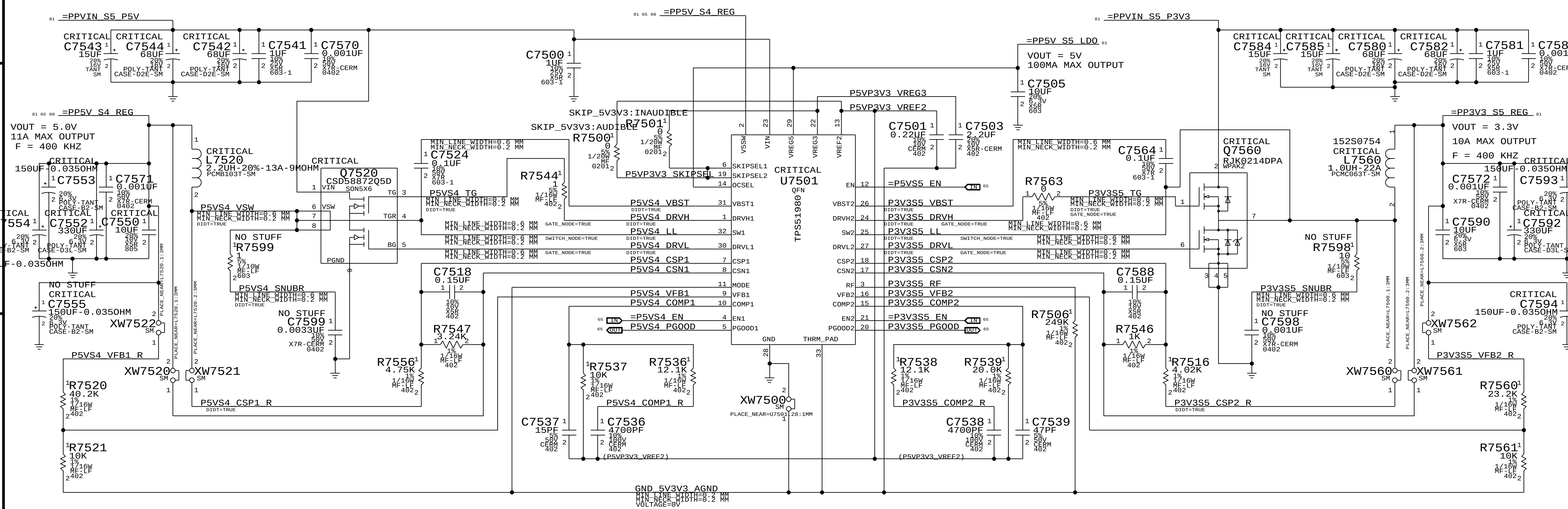
DDR3L (1V35 S3) REGULATOR



SYNC_MASTER=CLEAN_J45		SYNC_DATE=04/26/2013
PAGE TITLE: 1.35V DDR3L SUPPLY		
Apple Inc.	DRAWING NUMBER: 051-0675	SIZE: D
	REVISION: 6.0.0	
	BRANCH: dvt	
	PAGE: 74 OF 119	
	SHEET: 59 OF 94	



SYNC_MASTER=CLEAN_J45	SYNC_DATE=04/26/2013
PAGE TITLE	5V / 3.3V Power Supply
Apple Inc.	DRAWING NUMBER 051-0675 SIZE D
	REVISION 6.0.0
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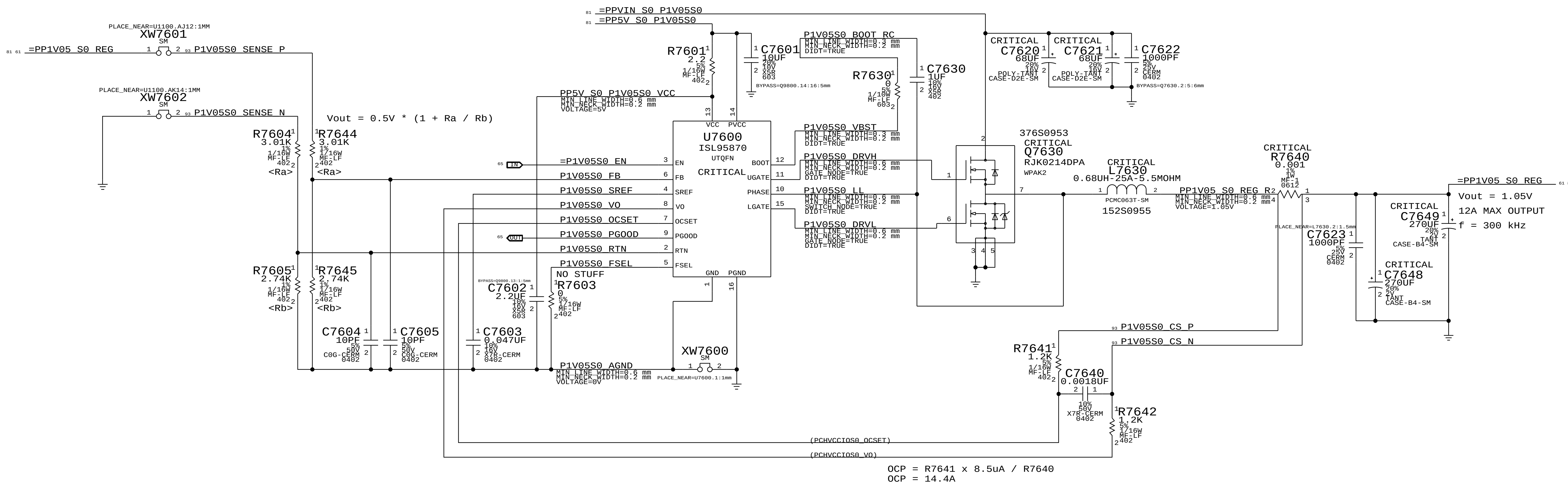
8 7 6 5 4 3 2 1

1V05 S0 REGULATOR



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: 1V05V POWER SUPPLY

Apple Inc.

DRAWING NUMBER: 051-0675 SIZE: D

REVISION: 6.0.0

BRANCH: dvt

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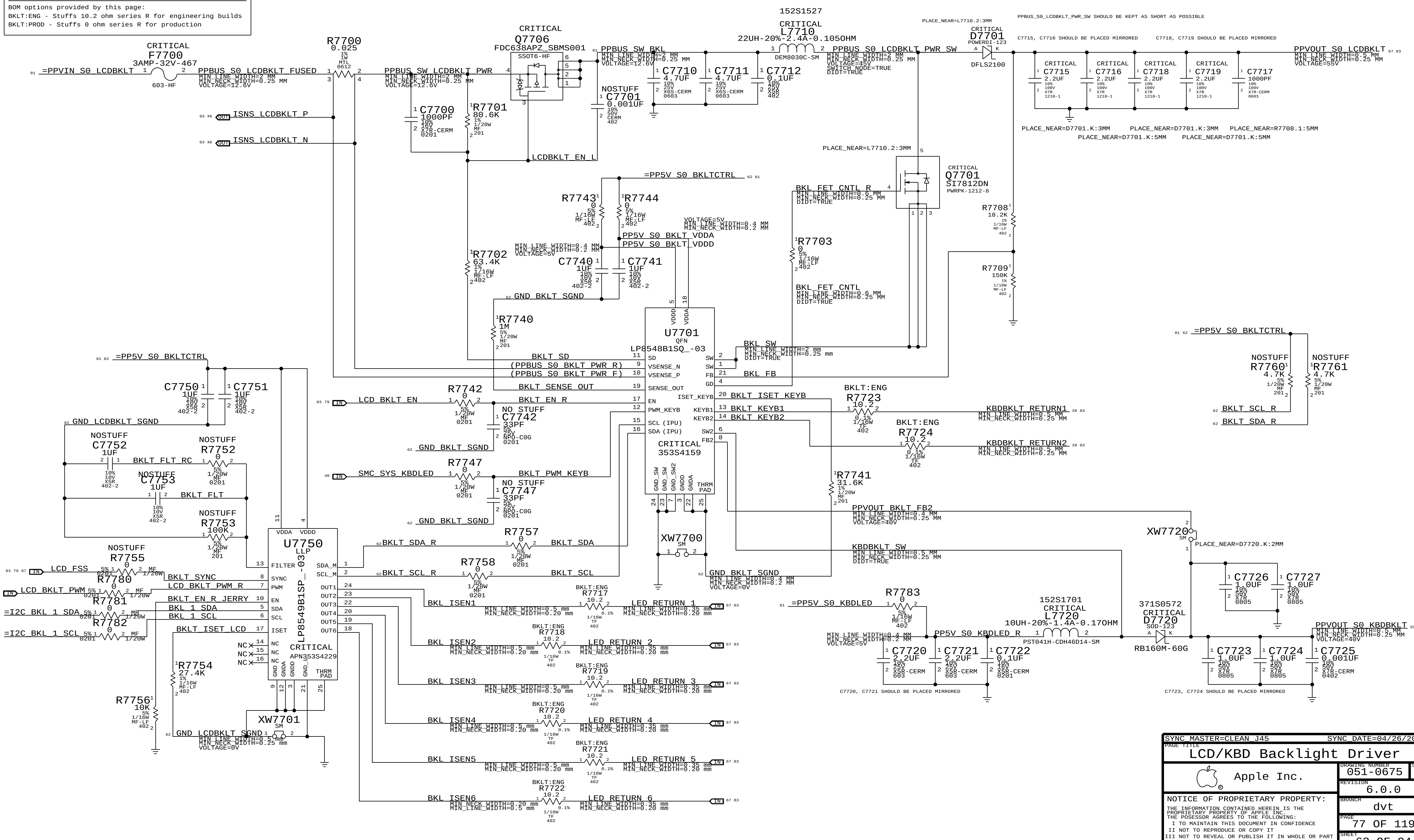
Page Notes

Power aliases required by this page:
- =PPVIN_S0_LCDBKLT (9-12.6V LCD Backlight Input)
- =PP5V_S0_BKLTCTRL (5V Backlight Driver Input)
- =PP5V_S0_KBDLED (5V Keyboard Backlight Input)

BOM options provided by this page:
BKLT:ENG - Stuffs 10.2 ohm series R for engineering builds
BKLT:PROD - Stuffs 0 ohm series R for production

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	8	RES,MTL FILM,0 OHM,1A MAX,0402,SMD	R7717,R7718,R7719,R7720,R7721,R7722,R7723,R7724		BKLT:PROD



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: LCD/KBD Backlight Driver

Apple Inc.

DRAWING NUMBER: 051-0675 SIZE: D

REVISION: 6.0.0

BRANCH: dvt

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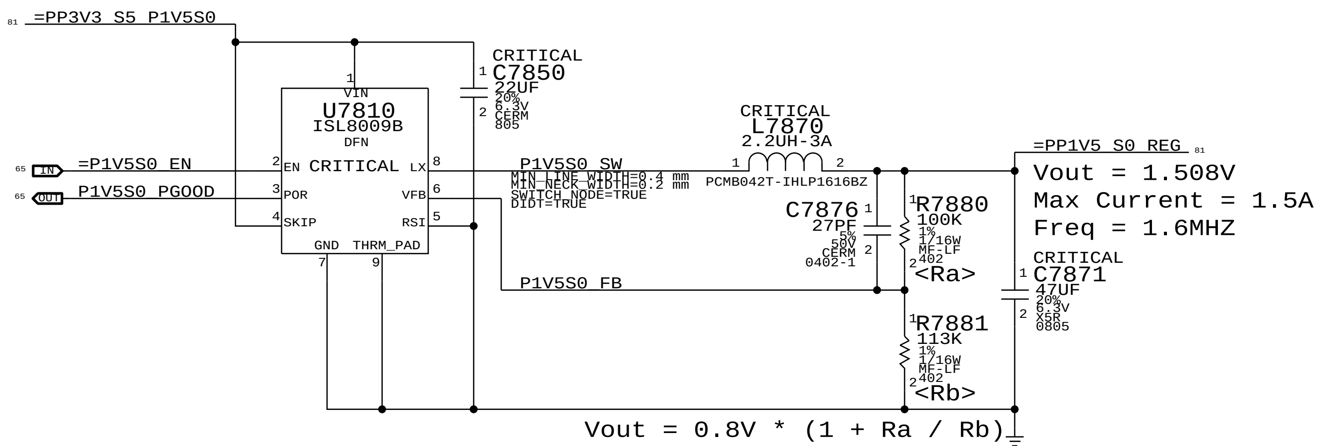
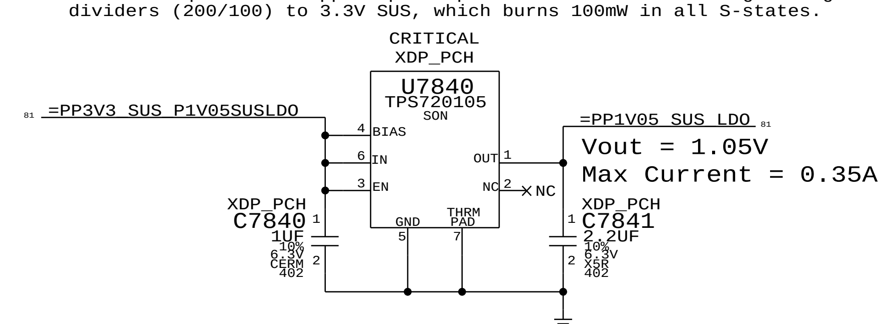
D C B A

1.5V S0 Regulator



1.05V SUS LDO

Lynx Point-H requires JTAG pull-ups to be powered at 1.05V in SUS.
Pull-ups (3) must be 51 ohms to support XDP (not required in production).
70mA is required to support pull-ups. Alternative is strong voltage
dividers (200/100) to 3.3V SUS, which burns 100mW in all S-states.



SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: Misc Power Supplies

Apple Inc.

DRAWING NUMBER: 051-0675 SIZE: D

REVISION: 6.0.0

BRANCH: dvt

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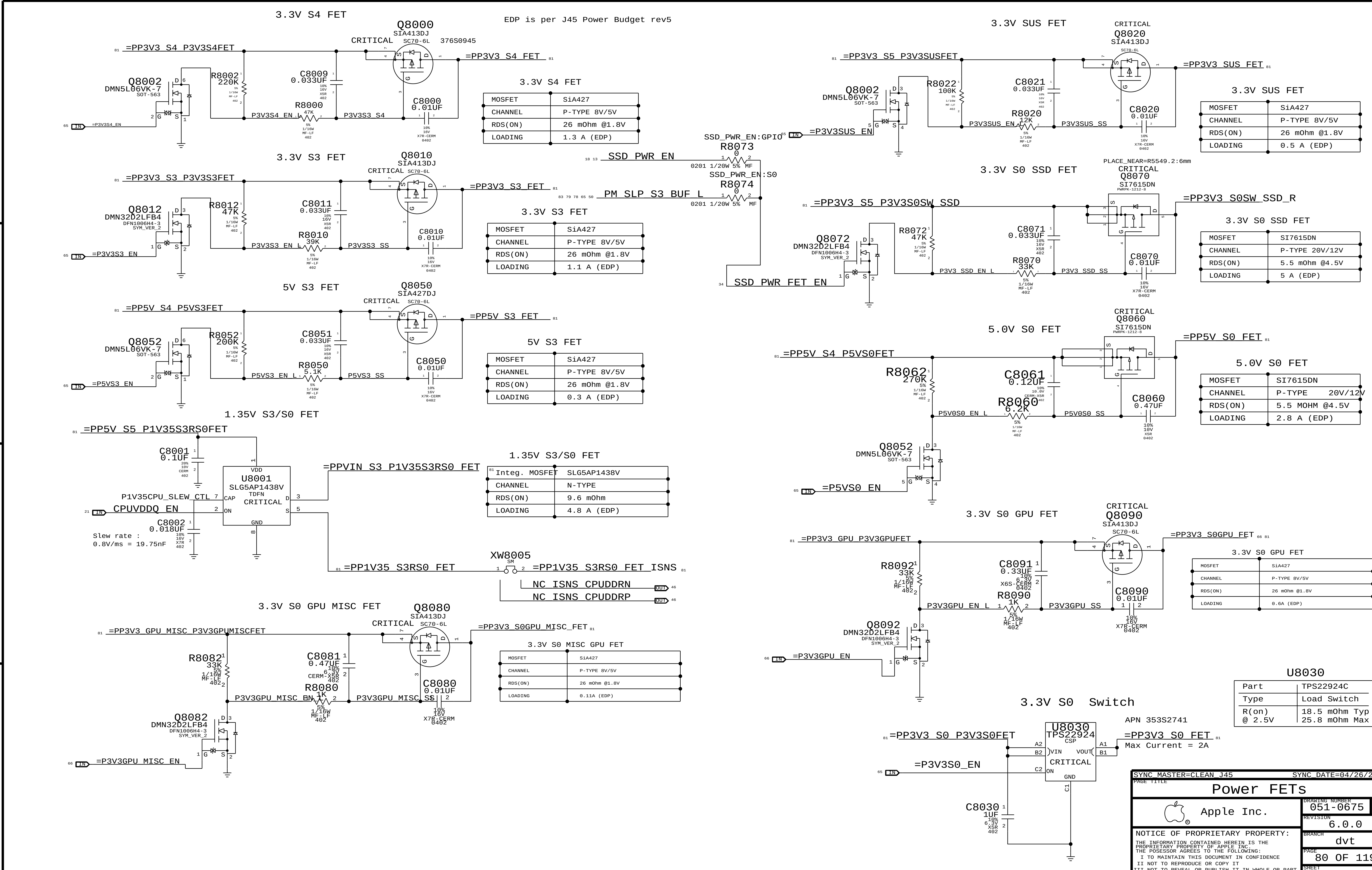

3.3V S4 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	1.3 A (EDP)

3.3V S3 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	1.1 A (EDP)

5V S3 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.3 A (EDP)

1.35V S3/S0 FET

Integ. MOSFET	SLG5AP1438V
CHANNEL	N-TYPE
RDS(ON)	9.6 mOhm
LOADING	4.8 A (EDP)

3.3V S0 MISC GPU FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.11A (EDP)

3.3V SUS FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.5 A (EDP)

3.3V S0 SSD FET

MOSFET	SI7615DN
CHANNEL	P-TYPE 20V/12V
RDS(ON)	5.5 mOhm @4.5V
LOADING	5 A (EDP)

5.0V S0 FET

MOSFET	SI7615DN
CHANNEL	P-TYPE 20V/12V
RDS(ON)	5.5 MOHM @4.5V
LOADING	2.8 A (EDP)

3.3V S0 GPU FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.6A (EDP)

U8030

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

Power FETs

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D

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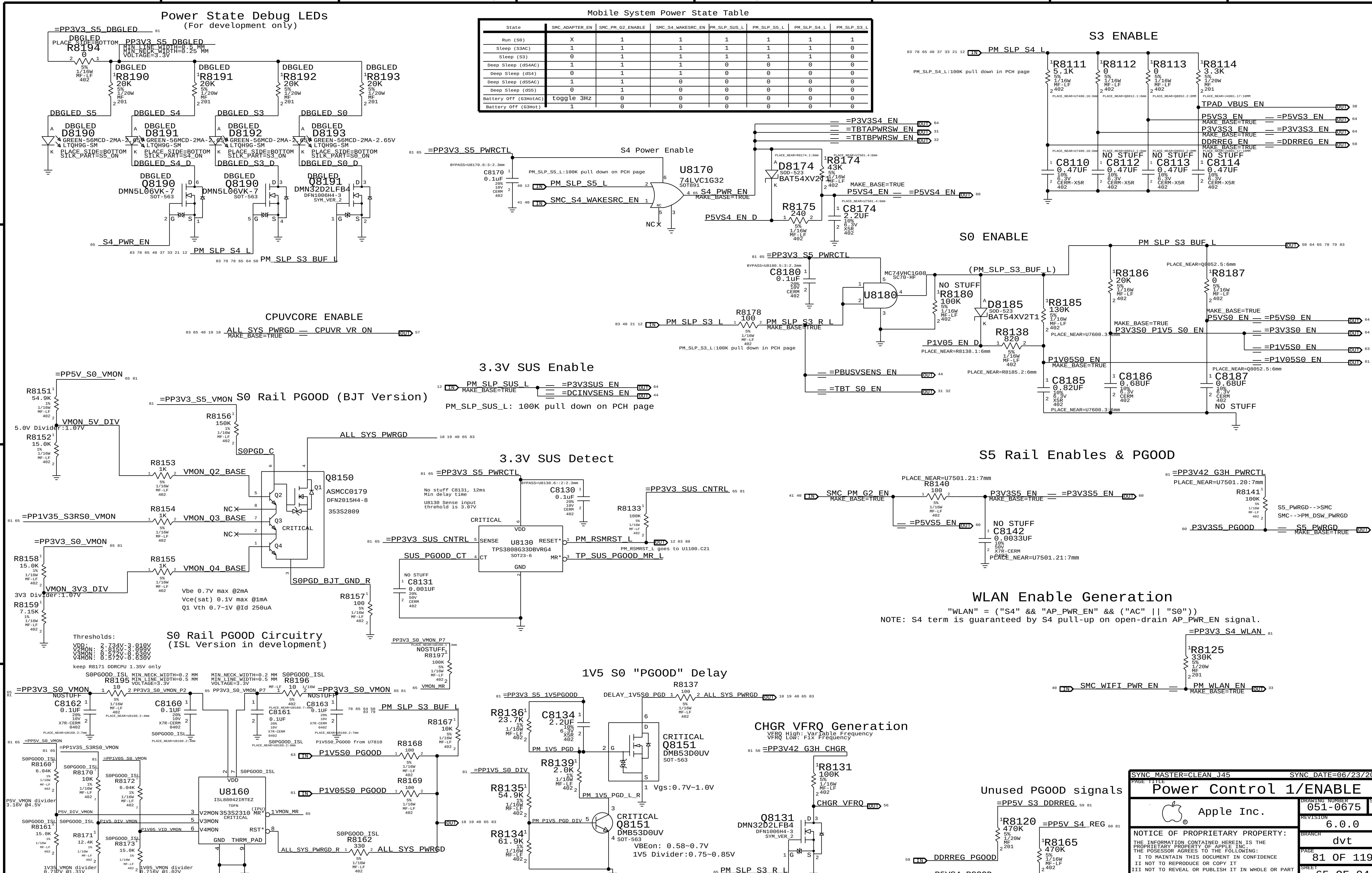

Mobile System Power State Table

State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	SMC_S4_WAKESRC_EN	PM_SLP_SUS_L	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	1	0
Deep Sleep (dS4AC)	1	1	1	0	0	0	0
Deep Sleep (dS4)	0	1	1	0	0	0	0
Deep Sleep (dS5AC)	1	1	0	0	0	0	0
Deep Sleep (dS5)	0	1	0	0	0	0	0
Battery Off (G3HotAC)	toggle 3Hz	0	0	0	0	0	0
Battery Off (G3Hot)	1	0	0	0	0	0	0

SYNC_MASTER=CLEAN_J45 SYNC_DATE=06/23/2013

PAGE TITLE
Power Control 1/ENABLE

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D

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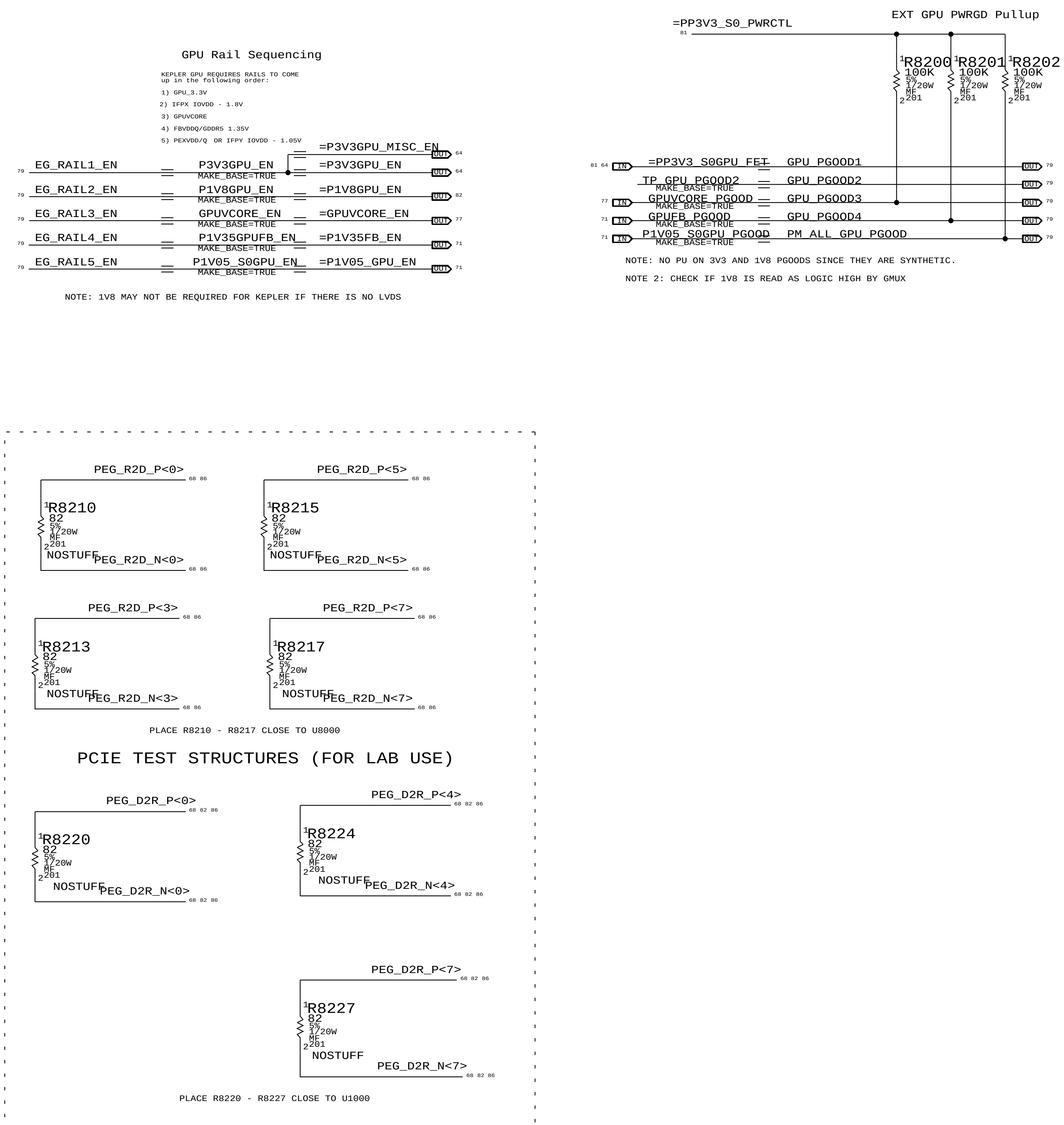

SYNC_MASTER=D2_KEPLER SYNC_DATE=01/13/2012

PAGE TITLE: Power Sequencing EG/PCH S0

Apple Inc.

DRAWING NUMBER: 051-0675 SIZE: D

REVISION: 6.0.0

BRANCH: dvt

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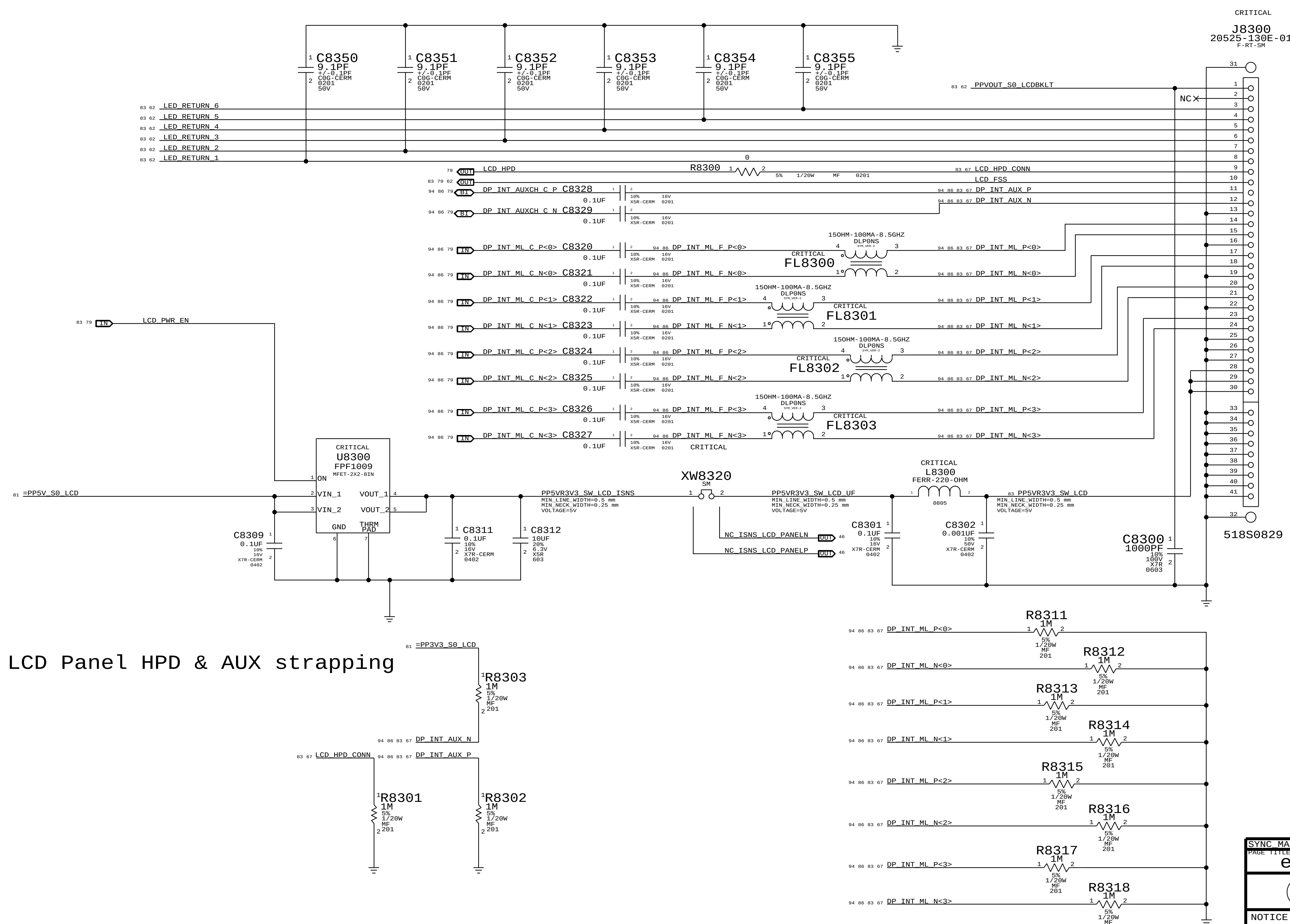
SHEET: 66 OF 94

8 7 6 5 4 3 2 1

LCD PANEL INTERFACE (eDP)



SYNC_MASTER=CLEAN_J45	SYNC_DATE=04/26/2013
PAGE TITLE: eDP Display Connector	
Apple Inc.	DRAWING NUMBER 051-0675 SIZE D
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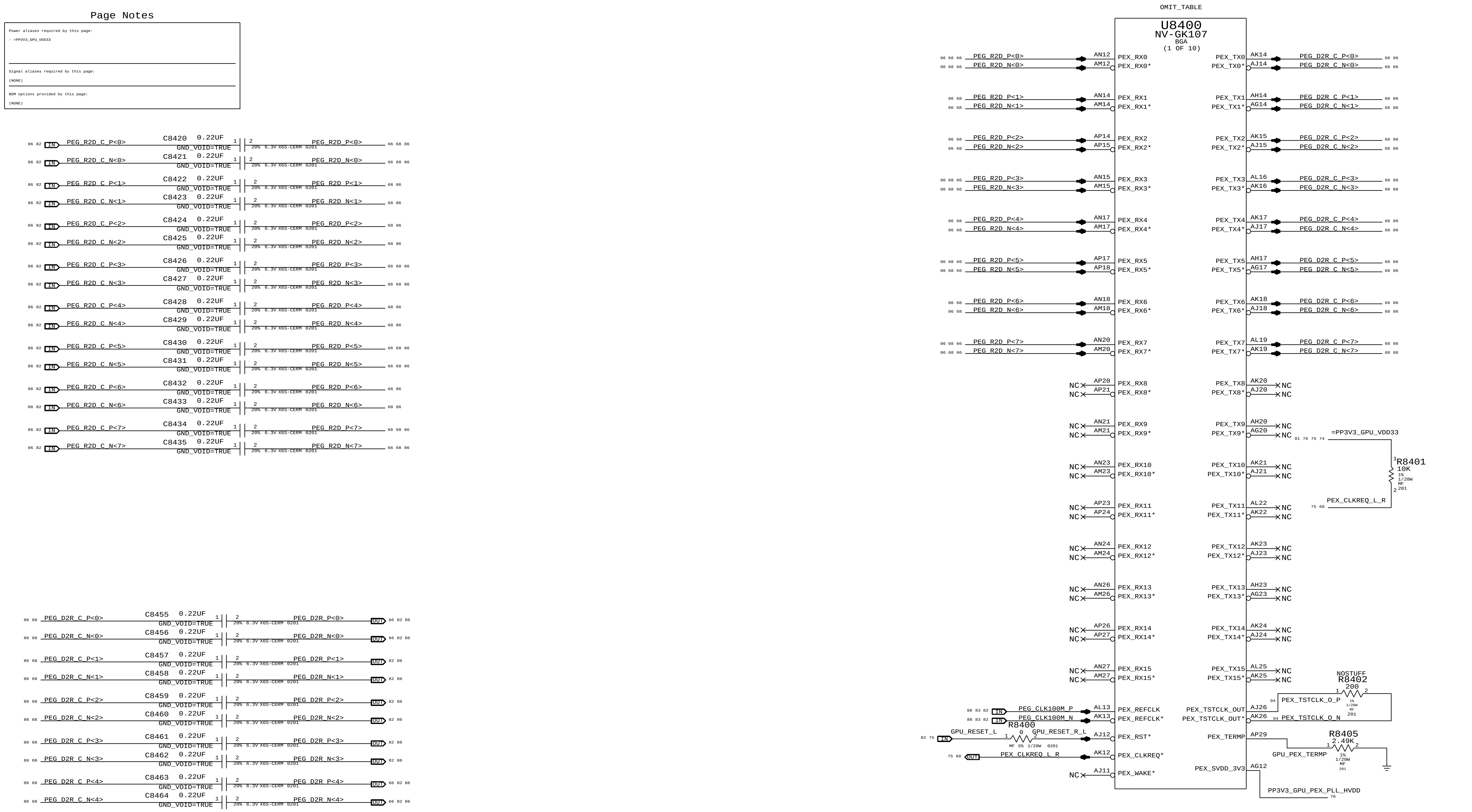
8 7 6 5 4 3 2 1

Page Notes

Power aliases required by this page:
- =PP3V3_GPU_VDD33

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



D C B A

SYNC_MASTER=D2_MLB	SYNC_DATE=07/31/2012
PAGE TITLE	KEPLER PCI-E
Apple Inc.	DRAWING NUMBER 051-0675 SIZE D
	REVISION 6.0.0
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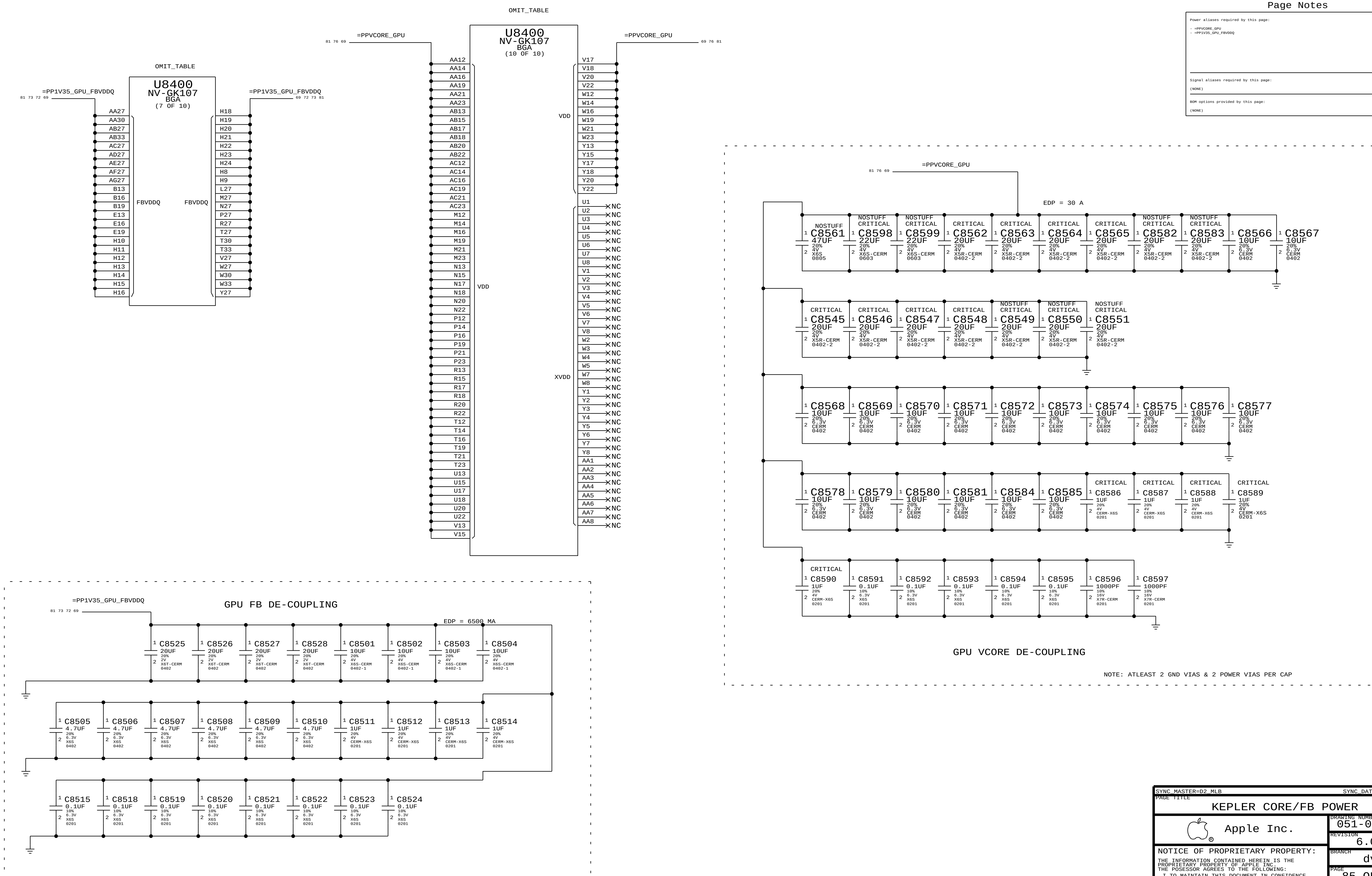
Page Notes

Power aliases required by this page:
- =PPVCORE_GPU
- =PP1V35_GPU_FBVDDQ

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



SYNC_MASTER=D2_MLB SYNC_DATE=07/31/2012

PAGE TITLE

KEPLER CORE/FB POWER

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D

REVISION 6.0.0

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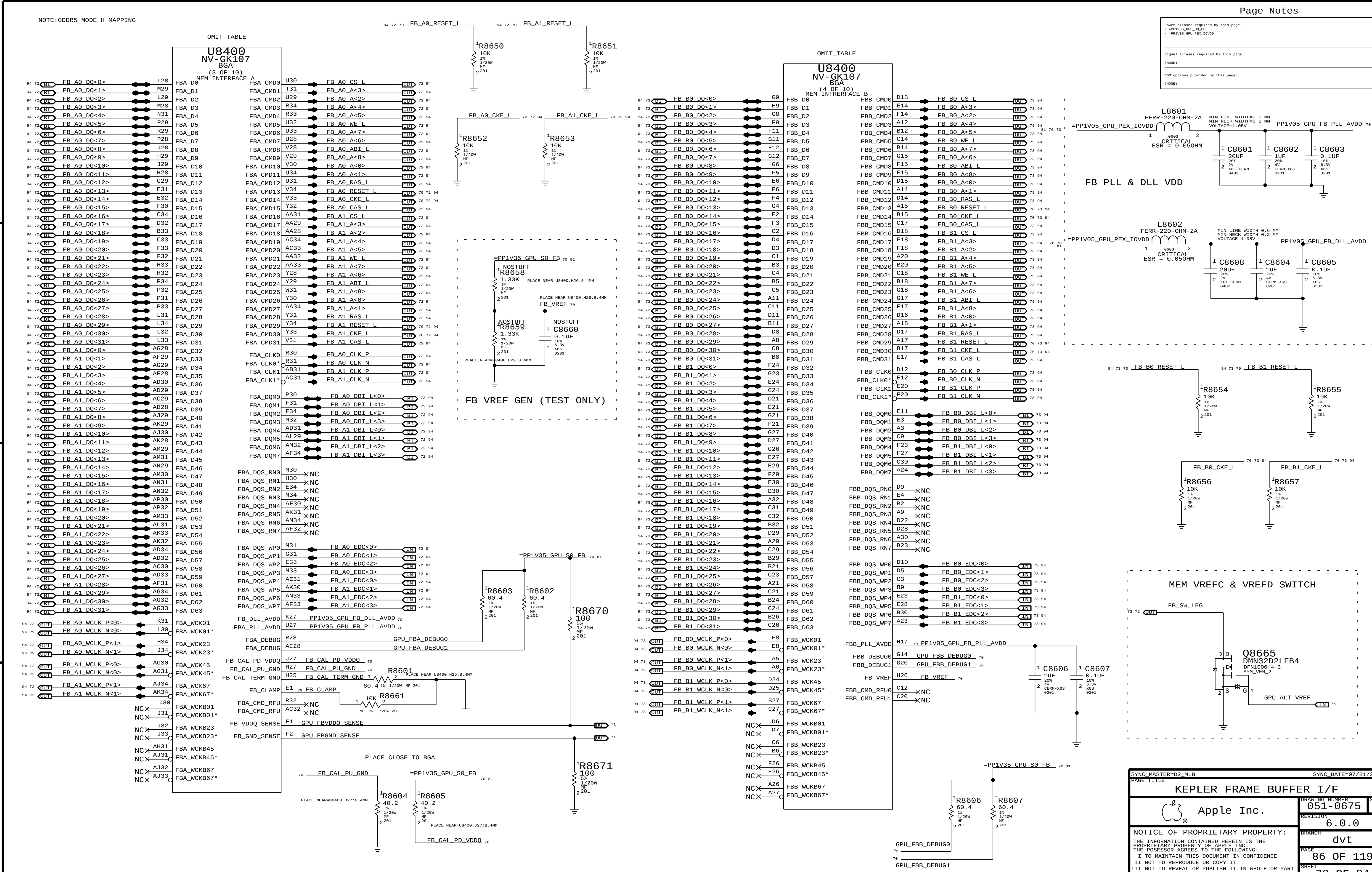

SYNC_MASTER=D2_MLB SYNC_DATE=07/31/2012

PAGE TITLE

KEPLER FRAME BUFFER I/F

Apple Inc.

DRAWING NUMBER	051-0675
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REVISION	6.0.0
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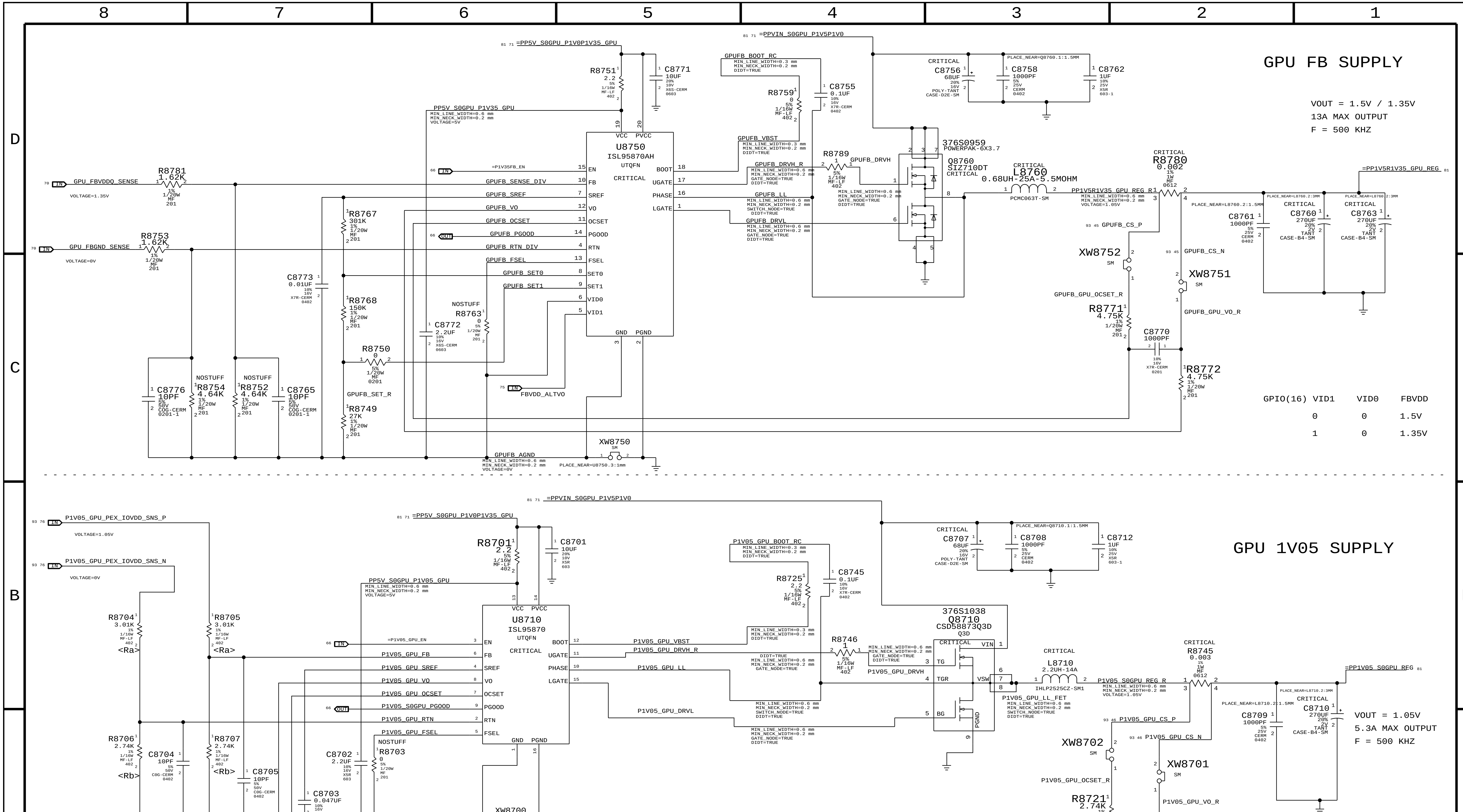

SYNC_MASTER=D2_MLB SYNC_DATE=07/31/2012

PAGE TITLE

1V05 GPU / 1V35 FB POWER SUPPLY

Apple Inc.

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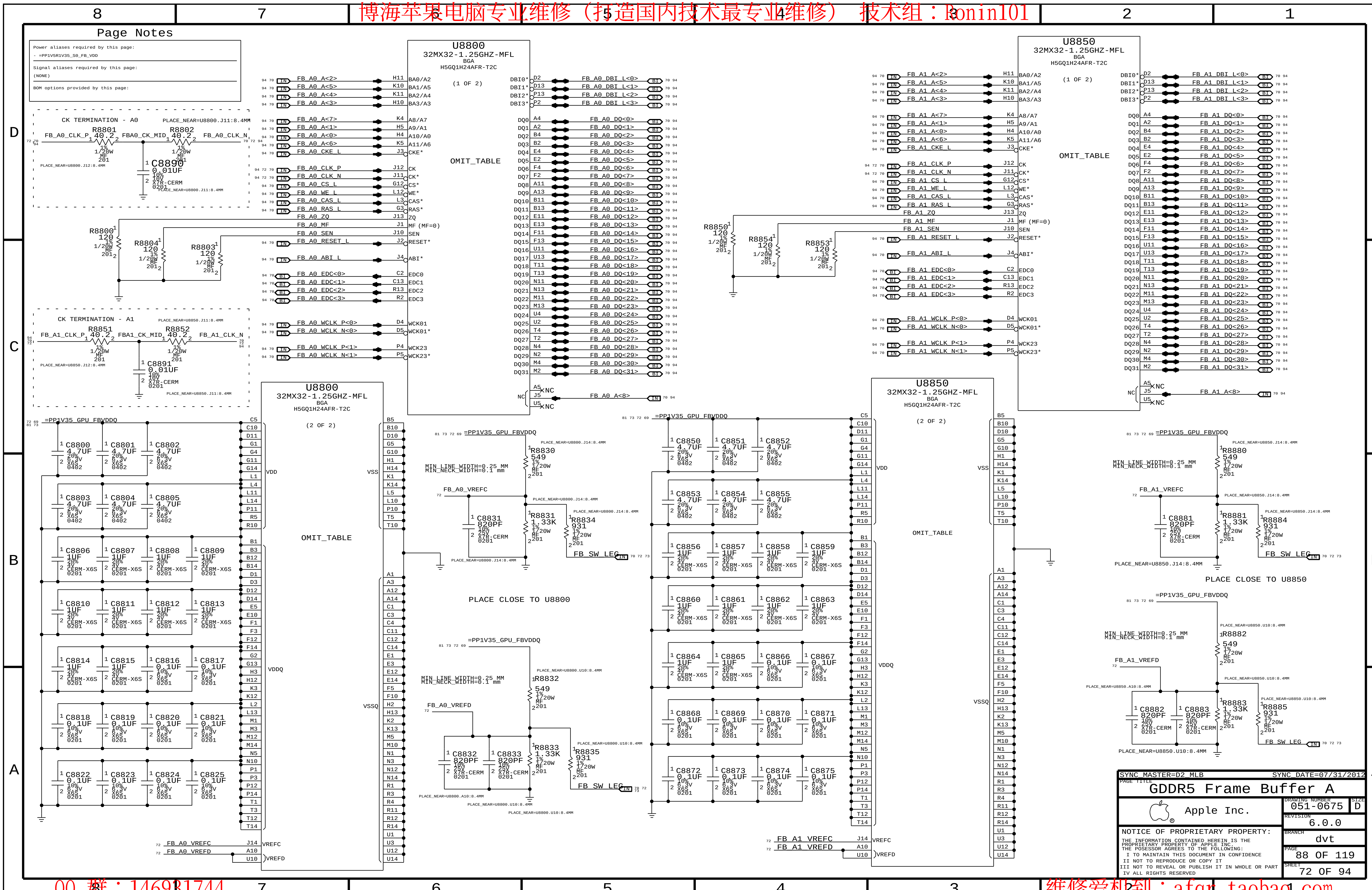

Page Notes
Power aliases required by this page:
- =PP1V5R1V35_S0_FB_VDD
Signal aliases required by this page:
(NONE)
BOM options provided by this page:
CK TERMINATION - A0
CK TERMINATION - A1
U8800
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C
(1 OF 2)
(2 OF 2)
U8850
OMIT_TABLE
=PP1V35_GPU_FBVDDQ
MIN_LINE_WIDTH=0.25 MM
MIN_NECK_WIDTH=0.1 mm
FB_A0_VREFC
FB_A1_VREFC
FB_SW_LEG
PLACE CLOSE TO U8800
PLACE CLOSE TO U8850
VDD
VSS
VDDQ
VSSQ
VREFC
VREFD
SYNC_MASTER=D2_MLB
SYNC_DATE=07/31/2012
PAGE TITLE
GDDR5 Frame Buffer A
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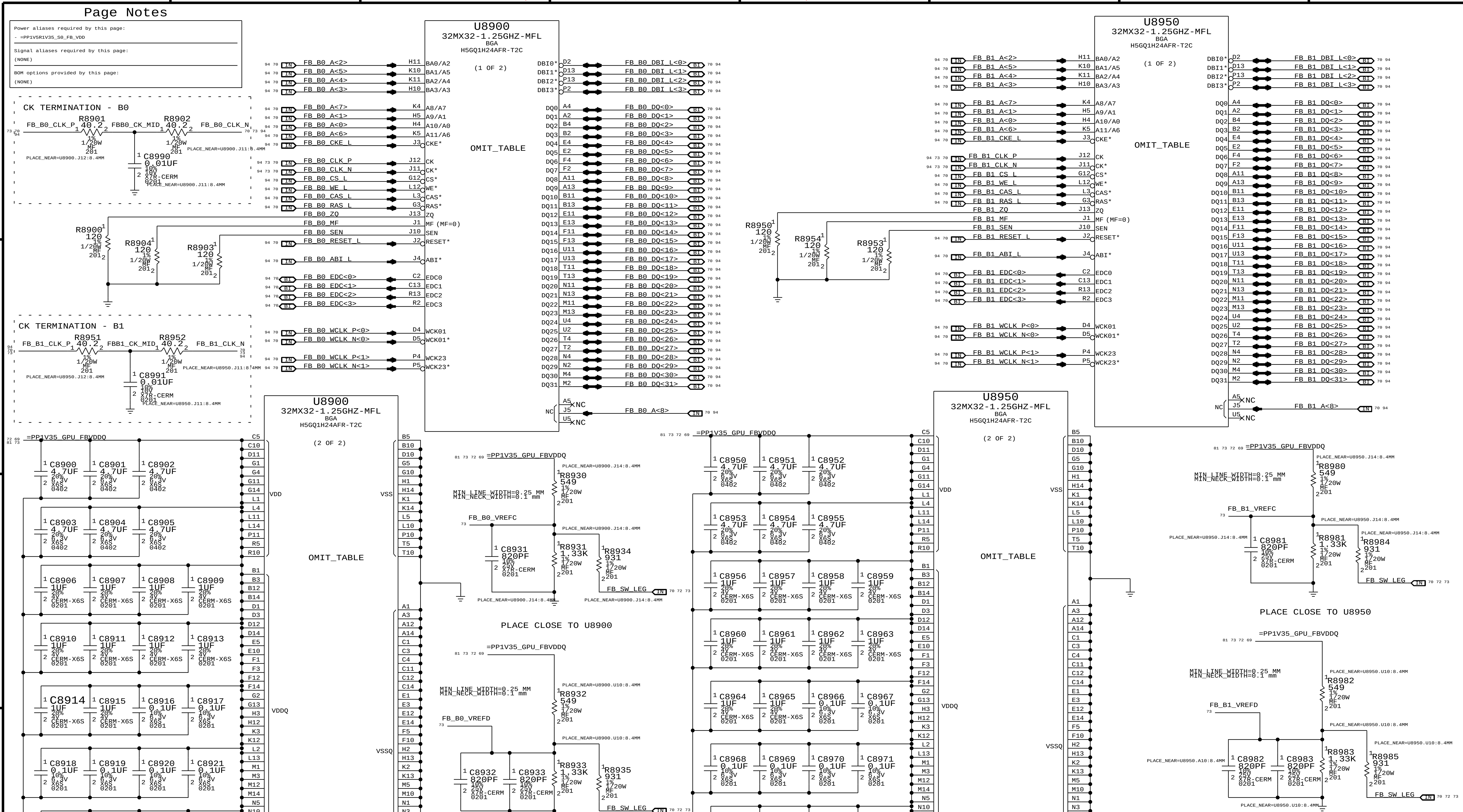

SYNC_MASTER=D2_MLB SYNC_DATE=07/31/2012

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GDDR5 Frame Buffer B

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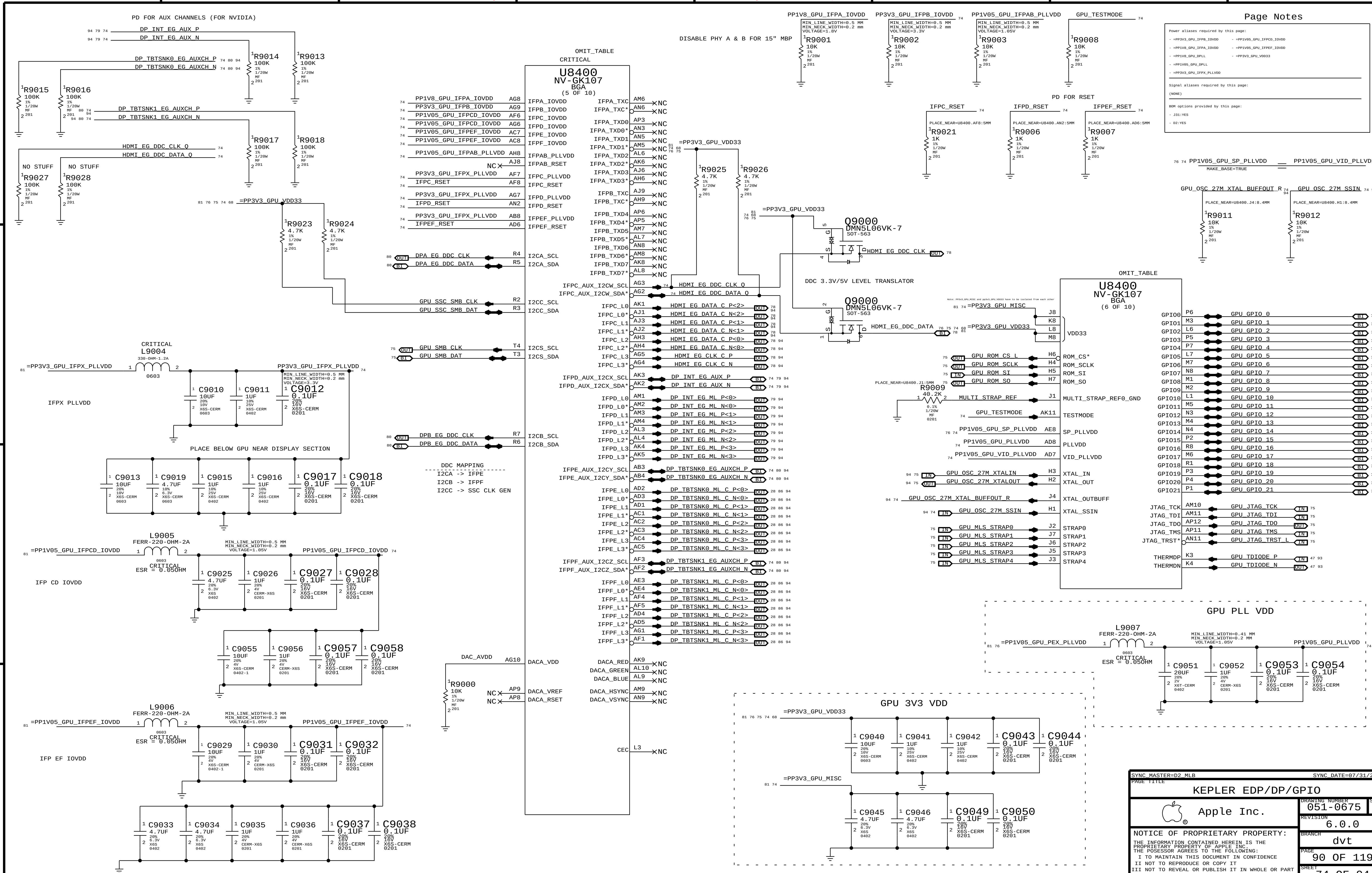

SYNC_MASTER=D2_MLB SYNC_DATE=07/31/2012

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KEPLER EDP/DP/GPIO

Apple Inc.

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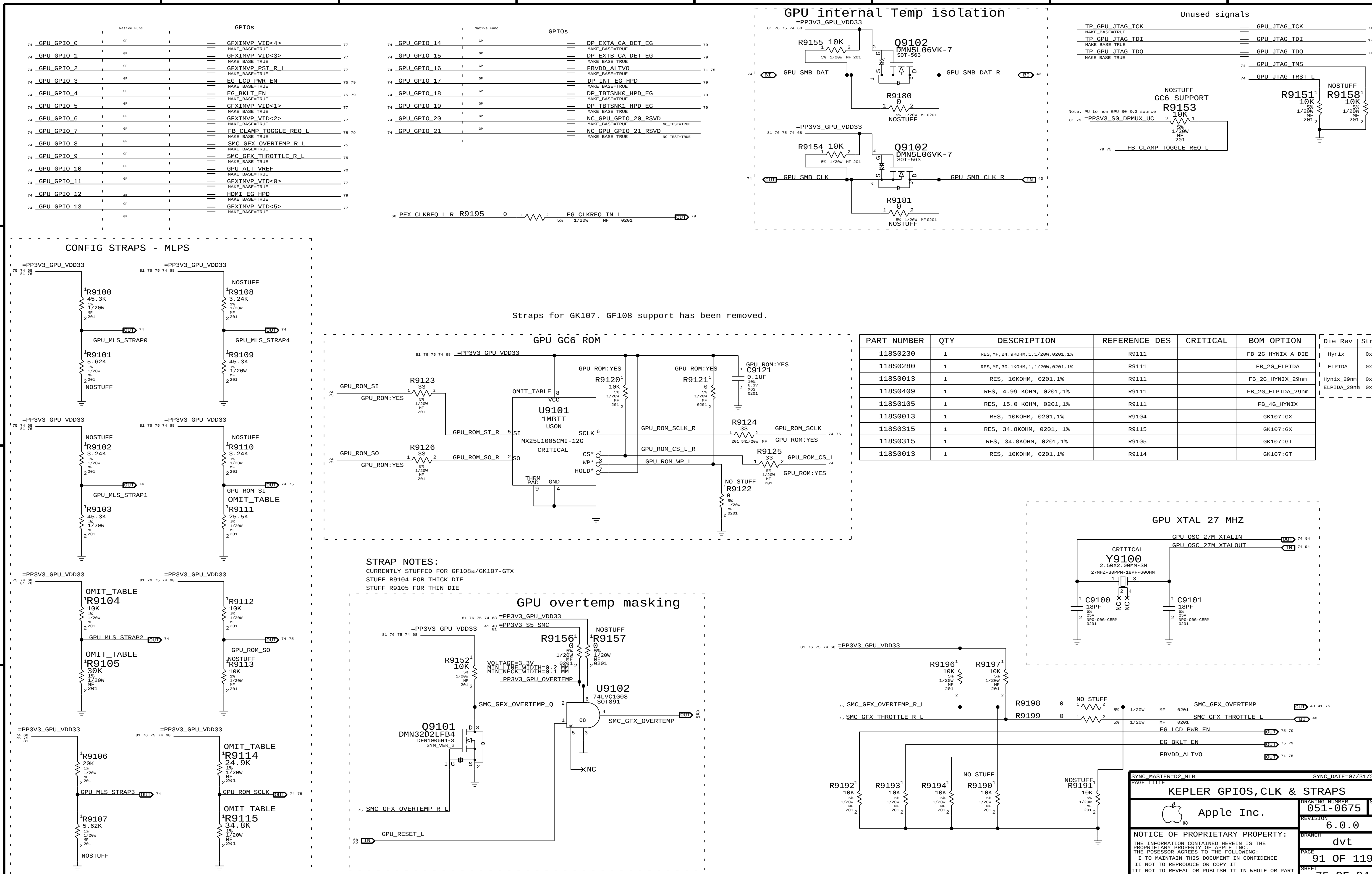

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0230	1	RES,MF,24.9KOHM,1,1/20W,0201,1%	R9111		FB_2G_HYNIX_A_DIE
118S0280	1	RES,MF,30.1KOHM,1,1/20W,0201,1%	R9111		FB_2G_ELPIDA
118S0013	1	RES, 10KOHM, 0201,1%	R9111		FB_2G_HYNIX_29nm
118S0409	1	RES, 4.99 KOHM, 0201,1%	R9111		FB_2G_ELPIDA_29nm
118S0105	1	RES, 15.0 KOHM, 0201,1%	R9111		FB_4G_HYNIX
118S0013	1	RES, 10KOHM, 0201,1%	R9104		GK107:GX
118S0315	1	RES, 34.8KOHM, 0201, 1%	R9115		GK107:GX
118S0315	1	RES, 34.8KOHM, 0201,1%	R9105		GK107:GT
118S0013	1	RES, 10KOHM, 0201,1%	R9114		GK107:GT

Die Rev	Strap
Hynix	0x4
ELPIDA	0x5
Hynix_29nm	0x1
ELPIDA_29nm	0x0

SYNC_MASTER=D2_MLB SYNC_DATE=07/31/2012

PAGE TITLE: KEPLER GPIOS,CLK & STRAPS

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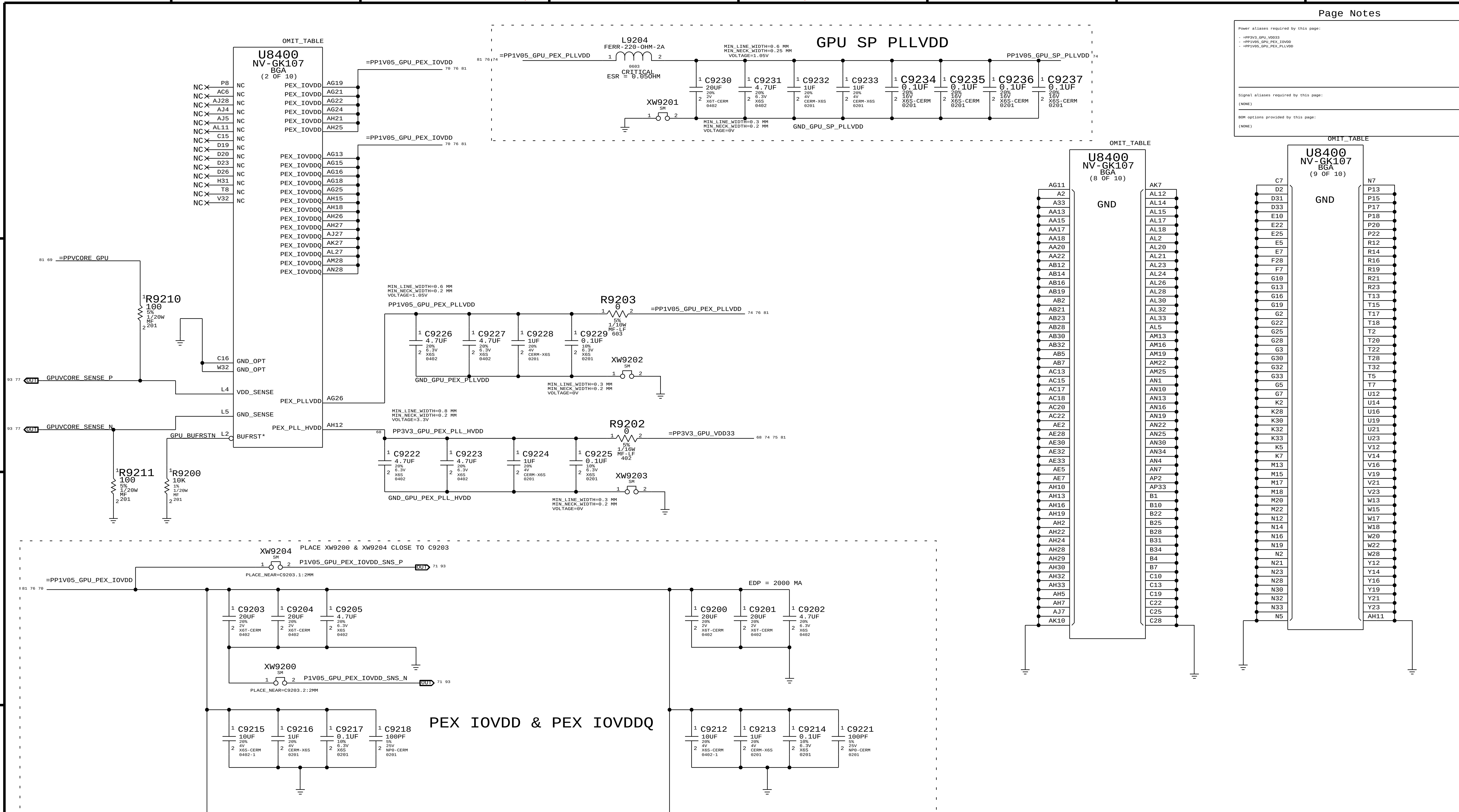

SYNC_MASTER=D2_MLB	SYNC_DATE=07/31/2012
PAGE TITLE	KEPLER PEX PWR/GNDS
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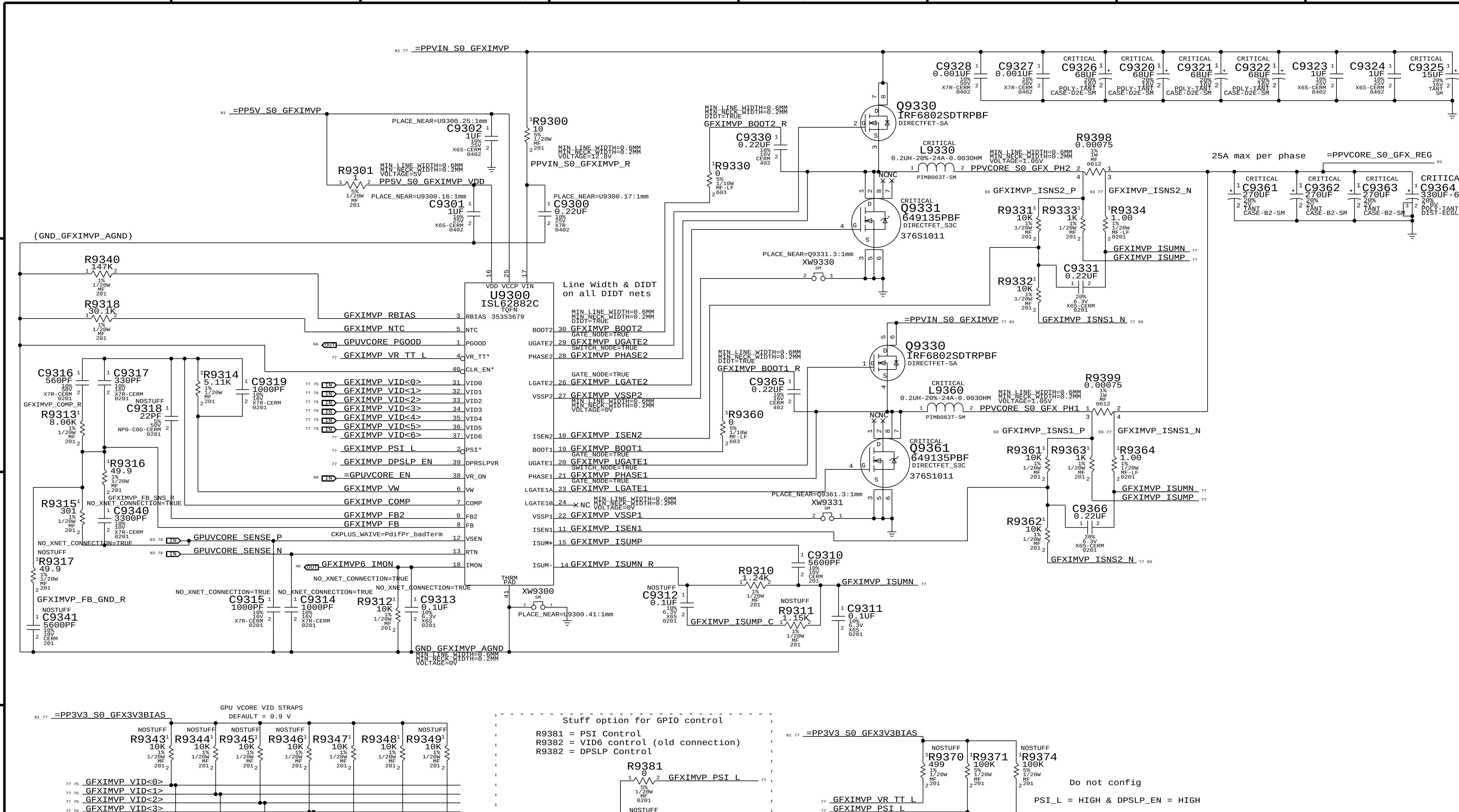

SYNC_MASTER=D2_MLB SYNC_DATE=07/31/2012

PAGE TITLE

GFX IMVP VCore Regulator

Apple Inc.

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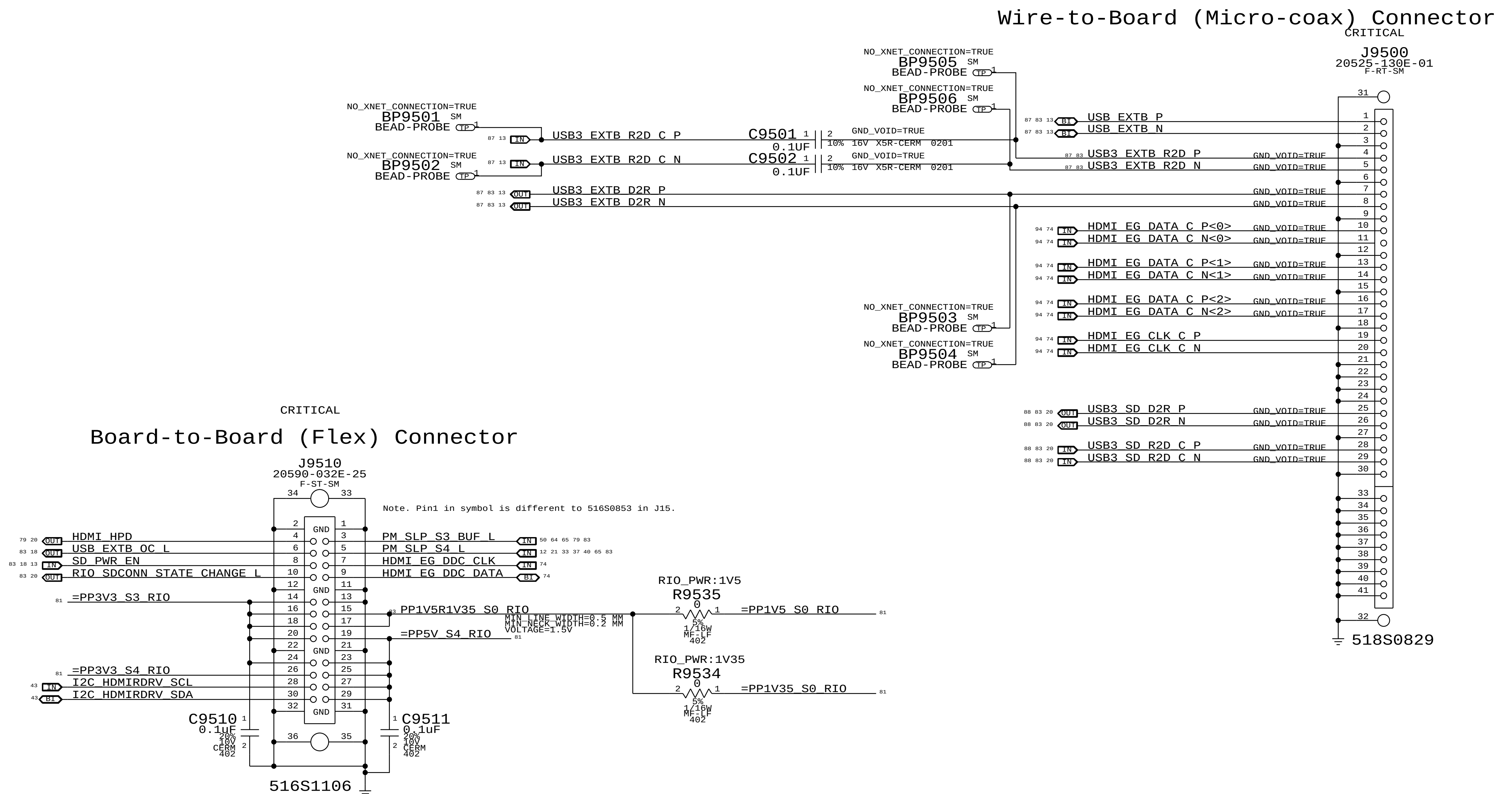

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE

RIO Connectors

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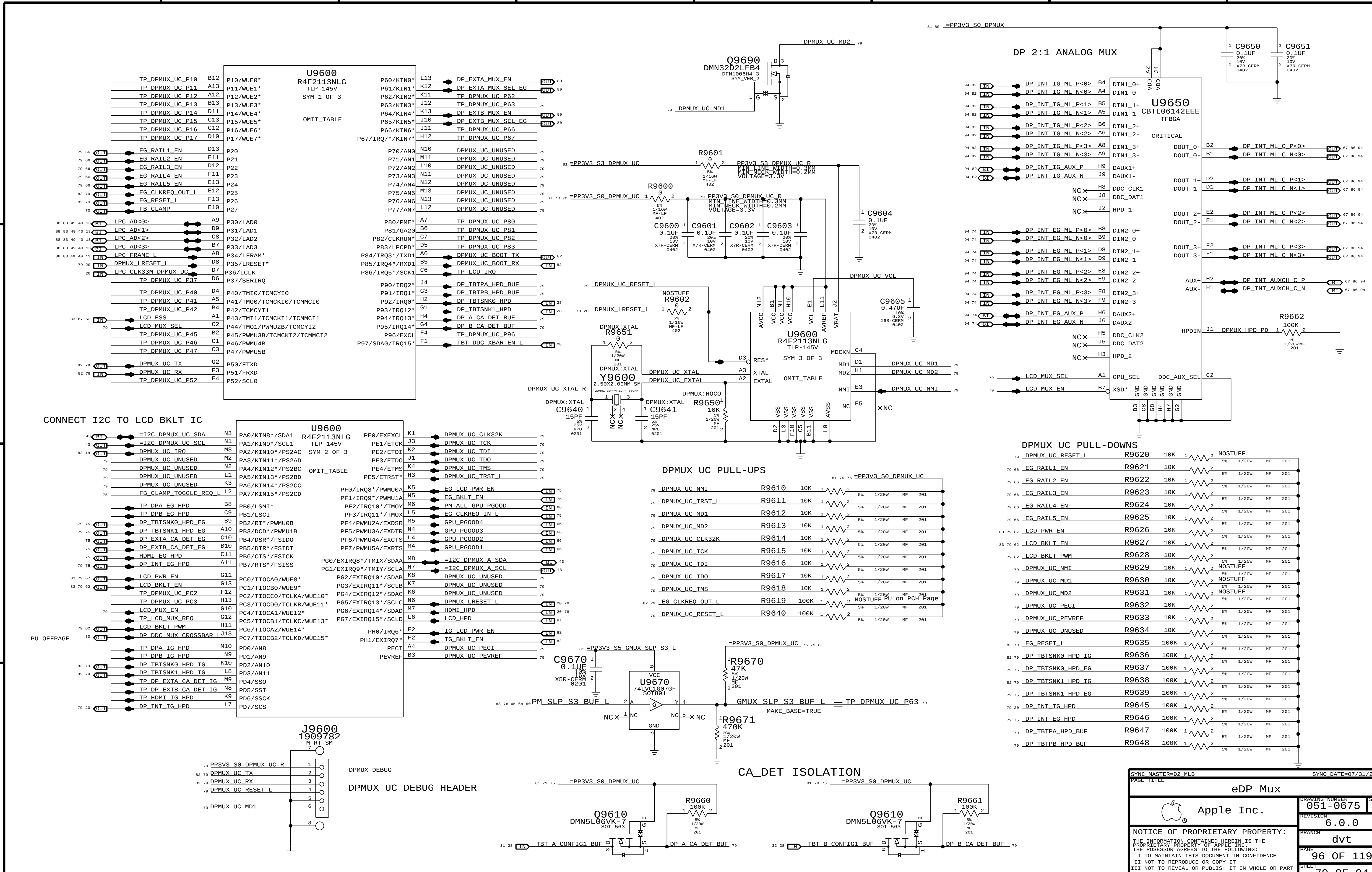
DP 2:1 ANALOG MUX
U9650
CBTL06142EEE
TFBGA
CRITICAL
C9650
0.1UF
C9651
0.1UF
=PP3V3_S0_DPMUX
U9600
R4F2113NLG
TLP-145V
SYM 1 OF 3
OMIT_TABLE
Q9690
DMN32D2LFB4
DFN1006H4-3
SYM_VER_2
DPMUX_UC_MD2
DPMUX_UC_MD1
R9601
R9600
=PP3V3_S3_DPMUX_UC
=PP3V3_S0_DPMUX_UC
PP3V3_S3_DPMUX_UC_R
PP3V3_S0_DPMUX_UC_R
C9600
C9601
C9602
C9603
C9604
C9605
0.47UF
R9602
NOSTUFF
R9651
Y9600
DPMUX:XTAL
C9640
15PF
C9641
15PF
R9650
10K
DPMUX:HOCO
U9600
R4F2113NLG
TLP-145V
SYM 3 OF 3
OMIT_TABLE
R9662
100K
DPMUX_HPD_PD
CONNECT I2C TO LCD BKLT IC
U9600
R4F2113NLG
TLP-145V
SYM 2 OF 3
OMIT_TABLE
PU OFFPAGE
DPMUX UC PULL-UPS
R9610 10K
R9611 10K
R9612 10K
R9613 10K
R9614 10K
R9615 10K
R9616 10K
R9617 10K
R9618 10K
R9619 100K
R9640 100K
NOSTUFF PU on PCH Page
DPMUX UC PULL-DOWNS
R9620 10K NOSTUFF
R9621 10K
R9622 10K
R9623 10K
R9624 10K
R9625 10K
R9626 10K
R9627 10K
R9628 10K
R9629 10K NOSTUFF
R9630 10K NOSTUFF
R9631 10K NOSTUFF
R9632 10K
R9633 10K
R9634 10K
R9635 100K
R9636 100K
R9637 100K
R9638 100K
R9639 100K
R9645 100K
R9646 100K
R9647 100K
R9648 100K
C9670
0.1UF
U9670
74LVC1G07GF
SOT891
=PP3V3_S5_GMUX_SLP_S3_L
PM_SLP_S3_BUF_L
GMUX_SLP_S3_BUF_L
TP_DPMUX_UC_P63
MAKE_BASE=TRUE
R9670
47K
R9671
470K
J9600
1909782
M-RT-SM
DPMUX_DEBUG
DPMUX UC DEBUG HEADER
PP3V3_S0_DPMUX_UC_R
DPMUX_UC_TX
DPMUX_UC_RX
DPMUX_UC_RESET_L
DPMUX_UC_MD1
CA_DET ISOLATION
Q9610
DMN5L06VK-7
SOT-563
R9660
100K
R9661
100K
TBT_A_CONFIG1_BUF
DP_A_CA_DET_BUF
TBT_B_CONFIG1_BUF
DP_B_CA_DET_BUF
SYNC_MASTER=D2_MLB
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eDP Mux
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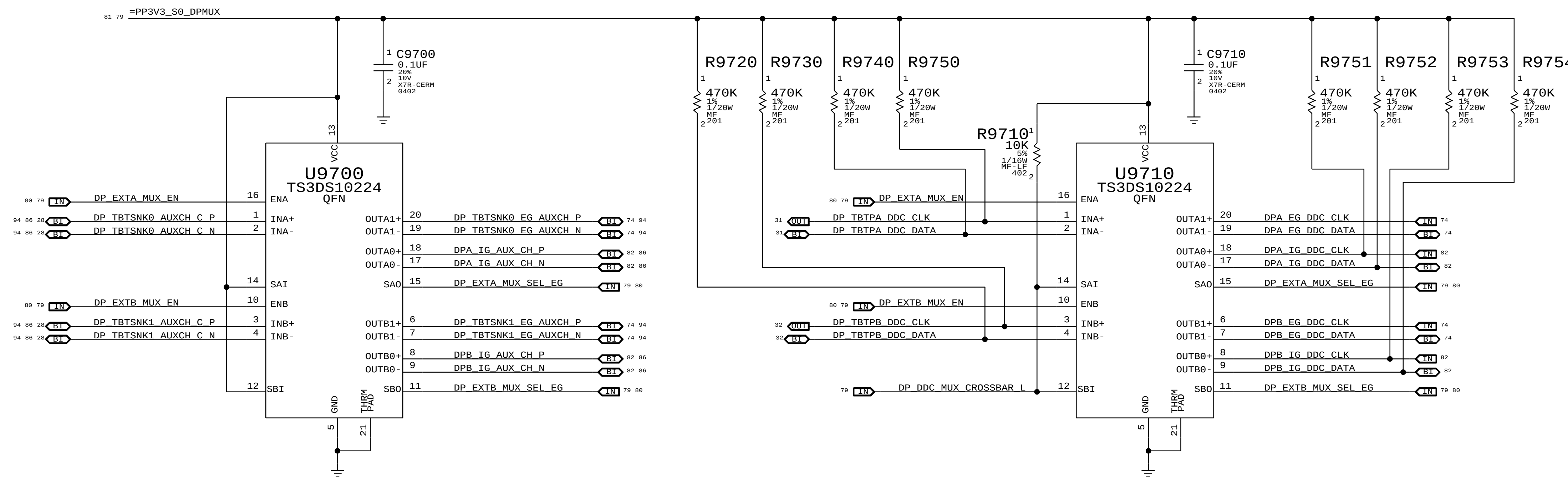
DP A & DP B AUX MUX

DP A & DP B DDC MUX



MUX TRUTH TABLE

SAI/SBI	SAO	SBO	INA	INB
0	0	0	OUTB0	OUTA0
0	0	1	OUTB1	OUTA0
0	1	0	OUTB0	OUTA1
0	1	1	OUTB1	OUTA1
1	0	0	OUTA0	OUTB0
1	0	1	OUTA0	OUTB1
1	1	0	OUTA1	OUTB0
1	1	1	OUTA1	OUTB1

SYNC_MASTER=CLEAN_D2 SYNC_DATE=08/14/2012
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eDP Muxed Graphics Support
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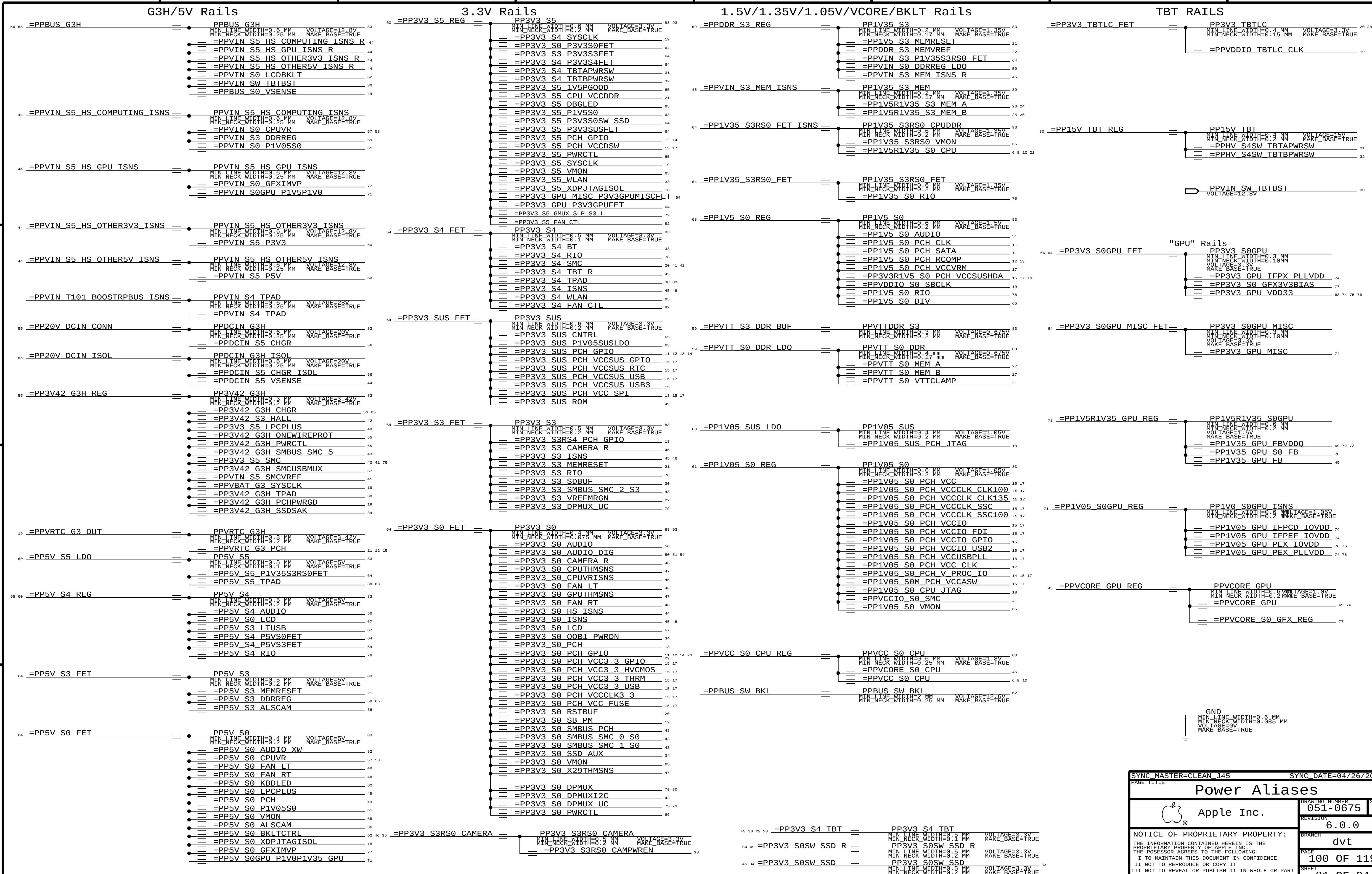

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: Power Aliases

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CPU signals

MEMVTT_EN = =DDRVTT_EN
MAKE_BASE=TRUE

Display Aliases

EDP_IG_PANEL_PWR (MAKE_BASE=TRUE)	=	IG_LCD_PWR_EN
EDP_IG_BKL_ON (MAKE_BASE=TRUE)	=	IG_BKLT_EN
DP_INT_IG_ML_P<3..0> (MAKE_BASE=TRUE)	=	TP_DP_IG_A_MLP<3..0>
DP_INT_IG_ML_N<3..0> (MAKE_BASE=TRUE)	=	TP_DP_IG_A_MLN<3..0>
DP_INT_IG_AUX_P (MAKE_BASE=TRUE)	=	TP_DP_IG_A_AUXCHP
DP_INT_IG_AUX_N (MAKE_BASE=TRUE)	=	TP_DP_IG_A_AUXCHN
DPA_IG_AUX_CH_P (MAKE_BASE=TRUE)	=	TP_DP_IG_B_AUXCHP
DPA_IG_AUX_CH_N (MAKE_BASE=TRUE)	=	TP_DP_IG_B_AUXCHN
DPB_IG_AUX_CH_P (MAKE_BASE=TRUE)	=	TP_DP_IG_C_AUXCHP
DPB_IG_AUX_CH_N (MAKE_BASE=TRUE)	=	TP_DP_IG_C_AUXCHN
DPA_IG_DDC_CLK (MAKE_BASE=TRUE)	=	TP_DP_IG_B_DDC_CLK
DPA_IG_DDC_DATA (MAKE_BASE=TRUE)	=	TP_DP_IG_B_DDC_DATA
DPB_IG_DDC_CLK (MAKE_BASE=TRUE)	=	TP_DP_IG_C_DDC_CLK
DPB_IG_DDC_DATA (MAKE_BASE=TRUE)	=	TP_DP_IG_C_DDC_DATA
DP_TBTSNK0_HPD_IG (MAKE_BASE=TRUE)	=	TP_DP_IG_B_HPD
DP_TBTSNK1_HPD_IG (MAKE_BASE=TRUE)	=	TP_DP_IG_C_HPD
DPMUX_UC_RX (MAKE_BASE=TRUE)	=	DPMUX_UC_BOOT_RX
DPMUX_UC_TX (MAKE_BASE=TRUE)	=	DPMUX_UC_BOOT_TX
EG_RESET_L (MAKE_BASE=TRUE)	=	GPU_RESET_L
PEG_CLKREQ_L (MAKE_BASE=TRUE)	=	EG_CLKREQ_OUT_L
DP_AUXCH_ISOL_L (MAKE_BASE=TRUE)	=	DP_AUXIO_EN

PEG_D2R_P<7..0> (MAKE_BASE=TRUE)	=	=PEG_D2R_P<7..0>
PEG_D2R_N<7..0> (MAKE_BASE=TRUE)	=	=PEG_D2R_N<7..0>
PEG_R2D_C_P<7..0> (MAKE_BASE=TRUE)	=	=PEG_R2D_C_P<7..0>
PEG_R2D_C_N<7..0> (MAKE_BASE=TRUE)	=	=PEG_R2D_C_N<7..0>
NC_PCIE_PEG_D2RP<15..12> (MAKE_BASE=TRUE, NO_TEST=TRUE)	=	=PEG_D2R_P<15..12>
NC_PCIE_PEG_D2RN<15..12> (MAKE_BASE=TRUE, NO_TEST=TRUE)	=	=PEG_D2R_N<15..12>
NC_PCIE_PEG_R2D_CP<15..12> (MAKE_BASE=TRUE, NO_TEST=TRUE)	=	=PEG_R2D_C_P<15..12>
NC_PCIE_PEG_R2D_CN<15..12> (MAKE_BASE=TRUE, NO_TEST=TRUE)	=	=PEG_R2D_C_N<15..12>

Thunderbolt Signals Through PEG

PCIE_TBT_D2R_P<3..0> (MAKE_BASE=TRUE)	=	=PEG_D2R_P<11..8>
PCIE_TBT_D2R_N<3..0> (MAKE_BASE=TRUE)	=	=PEG_D2R_N<11..8>
PCIE_TBT_R2D_C_P<3..0> (MAKE_BASE=TRUE)	=	=PEG_R2D_C_P<11..8>
PCIE_TBT_R2D_C_N<3..0> (MAKE_BASE=TRUE)	=	=PEG_R2D_C_N<11..8>
PEG_CLK100M_N (MAKE_BASE=TRUE)	=	TP_PCIE_CLK100M_GPUN
PEG_CLK100M_P (MAKE_BASE=TRUE)	=	TP_PCIE_CLK100M_GPUP

TP_P1V8GPU_EN = =P1V8GPU_EN

=PP3V3_S3_FAN_CTL = PP3V3_S3_FAN_CTL
MIN_LINE_WIDTH=0.4MM
MIN_NECK_WIDTH=0.2MM
VOLTAGE=3.3V
MAKE_BASE=TRUE

RA201 0 5% 1/20W MF 0201 → =PP3V3_S5_FAN_CTL

RA202 0 5% 1/20W MF 0201 NOSTUFF → =PP3V3_S4_FAN_CTL

Unused signals

BT_PWRRST_L
MEM_VDD_SEL_1V5_L
FW_PWR_EN_PCH
WOL_EN
FW_PME_L
DP_TBT_SEL
ENET_MEDIA_SENSE_RDIV
AUD_IPHS_SWITCH_EN_PCH
AUD_IP_PERIPHERAL_DET
AUD_I2C_INT_L
TBT_GO2SX_BIDIR
DPMUX_UC_IRQ
PEG_CLKREQ_L
ENET_CLKREQ_L
ENET_LOW_PWR_PCH
HDMITBTMUX_SEL_TBT
SDCONN_OC_L

		VOLTAGE	MAKE_BASE	
PPVREF_S3_MEM_VREFDQ_A	=	0.675V	TRUE	PP0V75_S3_MEM_VREFDQ_A
PPVREF_S3_MEM_VREFDQ_B	=	0.675V	TRUE	PP0V75_S3_MEM_VREFDQ_B
PPVREF_S3_MEM_VREFCA_A	=	0.675V	TRUE	PP0V75_S3_MEM_VREFCA_A
PPVREF_S3_MEM_VREFCA_B	=	0.675V	TRUE	PP0V75_S3_MEM_VREFCA_B

=PP5V_S0_AUDIO_XW

XWA202 SM → PP5V_S0_AUDIO_AMP_L
MIN_LINE_WIDTH=0.4MM
MIN_NECK_WIDTH=0.2MM
VOLTAGE=5V

XWA203 SM → PP5V_S0_AUDIO_AMP_R
MIN_LINE_WIDTH=0.4MM
MIN_NECK_WIDTH=0.2MM
VOLTAGE=5V

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Signal Aliases
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Functional Test Points

FUNC_TEST J3501 - airport

TRUE	AP_CLKREQ_Q_L	33
TRUE	AP_RESET_CONN_L	33
TRUE	PCIE_AP_D2R_PI_N	33 88
TRUE	PCIE_AP_D2R_PI_P	33 88
TRUE	PCIE_AP_R2D_N	33 88
TRUE	PCIE_AP_R2D_P	33 88
TRUE	PCIE_CLK100M_AP_CONN_N	33 88
TRUE	PCIE_CLK100M_AP_CONN_P	33 88
TRUE	PCIE_WAKE_L	12 33 35 88
TRUE	PP3V3_S3RS4_BT_F	33
TRUE	PP3V3_WLAN	33 41
TRUE	USB_BT_CONN_N	33 87
TRUE	USB_BT_CONN_P	33 87
TRUE	WIFI_EVENT_L	33 40 41
TRUE	GND	4X

J4002 - Camera

TRUE	MIPI_CLK_CONN_N	36 91
TRUE	MIPI_CLK_CONN_P	36 91
TRUE	CAM_SENSOR_WAKE_L_CONN	36
TRUE	MIPI_DATA_CONN_N	36 91
TRUE	MIPI_DATA_CONN_P	36 91
TRUE	=I2C_ALS_SDA	36 43
TRUE	=I2C_ALS_SCL	36 43
TRUE	I2C_CAM_SCK	35 36
TRUE	I2C_CAM_SDA	35 36
TRUE	PP5V_S3RS0_ALSCAM_F	36
TRUE	GND	

J9500 - rio coax

TRUE	HDMI_CLK_N	86
TRUE	HDMI_CLK_P	86
TRUE	HDMI_DATA_N<0>	86
TRUE	HDMI_DATA_N<1>	86
TRUE	HDMI_DATA_N<2>	86
TRUE	HDMI_DATA_P<0>	86
TRUE	HDMI_DATA_P<1>	86
TRUE	HDMI_DATA_P<2>	86
TRUE	USB3_SD_D2R_N	20 78 88
TRUE	USB3_SD_D2R_P	20 78 88
TRUE	USB3_SD_R2D_C_N	20 78 88
TRUE	USB3_SD_R2D_C_P	20 78 88
TRUE	USB3_EXTB_D2R_N	13 78 87
TRUE	USB3_EXTB_D2R_P	13 78 87
TRUE	USB3_EXTB_R2D_N	78 87
TRUE	USB3_EXTB_R2D_P	78 87
TRUE	USB_EXTB_N	13 78 87
TRUE	USB_EXTB_P	13 78 87
TRUE	GND	19X

J9510 - rio flex

TRUE	SD_PWR_EN	13 18 78
TRUE	PP1V5R1V35_S0_RIO	78
TRUE	HDMI_DDC_CLK	
TRUE	HDMI_DDC_DATA	
TRUE	HDMI_HPD_L	
TRUE	SMBUS_PCH_CLK	13 43 88
TRUE	SMBUS_PCH_DATA	13 43 88
TRUE	PM_SLP_S3_BUF_L	50 64 65 78 79
TRUE	PM_SLP_S4_L	12 21 33 37 40 65 78
TRUE	PP3V3_S3	3X 81 83
TRUE	PP3V3_S4	81 83
TRUE	PP5V_S4	5X 81
TRUE	RIO_SDCONN_STATE_CHANGE_L	20 78
TRUE	USB_EXTB_OC_L	18 78
TRUE	GND	10X

J5150 - hall effect

TRUE	PP3V42_G3H	81 83
TRUE	SMC_LID_R	42
TRUE	GND	

J6050 - left fan

TRUE	FAN_LT_PWM	48
TRUE	FAN_LT_TACH	48
TRUE	PP5V_S0	3X 81 83
TRUE	GND	5X

J6060 - right fan

TRUE	FAN_RT_PWM	48
TRUE	FAN_RT_TACH	48
TRUE	PP5V_S0	3X 81 83
TRUE	GND	5X

FUNC_TEST J6100 - lpc + spi

TRUE	LPCPLUS_GPIO	14 49
TRUE	LPCPLUS_RESET_L	20 49
TRUE	LPC_AD<0>	13 40 49 79 88
TRUE	LPC_AD<1>	13 40 49 79 88
TRUE	LPC_AD<2>	13 40 49 79 88
TRUE	LPC_AD<3>	13 40 49 79 88
TRUE	LPC_CLK33M_LPCPLUS	19 49 88
TRUE	LPC_FRAME_L	13 40 49 79 88
TRUE	LPC_PWRDWN_L	12 20 40 49
TRUE	LPC_SERIRQ	13 40 49
TRUE	PM_CLKRUN_L	12 40 49
TRUE	PP5V_S0	81 83
TRUE	SMC_RESET_L	40 41 49 56
TRUE	SMC_ROMBOOT	41 49
TRUE	SMC_RX_L	40 41 49
TRUE	SMC_TCK	40 41 49
TRUE	SMC_TDI	40 41 49
TRUE	SMC_TDO	40 41 49
TRUE	SMC_TMS	40 41 49
TRUE	SMC_TX_L	40 41 49
TRUE	SPIROM_USE_MLB	14 49
TRUE	SPI_ALT_CLK	49
TRUE	SPI_ALT_CS_L	49
TRUE	SPI_ALT_MISO	49
TRUE	SPI_ALT_MOSI	49
TRUE	TP_SMC_MD1	49
TRUE	TP_SMC_TRST_L	49
TRUE	GND	2X

J4800 - ipd flex

TRUE	Z2_CS_L	38
TRUE	Z2_MOSI	38
TRUE	Z2_MISO	38
TRUE	Z2_HOST_INTN	38
TRUE	Z2_CLKIN	38
TRUE	Z2_KEY_ACT_L	38
TRUE	PSOC_F_CS_L	38
TRUE	PICKB_L	38
TRUE	PSOC_MOSI	38
TRUE	PSOC_MISO	38
TRUE	PSOC_SCLK	38
TRUE	=I2C_TPAD_SCL	38 43
TRUE	=I2C_TPAD_SDA	38 43
TRUE	SMC_LID	38 40 41 42
TRUE	SMC_T101_COM_1	38
TRUE	=PP3V3_S4_TPAD	38 81
TRUE	=PP5V_S5_TPAD	38 81
TRUE	GND	2X

J4813 - keyboard

TRUE	PP3V3_S4	81 83
TRUE	PP3V42_G3H	81 83
TRUE	WS_CONTROL_KBD	38
TRUE	WS_KBD1	38
TRUE	WS_KBD10	38
TRUE	WS_KBD11	38
TRUE	WS_KBD12	38
TRUE	WS_KBD13	38
TRUE	WS_KBD14	38
TRUE	WS_KBD15_CAP	38
TRUE	WS_KBD16_NUM	38
TRUE	WS_KBD17	38
TRUE	WS_KBD18	38
TRUE	WS_KBD19	38
TRUE	WS_KBD2	38
TRUE	WS_KBD20	38
TRUE	WS_KBD21	38
TRUE	WS_KBD22	38
TRUE	WS_KBD23	38
TRUE	WS_KBD3	38
TRUE	WS_KBD4	38
TRUE	WS_KBD5	38
TRUE	WS_KBD6	38
TRUE	WS_KBD7	38
TRUE	WS_KBD8	38
TRUE	WS_KBD9	38
TRUE	WS_KBD_ONOFF_L	38
TRUE	WS_LEFT_OPTION_KBD	38
TRUE	WS_LEFT_SHIFT_KBD	38
TRUE	GND	2X

J4915 - kbd bklt

TRUE	KBDBKLT_RETURN1	2X 39 62
TRUE	KBDBKLT_RETURN2	2X 39 62
TRUE	PPVOUT_S0_KBDBKLT	39 62
TRUE	GND	4X

FUNC_TEST J6701 - audio flex

TRUE	AUD_HP_PORT_L	50 54
TRUE	AUD_HP_PORT_R	50 54
TRUE	AUD_SPDIF_OUT_JACK	
TRUE	AUD_TIPDET_INV	
TRUE	AUD_TYPEDET	50 54
TRUE	AUD_CONN_MIC_XW	4X
TRUE	CH_HS_MIC	
TRUE	PP3V3_S0	81 83 93
TRUE	AUD_CONN_SLEEVE_XW	4X 53 54
TRUE	US_HS_MIC	
TRUE	GND	2X GND

J6601 - mic

TRUE	DMIC_CLK3	51 54
TRUE	PP3V3_S0	81 83 93
TRUE	DMIC_SDA2	54
TRUE	DMIC_SDA3	51 54
TRUE	GND	

J6602 - L speaker

TRUE	SPKRCONN_L_ID	51 54
TRUE	SPKRCONN_L_OUT_N	52 54 93
TRUE	SPKRCONN_L_OUT_P	52 54 93
TRUE	SPKRCONN_SL_OUT_N	52 54 93
TRUE	SPKRCONN_SL_OUT_P	52 54 93
TRUE	GND	

J6603 - R speaker

TRUE	SPKRCONN_R_ID	51 54
TRUE	SPKRCONN_R_OUT_N	52 54 93
TRUE	SPKRCONN_R_OUT_P	52 54 93
TRUE	SPKRCONN_SR_OUT_N	52 54 93
TRUE	SPKRCONN_SR_OUT_P	52 54 93
TRUE	GND	

J7000 - DC PWR

TRUE	ADAPTER_SENSE	55
TRUE	PP20V_DCIN_FUSE	2X 55
TRUE	GND	2X

J7050 - battery

TRUE	PPVBAT_G3H_CONN	8X 55 56
TRUE	SMBUS_SMC_5_G3_SCL	40 43 92
TRUE	SMBUS_SMC_5_G3_SDA	40 43 92
TRUE	SYS_DETECT_L	55
TRUE	GND	8X

J8300 - eDP

TRUE	DP_INT_AUX_N	67 86 94
TRUE	DP_INT_AUX_P	67 86 94
TRUE	DP_INT_ML_N<0>	67 86 94
TRUE	DP_INT_ML_N<1>	67 86 94
TRUE	DP_INT_ML_N<2>	67 86 94
TRUE	DP_INT_ML_N<3>	67 86 94
TRUE	DP_INT_ML_P<0>	67 86 94
TRUE	DP_INT_ML_P<1>	67 86 94
TRUE	DP_INT_ML_P<2>	67 86 94
TRUE	DP_INT_ML_P<3>	67 86 94
TRUE	LCD_FSS	62 67 79
TRUE	LCD_HPD_CONN	67
TRUE	LED_RETURN_1	62 67
TRUE	LED_RETURN_2	62 67
TRUE	LED_RETURN_3	62 67
TRUE	LED_RETURN_4	62 67
TRUE	LED_RETURN_5	62 67
TRUE	LED_RETURN_6	62 67
TRUE	PP5VR3V3_SW_LCD	3X 67
TRUE	PPVOUT_S0_LCDBKLT	62 67
TRUE	GND	16X

Power Rails

TRUE	PM_SLP_S3_L	12 21 40 65
TRUE	PPVTT_S0_DDR	81
TRUE	PP3V3_S0	81 83 93
TRUE	PP3V3_S3	81 83
TRUE	PP3V3_S5	81 93
TRUE	PP3V3_S5_AVREF_SMC	40 41
TRUE	PP3V42_G3H	81 83
TRUE	PP5V_S0	81 83
TRUE	PP5V_S3	81
TRUE	PP5V_S5	81
TRUE	PPBUS_G3H	81
TRUE	PPDCIN_G3H	81
TRUE	PPVCC_S0_CPU	81
TRUE	PPVTTDDR_S3	81
TRUE	PP3V3_S0SW_SSD	81
TRUE	PP1V5_S0	81
TRUE	PP1V35_S3	81

FUNC_TEST XDP

TRUE	XDP_CPU_TCK	6 18 86
TRUE	XDP_PCH_TCK	11 18
TRUE	XDP_CPU_TDI	6 18 86
TRUE	XDP_CPU_TDO	6 18 86
TRUE	XDP_CPU_TRST_L	6 18 86
TRUE	XDP_CPU_TMS	6 18 86
TRUE	XDP_PCH_TMS	11 18
TRUE	XDP_PCH_TDI	11 18
TRUE	XDP_PCH_TDO	11 18
TRUE	XDP_CPU_PREQ_L	6 18 86
TRUE	XDP_CPU_PRDY_L	6 18 86
TRUE	PM_RSMRST_L	12 65 88
TRUE	PM_PCH_PWROK	12 19 88
TRUE	PM_SYSRST_L	12 19 40 88
TRUE	CPU_CFG<3>	6 18 86
TRUE	PP1V05_S0	81
TRUE	GND	2X GND

FUNC_TEST Power Sequence

TRUE	SMC_ONOFF_L	38 40 41
TRUE	PM_DSW_PWRGD	12 40 88
TRUE	ALL_SYS_PWRGD	18 19 40 65
TRUE	PM_PCH_SYS_PWROK	12 18 19 40 88
TRUE	PLT_RESET_L	12 18 20 21
TRUE	LCD_PWR_EN	67 79
TRUE	LCD_BKLT_EN	62 79

TPA401 A TP-P6 — PEG_CLK100M_P 68 82 88
TPA402 A TP-P6 — PEG_CLK100M_N 68 82 88

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NC NO_TESTs

PCH

Net	NO_TEST	MAKE_BASE	NC
TP_USB3_SPARE_D2RN	TRUE	TRUE	NC_USB3_SPARE_D2RN
TP_USB3_SPARE_D2RP	TRUE	TRUE	NC_USB3_SPARE_D2RP
TP_USB3_SPARE_R2D_CN	TRUE	TRUE	NC_USB3_SPARE_R2D_CN
TP_USB3_SPARE_R2D_CP	TRUE	TRUE	NC_USB3_SPARE_R2D_CP
USB3_EXTC_D2R_N	TRUE	TRUE	NC_USB3_EXTC_D2RN
USB3_EXTC_D2R_P	TRUE	TRUE	NC_USB3_EXTC_D2RP
USB3_EXTC_R2D_C_N	TRUE	TRUE	NC_USB3_EXTC_R2D_CN
USB3_EXTC_R2D_C_P	TRUE	TRUE	NC_USB3_EXTC_R2D_CP
USB3_EXTD_D2R_N	TRUE	TRUE	NC_USB3_EXTD_D2RN
USB3_EXTD_D2R_P	TRUE	TRUE	NC_USB3_EXTD_D2RP
USB3_EXTD_R2D_C_N	TRUE	TRUE	NC_USB3_EXTD_R2D_CN
USB3_EXTD_R2D_C_P	TRUE	TRUE	NC_USB3_EXTD_R2D_CP
PCIE_ENET_D2RN	TRUE	TRUE	NC_PCIE_ENET_D2RN
PCIE_ENET_D2RP	TRUE	TRUE	NC_PCIE_ENET_D2RP
PCIE_ENET_R2D_CN	TRUE	TRUE	NC_PCIE_ENET_R2D_CN
PCIE_ENET_R2D_CP	TRUE	TRUE	NC_PCIE_ENET_R2D_CP
SATA_A_D2R_N	TRUE	TRUE	NC_SATA_A_D2RN
SATA_A_D2R_P	TRUE	TRUE	NC_SATA_A_D2RP
SATA_A_R2D_C_N	TRUE	TRUE	NC_SATA_A_R2D_CN
SATA_A_R2D_C_P	TRUE	TRUE	NC_SATA_A_R2D_CP
SATA_B_D2R_N	TRUE	TRUE	NC_SATA_B_D2RN
SATA_B_D2R_P	TRUE	TRUE	NC_SATA_B_D2RP
SATA_B_R2D_C_N	TRUE	TRUE	NC_SATA_B_R2D_CN
SATA_B_R2D_C_P	TRUE	TRUE	NC_SATA_B_R2D_CP
TP_SATA_ODD_D2RN	TRUE	TRUE	NC_SATA_ODD_D2RN
TP_SATA_ODD_D2RP	TRUE	TRUE	NC_SATA_ODD_D2RP
TP_SATA_ODD_R2D_CN	TRUE	TRUE	NC_SATA_ODD_R2D_CN
TP_SATA_ODD_R2D_CP	TRUE	TRUE	NC_SATA_ODD_R2D_CP
TP_SATA_D_D2RN	TRUE	TRUE	NC_SATA_D_D2RN
TP_SATA_D_D2RP	TRUE	TRUE	NC_SATA_D_D2RP
TP_SATA_D_R2D_CN	TRUE	TRUE	NC_SATA_D_R2D_CN
TP_SATA_D_R2D_CP	TRUE	TRUE	NC_SATA_D_R2D_CP
TP_SATA_F_D2RN	TRUE	TRUE	NC_SATA_F_D2RN
TP_SATA_F_D2RP	TRUE	TRUE	NC_SATA_F_D2RP
TP_SATA_F_R2D_CN	TRUE	TRUE	NC_SATA_F_R2D_CN
TP_SATA_F_R2D_CP	TRUE	TRUE	NC_SATA_F_R2D_CP
USB_EXTC_N	TRUE	TRUE	NC_USB_EXTCN
USB_EXTC_P	TRUE	TRUE	NC_USB_EXTCP
TP_USB_SDN	TRUE	TRUE	NC_USB_SDN
TP_USB_SDP	TRUE	TRUE	NC_USB_SDP
TP_USB_WLANN	TRUE	TRUE	NC_USB_WLANN
TP_USB_WLANP	TRUE	TRUE	NC_USB_WLANP
TP_USB_6N	TRUE	TRUE	NC_USB_6N
TP_USB_6P	TRUE	TRUE	NC_USB_6P
TP_USB_7N	TRUE	TRUE	NC_USB_7N
TP_USB_7P	TRUE	TRUE	NC_USB_7P
USB_EXTD_N	TRUE	TRUE	NC_USB_EXTDN
USB_EXTD_P	TRUE	TRUE	NC_USB_EXTDP
TP_USB_PSOCN	TRUE	TRUE	NC_USB_PSOCN
TP_USB_PSOCP	TRUE	TRUE	NC_USB_PSOCP
USB_IR_N	TRUE	TRUE	NC_USB_IRN
USB_IR_P	TRUE	TRUE	NC_USB_IRP
ITPXDP_CLK100M_N	TRUE	TRUE	NC_ITPXDP_CLK100MN
ITPXDP_CLK100M_P	TRUE	TRUE	NC_ITPXDP_CLK100MP
TP_PCI_PME_L	TRUE	TRUE	NC_PCI_PME_L
TP_PCI_CLK33M_OUT2	TRUE	TRUE	NC_PCI_CLK33M_OUT2
TP_PCI_CLK33M_OUT3	TRUE	TRUE	NC_PCI_CLK33M_OUT3
TP_HDA_SDIN1	TRUE	TRUE	NC_HDA_SDIN1
TP_HDA_SDIN2	TRUE	TRUE	NC_HDA_SDIN2
TP_HDA_SDIN3	TRUE	TRUE	NC_HDA_SDIN3
TP_LPC_DREQ0_L	TRUE	TRUE	NC_LPC_DREQ0_L
TP_CLINK_CLK	TRUE	TRUE	NC_CLINK_CLK
TP_CLINK_DATA	TRUE	TRUE	NC_CLINK_DATA
TP_CLINK_RESET_L	TRUE	TRUE	NC_CLINK_RESET_L
EDP_IG_BKL_PWM	TRUE	TRUE	NC_EDP_IG_BKL_PWM
USB_SMC_P	TRUE	TRUE	NC_USB_SMCP
USB_SMC_N	TRUE	TRUE	NC_USB_SMCN
SMC_INTERFACE_2	TRUE	TRUE	NC_SMC_INTERFACE_2

Thunderbolt

Net	NO_TEST	MAKE_BASE	NC
TP_TBT_XTAL25OUT	TRUE	TRUE	NC_TBT_XTAL25OUT
TP_DP_TBTSRC_ML_CP<3..0>	TRUE	TRUE	NC_DP_TBTSRC_ML_CP<3..0>
TP_DP_TBTSRC_ML_CN<3..0>	TRUE	TRUE	NC_DP_TBTSRC_ML_CN<3..0>
TP_DP_TBTSRC_AUXCH_CP	TRUE	TRUE	NC_DP_TBTSRC_AUXCH_CP
TP_DP_TBTSRC_AUXCH_CN	TRUE	TRUE	NC_DP_TBTSRC_AUXCH_CN
TP_DP_IG_D_AUXCHN	TRUE	TRUE	NC_DP_IG_D_AUXCHN
TP_DP_IG_D_AUXCHP	TRUE	TRUE	NC_DP_IG_D_AUXCHP
TP_PCIE_CLK100M_PE5N	TRUE	TRUE	NC_PCIE_CLK100M_PE5N
TP_PCIE_CLK100M_PE5P	TRUE	TRUE	NC_PCIE_CLK100M_PE5P
PCIE_CLK100M_ENETSD_N	TRUE	TRUE	NC_PCIE_CLK100M_ENETSDN
PCIE_CLK100M_ENETSD_P	TRUE	TRUE	NC_PCIE_CLK100M_ENETSDP
TP_PCIE_CLK100M_ENETN	TRUE	TRUE	NC_PCIE_CLK100M_ENETN
TP_PCIE_CLK100M_ENETP	TRUE	TRUE	NC_PCIE_CLK100M_ENETP
TP_PCIE_CLK100M_PEGBN	TRUE	TRUE	NC_PCIE_CLK100M_PEGBN
TP_PCIE_CLK100M_PEGBP	TRUE	TRUE	NC_PCIE_CLK100M_PEGBP
TP_PCIE_CLK100M_SWN	TRUE	TRUE	NC_PCIE_CLK100M_SWN
TP_PCIE_CLK100M_SWP	TRUE	TRUE	NC_PCIE_CLK100M_SWP
TP_PCH_GPIO64_CLKOUTFLEX0	TRUE	TRUE	NC_PCH_GPIO64_CLKOUTFLEX0
TP_PCH_GPIO65_CLKOUTFLEX1	TRUE	TRUE	NC_PCH_GPIO65_CLKOUTFLEX1
TP_PCH_GPIO66_CLKOUTFLEX2	TRUE	TRUE	NC_PCH_GPIO66_CLKOUTFLEX2
TP_PCH_GPIO67_CLKOUTFLEX3	TRUE	TRUE	NC_PCH_GPIO67_CLKOUTFLEX3
TP_USB_4N	TRUE	TRUE	NC_USB_4N
TP_USB_4P	TRUE	TRUE	NC_USB_4P

TRUE	PCIE_TBT_R2D_P<3..0>
TRUE	PCIE_TBT_R2D_N<3..0>
TRUE	PCIE_TBT_D2R_C_P<3..0>
TRUE	PCIE_TBT_D2R_C_N<3..0>
TRUE	DMI_S2N_P<3..1>
TRUE	DMI_S2N_N<3..1>
TRUE	DMI_N2S_P<3..1>
TRUE	DMI_N2S_N<3..1>

PLACEABLE BEAD-PROBES FOR TBT

Net	Part	Ref	Property
TBT_A_R2D_C_P<1>	BEAD-PROBE	BPA535	NO_XNET_CONNECTION=TRUE
TBT_A_D2R_P<1>	BEAD-PROBE	BPA531	NO_XNET_CONNECTION=TRUE
TBT_A_D2R_N<1>	BEAD-PROBE	BPA532	NO_XNET_CONNECTION=TRUE

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J15 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,P65BGA	MM	16.2

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=45_OHM_SE	=45_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.095 MM	0.095 MM			
50_OHM_SE	*	Y	0.066 MM	0.066 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.116 MM	0.116 MM			
45_OHM_SE	*	Y	0.083 MM	0.083 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE_ADJ	TOP,BOTTOM	Y	0.116 MM	0.116 MM			
45_OHM_SE_ADJ	*	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.145 MM	0.095 MM			
40_OHM_SE	*	Y	0.102 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.095 MM			
37_OHM_SE	*	Y	0.118 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.265 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.186 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	ISL2,ISL11	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	TOP,BOTTOM	Y	0.146 MM	0.146 MM		0.120 MM	0.120 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.096 MM	0.096 MM		0.126 MM	0.126 MM
80_OHM_DIFF	ISL2,ISL11	Y	0.096 MM	0.096 MM		0.126 MM	0.126 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.120 MM	0.120 MM		0.160 MM	0.160 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	ISL2,ISL11	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.105 MM	0.105 MM		0.125 MM	0.125 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	ISL2,ISL11	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.101 MM	0.101 MM		0.180 MM	0.180 MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE
*	*	P65BGA	P075_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	0.1 MM	?
BGA_P2MM	*	0.2 MM	?
P072_SPACE	*	0.071 MM	?
P075_SPACE	*	0.075 MM	?

Stackup-Defined Spacing Rules

Note: Outer dielectric is 0.058 mm nominal,
Inner dielectric is 0.053 mm nominal.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1x_DIELECTRIC	TOP,BOTTOM	0.058 MM	?
1x_DIELECTRIC	ISL3,ISL4,ISL9,ISL10	0.053 MM	?
1X_DIELECTRIC	ISL2,ISL5,ISL6,ISL7,ISL8,ISL11	0.101 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
P65_BGA	*	Y	0.071MM	0.071MM		0.075MM	0.126MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	P65BGA	P65_BGA

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL
CPU_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_8MIL	*	8 MIL	?
CPU_COMP	*	20 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?
CPU_VID	*	0.457 MM	?
CPU_VREF	*	12 MIL	?

Most CPU signals with impedance requirements are 50-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: IVB PLATFORM DG , Tables 205-207

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_2SAME	*	=3X_DIELECTRIC	?
DMI_TXRX	*	=6X_DIELECTRIC	?
DMICLK2N2S	*	=6X_DIELECTRIC	?
DMICLK2S2N	*	=3X_DIELECTRIC	?
DMICLK2OTHER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
DMI_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
DMICLK2N2S	TOP,BOTTOM	=10X_DIELECTRIC	?
DMICLK2S2N	TOP,BOTTOM	=6X_DIELECTRIC	?
DMICLK2OTHER	TOP,BOTTOM	=4X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_*	=SAME	*	DMI_2SAME
DMI_N2S	DMI_S2N	*	DMI_TXRX
DMI_S2N	DMI_N2S	*	DMI_TXRX
CLK_DMI	DMI_N2S	*	DMICLK2N2S
CLK_DMI	DMI_S2N	*	DMICLK2S2N
CLK_DMI	*	*	DMICLK2OTHER

PEG - SSD & TBT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PEG_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_2SAME	*	=3X_DIELECTRIC	?
PEG_TXRX	*	=6X_DIELECTRIC	?
PEG_2OTHER	*	=4X_DIELECTRIC	?
PEG_2CLK	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
PEG_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
PEG_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
PEG_2CLK	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PEG_*	=SAME	*	PEG_2SAME
PEG_R2D	PEG_D2R	*	PEG_TXRX
PEG_*	*	*	PEG_2OTHER
PEG_*	CLK_*	*	PEG_2CLK

DIGITAL VIDEO SIGNAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	*	=3x_DIELECTRIC	?
DP_2OTHER	*	=4x_DIELECTRIC	?
HDMICLK_2CLK	*	=7x_DIELECTRIC	?
HDMICLK_2DP	*	=4x_DIELECTRIC	?
HDMICLK_2OTHER	*	=7x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_2OTHER	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMICLK_2CLK	TOP,BOTTOM	=10x_DIELECTRIC	?
HDMICLK_2DP	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMICLK_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	=SAME	*	DP_2SAME
DISPLAYPORT	*	*	DP_2OTHER
HDMI_CLK	CLK_*	*	HDMICLK_2CLK
HDMI_CLK	DISPLAYPORT	*	HDMICLK_2DP
HDMI_CLK	*	*	HDMICLK_2OTHER

DisplayPort/TMDS intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.
DIsplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.
SOURCE: Calpella SFF DG Rev 1.5 (407364) and Family GPU DG-04202-001-v04.
MAX LENGTH OF DISPLAYPORT/TMDS TRACES: 13 INCHES.

CPU Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
DMI_S2N	CPU_85D	DMI_S2N	DMI_S2N_P<3..0>	5 12 84
DMI_S2N	CPU_85D	DMI_S2N	DMI_S2N_N<3..0>	5 12 84
DMI_N2S	CPU_85D	DMI_N2S	DMI_N2S_P<3..0>	5 12 84
DMI_N2S	CPU_85D	DMI_N2S	DMI_N2S_N<3..0>	5 12 84
FDI_INT	CPU_50S	CPU_AGTL	FDI_INT	5 12
FDI_CSYNC	CPU_50S	CPU_AGTL	FDI_CSYNC	5 12
DMI_CLK	CPU_85D	CLK_DMI	DMI_CLK100M_CPU_P	6 11
DMI_CLK	CPU_85D	CLK_DMI	DMI_CLK100M_CPU_N	6 11
CPU_CLK135_PLL	CPU_85D	CLK_PCIE	CPU_CLK135M_DPLLREF_N	6 11
CPU_CLK135_PLL	CPU_85D	CLK_PCIE	CPU_CLK135M_DPLLREF_P	6 11
CPU_CLK135_PLL	CPU_85D	CLK_PCIE	CPU_CLK135M_DPLLSS_N	6 11
CPU_CLK135_PLL	CPU_85D	CLK_PCIE	CPU_CLK135M_DPLLSS_P	6 11
CPU_EDP_COMP	CPU_27P4S	CPU_COMP	CPU_EDP_RCOMP	5
CPU_PEG_COMP	CPU_27P4S	CPU_COMP	CPU_PEG_RCOMP	5
CPU_CFG	CPU_45S	CPU_ITP	CPU_CFG<19..0>	6 18 83
XDP_CLK_PCH	CLK_PCIE_85D	CLK_PCIE	ITPXDP_CLK100M_P	11 84
XDP_CLK_PCH	CLK_PCIE_85D	CLK_PCIE	ITPXDP_CLK100M_N	11 84
XDP_TDI	CPU_45S	CPU_ITP	XDP_CPU_TDI	6 18 83
XDP_TDO	CPU_45S	CPU_ITP	XDP_CPU_TDO	6 18 83
XDP_TMS	CPU_45S	CPU_ITP	XDP_CPU_TMS	6 18 83
XDP_TCK	CPU_45S	CPU_ITP	XDP_CPU_TCK	6 18 83
XDP_TRST_L	CPU_45S	CPU_ITP	XDP_CPU_TRST_L	6 18 83
XDP_BPM	CPU_45S	CPU_ITP	XDP_BPM_L<3..0>	6 18
XDP_BPM_L	CPU_45S	CPU_ITP	XDP_BPM_L<7..4>	6 18
XDP_BDRESET_L	CPU_45S	CPU_ITP	XDP_DBRESET_L	6 18 19
XDP_PRDY_L	CPU_45S	CPU_ITP	XDP_CPU_PRDY_L	6 18 83
XDP_PREQ_L	CPU_45S	CPU_ITP	XDP_CPU_PREQ_L	6 18 83
CPU_CATERR_L	CPU_45S	CPU_AGTL	CPU_CATERR_L	6 40
CPU_PECI	CPU_45S	CPU_VID	CPU_PECI	6 14 41
CPU_PROCHOT_L	CPU_45S	CPU_AGTL	CPU_PROCHOT_L	6 40 41 57
CPU_PWRGD	CPU_45S	CPU_AGTL	CPU_PWRGD	6 14 18
PM_THRMTRIP_L	CPU_45S	CPU_8MIL	PM_THRMTRIP_L	6 14 41
PM_MEM_PWRGD	CPU_45S	CPU_AGTL	PM_MEM_PWRGD	6 12 21
PM_SYNC	CPU_45S	CPU_AGTL	PM_SYNC	6 12
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<2..0>	6
CPU_VID	CPU_45S	CPU_VID	CPU_VIDSOUT	8 57
CPU_VID	CPU_45S	CPU_VID	CPU_VIDSCLK	8 57
CPU_VID	CPU_45S	CPU_VID	CPU_VIDALERT_L	8 57
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	8 57
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	9 57
CPU_MEM_VREF		CPU_VREF	CPU_DIMMA_VREFDQ	7 22
CPU_MEM_VREF		CPU_VREF	CPU_DIMMB_VREFDQ	7 22
CPU_MEM_VREF		MEM_PWR	PPVREF_S3_MEM_VREFDQ_A	22 23 24 82 89
CPU_MEM_VREF		CPU_VREF	PPVREF_S3_MEM_VREFDQ_B	22 25 26 82
CPU_MEM_VREF		MEM_PWR	PPVREF_S3_MEM_VREFCA_A	22 23 24 82 89
CPU_MEM_VREF		CPU_VREF	PPVREF_S3_MEM_VREFCA_B	22 25 26 82
PEG_D2R	PEG_80D	PEG_D2R	PEG_D2R_P<7..0>	66 68 82
PEG_D2R	PEG_80D	PEG_D2R	PEG_D2R_N<7..0>	66 68 82
	PEG_80D	PEG_D2R	PEG_D2R_C_P<7..0>	68
	PEG_80D	PEG_D2R	PEG_D2R_C_N<7..0>	68
	PEG_80D	PEG_R2D	PEG_R2D_P<7..0>	66 68
	PEG_80D	PEG_R2D	PEG_R2D_N<7..0>	66 68
PEG_R2D	PEG_80D	PEG_R2D	PEG_R2D_C_P<7..0>	68 82
PEG_R2D	PEG_80D	PEG_R2D	PEG_R2D_C_N<7..0>	68 82
PEG_D2R_TBT	CPU_85D	PEG_D2R	PCIE_TBT_D2R_P<3..0>	28 82
PEG_D2R_TBT	CPU_85D	PEG_D2R	PCIE_TBT_D2R_N<3..0>	28 82
	CPU_85D	PEG_D2R	PCIE_TBT_D2R_C_P<3..0>	28 84
	CPU_85D	PEG_D2R	PCIE_TBT_D2R_C_N<3..0>	28 84
PEG_R2D_TBT	CPU_85D	PEG_R2D	PCIE_TBT_R2D_P<3..0>	28 84
PEG_R2D_TBT	CPU_85D	PEG_R2D	PCIE_TBT_R2D_N<3..0>	28 84
	CPU_85D	PEG_R2D	PCIE_TBT_R2D_C_P<3..0>	28 82
	CPU_85D	PEG_R2D	PCIE_TBT_R2D_C_N<3..0>	28 82

DP AUX NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
DP_INT_IG_ML	DP_85D	DISPLAYPORT	DP_INT_ML_C_P<3..0>	67 79 94
DP_INT_IG_ML	DP_85D	DISPLAYPORT	DP_INT_ML_C_N<3..0>	67 79 94
	DP_85D	DISPLAYPORT	DP_INT_ML_P<3..0>	67 83 86 94
	DP_85D	DISPLAYPORT	DP_INT_ML_N<3..0>	67 83 86 94
	DP_85D	DISPLAYPORT	DP_INT_ML_F_P<3..0>	67 94
	DP_85D	DISPLAYPORT	DP_INT_ML_F_N<3..0>	67 94
	DP_85D	DISPLAYPORT	DP_INT_ML_P<3..0>	67 83 86 94
	DP_85D	DISPLAYPORT	DP_INT_ML_N<3..0>	67 83 86 94
DP_INT_IG_AUX	DP_85D	DISPLAYPORT	DP_INT_AUXCH_C_P	67 79 94
DP_INT_IG_AUX	DP_85D	DISPLAYPORT	DP_INT_AUXCH_C_N	67 79 94
DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_P	67 83 94
DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_N	67 83 94
DP_INT_IG_AUX	DP_85D	DISPLAYPORT	DPA_IG_AUX_CH_P	80 82
DP_INT_IG_AUX	DP_85D	DISPLAYPORT	DPA_IG_AUX_CH_N	80 82
DP_INT_IG_AUX	DP_85D	DISPLAYPORT	DPB_IG_AUX_CH_P	80 82
DP_INT_IG_AUX	DP_85D	DISPLAYPORT	DPB_IG_AUX_CH_N	80 82

DP / HDMI NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
HDMI_DATA	DP_85D	DISPLAYPORT	HDMI_DATA_P<2..0>	83
HDMI_DATA	DP_85D	DISPLAYPORT	HDMI_DATA_N<2..0>	83
HDMI_CLK	DP_85D	HDMI_CLK	HDMI_CLK_P	83
HDMI_CLK	DP_85D	HDMI_CLK	HDMI_CLK_N	83
DP_TBT_ML0	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	28 74 94
DP_TBT_ML0	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	28 74 94
	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	28 94
	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	28 94
DP_TBT_ML1	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	28 74 94
DP_TBT_ML1	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	28 74 94
	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	28 94
	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	28 94
TBTSNK0_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_P	28 94
TBTSNK0_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_N	28 94
	DP_85D		DP_TBTSNK0_AUXCH_C_P	28 80 94
	DP_85D		DP_TBTSNK0_AUXCH_C_N	28 80 94
TBTSNK1_AUXCH1	DP_85D		DP_TBTSNK1_AUXCH_P	28 94
TBTSNK1_AUXCH1	DP_85D		DP_TBTSNK1_AUXCH_N	28 94
	DP_85D		DP_TBTSNK1_AUXCH_C_P	28 80 94
	DP_85D		DP_TBTSNK1_AUXCH_C_N	28 80 94

SYNC_MASTER=CLEAN_J45 SYNC_DATE=04/26/2013

PAGE TITLE: CPU Constraints

Apple Inc.

DRAWING NUMBER: 051-0675 SIZE: D
REVISION: 6.0.0
BRANCH: dvt
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SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_37SE	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE
SATA_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	*	=3X_DIELECTRIC	?
SATA_TXRX	*	=6X_DIELECTRIC	?
SATA_2OTHER	*	=4X_DIELECTRIC	?
SATA_RCOMP	*	=6X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
SATA_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
SATA_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
SATA_RCOMP	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA_*	=SAME	*	SATA_2SAME
SATA_R2D	SATA_D2R	*	SATA_TXRX
SATA_*	*	*	SATA_2OTHER

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=4X_DIELECTRIC	?
USB_RBIAS	*	=6X_DIELECTRIC	?
BT_WAKE	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=6X_DIELECTRIC	?
USB_RBIAS	TOP,BOTTOM	=10X_DIELECTRIC	?
BT_WAKE	TOP,BOTTOM	=6X_DIELECTRIC	?

USB 3.0 INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB3_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	*	=3X_DIELECTRIC	?
USB3_TXRX	*	=6X_DIELECTRIC	?
USB3_2OTHER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
USB3_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
USB3_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_*	=SAME	*	USB3_2SAME
USB3_R2D	USB3_D2R	*	USB3_TXRX
USB3_*	*	*	USB3_2OTHER

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=4x_DIELECTRIC	?
CLK_25M	*	=5x_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.
NOTE: Latest Intel DG calls out 50ohms SE for sys clocks

PCH Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
		SATA_85D	SATA_R2D	NC_SATA_A_R2D_CP	84
		SATA_85D	SATA_R2D	NC_SATA_A_R2D_CN	84
I272		SATA_85D	SATA_D2R	NC_SATA_A_D2RP	84
I271		SATA_85D	SATA_D2R	NC_SATA_A_D2RN	84
I299		SATA_85D	SATA_R2D	NC_SATA_B_R2D_CP	84
I298		SATA_85D	SATA_R2D	NC_SATA_B_R2D_CN	84
I300		SATA_85D	SATA_D2R	NC_SATA_B_D2RP	84
I301		SATA_85D	SATA_D2R	NC_SATA_B_D2RN	84
I213	PCH_SATA_RCOMP	SATA_45SE	SATA_RCOMP	PCH_SATA_RCOMP	11
	USB_EXTA	USB_85D	USB	USB_EXTA_P	13 37
	USB_EXTA	USB_85D	USB	USB_EXTA_N	13 37
I245	USB_EXTA	USB_85D	USB	USB_EXTA_MUXED_P	37
I244	USB_EXTA	USB_85D	USB	USB_EXTA_MUXED_N	37
I248	USB_EXTA	USB_85D	USB	USB_LT1_P	37
I249	USB_EXTA	USB_85D	USB	USB_LT1_N	37
	USB_NC	USB_85D	USB	NC_USB_EXTCP	84
	USB_NC	USB_85D	USB	NC_USB_EXTCN	84
I238	USB_NC	USB_85D	USB	NC_USB_SDP	84
I239	USB_NC	USB_85D	USB	NC_USB_SDN	84
I273		CPU_45S	CPU_ITP	SMC_DEBUGPRT_RX_L	37 40 41
I274		CPU_45S	CPU_ITP	SMC_DEBUGPRT_TX_L	37 40 41
	USB_SMC	USB_85D	USB	USB_SMC_P	84
	USB_SMC	USB_85D	USB	USB_SMC_N	84
I281	USB_NC	USB_85D	USB	NC_USB_6P	84
I282	USB_NC	USB_85D	USB	NC_USB_6N	84
I283	USB_NC	USB_85D	USB	NC_USB_7P	84
I284	USB_NC	USB_85D	USB	NC_USB_7N	84
	USB_EXTB	USB_85D	USB	USB_EXTB_P	13 78 83
	USB_EXTB	USB_85D	USB	USB_EXTB_N	13 78 83
I260	USB_NC	USB_85D	USB	NC_USB_EXTDP	84
I259	USB_NC	USB_85D	USB	NC_USB_EXTDN	84
	USB_BT	USB_85D	USB	USB_BT_P	13 33
	USB_BT	USB_85D	USB	USB_BT_N	13 33
		USB_85D	USB	USB_BT_CONN_P	33 83
		USB_85D	USB	USB_BT_CONN_N	33 83
I285	USB_NC	USB_85D	USB	NC_USB_IRP	84
I286	USB_NC	USB_85D	USB	NC_USB_IRN	84
I288	USB_TPAD	USB_85D	USB	USB_TPAD_P	13 38
I287	USB_TPAD	USB_85D	USB	USB_TPAD_N	13 38
I302		USB_85D	USB	USB_TPAD_R_P	38
I303		USB_85D	USB	USB_TPAD_R_N	38
	PCH_USB_RBIAS	PCH_USB_RBIAS	USB_RBIAS	PCH_USB_RBIAS	13
I228	USB3_EXTA_RX	USB_85D	USB3_D2R	USB3_EXTA_D2R_P	13 37
I231	USB3_EXTA_RX	USB_85D	USB3_D2R	USB3_EXTA_D2R_N	13 37
I267		USB_85D	USB3_D2R	USB3_EXTA_D2R_C_P	
I265		USB_85D	USB3_D2R	USB3_EXTA_D2R_C_N	
I230	USB3_EXTA_TX	USB_85D	USB3_R2D	USB3_EXTA_R2D_P	37
I229	USB3_EXTA_TX	USB_85D	USB3_R2D	USB3_EXTA_R2D_N	37
I268		USB_85D	USB3_R2D	USB3_EXTA_R2D_C_P	13 37
I266		USB_85D	USB3_R2D	USB3_EXTA_R2D_C_N	13 37
I222	USB3_EXTB_RX	USB_85D	USB3_D2R	USB3_EXTB_D2R_P	13 78 83
I223	USB3_EXTB_RX	USB_85D	USB3_D2R	USB3_EXTB_D2R_N	13 78 83
I264		USB_85D	USB3_D2R	USB3_EXTB_D2R_C_P	
I262		USB_85D	USB3_D2R	USB3_EXTB_D2R_C_N	
I220	USB3_EXTB_TX	USB_85D	USB3_R2D	USB3_EXTB_R2D_P	78 83
I221	USB3_EXTB_TX	USB_85D	USB3_R2D	USB3_EXTB_R2D_N	78 83
I261		USB_85D	USB3_R2D	USB3_EXTB_R2D_C_P	13 78
I263		USB_85D	USB3_R2D	USB3_EXTB_R2D_C_N	13 78
I290	NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTC_D2RP	84
I291	NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTC_D2RN	84
I293		USB_85D	USB3_R2D	NC_USB3_EXTC_R2D_CP	84
I292		USB_85D	USB3_R2D	NC_USB3_EXTC_R2D_CN	84
I295	NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTD_D2RP	84
I294	NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTD_D2RN	84
I296		USB_85D	USB3_R2D	NC_USB3_EXTD_R2D_CP	84
I297		USB_85D	USB3_R2D	NC_USB3_EXTD_R2D_CN	84

Clock Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
I256	SYSCLK_CLK32K_RTC	CLK_SLOW_45S	CLK_SLOW	SYSCLK_CLK32K_RTC	11 19
I255	SYSCLK_CLK25M_SB	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_SB	11 19
I254		CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_SB_R	11
I253	SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA	19 36
I251	SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT	19 28
I250		CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R	28

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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Single Net Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_SE	*	=2x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_SE	TOP,BOTTOM	=3x_DIELECTRIC	?

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_2SAME	*	=2X_DIELECTRIC	?
PCIE_TXRX	*	=6X_DIELECTRIC	?
PCIE_2OTHER	*	=4X_DIELECTRIC	?
PCIE_2CLK	*	=7X_DIELECTRIC	?
PCIECLK_2OTHER	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
PCIE_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIE_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
PCIE_2CLK	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIECLK_2OTHER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_*	=SAME	*	PCIE_2SAME
PCIE_R2D	PCIE_D2R	*	PCIE_TXRX
PCIE_*	*	*	PCIE_2OTHER
PCIE_*	CLK_*	*	PCIE_2CLK
CLK_PCIE	*	*	PCIECLK_2OTHER

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
LPC_AD	LPC_45S	LPC	LPC_AD<3..0>	13 40 49 79 83
LPC_FRAME_L	LPC_45S	LPC	LPC_FRAME_L	13 40 49 79 83
LPC_RESET_L	LPC_45S	LPC	LPC_RESET_L	20
SMBUS_PCH_CLK	SMB_45S	SMB	SMBUS_PCH_CLK	13 43 83
SMBUS_PCH_DATA	SMB_45S	SMB	SMBUS_PCH_DATA	13 43 83
SMBUS_PCH_0_CLK	SMB_45S	SMB	SML_PCH_0_CLK	13 43
SMBUS_PCH_0_DATA	SMB_45S	SMB	SML_PCH_0_DATA	13 43
SMBUS_PCH_1_CLK	SMB_45S	SMB	SML_PCH_1_CLK	13 43
SMBUS_PCH_1_DATA	SMB_45S	SMB	SML_PCH_1_DATA	13 43
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	11 51
	HDA_45S	HDA	HDA_BIT_CLK_R	11
HDA_SYNC	HDA_45S	HDA	HDA_SYNC	11 51
	HDA_45S	HDA	HDA_SYNC_R	11
	HDA_45S	HDA	HDA_RST_R_L	11
HDA_RST_L	HDA_45S	HDA	HDA_RST_L	11 51
HDA_SDIN0	HDA_45S	HDA	HDA_SDIN0	11 51
HDA_SDIN0_R	HDA_45S	HDA	CS4208_HDA_SDOUT0_R	51
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	11 51
	HDA_45S	HDA	HDA_SDOUT_R	11 19
	SPI_45S	SPI	SPI_CLK_R	13 49
SPI_CLK	SPI_45S	SPI	SPI_CLK	49
	SPI_45S	SPI	SPI_MOSI_R	13 49
SPI_MOSI	SPI_45S	SPI	SPI_MOSI	49
SPI_MISO	SPI_45S	SPI	SPI_MISO	13 49
	SPI_45S	SPI	SPI_CS0_R_L	13 49
SPI_CS0	SPI_45S	SPI	SPI_CS0_L	49
USB3_SD_R2D	USB3_85D	USB3_R2D	USB3_SD_R2D_C_P	20 78 83
USB3_SD_R2D	USB3_85D	USB3_R2D	USB3_SD_R2D_C_N	20 78 83
USB3_SD_D2R	USB3_85D	USB3_D2R	USB3_SD_D2R_P	20 78 83
USB3_SD_D2R	USB3_85D	USB3_D2R	USB3_SD_D2R_N	20 78 83
PCIE_AP_R2D	PCIE_85D	PCIE_R2D	PCIE_AP_R2D_P	33 83
PCIE_AP_R2D	PCIE_85D	PCIE_R2D	PCIE_AP_R2D_N	33 83
	PCIE_85D	PCIE_R2D	PCIE_AP_R2D_C_P	13 33
I304	PCIE_85D	PCIE_R2D	PCIE_AP_R2D_C_N	13 33
	PCIE_85D	PCIE_R2D	PCIE_AP_R2D_PI_P	33
I303	PCIE_85D	PCIE_R2D	PCIE_AP_R2D_PI_N	33
PCIE_AP_D2R	PCIE_85D	PCIE_D2R	PCIE_AP_D2R_P	13 33
PCIE_AP_D2R	PCIE_85D	PCIE_D2R	PCIE_AP_D2R_N	13 33
I275	PCIE_85D	PCIE_D2R	PCIE_AP_D2R_PI_P	33 83
I276	PCIE_85D	PCIE_D2R	PCIE_AP_D2R_PI_N	33 83
I325 PCIE_CAMERA_R2D	PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_P	35 36
I326 PCIE_CAMERA_R2D	PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_N	35 36
I327	PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_C_P	13 36
I328	PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_C_N	13 36
I329 PCIE_CAMERA_D2R	PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_P	13 36
I330 PCIE_CAMERA_D2R	PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_N	13 36
I331	PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_C_P	35 36
I332	PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_C_N	35 36
	CLK_LPC_45S	CLK_LPC	LPC_CLK33M_SMC_R	11 19
PCH_LPC_CLK0	CLK_LPC_45S	CLK_LPC	LPC_CLK33M_SMC	19 40
PCH_LPC_CLK0	CLK_LPC_45S	CLK_LPC	LPC_CLK33M_LPCPLUS	19 49 83
I293	CLK_LPC_45S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	11 19
I298 PCIE_CLK100M	CPU_45S	CLK_PCIE	PCH_CLK33M_PCIIN	11 19
I258 PCIE_CLK100M	CPU_45S	CLK_PCIE	PCH_CLK14P3M_REFCLK	11
I260 PCIE_CLK100M	CPU_45S	CLK_PCIE	PCH_CLK33M_PCIOUT	11 19
I253 PCIE_CLK100M_PCH	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_PCH_P	11
I254 PCIE_CLK100M_PCH	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_PCH_N	11
I262 PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_P	11 28
I261 PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_N	11 28
I255 PCIE_CLK100M_DOT	CLK_PCIE_85D	CLK_PCIE	PCH_CLK96M_DOT_P	11
I257 PCIE_CLK100M_DOT	CLK_PCIE_85D	CLK_PCIE	PCH_CLK96M_DOT_N	11
I256 PCIE_CLK100M_SATA	PCIE_85D	CLK_PCIE	PCH_CLK100M_SATA_P	11
I259 PCIE_CLK100M_SATA	PCIE_85D	CLK_PCIE	PCH_CLK100M_SATA_N	11
PEG_CLK100M	PEG_80D	CLK_PCIE	PEG_CLK100M_P	68 82 83
PEG_CLK100M	PEG_80D	CLK_PCIE	PEG_CLK100M_N	68 82 83
PCIE_CLK100M_AP	PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_P	11 33
PCIE_CLK100M_AP	PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_N	11 33
I300	PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_P	33 83
I299	PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_N	33 83
PCIE_CLK100M_S2	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_P	11 36
PCIE_CLK100M_S2	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_N	11 36
I333	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_P	35 36
I334	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_N	35 36
I281 PCIE_CLK100M_ENET	PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_P	11 34
I282	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_N	11 34

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I313 PCH_PM_NET	PCH_45S	PCH_SE	PCH_INTRUDER_L	11
I314 PCH_PM_NET	PCH_45S	PCH_SE	PCH_INTVRMEN_L	11
I315 PCH_PM_NET	PCH_45S	PCH_SE	PCH_DSWVRMEN	12
I316 PCH_PM_NET	PCH_45S	PCH_SE	PCH_SRTCRST_L	11
I308 PCH_PM_NET	PCH_45S	PCH_SE	PM_RSMRST_L	12 65 83
I307 PCH_PM_NET	PCH_45S	PCH_SE	PM_SYSRST_L	12 19 40 83
I318 PCH_PM_NET	PCH_45S	PCH_SE	PM_PCH_PWROK	12 19 83
I317 PCH_PM_NET	PCH_45S	PCH_SE	PM_PCH_APWROK	12 19
I320 PCH_PM_NET	PCH_45S	PCH_SE	PM_DSW_PWRGD	12 40 83
I319 PCH_PM_NET	PCH_45S	PCH_SE	PM_PCH_SYS_PWROK	12 18 19 40 83
I321 PCH_PM_NET	PCH_45S	PCH_SE	PM_PWRBTN_L	12 18 40
I322 PCH_PM_NET	PCH_45S	PCH_SE	PM_THRMTRIP_L_R	14 41
I323 PCH_PCIE_WAKE	PCH_45S	PCH_SE	PCIE_WAKE_L	12 33 35 83
I324 PCH_PM_NET	PCH_45S	PCH_SE	PCH_RCIN_L	14
I335 PCIE_D2R_SSD	PCIE_85D	PCIE_D2R	PCIE_SSD_D2R_P<3..0>	13 34
I336 PCIE_D2R_SSD	PCIE_85D	PCIE_D2R	PCIE_SSD_D2R_N<3..0>	13 34
I338 PCIE_R2D_SSD	PCIE_85D	PCIE_R2D	PCIE_SSD_R2D_C_P<3..0>	13 34
I337 PCIE_R2D_SSD	PCIE_85D	PCIE_R2D	PCIE_SSD_R2D_C_N<3..0>	13 34
I339	PCIE_85D	PCIE_R2D	PCIE_SSD_R2D_P<3..0>	34
I340	PCIE_85D	PCIE_R2D	PCIE_SSD_R2D_N<3..0>	34

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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_37S	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=STANDARD	=STANDARD
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=2x_DIELECTRIC	?
MEM_CMD2CMD	*	=2x_DIELECTRIC	?
MEM_CMD2CTRL	*	=2x_DIELECTRIC	?
MEM_CTRL2CTRL	*	=2x_DIELECTRIC	?
MEM_CLK2CLK	*	=4x_DIELECTRIC	?
MEM_2OTHERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	?
MEM_2GND	*	=2x_DIELECTRIC	?
MEM_2OTHER	*	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_DQS2OWNDATA	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CMD	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CTRL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CTRL2CTRL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CLK2CLK	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2OTHERMEM	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	*	*	MEM_2OTHER
MEM_*_DQS_*	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_CTRL	*	*	MEM_2OTHER
MEM_CLK	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTRL	*	MEM_CMD2CTRL
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK
MEM_*	MEM_*	*	MEM_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DATA_0	*	MEM_DQS2OWNDATA
MEM_A_DQS_1	MEM_A_DATA_1	*	MEM_DQS2OWNDATA
MEM_A_DQS_2	MEM_A_DATA_2	*	MEM_DQS2OWNDATA
MEM_A_DQS_3	MEM_A_DATA_3	*	MEM_DQS2OWNDATA
MEM_A_DQS_4	MEM_A_DATA_4	*	MEM_DQS2OWNDATA
MEM_A_DQS_5	MEM_A_DATA_5	*	MEM_DQS2OWNDATA
MEM_A_DQS_6	MEM_A_DATA_6	*	MEM_DQS2OWNDATA
MEM_A_DQS_7	MEM_A_DATA_7	*	MEM_DQS2OWNDATA
MEM_B_DQS_0	MEM_B_DATA_0	*	MEM_DQS2OWNDATA
MEM_B_DQS_1	MEM_B_DATA_1	*	MEM_DQS2OWNDATA
MEM_B_DQS_2	MEM_B_DATA_2	*	MEM_DQS2OWNDATA
MEM_B_DQS_3	MEM_B_DATA_3	*	MEM_DQS2OWNDATA
MEM_B_DQS_4	MEM_B_DATA_4	*	MEM_DQS2OWNDATA
MEM_B_DQS_5	MEM_B_DATA_5	*	MEM_DQS2OWNDATA
MEM_B_DQS_6	MEM_B_DATA_6	*	MEM_DQS2OWNDATA
MEM_B_DQS_7	MEM_B_DATA_7	*	MEM_DQS2OWNDATA

DDR3 (Memory Down):

DQ signals should be matched within 0.508mm of associated DQS pair
DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.
DQS to clock matching should be within [CLK-139.73mm] and [CLK-30.48mm].
CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.508mm.
CONTROL signals should be matched within [CLK-2.54mm] to [CLK+0mm] of CLK pairs.
A/BA/CMD signals should be matched within [CLK-2.54mm] to [CLK+2.54mm] of CLK pairs.
DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.
Maximum length of any signal from die pad to first DRAM device is 139.7mm max, to last DRAM device is 194.31mm max.

SOURCE: Double checked with Doc#486985 Chief River SFF Platform DG: Memory Down
SOURCE: Need to re-confirm CRW DG for memory down (Intel not yet provided)

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	MEM_A_CLK0	MEM_72D	MEM_CLK	MEM_A_CLK_P<0>	7 23 27
	MEM_A_CLK0	MEM_72D	MEM_CLK	MEM_A_CLK_N<0>	7 23 27
I130	MEM_A_CLK1	MEM_72D	MEM_CLK	MEM_A_CLK_P<1>	7 24 27
I129	MEM_A_CLK1	MEM_72D	MEM_CLK	MEM_A_CLK_N<1>	7 24 27
	MEM_A_CNTL0	MEM_40S	MEM_CTRL	MEM_A_CKE<0>	7 23 27
I124	MEM_A_CNTL1	MEM_40S	MEM_CTRL	MEM_A_CKE<1>	7 24 27
I125	MEM_A_CNTL0	MEM_40S	MEM_CTRL	MEM_A_CS_L<0>	7 23 27
I101	MEM_A_CNTL1	MEM_40S	MEM_CTRL	MEM_A_CS_L<1>	7 24 27
I126	MEM_A_CNTL0	MEM_40S	MEM_CTRL	MEM_A_ODT<0>	7 23 27
I103	MEM_A_CNTL1	MEM_40S	MEM_CTRL	MEM_A_ODT<1>	7 24 27
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_A<15..0>	7 23 24 27
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_BA<2..0>	7 23 24 27
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_RAS_L	7 23 24 27
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_CAS_L	7 23 24 27
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_WE_L	7 23 24 27
	MEM_A_DATA_0	MEM_45S	MEM_A_DATA_0	MEM_A_DQ<7..0>	7 23 24
	MEM_A_DATA_1	MEM_45S	MEM_A_DATA_1	MEM_A_DQ<15..8>	7 23 24
	MEM_A_DATA_2	MEM_45S	MEM_A_DATA_2	MEM_A_DQ<23..16>	7 23 24
	MEM_A_DATA_3	MEM_45S	MEM_A_DATA_3	MEM_A_DQ<31..24>	7 23 24
	MEM_A_DATA_4	MEM_45S	MEM_A_DATA_4	MEM_A_DQ<39..32>	7 23 24
	MEM_A_DATA_5	MEM_45S	MEM_A_DATA_5	MEM_A_DQ<47..40>	7 23 24
	MEM_A_DATA_6	MEM_45S	MEM_A_DATA_6	MEM_A_DQ<55..48>	7 23 24
I105	MEM_A_DATA_7	MEM_45S	MEM_A_DATA_7	MEM_A_DQ<63..56>	7 23 24
I106	MEM_A_DQS0	MEM_85D	MEM_A_DQS_0	MEM_A_DQS_P<0>	7 23 24
	MEM_A_DQS0	MEM_85D	MEM_A_DQS_0	MEM_A_DQS_N<0>	7 23 24
	MEM_A_DQS1	MEM_85D	MEM_A_DQS_1	MEM_A_DQS_P<1>	7 23 24
	MEM_A_DQS1	MEM_85D	MEM_A_DQS_1	MEM_A_DQS_N<1>	7 23 24
	MEM_A_DQS2	MEM_85D	MEM_A_DQS_2	MEM_A_DQS_P<2>	7 23 24
	MEM_A_DQS2	MEM_85D	MEM_A_DQS_2	MEM_A_DQS_N<2>	7 23 24
	MEM_A_DQS3	MEM_85D	MEM_A_DQS_3	MEM_A_DQS_P<3>	7 23 24
	MEM_A_DQS3	MEM_85D	MEM_A_DQS_3	MEM_A_DQS_N<3>	7 23 24
	MEM_A_DQS4	MEM_85D	MEM_A_DQS_4	MEM_A_DQS_P<4>	7 23 24
	MEM_A_DQS4	MEM_85D	MEM_A_DQS_4	MEM_A_DQS_N<4>	7 23 24
	MEM_A_DQS5	MEM_85D	MEM_A_DQS_5	MEM_A_DQS_P<5>	7 23 24
	MEM_A_DQS5	MEM_85D	MEM_A_DQS_5	MEM_A_DQS_N<5>	7 23 24
I117	MEM_A_DQS6	MEM_85D	MEM_A_DQS_6	MEM_A_DQS_P<6>	7 23 24
I118	MEM_A_DQS6	MEM_85D	MEM_A_DQS_6	MEM_A_DQS_N<6>	7 23 24
I116	MEM_A_DQS7	MEM_85D	MEM_A_DQS_7	MEM_A_DQS_P<7>	7 23 24
I119	MEM_A_DQS7	MEM_85D	MEM_A_DQS_7	MEM_A_DQS_N<7>	7 23 24
	MEM_B_CLK0	MEM_72D	MEM_CLK	MEM_B_CLK_P<0>	7 25 27
	MEM_B_CLK0	MEM_72D	MEM_CLK	MEM_B_CLK_N<0>	7 25 27
I127	MEM_B_CLK1	MEM_72D	MEM_CLK	MEM_B_CLK_P<1>	7 26 27
I128	MEM_B_CLK1	MEM_72D	MEM_CLK	MEM_B_CLK_N<1>	7 26 27
	MEM_B_CNTL0	MEM_40S	MEM_CTRL	MEM_B_CKE<0>	7 25 27
I120	MEM_B_CNTL1	MEM_40S	MEM_CTRL	MEM_B_CKE<1>	7 26 27
I121	MEM_B_CNTL0	MEM_40S	MEM_CTRL	MEM_B_CS_L<0>	7 25 27
I122	MEM_B_CNTL1	MEM_40S	MEM_CTRL	MEM_B_CS_L<1>	7 26 27
I123	MEM_B_CNTL0	MEM_40S	MEM_CTRL	MEM_B_ODT<0>	7 25 27
	MEM_B_CNTL1	MEM_40S	MEM_CTRL	MEM_B_ODT<1>	7 26 27
I108	MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_A<15..0>	7 25 26 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_BA<2..0>	7 25 26 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_RAS_L	7 25 26 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_CAS_L	7 25 26 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_WE_L	7 25 26 27
	MEM_B_DATA_0	MEM_45S	MEM_B_DATA_0	MEM_B_DQ<7..0>	7 25 26
	MEM_B_DATA_1	MEM_45S	MEM_B_DATA_1	MEM_B_DQ<15..8>	7 25 26
	MEM_B_DATA_2	MEM_45S	MEM_B_DATA_2	MEM_B_DQ<23..16>	7 25 26
	MEM_B_DATA_3	MEM_45S	MEM_B_DATA_3	MEM_B_DQ<31..24>	7 25 26
	MEM_B_DATA_4	MEM_45S	MEM_B_DATA_4	MEM_B_DQ<39..32>	7 25 26
	MEM_B_DATA_5	MEM_45S	MEM_B_DATA_5	MEM_B_DQ<47..40>	7 25 26
	MEM_B_DATA_6	MEM_45S	MEM_B_DATA_6	MEM_B_DQ<55..48>	7 25 26
	MEM_B_DATA_7	MEM_45S	MEM_B_DATA_7	MEM_B_DQ<63..56>	7 25 26
	MEM_B_DQS0	MEM_85D	MEM_B_DQS_0	MEM_B_DQS_P<0>	7 25 26
	MEM_B_DQS0	MEM_85D	MEM_B_DQS_0	MEM_B_DQS_N<0>	7 25 26
	MEM_B_DQS1	MEM_85D	MEM_B_DQS_1	MEM_B_DQS_P<1>	7 25 26
	MEM_B_DQS1	MEM_85D	MEM_B_DQS_1	MEM_B_DQS_N<1>	7 25 26
	MEM_B_DQS2	MEM_85D	MEM_B_DQS_2	MEM_B_DQS_P<2>	7 25 26
	MEM_B_DQS2	MEM_85D	MEM_B_DQS_2	MEM_B_DQS_N<2>	7 25 26
	MEM_B_DQS3	MEM_85D	MEM_B_DQS_3	MEM_B_DQS_P<3>	7 25 26
	MEM_B_DQS3	MEM_85D	MEM_B_DQS_3	MEM_B_DQS_N<3>	7 25 26
	MEM_B_DQS4	MEM_85D	MEM_B_DQS_4	MEM_B_DQS_P<4>	7 25 26
	MEM_B_DQS4	MEM_85D	MEM_B_DQS_4	MEM_B_DQS_N<4>	7 25 26
	MEM_B_DQS5	MEM_85D	MEM_B_DQS_5	MEM_B_DQS_P<5>	7 25 26
	MEM_B_DQS5	MEM_85D	MEM_B_DQS_5	MEM_B_DQS_N<5>	7 25 26
I113	MEM_B_DQS6	MEM_85D	MEM_B_DQS_6	MEM_B_DQS_P<6>	7 25 26
I112	MEM_B_DQS6	MEM_85D	MEM_B_DQS_6	MEM_B_DQS_N<6>	7 25 26
I115	MEM_B_DQS7	MEM_85D	MEM_B_DQS_7	MEM_B_DQS_P<7>	7 25 26
I114	MEM_B_DQS7	MEM_85D	MEM_B_DQS_7	MEM_B_DQS_N<7>	7 25 26
I110			MEM_PWR	PPVREF_S3_MEM_VREFDQ_A	22 23 24 82 86
I109			MEM_PWR	PPVREF_S3_MEM_VREFCA_A	22 23 24 82 86
I111			MEM_PWR	PP1V35_S3_MEM	81

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PAGE TITLE: Memory Constraints

Apple Inc.

DRAWING NUMBER 051-0675 SIZE D

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BRANCH dvt

DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt/DP Connector Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTDP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SOURCE: Bill Cornelius’s Thunderbolt Routing Notes

TBT_DP Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTDP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_2SAME	*	=3X_DIELECTRIC	?
TBTDP_TXRX	*	=6X_DIELECTRIC	?
TBTDP_2OTHER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
TBTDP_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
TBTDP_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBTDP_*	=SAME	*	TBTDP_2SAME
TBTDP_R2D	TBTDP_D2R	*	TBTDP_TXRX
TBTDP_*	*	*	TBTDP_2OTHER

Thunderbolt/DP Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	Pages
	TBT_A_R2D	TBTDP_85D	TBTDP_R2D	TBT_A_R2D_C_P<1..0>	28 31 84
	TBT_A_R2D	TBTDP_85D	TBTDP_R2D	TBT_A_R2D_C_N<1..0>	28 31
	TBT_A_R2D	TBTDP_85D	TBTDP_R2D	TBT_A_R2D_P<1..0>	31
	TBT_A_R2D	TBTDP_85D	TBTDP_R2D	TBT_A_R2D_N<1..0>	31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	28 31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	28 31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<1>	31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<1>	31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_A_LSX_ML_P<1>	31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_A_LSX_ML_N<1>	31
I308	DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	28 31
I310	DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	28 31
I311	DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3>	31
I309	DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3>	31
	TBT_A_D2R0	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_C_P<0>	31
	TBT_A_D2R0	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_C_N<0>	31
	TBT_A_D2R0	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_P<0>	28 31
	TBT_A_D2R0	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_N<0>	28 31
I312	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_C_P<1>	31
I313	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_C_N<1>	31
I314	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_P<1>	28 31 84
I315	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_N<1>	28 31 84
	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R1_AUXDDC_P	31
	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R1_AUXDDC_N	31
	TBT_A_AUXCH	DP_85D		DP_TBTPA_AUXCH_C_P	28 31
	TBT_A_AUXCH	DP_85D		DP_TBTPA_AUXCH_C_N	28 31
	TBT_A_AUXCH	DP_85D		DP_TBTPA_AUXCH_P	31
	TBT_A_AUXCH	DP_85D		DP_TBTPA_AUXCH_N	31
I316	TBT_B_R2D	TBTDP_85D	TBTDP_R2D	TBT_B_R2D_C_P<1..0>	28 32
I317	TBT_B_R2D	TBTDP_85D	TBTDP_R2D	TBT_B_R2D_C_N<1..0>	28 32
I318	TBT_B_R2D	TBTDP_85D	TBTDP_R2D	TBT_B_R2D_P<1..0>	32
I319	TBT_B_R2D	TBTDP_85D	TBTDP_R2D	TBT_B_R2D_N<1..0>	32
I321	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<1>	28 32
I322	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<1>	28 32
I320	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<1>	32
I324	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<1>	32
I325	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_B_LSX_ML_P<1>	32
I323	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_B_LSX_ML_N<1>	32
I327	DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3>	28 32
I326	DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3>	28 32
I328	DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3>	32
I329	DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3>	32
I331	TBT_B_D2R0	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_C_P<0>	32
I332	TBT_B_D2R0	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_C_N<0>	32
I330	TBT_B_D2R0	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_P<0>	28 32
I334	TBT_B_D2R0	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_N<0>	28 32
I333	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_C_P<1>	32
I336	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_C_N<1>	32
I337	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_P<1>	28 32
I335	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_N<1>	28 32
I339	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R1_AUXDDC_P	32
I338	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R1_AUXDDC_N	32
I341	TBT_B_AUXCH	DP_85D		DP_TBTPB_AUXCH_C_P	28 32
I340	TBT_B_AUXCH	DP_85D		DP_TBTPB_AUXCH_C_N	28 32
I342	TBT_B_AUXCH	DP_85D		DP_TBTPB_AUXCH_P	32
I343	TBT_B_AUXCH	DP_85D		DP_TBTPB_AUXCH_N	32

Only used on dual-port hosts.

Thunderbolt IC Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	Pages
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_P<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_N<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_P	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_N	
TBT_SPI_CLK	TBT_SPI_45S	TBT_SPI	TBT_SPI_CLK	28
TBT_SPI_MOSI	TBT_SPI_45S	TBT_SPI	TBT_SPI_MOSI	28
TBT_SPI_MISO	TBT_SPI_45S	TBT_SPI	TBT_SPI_MISO	28
TBT_SPI_CS_L	TBT_SPI_45S	TBT_SPI	TBT_SPI_CS_L	28

Only used on hosts supporting Thunderbolt video-in

8 7 6 5 4 3 2 1

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MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	*	=4X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?
MIPICLK_2OTHER	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
MIPI_2CLK	TOP,BOTTOM	=8X_DIELECTRIC	?
MIPICLK_2OTHER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_2OTHER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICLK_2OTHER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	*	=2x_DIELECTRIC	?
S2_DQS2OWNDATA	*	=2x_DIELECTRIC	?
S2_CMD2CMD	*	=2x_DIELECTRIC	?
S2_CMD2CTRL	*	=2x_DIELECTRIC	?
S2_CTRL2CTRL	*	=2x_DIELECTRIC	?
S2_2OTHERMEM	*	=4x_DIELECTRIC	?
S2MEM_2PWR	*	=2x_DIELECTRIC	?
S2MEM_2GND	*	=2x_DIELECTRIC	?
S2MEM_2OTHER	*	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_DQS2OWNDATA	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CTRL2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_2OTHERMEM	TOP,BOTTOM	=6x_DIELECTRIC	?
S2MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_2OTHER
S2_MEM_DQS*	*	*	S2MEM_2OTHER
S2_MEM_CMD	*	*	S2MEM_2OTHER
S2_MEM_CTRL	*	*	S2MEM_2OTHER
S2_MEM_CLK	*	*	S2MEM_2OTHER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS2OWNDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS2OWNDATA

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_P	35 36
	S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N	35 36
	S2_MEM_CNTL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE	35 36
I101	S2_MEM_CNTL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CS_L	35 36
I102		S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_ODT	36
I103	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L	35 36
I104	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L	35 36
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L	35 36
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>	35 36
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>	35 36
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>	35 36
I106	S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>	35 36
	S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>	35 36
I108	S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>	35 36
I107	S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>	35 36
I109	S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>	35 36
I110	S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>	35 36
I147	S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>	35 36
	S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7..0>	35 36
	S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15..8>	35 36
I127	MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P	35 36
I128	MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N	35 36
I129		MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P	36 83
I130		MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N	36 83
I134	MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P	35 36
I133	MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N	35 36
I132		MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P	36 83
I131		MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N	36 83
I145			S2_MEM_PWR	PP1V35_CAM	35 36
I146			S2_MEM_PWR	PP0V675_CAM_VREF	35 36
I148			S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA	36
I149			S2_MEM_PWR	PP0V675_MEM_CAM_VREFDQ	36

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SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	Pages
SMBUS_SMC_2_S3_SCL	SMB_45S	SMB	SMBUS_SMC_2_S3_SCL	40 43
SMBUS_SMC_2_S3_SDA	SMB_45S	SMB	SMBUS_SMC_2_S3_SDA	40 43
SMBUS_SMC_1_S0_SCL	SMB_45S	SMB	SMBUS_SMC_1_S0_SCL	40 43
SMBUS_SMC_1_S0_SDA	SMB_45S	SMB	SMBUS_SMC_1_S0_SDA	40 43
SMBUS_SMC_0_S0_SCL	SMB_45S	SMB	SMBUS_SMC_0_S0_SCL	40 43
SMBUS_SMC_0_S0_SDA	SMB_45S	SMB	SMBUS_SMC_0_S0_SDA	40 43
SMBUS_SMC_5_SCL	SMB_45S	SMB	SMBUS_SMC_5_g3_SCL	40 43 83
SMBUS_SMC_5_SDA	SMB_45S	SMB	SMBUS_SMC_5_G3_SDA	40 43 83
SMBUS_SMC_3_SCL	SMB_45S	SMB	SMBUS_SMC_3_SCL	40 42
SMBUS_SMC_3_SDA	SMB_45S	SMB	SMBUS_SMC_3_SDA	40 42

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	Pages
CHGR_CSI	1TO1_DIFFPAIR		CHGR_CSI_P	56
	1TO1_DIFFPAIR		CHGR_CSI_N	56
CHGR_CSO	1TO1_DIFFPAIR		CHGR_CSO_P	56
	1TO1_DIFFPAIR		CHGR_CSO_N	56

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1TO1_50S	*	=1:1_DIFFPAIR	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1TO1_50S	*	=1:1_DIFFPAIR	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR
AUDIODIFF	*	=1:1_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
THERM_45S_CPUVRISNS1	*	=1:1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.2 MM	0.2 MM
THERM_1TO1_45S	*	=1:1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
SENSE_1TO1_45S	*	=1:1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2X_DIELECTRIC	?
THERM	*	=2X_DIELECTRIC	?
AUDIO	*	=2X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
GND	PCIE_*	*	GND_P2MM
GND	SATA_*	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SB_POWER	SATA_*	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_37S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D	TOP			0.1 MM	500 MIL		
CPU_27P4S	BOTTOM			0.23 MM	100 MIL		
USB3_85D	TOP			0.1 MM	500 MIL		
USB3_85D	ISL10			0.075 MM			0.090 MM
DP_85D	ISL9			0.075 MM			0.090 MM
PCIE_85D	ISL10			0.075 MM			0.090 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
1TO1_DIFFPAIR	*	1:1_DIFFPAIR

J15 Specific Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
I426	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	ISNS_CPUDDR_P	
I425	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	ISNS_CPUDDR_N	
I421		THERM_1TO1_45S	THERM	ISNS_CPU_DDR_R_P	46
I423		THERM_1TO1_45S	THERM	ISNS_CPU_DDR_R_N	46
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	CPUTHMSNS_D2_P	47
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	CPUTHMSNS_D2_N	47
I422	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	ISNS_LCD_PANEL_P	
I424	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	ISNS_LCD_PANEL_N	
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	DDR3THMSNS_D1_P	47
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	DDR3THMSNS_D1_N	47
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	GPUTHMSNS_D_P	47
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	GPUTHMSNS_D_N	47
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_1V35_MEM_P	45
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_1V35_MEM_N	45
I419		SENSE_1TO1_45S	SENSE	ISNS_1V35_MEM_R_P	45
I420		SENSE_1TO1_45S	SENSE	ISNS_1V35_MEM_R_N	45
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_AIRPORT_P	
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_AIRPORT_N	
		SENSE_1TO1_45S	SENSE	ISNS_AIRPORT_R_P	46
		SENSE_1TO1_45S	SENSE	ISNS_AIRPORT_R_N	46
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_LCDBKLT_N	46 62
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_LCDBKLT_P	46 62
I520	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_GPUFB_R_P	45
I522	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_GPUFB_R_N	45
I521	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GPUFB_CS_P	45 71
I523	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GPUFB_CS_N	45 71
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_OTHER5V_P	44
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_OTHER5V_N	44
I514	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_OTHER3V3_P	44
I515	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_OTHER3V3_N	44
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_COMPUTING_P	44
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_HS_COMPUTING_N	44
I524	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	P1V05_GPU_CS_P	46 71
I525	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	P1V05_GPU_CS_N	46 71
I526	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_HS_GPU_P	44
I527	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_HS_GPU_N	44
		SENSE_1TO1_45S	SENSE	CPUVR_ISNS_P	45
		SENSE_1TO1_45S	SENSE	CPUVR_ISNS_N	45
I529	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_PP1V0_S0GPU_R_P	46
I528	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_PP1V0_S0GPU_R_N	46
I510	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	P1V05_GPU_PEX_IOVDD_SNS_P	71 76
I511	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	P1V05_GPU_PEX_IOVDD_SNS_N	71 76
I429	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS1_P	45 58
I430	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS1_N	45 58
I433	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS2_P	45 58
I434	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS2_N	45 58
I435	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS3_P	45 58
I431	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS3_N	45 58
I432	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	CPUVR_ISUM_R_P	45
I436	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	CPUVR_ISUM_R_N	45
I530	SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	GFXIMVP_ISNS2_P	77
I531	SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	GFXIMVP_ISNS2_N	77
I532	SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	GFXIMVP_ISNS1_P	77
I533	SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	GFXIMVP_ISNS1_N	77
I534	SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	GPU_TDIODE_P	47 74
I535	SENSE_DIFFPAIR	THERM_1TO1_55S	THERM	GPU_TDIODE_N	47 74
I536	SENSE_DIFFPAIR	40_OHM_SE	THERM	GPUVCORE_SENSE_P	76 77
I537	SENSE_DIFFPAIR	40_OHM_SE	THERM	GPUVCORE_SENSE_N	76 77
I405	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	ISNS_TBT_N	45
I406	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	ISNS_TBT_P	45
I407		AUDIODIFF	AUDIO	ISNS_TBT_R_N	45
I408		AUDIODIFF	AUDIO	ISNS_TBT_R_P	45
I446	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	ISNS_SSD_P	45
I445		THERM_1TO1_45S	THERM	ISNS_SSD_N	45
I447	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	ISNS_SSD_R_P	45
I448		THERM_1TO1_45S	THERM	ISNS_SSD_R_N	45
I449	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	P1V05S0_CS_P	61
I450		THERM_1TO1_45S	THERM	P1V05S0_CS_N	61
I451	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	P1V05S0_SENSE_P	61
I452		THERM_1TO1_45S	THERM	P1V05S0_SENSE_N	61
I453	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	TBT_THERMD_P	47
I454		THERM_1TO1_45S	THERM	TBT_THERMD_N	47
I455	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	CHGR_CSI_R_P	56
I456		AUDIODIFF	AUDIO	CHGR_CSI_R_N	56
I457	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	CHGR_CSO_R_P	56
I458		AUDIODIFF	AUDIO	CHGR_CSO_R_N	56
I517	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_S2_P	
I516	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_S2_N	
I519	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_S2_R_P	46
I518	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_S2_R_N	46

J15 Specific Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
I361		AUDIODIFF	AUDIO	AUD_SPKRAMP_RSUBIN_P	52
I362		AUDIODIFF	AUDIO	AUD_SPKRAMP_RSUBIN_N	52
		AUDIODIFF	AUDIO	AUD_SPKRAMP_LSUBIN_P	52
		AUDIODIFF	AUDIO	AUD_SPKRAMP_LSUBIN_N	52
I459		AUDIODIFF	AUDIO	RSUBIN_P	52
I440		AUDIODIFF	AUDIO	RSUBIN_N	52
I441		AUDIODIFF	AUDIO	LSUBIN_P	52
I442		AUDIODIFF	AUDIO	LSUBIN_N	52
	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_P	50 52
	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_N	50 52
	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_P	50 52
	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_N	50 52
		AUDIODIFF	AUDIO	AUD_SPKRAMP_RIN_P	52
		AUDIODIFF	AUDIO	AUD_SPKRAMP_RIN_N	52
		AUDIODIFF	AUDIO	AUD_SPKRAMP_LIN_P	52
		AUDIODIFF	AUDIO	AUD_SPKRAMP_LIN_N	52
I461		AUDIODIFF	AUDIO	SPKRAMP_RIN_P	52
I462		AUDIODIFF	AUDIO	SPKRAMP_RIN_N	52
I460		AUDIODIFF	AUDIO	SPKRAMP_LIN_P	52
I463		AUDIODIFF	AUDIO	SPKRAMP_LIN_N	52
I358	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_SL_OUT_P	52 54 83
I357	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_SL_OUT_N	52 54 83
I360	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_SR_OUT_P	52 54 83
I359	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_SR_OUT_N	52 54 83
	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_L_OUT_P	52 54 83
	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_L_OUT_N	52 54 83
	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_R_OUT_P	52 54 83
	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_R_OUT_N	52 54 83
I480	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	AUD_MIC_IN1_R_P	
I481	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	AUD_MIC_IN1_R_N	
I482	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	CODEC_HS_MIC_P	50
I483	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	CODEC_HS_MIC_N	50
I484	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	AUD_MIC_IN1_L_P	
I485		DIFFPAIR	AUDIO	AUD_MIC_IN1_L_N	
I486		DIFFPAIR	AUDIO	AUD_HS_MIC_P	53 54
I487		DIFFPAIR	AUDIO	AUD_HS_MIC_N	53 54
I489		DIFFPAIR	AUDIO	HS_MIC_P	50 53
I488		DIFFPAIR	AUDIO	HS_MIC_N	50 53
I490		DIFFPAIR	AUDIO	AUD_CONN_HS_MIC_P	
I491		DIFFPAIR	AUDIO	AUD_CONN_HS_MIC_N	
I494	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO3_R_P	50 52
I495	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO3_R_N	50 52
I493	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO3_L_P	50 52
I492	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO3_L_N	50 52
			SB_POWER	PP3V3_S5	81 83
			SB_POWER	PP3V3_S0	81 83
			SB_POWER	PP1V35_S3RS0_CPUDDR	81
			GND	GND	

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GDDR5 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR5_45R50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12.7 MM	=STANDARD	=STANDARD
GDDR5_45SE	*	=45_OHM_SE_ADJ	=45_OHM_SE_ADJ	=45_OHM_SE_ADJ	=45_OHM_SE_ADJ	=STANDARD	=STANDARD
GDDR5_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	*	=5x_DIELECTRIC	?
GDDR5_CMD	*	=3x_DIELECTRIC	?
GDDR5_DATA	*	=3x_DIELECTRIC	?
GDDR5_EDC	*	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR5_CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
GDDR5_DATA	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR5_EDC	TOP,BOTTOM	=5x_DIELECTRIC	?

D C B A

GDDR5 FB A Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I453	FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB_A0_CLK_P	70 72
I454	FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB_A0_CLK_N	70 72
I452	FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB_A1_CLK_P	70 72
I451	FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB_A1_CLK_N	70 72
I450	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_A<8..0>	70 72
I448	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_A<8..0>	70 72
I449	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_ABI_L	70 72
I447	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_ABI_L	70 72
I446	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_RAS_L	70 72
I445	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_RAS_L	70 72
I444	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_CAS_L	70 72
I442	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_CAS_L	70 72
I443	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_WE_L	70 72
I440	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_WE_L	70 72
I441	FB_A0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_A0_CKE_L	70 72
I439	FB_A1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_A1_CKE_L	70 72
I437	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_CS_L	70 72
I438	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_CS_L	70 72
I435	FB_A0_EDC0	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<0>	70 72
I436	FB_A0_EDC1	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<1>	70 72
I402	FB_A0_EDC2	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<2>	70 72
I400	FB_A0_EDC3	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<3>	70 72
I401	FB_A1_EDC0	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<0>	70 72
I398	FB_A1_EDC1	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<1>	70 72
I399	FB_A1_EDC2	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<2>	70 72
I397	FB_A1_EDC3	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<3>	70 72
I395	FB_A0_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<0>	70 72
I396	FB_A0_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<1>	70 72
I394	FB_A0_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<2>	70 72
I392	FB_A0_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<3>	70 72
I393	FB_A1_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<0>	70 72
I390	FB_A1_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<1>	70 72
I391	FB_A1_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<2>	70 72
I389	FB_A1_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<3>	70 72
I388	FB_A0_WCLK0	GDDR5_80D	GDDR5_CMD	FB_A0_WCLK_P<0>	70 72
I387	FB_A0_WCLK0	GDDR5_80D	GDDR5_CMD	FB_A0_WCLK_N<0>	70 72
I385	FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB_A0_WCLK_P<1>	70 72
I386	FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB_A0_WCLK_N<1>	70 72
I384	FB_A1_WCLK0	GDDR5_80D	GDDR5_CMD	FB_A1_WCLK_P<0>	70 72
I383	FB_A1_WCLK0	GDDR5_80D	GDDR5_CMD	FB_A1_WCLK_N<0>	70 72
I382	FB_A1_WCLK1	GDDR5_80D	GDDR5_CMD	FB_A1_WCLK_P<1>	70 72
I380	FB_A1_WCLK1	GDDR5_80D	GDDR5_CMD	FB_A1_WCLK_N<1>	70 72
I381	FB_A0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<7..0>	70 72
I379	FB_A0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<15..8>	70 72
I378	FB_A0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<23..16>	70 72
I377	FB_A0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<31..24>	70 72
I376	FB_A1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<7..0>	70 72
I375	FB_A1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<15..8>	70 72
I374	FB_A1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<23..16>	70 72
I373	FB_A1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<31..24>	70 72
I372	FB_A0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_A0_RESET_L	70 72
I371	FB_A1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_A1_RESET_L	70 72

GDDR5 FB B Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I473	FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB_B0_CLK_P	70 73
I474	FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB_B0_CLK_N	70 73
I472	FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB_B1_CLK_P	70 73
I470	FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB_B1_CLK_N	70 73
I471	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_A<8..0>	70 73
I468	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_A<8..0>	70 73
I469	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_ABI_L	70 73
I467	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_ABI_L	70 73
I466	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_RAS_L	70 73
I465	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_RAS_L	70 73
I464	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_CAS_L	70 73
I463	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_CAS_L	70 73
I462	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_WE_L	70 73
I460	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_WE_L	70 73
I461	FB_B0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_B0_CKE_L	70 73
I459	FB_B1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_B1_CKE_L	70 73
I457	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_CS_L	70 73
I458	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_CS_L	70 73
I455	FB_B0_EDC0	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<0>	70 73
I456	FB_B0_EDC1	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<1>	70 73
I434	FB_B0_EDC2	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<2>	70 73
I433	FB_B0_EDC3	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<3>	70 73
I432	FB_B1_EDC0	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<0>	70 73
I430	FB_B1_EDC1	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<1>	70 73
I431	FB_B1_EDC2	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<2>	70 73
I429	FB_B1_EDC3	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<3>	70 73
I428	FB_B0_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<0>	70 73
I427	FB_B0_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<1>	70 73
I426	FB_B0_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<2>	70 73
I424	FB_B0_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<3>	70 73
I425	FB_B1_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<0>	70 73
I423	FB_B1_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<1>	70 73
I422	FB_B1_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<2>	70 73
I421	FB_B1_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<3>	70 73
I420	FB_B0_WCLK0	GDDR5_80D	GDDR5_CMD	FB_B0_WCLK_P<0>	70 73
I419	FB_B0_WCLK0	GDDR5_80D	GDDR5_CMD	FB_B0_WCLK_N<0>	70 73
I418	FB_B0_WCLK1	GDDR5_80D	GDDR5_CMD	FB_B0_WCLK_P<1>	70 73
I417	FB_B0_WCLK1	GDDR5_80D	GDDR5_CMD	FB_B0_WCLK_N<1>	70 73
I416	FB_B1_WCLK0	GDDR5_80D	GDDR5_CMD	FB_B1_WCLK_P<0>	70 73
I414	FB_B1_WCLK0	GDDR5_80D	GDDR5_CMD	FB_B1_WCLK_N<0>	70 73
I415	FB_B1_WCLK1	GDDR5_80D	GDDR5_CMD	FB_B1_WCLK_P<1>	70 73
I412	FB_B1_WCLK1	GDDR5_80D	GDDR5_CMD	FB_B1_WCLK_N<1>	70 73
I413	FB_B0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<7..0>	70 73
I411	FB_B0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<15..8>	70 73
I410	FB_B0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<23..16>	70 73
I409	FB_B0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<31..24>	70 73
I408	FB_B1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<7..0>	70 73
I407	FB_B1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<15..8>	70 73
I406	FB_B1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<23..16>	70 73
I404	FB_B1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<31..24>	70 73
I405	FB_B0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_B0_RESET_L	70 73
I403	FB_B1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_B1_RESET_L	70 73

MUXGFX & DP AUX MUX NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	DP_85D	DISPLAYPORT	DP_INT_ML_C_P<3..0>	67 79 86
	DP_85D	DISPLAYPORT	DP_INT_ML_C_N<3..0>	67 79 86
	DP_85D	DISPLAYPORT	DP_INT_ML_F_P<3..0>	67 86
	DP_85D	DISPLAYPORT	DP_INT_ML_F_N<3..0>	67 86
DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_P<3..0>	67 83 86
DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_N<3..0>	67 83 86
	DP_85D	DISPLAYPORT	DP_INT_AUXCH_C_P	67 79 86
	DP_85D	DISPLAYPORT	DP_INT_AUXCH_C_N	67 79 86
DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_P	67 83 86
DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_N	67 83 86
DP_INT_IG_AUX	DP_85D	DISPLAYPORT	DP_INT_IG_AUX_P	79 82
	DP_85D	DISPLAYPORT	DP_INT_IG_AUX_N	79 82
DP_INT_IG_ML	DP_85D	DISPLAYPORT	DP_INT_IG_ML_P<3..0>	79 82
	DP_85D	DISPLAYPORT	DP_INT_IG_ML_N<3..0>	79 82
DP_INT_EG_AUX	DP_85D	DISPLAYPORT	DP_INT_EG_AUX_P	74 79
	DP_85D	DISPLAYPORT	DP_INT_EG_AUX_N	74 79
DP_INT_EG_ML	DP_85D	DISPLAYPORT	DP_INT_EG_ML_P<3..0>	74 79
	DP_85D	DISPLAYPORT	DP_INT_EG_ML_N<3..0>	74 79
DP_INT_EG_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_EG_AUXCH_P	74 80
DP_INT_EG_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_EG_AUXCH_N	74 80
DP_INT_EG_AUX1	DP_85D	DISPLAYPORT	DP_TBTSNK1_EG_AUXCH_P	74 80
DP_INT_EG_AUX1	DP_85D	DISPLAYPORT	DP_TBTSNK1_EG_AUXCH_N	74 80
TBTSNK0_AUXCH	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_P	28 86
TBTSNK0_AUXCH	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_N	28 86
	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_P	28 80 86
	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_N	28 80 86
TBTSNK1_AUXCH1	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_P	28 86
TBTSNK1_AUXCH1	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_N	28 86
	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_P	28 80 86
	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_N	28 80 86
DP_TBT_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	28 86
DP_TBT_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	28 86
	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	28 74 86
	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	28 74 86
DP_TBT_ML1	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	28 86
DP_TBT_ML1	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	28 86
	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	28 74 86
	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	28 74 86

Kepler Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
GPU_CLK27M	CLK_SLOW_50S	CLK_SLOW	GPU_OSC_27M_XTALIN	74 75
GPU_CLK27M	CLK_SLOW_50S	CLK_SLOW	GPU_OSC_27M_XTALOUT	74 75
GPU_CLK27M	CLK_SLOW_50S	CLK_SLOW	GPU_OSC_27M_XTAL_BUFFOUT_R	74
GPU_CLK27M	CLK_SLOW_50S	CLK_SLOW	GPU_OSC_27M_SSIN	74
	1TO1_DIFFPAIR		PEX_TSTCLK_O_P	68
	1TO1_DIFFPAIR		PEX_TSTCLK_O_N	68
HDMI_DATA	DP_85D	DISPLAYPORT	HDMI_EG_DATA_C_P<2..0>	74 78
HDMI_DATA	DP_85D	DISPLAYPORT	HDMI_EG_DATA_C_N<2..0>	74 78
HDMI_CLK	DP_85D	HDMI_CLK	HDMI_EG_CLK_C_P	74 78
HDMI_CLK	DP_85D	HDMI_CLK	HDMI_EG_CLK_C_N	74 78

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PAGE TITLE

GPU (Kepler) Constraints

Apple Inc.

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